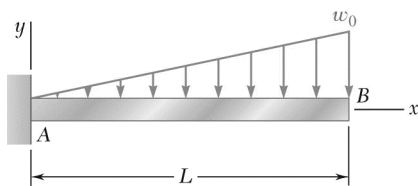


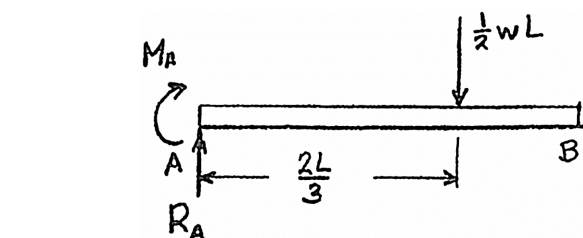
CHAPTER 9



PROBLEM 9.1

For the loading shown, determine (a) the equation of the elastic curve for the cantilever beam AB , (b) the deflection at the free end, (c) the slope at the free end.

SOLUTION



$$+\uparrow \Sigma F_y = 0:$$

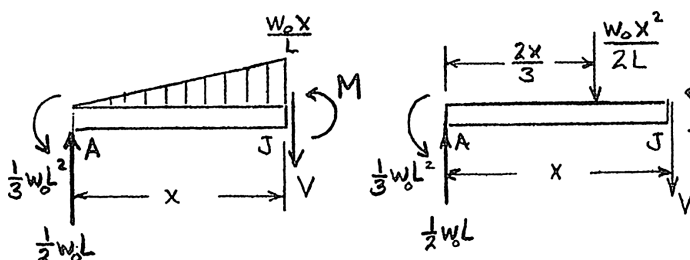
$$R_A - \frac{1}{2}wL = 0$$

$$R_A = \frac{1}{2}w_0L$$

$$+\circlearrowleft \Sigma M_A = 0:$$

$$-M_A + \frac{2L}{3} \cdot \frac{wL}{2} = 0$$

$$M_A = -\frac{1}{3}w_0L^2$$



$$+\circlearrowleft \Sigma M_J = 0: \quad \frac{1}{3}w_0L^2 - \frac{1}{2}w_0Lx + \frac{w_0x^2}{2L} \cdot \frac{x}{3} + M = 0$$

$$M = -\frac{1}{3}w_0L^2 + \frac{1}{2}w_0Lx - \frac{w_0x^3}{6L}$$

$$EI \frac{d^2y}{dx^2} = -\frac{1}{3}w_0L^2 + \frac{1}{2}w_0Lx - \frac{w_0x^3}{6L}$$

$$EI \frac{dy}{dx} = -\frac{1}{3}w_0L^2x + \frac{1}{4}w_0Lx^2 - \frac{w_0x^4}{24L} + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: \quad 0 = -0 + 0 - 0 + C_1 \quad C_1 = 0$$

$$EIy = -\frac{1}{6}w_0L^2x^2 + \frac{1}{12}w_0Lx^3 - \frac{w_0x^5}{120L} + C_2$$

$$[x = 0, y = 0]: \quad 0 = -0 + 0 - 0 + 0 + C_2 \quad C_2 = 0$$

(a) Elastic curve:

$$y = -\frac{w_0}{EIL} \left(\frac{1}{6}L^3x^2 - \frac{1}{12}Lx^4 + \frac{1}{120}x^5 \right) \quad \blacktriangleleft$$

PROBLEM 9.1 (Continued)

(b) y at $x = L$

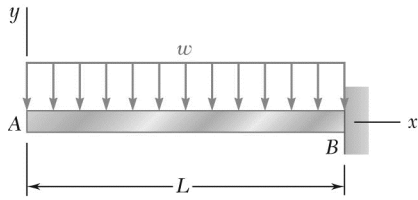
$$y_B = -\frac{w_0 L^4}{EI} \left(\frac{1}{6} - \frac{1}{12} + \frac{1}{120} \right) = -\frac{11}{120} \frac{w_0 L^4}{EI}$$

$$y_B = \frac{11}{120} \frac{w_0 L^4}{EI} \downarrow \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at $x = L$

$$\left. \frac{dy}{dx} \right|_B = -\frac{w_0 L^3}{EI} \left(\frac{1}{3} - \frac{1}{4} + \frac{1}{24} \right) = -\frac{1}{8} \frac{w_0 L^3}{EI}$$

$$\theta_B = \frac{1}{8} \frac{w_0 L^3}{EI} \nearrow \blacktriangleleft$$



PROBLEM 9.2

For the loading shown, determine (a) the equation of the elastic curve for the cantilever beam AB , (b) the deflection at the free end, (c) the slope at the free end.

SOLUTION

$$+\circlearrowleft \Sigma M_J = 0: (wx) \frac{x}{2} + M = 0$$

$$M = -\frac{1}{2}wx^2$$

$$EI \frac{d^2 y}{dx^2} = M = -\frac{1}{2}wx^2$$

$$EI \frac{dy}{dx} = -\frac{1}{6}wx^3 + C_1$$

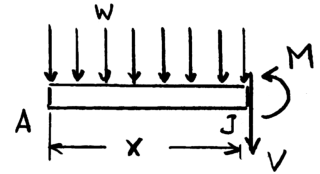
$$\left[x = L, \frac{dy}{dx} = 0 \right]: 0 = -\frac{1}{6}wL^3 + C_1 \quad C_1 = \frac{1}{6}wL^3$$

$$EI \frac{dy}{dx} = -\frac{1}{6}wx^3 + \frac{1}{6}wL^3$$

$$EI y = -\frac{1}{24}wx^4 + \frac{1}{6}wL^3 x + C_2$$

$$[x = L, y = 0] \quad 0 = -\frac{1}{24}wL^4 + \frac{1}{6}wL^4 + C_2 = 0$$

$$C_2 = \left(\frac{1}{24} - \frac{1}{6} \right) wL^4 = -\frac{3}{24}wL^4$$



(a) Elastic curve.

$$y = -\frac{w}{24EI}(x^4 - 4L^3x + 3L^4) \quad \blacktriangleleft$$

(b) y at $x = 0$.

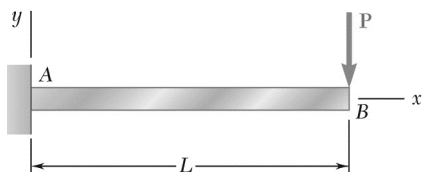
$$y_A = -\frac{3wL^4}{24EI} = -\frac{wL^4}{8EI}$$

$$y_A = \frac{wL^4}{8EI} \downarrow \quad \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at $x = 0$.

$$\left. \frac{dy}{dx} \right|_A = \frac{wL^3}{6EI}$$

$$\theta_A = \frac{wL^3}{6EI} \nearrow \quad \blacktriangleleft$$



PROBLEM 9.3

For the loading shown, determine (a) the equation of the elastic curve for the cantilever beam AB , (b) the deflection at the free end, (c) the slope at the free end.

SOLUTION

$$+\circlearrowleft \Sigma M_J = 0: -M - P(L - x) = 0$$

$$M = -P(L - x)$$

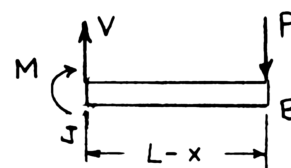
$$EI \frac{d^2 y}{dx^2} = -P(L - x) = -PL + Px$$

$$EI \frac{dy}{dx} = -PLx + \frac{1}{2}Px^2 + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: 0 = -0 + 0 + C_1 \quad C_1 = 0$$

$$EIy = -\frac{1}{2}PLx^2 + \frac{1}{6}Px^3 + C_1x + C_2$$

$$[x = 0, y = 0]: 0 = -0 + 0 + 0 + C_2 \quad C_2 = 0$$



(a) Elastic curve.

$$y = -\frac{Px^2}{6EI}(3L - x) \quad \blacktriangleleft$$

$$\frac{dy}{dx} = -\frac{Px}{2EI}(2L - x)$$

(b) y at $x = L$.

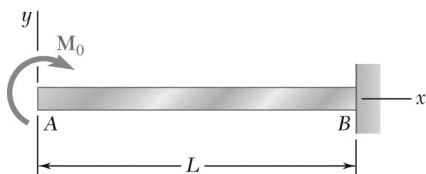
$$y_B = -\frac{PL^2}{6EI}(3L - L) = -\frac{PL^3}{3EI}$$

$$y_B = \frac{PL^3}{3EI} \downarrow \quad \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at $x = L$.

$$\left. \frac{dy}{dx} \right|_B = -\frac{PL}{2EI}(2L - L) = -\frac{PL^2}{2EI}$$

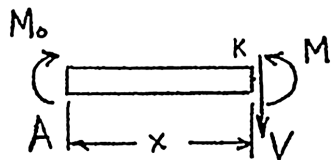
$$\theta_B = \frac{PL^2}{2EI} \quad \nwarrow \quad \blacktriangleleft$$



PROBLEM 9.4

For the loading shown, determine (a) the equation of the elastic curve for the cantilever beam AB , (b) the deflection at the free end, (c) the slope at the free end.

SOLUTION



$$+\circlearrowleft \sum M_K = 0: -M_0 + M = 0$$

$$M = M_0$$

$$EI \frac{d^2 y}{dx^2} = M = M_0$$

$$EI \frac{dy}{dx} = M_0 x + C_1$$

$$\left[x = L, \frac{dy}{dx} = 0 \right]: 0 = M_0 L + C_1 \quad C_1 = -M_0 L$$

$$EI y = \frac{1}{2} M_0 x^2 + C_1 x + C_2$$

$$[x = L, y = 0] \quad 0 = \frac{1}{2} M_0 L^2 - M_0 L^2 + C_2 \quad C_2 = \frac{1}{2} M_0 L^2$$

(a) Elastic curve:

$$y = \frac{M_0}{2EI} (x^2 - 2Lx + L^2) \quad \blacktriangleleft$$

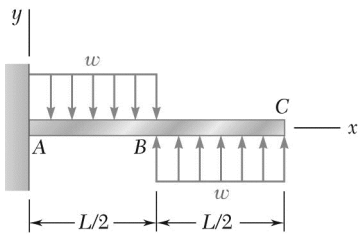
$$y = \frac{M_0}{2EI} (L - x)^2 \quad \blacktriangleleft$$

(b) y at $x = 0$: $y_A = \frac{M_0}{2EI} (L - 0)^2$

$$y_A = \frac{M_0 L^2}{2EI} \uparrow \quad \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at $x = 0$: $\frac{dy}{dx} = -\frac{M_0}{EI} (L - x) = -\frac{M_0}{EI} (L - 0) = -\frac{M_0 L}{EI}$

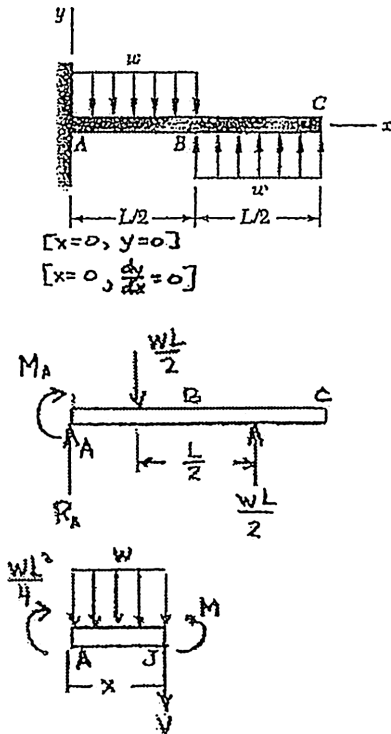
$$\theta_A = \frac{w_0 L}{EI} \searrow \quad \blacktriangleleft$$



PROBLEM 9.5

For the cantilever beam and loading shown, determine (a) the equation of the elastic curve for portion AB of the beam, (b) the deflection at B, (c) the slope at B.

SOLUTION



Using ABC as a free body,

$$+\uparrow \Sigma F_y = 0: R_A - \frac{wL}{2} + \frac{wL}{2} = 0 \quad R_A = 0$$

$$+\curvearrowright M_A = 0: -M_A + \left(\frac{wL}{2}\right)\left(\frac{L}{2}\right) = 0 \quad M_A = \frac{wL^2}{4}$$

Using AJ as a free body (J between A and B),

$$+\curvearrowright \Sigma M_J = 0: -\frac{wL^2}{4} + (wx)\frac{x}{2} + M = 0$$

$$M = \frac{1}{4}wL^2 - \frac{1}{2}wx^2$$

$$EI \frac{d^2 y}{dx^2} = \frac{1}{4}wL^2 - \frac{1}{2}wx^2$$

$$EI \frac{dy}{dx} = \frac{1}{4}wL^2 x - \frac{1}{6}wx^3 + C_1$$

$$\left[x=0, \frac{dy}{dx}=0 \right]: 0 = 0 - 0 + C_1 \quad C_1 = 0$$

$$EI y = \frac{1}{8}wL^2 x^2 - \frac{1}{24}wx^4 + C_1 x + C_2$$

$$[x=0, y=0]: 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$y = \frac{w}{EI} \left(\frac{1}{8}L^2 x^2 - \frac{1}{24}x^4 \right) \quad \blacktriangleleft$$

(a) Elastic curve.

$$\frac{dy}{dx} = \frac{w}{EI} \left(\frac{1}{4}L^2 x - \frac{1}{6}x^3 \right)$$

(b) y at $x = \frac{L}{2}$.

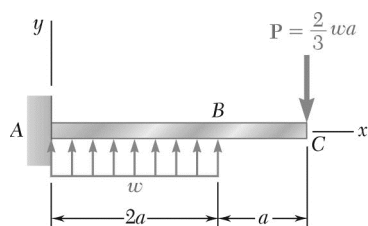
$$y_B = \frac{w}{EI} \left\{ \frac{1}{8}L^2 \left(\frac{L}{2} \right)^2 - \frac{1}{24} \left(\frac{L}{2} \right)^4 \right\} = \frac{wL^4}{EI} \left\{ \frac{1}{32} - \frac{1}{384} \right\}$$

$$= \frac{11wL^4}{384EI}$$

$$y_B = \frac{11wL^4}{384EI} \quad \uparrow \quad \blacktriangleleft$$

PROBLEM 9.5 (Continued)

$$\begin{aligned} (c) \quad \frac{dy}{dx} \text{ at } x = \frac{L}{2} \quad \theta_B &= \frac{w}{EI} \left\{ \frac{1}{4} L^2 \left(\frac{L}{2} \right) - \frac{1}{6} \left(\frac{L}{2} \right)^3 \right\} = \frac{wL^3}{EI} \left\{ \frac{1}{8} - \frac{1}{48} \right\} \\ &= \frac{5wL^3}{48EI} \quad \theta_B = \frac{5wL^3}{48EI} \curvearrowleft \blacktriangleleft \end{aligned}$$



PROBLEM 9.6

For the cantilever beam and loading shown, determine (a) the equation of the elastic curve for portion AB of the beam, (b) the deflection at B, (c) the slope at B.

SOLUTION

Using ABC as a free body,

$$+\uparrow \Sigma F_y = 0: R_A + 2wa - \frac{2}{3}wa = 0$$

$$R_A = -\frac{4}{3}wa = \frac{4}{3}wa \downarrow$$

$$+\circlearrowleft \Sigma M_A = 0: -M_A + (2wa)(a) - \left(\frac{2}{3}wa\right)(3a) = 0$$

$$M_A = 0$$

Using AJ as a free body,

$$+\circlearrowleft \Sigma M_J = 0: M + \left(\frac{4}{3}wa\right)(x) - (wx)\left(\frac{x}{2}\right) = 0$$

$$M = \frac{1}{2}wx^2 - \frac{4}{3}wax$$

$$EI \frac{d^2y}{dx^2} = \frac{1}{2}wx^2 - \frac{4}{3}wax$$

$$EI \frac{dy}{dx} = \frac{1}{6}wx^3 - \frac{2}{3}wax^2 + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: 0 = 0 - 0 + C_1 \quad \therefore C_1 = 0$$

$$EIy = \frac{1}{24}wx^4 - \frac{2}{9}wax^3 + C_2$$

$$[x = 0, y = 0]: 0 = 0 - 0 + C_2 \quad \therefore C_2 = 0$$

(a) Elastic curve over AB.

$$y = \frac{w}{72EI}(3x^4 - 16ax^3) \quad \blacktriangleleft$$

(b) y at x = 2a.

$$\frac{dy}{dx} = \frac{w}{6EI}(x^3 - 4ax^2)$$

$$y_B = -\frac{10wa^4}{9EI}$$

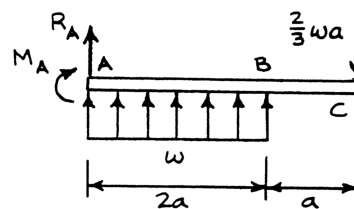
$$y_B = \frac{10wa^4}{9EI} \downarrow \quad \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at x = 2a.

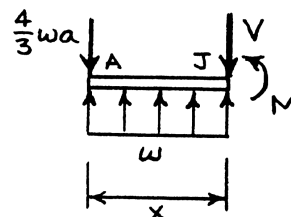
$$\left(\frac{dy}{dx}\right)_B = -\frac{4wa^3}{3EI}$$

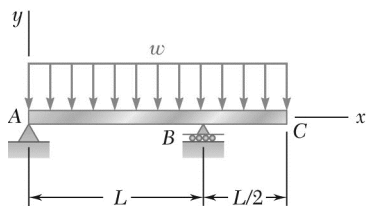
$$\theta_B = \frac{4wa^3}{3EI} \quad \blacktriangleleft$$

FBD ABC:



FBD AJ:

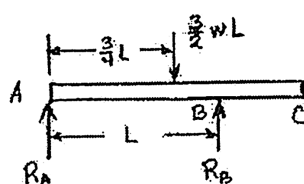
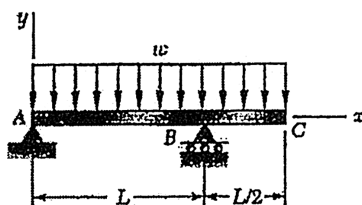




PROBLEM 9.7

For the beam and loading shown, determine (a) the equation of the elastic curve for portion AB of the beam, (b) the slope at A, (c) the slope at B.

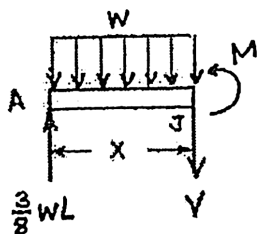
SOLUTION



$$+\circlearrowleft \Sigma M_B = 0:$$

$$-R_A L + \left(\frac{3}{2}wL\right)\left(\frac{1}{4}L\right) = 0$$

$$R_A = \frac{3}{8}wL$$



For portion AB only, $(0 \leq x < L)$

$$+\circlearrowleft \Sigma M_J = 0: -\frac{3}{8}wLx + (wx)\frac{x}{2} + M = 0$$

$$M = \frac{3}{8}wLx - \frac{1}{2}wx^2$$

$$EI \frac{d^2 y}{dx^2} = \frac{3}{8}wLx - \frac{1}{2}wx^2$$

$$EI \frac{dy}{dx} = \frac{3}{16}wLx^2 - \frac{1}{6}wx^3 + C_1$$

$$EIy = \frac{1}{16}wLx^3 - \frac{1}{24}wx^4 + C_1x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, y = 0] \quad 0 = \frac{1}{16}wL^3 - \frac{1}{24}wL^4 + C_1L \quad C_1 = -\frac{1}{48}wL^3$$

(a) Elastic curve.

$$y = \frac{w}{EI} \left(\frac{1}{16}Lx^3 - \frac{1}{24}x^4 - \frac{1}{48}L^3x \right) \quad \blacktriangleleft$$

$$\frac{dy}{dx} = \frac{w}{EI} \left(\frac{3}{16}Lx^2 - \frac{1}{6}x^3 - \frac{1}{48}L^3 \right)$$

(b) $\frac{dy}{dx}$ at $x = 0$.

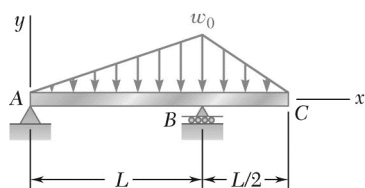
$$\left. \frac{dy}{dx} \right|_A = \frac{w}{EI} \left(0 - 0 - \frac{1}{48}L^3 \right) = -\frac{wL^3}{48EI}$$

$$\theta_A = \frac{wL^3}{48EI} \quad \nwarrow \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at $x = L$.

$$\left. \frac{dy}{dx} \right|_B = \frac{w}{EI} \left(\frac{3}{16}L^3 - \frac{1}{6}L^3 - \frac{1}{48}L^3 \right) = 0$$

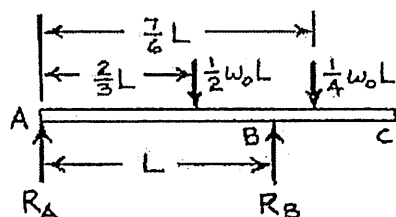
$$\theta_B = 0 \quad \blacktriangleleft$$



PROBLEM 9.8

For the beam and loading shown, determine (a) the equation of the elastic curve for portion AB of the beam, (b) the deflection at midspan, (c) the slope at B.

SOLUTION



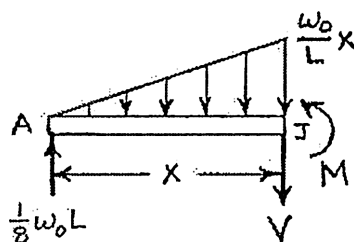
Reactions:

$$+\circlearrowleft \Sigma M_B = 0: -R_A L + \left(\frac{1}{2} w_0 L\right) \left(\frac{1}{3} L\right) - \left(\frac{1}{4} w_0 L\right) \left(\frac{1}{6} L\right) = 0$$

$$R_A = \frac{1}{8} w_0 L$$

Boundary conditions: $[x = 0, y = 0] \quad [x = L, y = 0]$

For portion AB only, $(0 \leq x < L)$



$$+\circlearrowleft \Sigma M_J = 0: -\frac{1}{8} w_0 L x + \frac{1}{2} \left(\frac{w_0}{L} x\right) (x) \left(\frac{x}{3}\right) + M = 0$$

$$M = \frac{1}{8} w_0 L x - \frac{1}{6} \frac{w_0}{L} x^3$$

$$EI \frac{d^2 y}{dx^2} = \frac{1}{8} w_0 L x - \frac{1}{6} \frac{w_0}{L} x^3$$

$$EI \frac{dy}{dx} = \frac{1}{16} w_0 L x^2 - \frac{1}{24} \frac{w_0}{L} x^4 + C_1$$

$$EI y = \frac{1}{48} w_0 L x^3 - \frac{1}{120} \frac{w_0}{L} x^5 + C_1 x + C_2$$

$$[x = 0, y = 0]: 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, y = 0]: 0 = \frac{1}{48} w_0 L^4 - \frac{1}{120} w_0 L^4 + C_1 L \quad C_1 = -\frac{1}{80} w_0 L^3$$

(a) Elastic curve.

$$y = \frac{w_0}{EIL} \left(\frac{1}{48} L^2 x^3 - \frac{1}{120} x^5 - \frac{1}{80} L^4 x \right) \blacktriangleleft$$

$$\frac{dy}{dx} = \frac{w_0}{EIL} \left(\frac{1}{16} L^2 x^2 - \frac{1}{24} x^4 - \frac{1}{80} L^4 \right)$$

(b) y at $x = \frac{L}{2}$.

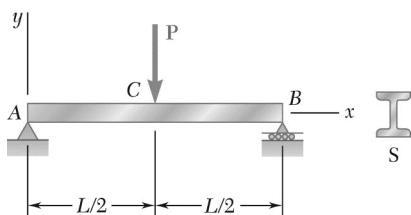
$$y_{L/2} = \frac{w_0}{EIL} \left(\frac{L^5}{384} - \frac{L^5}{3840} - \frac{L^5}{160} \right) = -\frac{15 w_0 L^4}{3840 EI}$$

$$y_{L/2} = \frac{w_0 L^4}{256 EI} \downarrow \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at $x = L$.

$$\left. \frac{dy}{dx} \right|_B = \frac{w_0}{EIL} \left(\frac{L^4}{16} - \frac{L^4}{24} - \frac{L^4}{80} \right) = +\frac{2 w_0 L^3}{240 EI}$$

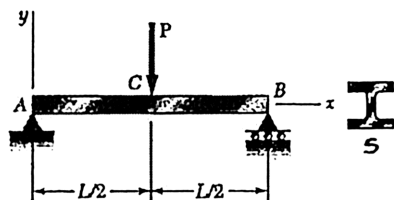
$$\theta_B = \frac{w_0 L^3}{120 EI} \nearrow \blacktriangleleft$$



PROBLEM 9.9

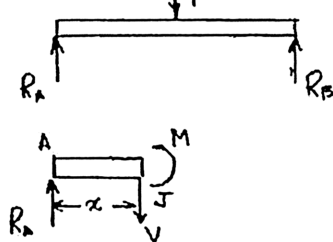
Knowing that beam AB is an S200 $\times$ 34 rolled shape and that $P = 60$ kN, $L = 2$ m, and $E = 200$ GPa, determine (a) the slope at A , (b) the deflection at C .

SOLUTION



$$[x = 0, y = 0] \quad [x = L, y = 0]$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right]$$



Use symmetry boundary conditions at C .

$$\text{By symmetry, } R_A = R_B = \frac{1}{2}P$$

$$\text{Using free body } AJ, \quad \left(0 \leq x \leq \frac{L}{2} \right)$$

$$+\circlearrowleft \sum M_J = 0 : M - R_A x = 0$$

$$M = R_A x = \frac{1}{2}Px$$

$$EI \frac{d^2 y}{dx^2} = \frac{1}{2}Px$$

$$EI \frac{dy}{dx} = \frac{1}{4}Px^2 + C_1$$

$$EI y = \frac{1}{12}Px^3 + C_1 x + C_2$$

$$[x = 0, y = 0]$$

$$0 = 0 + 0 + C_2$$

$$C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right]$$

$$0 = \frac{1}{4}P \left(\frac{L}{2} \right)^2 + C_1$$

$$C_1 = -\frac{1}{16}PL^2$$

Elastic curve.

$$y = \frac{PL}{48EI} (4x^3 - 3L^2 x)$$

$$\frac{dy}{dx} = \frac{PL}{16EI} (4x^2 - L^2)$$

Slope at $x = 0$.

$$\left. \frac{dy}{dx} \right|_A = -\frac{PL^2}{16EI}$$

$$\theta_A = \frac{PL^2}{16EI} \quad \swarrow$$

Deflection at $x = \frac{L}{2}$.

$$y_C = -\frac{PL^3}{48EI}$$

$$y_C = \frac{PL^3}{48EI} \quad \downarrow$$

PROBLEM 9.9 (Continued)

Data:

$$P = 60 \times 10^3 \text{ N},$$

$$I = 26.9 \times 10^6 \text{ mm}^4 = 26.9 \times 10^{-6} \text{ m}^4$$

$$E = 200 \times 10^9 \text{ Pa}$$

$$EI = 5.38 \times 10^6 \text{ N} \cdot \text{m}^2 \quad L = 2 \text{ m}$$

(a)

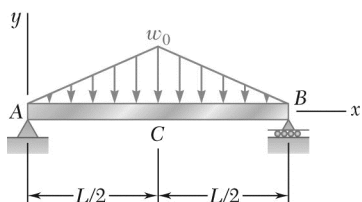
$$\theta_A = \frac{(60 \times 10^3)(2)^2}{(16)(5.38 \times 10^6)}$$

$$\theta_A = 2.79 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

(b)

$$y_C = \frac{(60 \times 10^3)(2)^3}{(48)(5.38 \times 10^6)} = 1.859 \times 10^{-3} \text{ m}$$

$$y_C = 1.859 \text{ mm} \quad \downarrow \blacktriangleleft$$

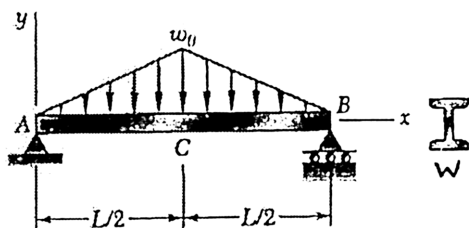


PROBLEM 9.10

Knowing that beam AB is a $W10 \times 33$ rolled shape and that $w_0 = 3$ kips/ft, $L = 12$ ft, and $E = 29 \times 10^6$ psi, determine (a) the slope at A , (b) the deflection at C .

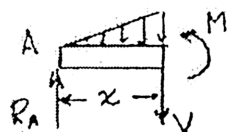
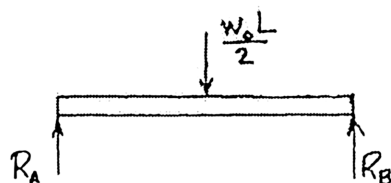


SOLUTION



$$[x = 0, y = 0] \quad [x = L, y = 0]$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right]$$



$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right]$$

$$[x = 0, y = 0]$$

Elastic curve.

Use symmetry boundary conditions at C .

Using free body ACB and symmetry,

$$R_A = R_B = \frac{1}{4} w_0 L$$

$$\text{For } 0 < x < \frac{L}{2}, \quad w = \frac{2w_0 x}{L}$$

$$\frac{dV}{dx} = -w = -\frac{2w_0 x}{L}$$

$$\frac{dM}{dx} = V = -\frac{w_0 x^2}{L} + R_A = \frac{w_0}{L} \left(\frac{1}{4} L^2 - x^2 \right)$$

$$M = \frac{w_0}{L} \left(\frac{1}{4} L^2 x - \frac{1}{3} x^3 \right) + C_M$$

But $M = 0$ at $x = 0$; hence $C_M = 0$

$$EI \frac{d^2 y}{dx^2} = \frac{w_0}{L} \left(\frac{1}{4} L^2 x - \frac{1}{3} x^3 \right)$$

$$EI \frac{dy}{dx} = \frac{w_0}{L} \left(\frac{1}{8} L^2 x^2 - \frac{1}{12} x^4 \right) + C_1$$

$$0 = \frac{w_0}{L} \left(\frac{1}{32} L^4 - \frac{1}{192} L^4 \right) + C_1 = 0 \quad C_1 = -\frac{5}{192} w_0 L^3$$

$$EI y = \frac{w_0}{L} \left(\frac{1}{24} L^2 x^3 - \frac{1}{120} x^5 \right) - \frac{5}{192} w_0 L^3 x + C_2$$

$$0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$y = \frac{w_0}{EIL} \left(\frac{1}{24} L^2 x^3 - \frac{1}{60} x^5 - \frac{5}{192} L^4 x \right)$$

$$\frac{dy}{dx} = \frac{w_0}{EIL} \left(\frac{1}{8} L^2 x^2 - \frac{1}{12} x^4 - \frac{5}{192} L^4 \right)$$

PROBLEM 9.10 (Continued)

Data:

$$w_0 = 3 \text{ kips/ft}, \quad E = 29 \times 10^6 \text{ psi}, \quad I = 171 \text{ in}^4$$

$$EI = (29 \times 10^6)(171) = 4.959 \times 10^9 \text{ lb} \cdot \text{in}^2 = 34.438 \times 10^3 \text{ kip} \cdot \text{ft},$$

$$L = 12 \text{ ft}$$

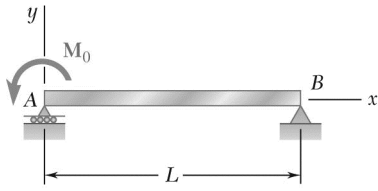
(a) Slope at $x = 0$. $\frac{dy}{dx} = \frac{3}{(34.438 \times 10^3)(12)} \left[-\left(\frac{5}{192} \right) (12)^4 \right] = -3.92 \times 10^3$

$$\theta_A = 3.92 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at $x = 6 \text{ ft}$.

$$y_C = \frac{3}{(34.438 \times 10^3)(12)} \left[\left(\frac{1}{24} \right) (12)^2 (6)^3 - \frac{1}{60} (6)^5 - \frac{5}{192} (12)^4 (6) \right] = -15.0531 \times 10^{-3} \text{ ft}$$

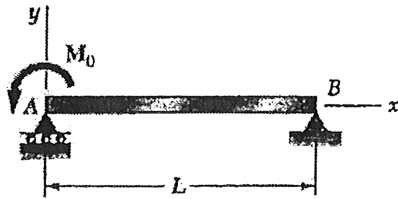
$$y_C = 0.1806 \text{ in.} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.11

(a) Determine the location and magnitude of the maximum deflection of beam AB . (b) Assuming that beam AB is a $W360 \times 64$, $L = 3.5$ m, and $E = 200$ GPa, calculate the maximum allowable value of the applied moment M_0 if the maximum deflection is not to exceed 1 mm.

SOLUTION



Using entire beam as a free body,

$$+\circlearrowleft \Sigma M_B = 0: M_0 - R_A L = 0 \quad R_A = \frac{M_0}{L}$$

Using portion AJ ,

$$[x = 0, y = 0] \quad [x = L, y = 0] \quad +\circlearrowleft \Sigma M_J = 0: M_0 - \frac{M_0}{L}x + M = 0$$

$$M = \frac{M_0}{L}(x - L)$$

$$EI \frac{d^2 y}{dx^2} = \frac{M_0}{L}(x - L)$$

$$EI \frac{dy}{dx} = \frac{M_0}{L} \left(\frac{1}{2}x^2 - Lx \right) + C_1$$

$$EI y = \frac{M_0}{L} \left(\frac{1}{6}x^3 - \frac{1}{2}Lx^2 \right) + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, y = 0] \quad 0 = \frac{M_0}{L} \left(\frac{1}{6}L^3 - \frac{1}{2}L^3 \right) + C_1 L + 0 \quad C_1 = \frac{1}{3}M_0 L$$

$$y = \frac{M_0}{EIL} \left(\frac{1}{6}x^3 - \frac{1}{2}Lx^2 + \frac{1}{3}L^2 x \right) \quad \frac{dy}{dx} = \frac{M_0}{EIL} \left(\frac{1}{2}x^2 - Lx + \frac{1}{3}L^2 \right)$$

(a) To find location maximum deflection, set $\frac{dy}{dx} = 0$.

$$\frac{1}{2}x_m^2 - Lx_m + \frac{1}{3}L^2 = 0 \quad x_m = L - \sqrt{L^2 - (4)\left(\frac{1}{2}\right)\left(\frac{1}{3}L^2\right)} = \left(1 - \sqrt{\frac{1}{3}}\right)L$$

$$= 0.42265L \quad x_m = 0.423L \quad \blacktriangleleft$$

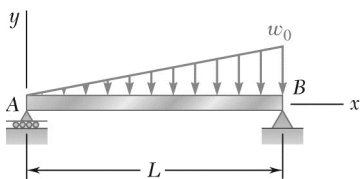
$$y_m = \frac{M_0 L^2}{EI} \left\{ \left(\frac{1}{6}\right)(0.42265)^3 - \left(\frac{1}{2}\right)(0.42265)^2 + \left(\frac{1}{3}\right)(0.42265) \right\} \quad y_m = 0.06415 \frac{M_0 L^2}{EI} \quad \blacktriangleleft$$

PROBLEM 9.11 (Continued)

Solving for M_0 ,
$$M_0 = \frac{EIy_m}{0.06415L^2}$$

(b) Data: $E = 200 \times 10^9 \text{ Pa}$, $I = 178 \times 10^6 \text{ mm}^4 = 178 \times 10^{-6} \text{ m}^4$
 $L = 3.5 \text{ m}$ $y_m = 1 \text{ mm} = 10^{-3} \text{ m}$
$$M_0 = \frac{(200 \times 10^9)(178 \times 10^{-6})(10^{-3})}{(0.06415)(3.5)^2} = 45.3 \times 10^3 \text{ N} \cdot \text{m}$$

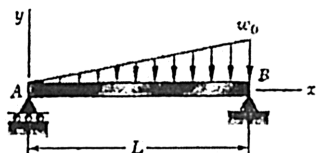
$$M_0 = 45.3 \text{ kN} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 9.12

For the beam and loading shown, (a) express the magnitude and location of the maximum deflection in terms of w_0 , L , E , and I . (b) Calculate the value of the maximum deflection, assuming that beam AB is a W18 $\times$ 50 rolled shape and that $w_0 = 4.5$ kips/ft, $L = 18$ ft, and $E = 29 \times 10^6$ psi.

SOLUTION



$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

Using entire beam as a free body,

$$+\circlearrowleft \Sigma M_B = 0: -R_A L + \left(\frac{1}{2} w_0 L \right) \left(\frac{L}{3} \right) = 0$$

$$R_A = \frac{1}{6} w_0 L$$

Using AJ as a free body,

$$+\circlearrowleft \Sigma M_J = 0: -\frac{1}{6} w_0 L x + \left(\frac{1}{2} \frac{w_0 x^2}{L} \right) \left(\frac{x}{3} \right) + M = 0$$

$$M = \frac{1}{6} w_0 L x - \frac{1}{6} \frac{w_0}{L} x^3$$

$$EI \frac{d^2 y}{dx^2} = \frac{1}{6} w_0 L x - \frac{1}{6} \frac{w_0}{L} x^3$$

$$EI \frac{dy}{dx} = \frac{1}{12} w_0 L x^2 - \frac{1}{24} \frac{w_0}{L} x^4 + C_1$$

$$EI y = \frac{1}{36} w_0 L x^3 - \frac{1}{120} \frac{w_0}{L} x^5 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, y = 0] \quad 0 = \frac{1}{36} w_0 L^4 - \frac{1}{120} w_0 L^4 + C_1 L + 0 \quad C_1 = -\frac{7w_0 L^3}{360}$$

Elastic curve.

$$y = \frac{w_0}{EI} \left\{ \frac{1}{36} L x^3 - \frac{1}{120} \frac{x^5}{L} - \frac{7}{360} L^3 x \right\}$$

$$\frac{dy}{dx} = \frac{w_0}{EI} \left\{ \frac{1}{12} L x^2 - \frac{1}{24} \frac{x^4}{L} - \frac{7}{360} L^3 \right\}$$

PROBLEM 9.12 (Continued)

(a) To find location of maximum deflection, set $\frac{dy}{dx} = 0$.

$$15x_m^4 - 30L^2x_m^2 + 7L^4 = 0 \quad x_m^2 = \frac{30L^2 - \sqrt{900L^4 - 420L^4}}{30}$$

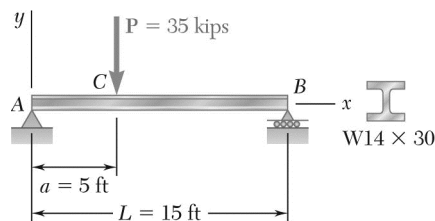
$$x_m^2 = \left(1 - \sqrt{\frac{8}{15}}\right)L^2 = 0.2697L^2 \quad x_m = 0.5193L \quad \blacktriangleleft$$

$$y_m = \frac{w_0}{EI} \left\{ \frac{1}{36} L(0.5193L)^3 - \frac{1}{120} \frac{(0.5193L^5)}{L} - \frac{7}{360} L^3(0.5193L) \right\}$$
$$= -0.00652 \frac{w_0 L^4}{EI} \quad y_m = 0.00652 \frac{w_0 L^4}{EI} \downarrow \quad \blacktriangleleft$$

(b) Data: $w_0 = 4.5 \text{ kips/ft} = \frac{4500}{12} = 375 \text{ lb/in.}$, $L = 18 \text{ ft} = 216 \text{ in.}$,

$I = 800 \text{ in}^4$ for W18 $\times$ 50.

$$y_m = \frac{(0.00652)(375)(216)^4}{(29 \times 10^6)(800)} \quad y_m = 0.229 \text{ in.} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.13

For the beam and loading shown, determine the deflection at point C. Use $E = 29 \times 10^6$ psi.

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

$$[x = a, y = y]$$

$$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right]$$

SOLUTION

Let $b = L - a$.

Reactions: $R_A = \frac{Pb}{L} \uparrow$,

$$R_B = \frac{Pa}{L} \uparrow$$

Bending moments:

$$0 < x < a: M = \frac{Pb}{L}x$$

$$a < x < L: M = \frac{P}{L}[bx - L(x - a)]$$

$$\begin{aligned} 0 < x < a \\ EI \frac{d^2 y}{dx^2} &= \frac{P}{L}(bx) \\ EI \frac{dy}{dx} &= \frac{P}{L} \left(\frac{1}{2}bx^2 \right) + C_1 \end{aligned} \quad (1)$$

$$EIy = \frac{P}{L} \left(\frac{1}{6}bx^3 \right) + C_1x + C_2 \quad (2)$$

$$\begin{aligned} a < x < L \\ EI \frac{d^2 y}{dx^2} &= \frac{P}{L}[bx - L(x - a)] \\ EI \frac{dy}{dx} &= \frac{P}{L} \left[\frac{1}{2}bx^2 - \frac{1}{2}L(x - a)^2 \right] + C_3 \end{aligned} \quad (3)$$

$$EIy = \frac{P}{L} \left[\frac{1}{6}bx^3 - \frac{1}{6}L(x - a)^3 \right] + C_3x + C_4 \quad (4)$$

$$[x = 0, y = 0] \quad \text{Eq. (2)} \quad 0 = 0 + 0 + C_2 \quad C_2 = 0$$

$$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right] \quad \text{Eqs. (1) and (3)} \quad \frac{P}{L} \left(\frac{1}{2}ba^2 \right) + C_1 = \frac{P}{L} \left[\frac{1}{2}ba^2 + 0 \right] + C_3 \quad \therefore C_3 = C_1$$

$$[x = a, y = y] \quad \text{Eqs. (2) and (4)} \quad \frac{P}{L} \left(\frac{1}{6}ba^3 \right) + C_1a + C_2$$

$$= \frac{P}{L} \left[\frac{1}{6}ba^3 + 0 \right] + C_1a + C_4 \quad C_4 = C_2 = 0$$

PROBLEM 9.13 (Continued)

$$[x = L, y = 0] \quad \text{Eq. (4):} \quad \frac{P}{L} \left[\frac{1}{6} bL^3 - \frac{1}{6} L(L-a)^3 \right] + C_3 L = 0$$

$$C_1 = C_3 = \frac{P}{L} \left[\frac{1}{6} (L-a)^3 - \frac{1}{6} bL^2 \right] = \frac{P}{L} \left(\frac{1}{6} b^3 - \frac{1}{6} bL^2 \right)$$

Make $x = a$ in Eq. (2).

$$y_C = \frac{P}{EIL} \left[\frac{1}{6} ba^3 + \frac{1}{6} b^3 a - \frac{1}{6} bL^2 a \right] = \frac{P(ba^3 + b^3 a - L^2 ab)}{6EIL}$$

Data: $P = 35$ kips, $E = 29 \times 10^6$ psi

$$= 29 \times 10^3 \text{ kips/in}^2$$

$$L = 15 \text{ ft}, \quad a = 5 \text{ ft}, \quad b = 10 \text{ ft}$$

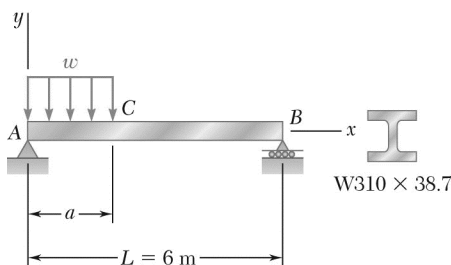
$$I = 291 \text{ in}^4, \quad EI = 8.439 \times 10^6 \text{ kip} \cdot \text{in}^2$$

$$= 58.604 \times 10^3 \text{ kip} \cdot \text{ft}^2$$

$$y_C = \frac{35}{(6)(58.604 \times 10^3)(15)} [(10)(5)^3 + (10^3)(5) - (15)^2(5)(10)]$$

$$= -33.179 \times 10^{-3} \text{ ft} = -0.398 \text{ in.}$$

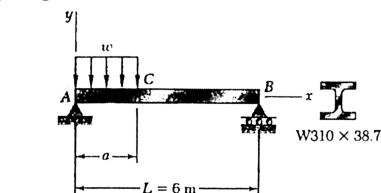
$$y_C = 0.398 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.14

For the beam and loading shown, knowing that $a = 2$ m, $w = 50$ kN/m, and $E = 200$ GPa, determine (a) the slope at support A, (b) the deflection at point C.

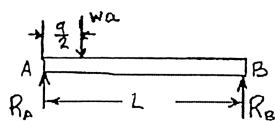
SOLUTION



$$[x = 0, y = 0]$$

$$[x = a, y = y]$$

$$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right]$$



Using ACB as a free body and noting that $L = 3a$,

$$+\circlearrowleft \sum M_A = 0: R_B L - (wa) \left(\frac{a}{2} \right) = 0$$

$$R_B = (wa) \frac{a}{2L} = \frac{1}{6} wa$$

$$+\uparrow \sum F_y = 0: R_A + R_B - wa = 0 \quad R_A = \frac{5}{6} wa$$

$0 \leq x \leq a$

$$+\circlearrowleft \sum M_J = 0: M - R_A x + (wx) \left(\frac{x}{2} \right) = 0$$

$$M = R_A x - \frac{1}{2} wx^2$$

$$EI \frac{d^2 y}{dx^2} = R_A x - \frac{1}{2} wx^2$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{6} wx^3 + C_1$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{24} wx^4 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{24} wx^4 + C_1 x$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{6} wx^3 + C_1$$

$a \leq x \leq L$

$$+\circlearrowleft \sum M_K = 0: -M + R_B(L - x) = 0$$

$$M = R_B(L - x)$$

$$EI \frac{d^2 y}{dx^2} = R_B(L - x)$$

$$EI \frac{dy}{dx} = -\frac{1}{2} R_B(L - x)^2 + C_3$$

$$EI y = \frac{1}{6} R_B(L - x)^3 + C_3 x + C_4$$

$$[x = L, y = 0] \quad 0 = 0 + C_3 L + C_4 \quad C_4 = -C_3 L$$

$$EI y = \frac{1}{6} R_B(L - x)^3 - C_3(L - x)$$

$$EI \frac{dy}{dx} = -\frac{1}{2} R_B(L - x)^2 + C_3$$

PROBLEM 9.14 (Continued)

$$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right] \quad \frac{1}{2} R_A a^2 - \frac{1}{6} w a^3 + C_1 = -\frac{1}{2} R_B (2a)^2 + C_3$$

$$C_3 = C_1 + \frac{1}{2} R_A a^2 - \frac{1}{6} w a^3 + \frac{1}{2} R_B (2a)^2 = C_1 + \frac{7}{12} w a^3$$

$$[x = a, y = y] \quad \frac{1}{6} R_A a^3 - \frac{1}{24} w a^4 + C_1 a = \frac{1}{6} R_B (2a)^3 - \left(C_1 + \frac{7}{12} w a^3 \right) (2a)$$

$$3C_1 a = -\frac{1}{6} R_A a^3 + \frac{1}{24} w a^4 + \frac{1}{6} R_B (2a)^3 - \frac{7}{12} w a^2 (2a) = -\frac{25}{24} w a^4$$

$$C_1 = -\frac{25}{72} w a^3$$

$$\text{For } 0 \leq x \leq a, \quad EIy = \frac{5}{36} w a x^3 - \frac{1}{24} w x^4 - \frac{25}{72} w a^3 x$$

$$EI \frac{dy}{dx} = \frac{5}{12} w a x^2 - \frac{1}{6} w x^3 - \frac{25}{72} w a^3$$

Data: $w = 50 \times 10^3 \text{ N/m}, \quad a = 2 \text{ m}, \quad E = 200 \times 10^9 \text{ Pa}$

$$I = 84.9 \times 10^6 \text{ mm}^4 = 84.9 \times 10^{-6} \text{ m}^4, \quad EI = 16.98 \times 10^6 \text{ N} \cdot \text{m}^2$$

(a) Slope at $x = 0$.

$$16.98 \times 10^6 \left. \frac{dy}{dx} \right|_A = 0 - 0 - \frac{25}{72} (50 \times 10^3) (2)^3$$

$$\left. \frac{dy}{dx} \right|_A = \theta_A = -8.18 \times 10^{-3}$$

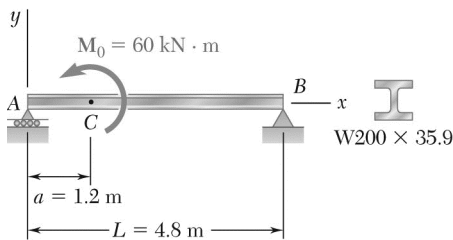
$$\theta_A = 8.18 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at $x = 2 \text{ m}$.

$$EI y_C = \frac{5}{36} w a^4 - \frac{1}{24} w a^4 - \frac{25}{72} w a^4 = -\frac{1}{4} w a^4$$

$$16.98 \times 10^6 y_C = -\frac{1}{4} (50 \times 10^3) (2)^4 \quad y_C = -11.78 \times 10^{-3} \text{ m}$$

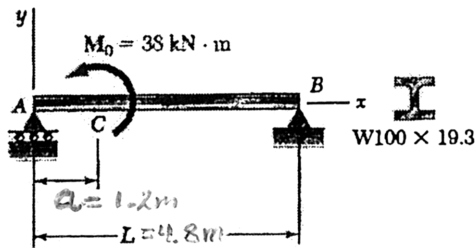
$$y_C = 11.78 \text{ mm} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.15

For the beam and loading shown, determine the deflection at point C. Use $E = 200 \text{ GPa}$.

SOLUTION



Reactions: $R_A = M_0/L \uparrow$, $R_B = M_0/L \downarrow$

$0 < x < a$: $\sum M_J = 0$:

$$-\frac{M_0}{L}x + M = 0$$

$$M = \frac{M_0}{L}x$$

$a < x < L$: $\sum M_K = 0$:

$$-\frac{M_0}{L}x + M_0 + M = 0$$

$$M = \frac{M_0}{L}(x - L)$$

$0 < x < a$

$$EI \frac{d^2 y}{dx^2} = \frac{M_0}{L} x$$

$$EI \frac{dy}{dx} = \frac{M_0}{L} \left(\frac{1}{2} x^2 \right) + C_1 \quad (1)$$

$$EI y = \frac{M_0}{L} \left(\frac{1}{6} x^3 \right) + C_1 x + C_2 \quad (2)$$

$a < x < L$

$$EI \frac{d^2 y}{dx^2} = \frac{M_0}{L} (x - L)$$

$$EI \frac{dy}{dx} = \frac{M_0}{L} \left(\frac{1}{2} x^2 - Lx \right) + C_3 \quad (3)$$

$$EI y = \frac{M_0}{L} \left(\frac{1}{6} x^3 - \frac{1}{2} Lx^2 \right) + C_3 x + C_4 \quad (4)$$

$[x = 0, y = 0]$ Eq. (2): $0 = 0 + 0 + C_2$ $C_2 = 0$

$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right]$ Eqs. (1) and (3): $\frac{M_0}{L} \left(\frac{1}{2} a^2 \right) + C_1 = \frac{M_0}{L} \left(\frac{1}{2} a^2 - La \right) + C_3$

$$C_3 = C_1 + M_0 a$$

PROBLEM 9.15 (Continued)

$$[x = a, y = y] \quad \text{Eqs. (2) and (4):} \quad \frac{M_0}{L} \left(\frac{1}{6} a^3 \right) + \cancel{C_1 a} = \frac{M_0}{L} \left(\frac{1}{6} a^3 - \frac{1}{2} L a^2 \right) + (\cancel{C_1} + M_0 a) a + C_4$$

$$C_4 = -\frac{1}{2} M_0 a^2$$

$$[x = L, y = 0] \quad \text{Eq. (4):} \quad \frac{M_0}{L} \left(\frac{1}{6} L^3 - \frac{1}{2} L^3 \right) + (C_1 + M_0 a) L - \frac{1}{2} M_0 a^2 = 0$$

$$C_1 = \frac{M_0}{L} \left(\frac{1}{3} L^2 + \frac{1}{2} a^2 - a L \right)$$

Elastic curve for $0 < x < a$.
$$y = \frac{M_0}{EIL} \left[\frac{1}{6} x^3 + \left(\frac{1}{3} L^2 + \frac{1}{2} a^2 - a L \right) x \right]$$

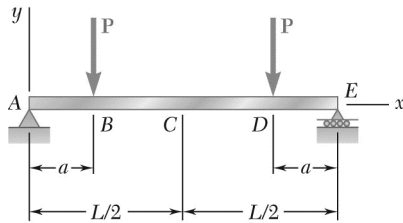
Make $x = a$.
$$y_C = \frac{M_0}{EIL} \left[\frac{1}{6} a^3 + \frac{1}{3} L^2 a + \frac{1}{2} a^3 - a^2 L \right] = \frac{M_0}{EIL} \left[\frac{2}{3} a^3 + \frac{1}{3} L^2 a - L a^2 \right]$$

Data: $E = 200 \times 10^9 \text{ Pa}, \quad I = 34.4 \times 10^6 \text{ mm}^4 = 34.4 \times 10^{-6} \text{ m}^4, \quad M_0 = 60 \times 10^3 \text{ N} \cdot \text{m}$

$a = 1.2 \text{ m}, \quad L = 4.8 \text{ m}$

$$y_C = \frac{(60 \times 10^3) \left[(2)(1.2)^3 / 3 + (4.8)^2 (1.2) / 3 - (4.8)(1.2)^2 \right]}{(200 \times 10^9)(34.4 \times 10^{-6})(4.8)} = 6.28 \times 10^{-3} \text{ m}$$

$y_C = 6.28 \text{ mm} \uparrow \blacktriangleleft$



PROBLEM 9.16

Knowing that beam AE is an $S200 \times 27.4$ rolled shape and that $P = 17.5$ kN, $L = 2.5$ m, $a = 0.8$ m and $E = 200$ GPa, determine (a) the equation of the elastic curve for portion BD , (b) the deflection at the center C of the beam.

SOLUTION

Consider portion ABC only. Apply symmetry about C .

Reactions: $R_A = R_E = P$

Boundary conditions: $[x = 0, y = 0]$, $[x = a, y = y]$, $\left[x = a, \frac{dy}{dx} = \frac{dy}{dx}\right]$, $\left[x = \frac{L}{2}, \frac{dy}{dx} = 0\right]$

$$\begin{aligned} 0 < x < a \\ EI \frac{d^2 y}{dx^2} &= M = Px \\ EI \frac{dy}{dx} &= \frac{1}{2} Px^2 + C_1 \quad (1) \\ EI y &= \frac{1}{6} Px^3 + C_1 x + C_2 \quad (2) \\ [x = 0, y = 0] &\rightarrow C_2 = 0 \end{aligned}$$

$$\begin{aligned} a < x < L - a \\ EI \frac{d^2 y}{dx^2} &= M = Pa \\ EI \frac{dy}{dx} &= Pax + C_3 \\ EI y &= \frac{1}{2} Pax^2 + C_3 x + C_4 \\ \left[x = \frac{L}{2}, \frac{dy}{dx} = 0\right] &\rightarrow C_3 = -\frac{1}{2} PaL \end{aligned}$$

$$\begin{aligned} \left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx}\right] \quad \frac{1}{2} Pa^2 + C_1 &= Pa^2 - \frac{1}{2} PaL \quad C_1 = \frac{1}{2} Pa^2 - \frac{1}{2} PaL \\ \left[x = \frac{L}{2}, y = y\right] \quad \frac{1}{6} Pa^3 + \left(\frac{1}{2} \cancel{Pa^2} - \frac{1}{2} \cancel{PaL}\right)a &= \frac{1}{2} \cancel{Pa^3} - \frac{1}{2} \cancel{Pa^2} L + C_4 \\ C_4 &= \frac{1}{6} Pa^3 \end{aligned}$$

(a) Elastic curve for portion BD .

$$y = \frac{1}{EI} \left(\frac{1}{2} Pax^2 + C_3 x + C_4 \right)$$

$$y = \frac{P}{EI} \left(\frac{1}{2} ax^2 - \frac{1}{2} aLx + \frac{1}{6} a^3 \right) \blacktriangleleft$$

PROBLEM 9.16 (Continued)

For deflection at C,

set

$$x = \frac{L}{2}.$$

$$\begin{aligned} y_C &= \frac{P}{EI} \left(\frac{1}{8} aL^2 - \frac{1}{4} aL^2 + \frac{1}{6} a^3 \right) \\ &= -\frac{Pa}{EI} \left(\frac{1}{8} L^2 - \frac{1}{6} a^2 \right) \end{aligned}$$

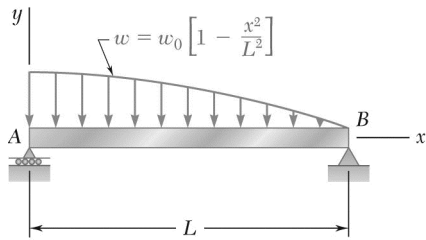
Data: $I = 23.9 \times 10^6 \text{ mm}^4 = 23.9 \times 10^{-6} \text{ m}^4,$

$$E = 200 \times 10^9 \text{ Pa}$$

$$P = 17.5 \times 10^3 \text{ N},$$

$$L = 2.5 \text{ m}, \quad a = 0.8 \text{ m}$$

$$(b) \quad y_C = -\frac{(17.5 \times 10^3)(0.8)}{(200 \times 10^9)(23.9 \times 10^6)} \left\{ \frac{2.5^2}{8} - \frac{0.8^2}{6} \right\} = -1.976 \times 10^{-3} \text{ m} \quad y_C = 1.976 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.17

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the slope at end A, (c) the deflection at the midpoint of the span.

SOLUTION

$$w = w_0 \left[1 - \frac{x^2}{L^2} \right] = \frac{w_0}{L^2} [L^2 - x^2]$$

$$\frac{dV}{dx} = -w = \frac{w_0}{L^2} [x^2 - L^2]$$

$$\frac{dM}{dx} = V = \frac{w_0}{L^2} \left[\frac{x^3}{3} - L^2 x \right] + C_1$$

$$M = \frac{w_0}{L^2} \left[\frac{1}{12} x^4 - \frac{1}{2} L^2 x^2 \right] + C_1 x + C_2$$

$$[x = 0, M = 0]: \quad 0 = 0 - 0 + 0 + C_2 \quad \therefore \quad C_2 = 0$$

$$[x = L, M = 0]: \quad 0 = \frac{w_0}{L^2} \left[\frac{1}{12} L^4 - \frac{1}{2} L^4 \right] + C_1 L \quad \therefore \quad C_1 = \frac{5}{12} w_0 L$$

$$EI \frac{d^2 y}{dx^2} = M = \frac{w_0}{L^2} \left[\frac{1}{12} x^4 - \frac{1}{2} L^2 x^2 + \frac{5}{12} L^3 x \right]$$

$$EI \frac{dy}{dx} = \frac{w_0}{L^2} \left[\frac{1}{60} x^5 - \frac{1}{6} L^2 x^3 + \frac{5}{24} L^3 x^2 \right] + C_3$$

$$EI y = \frac{w_0}{L^2} \left[\frac{1}{360} x^6 - \frac{1}{24} L^2 x^4 + \frac{5}{72} L^3 x^3 \right] + C_3 x + C_4$$

$$[x = 0, y = 0]: \quad 0 = 0 - 0 + 0 + 0 + C_4 \quad \therefore \quad C_4 = 0$$

$$[x = L, y = 0]: \quad 0 = \frac{w_0}{L^2} \left[\frac{1}{360} L^6 - \frac{1}{24} L^6 + \frac{5}{72} L^6 \right] + C_3 L \quad \therefore \quad C_3 = -\frac{11}{360} w_0 L^3$$

(a) Elastic curve. $y = w_0 (x^6 - 15L^2 x^4 + 25L^3 x^3 - 11L^5 x) / 360 EIL^2 \quad \blacktriangleleft$

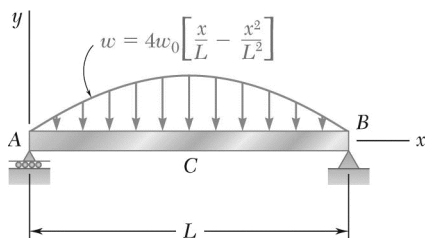
$$\frac{dy}{dx} = w_0 (6x^5 - 60L^2 x^3 + 75L^3 x^2 - 11L^5) / 360 EIL^2$$

(b) Slope at end A. Set $x = 0$ in $\frac{dy}{dx}$. $\frac{dy}{dx} \bigg|_A = -\frac{11}{360} \frac{w_0 L^3}{EI} \quad \theta_A = \frac{11}{360} \frac{w_0 L^3}{EI} \quad \blacktriangleleft$

PROBLEM 9.17 (Continued)

(c) Deflection at midpoint (say, point C). Set $x = \frac{L}{2}$ in. y .

$$\begin{aligned} y_C &= w_0 \left(\frac{1}{64} L^6 - \frac{15}{16} L^6 + \frac{25}{8} L^6 - \frac{11}{2} L^6 \right) \bigg/ 360 EIL^2 \\ &= w_0 \left(\frac{1}{64} L^6 - \frac{60}{64} L^6 + \frac{200}{64} L^6 - \frac{352}{64} L^6 \right) \bigg/ 360 EIL^2 \\ &= -\frac{211}{23040} \frac{w_0 L^4}{EI} \end{aligned} \qquad y_C = 0.00916 \frac{w_0 L^4}{EI} \downarrow \blacktriangleleft$$



PROBLEM 9.18

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the slope at end A, (c) the deflection at the midpoint of the span.

$$[x = 0, M = 0]$$

$$[x = L, M = 0]$$

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

SOLUTION

Boundary conditions at A and B are noted.

$$w = \frac{w_0}{L^2} (4Lx - 4x^2)$$

$$\frac{dV}{dx} = -w = \frac{w_0}{L^2} (4x^2 - 4Lx)$$

$$\frac{dM}{dx} = V = \frac{w_0}{L^2} \left(\frac{4}{3}x^3 - 2Lx^2 \right) + C_1$$

$$M = \frac{w_0}{L^2} \left(\frac{1}{3}x^4 - \frac{2}{3}Lx^3 \right) + C_1x + C_2$$

$$[x = 0, M = 0] \quad 0 = 0 + 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, M = 0] \quad 0 = \frac{w_0}{L^2} \left(\frac{1}{3}L^4 - \frac{2}{3}L^4 \right) + C_1L + 0 \quad C_1 = \frac{1}{3}w_0L$$

$$EI \frac{d^2y}{dx^2} = M = \frac{w_0}{L^2} \left(\frac{1}{3}x^4 - \frac{2}{3}Lx^3 + \frac{1}{3}L^3x \right)$$

$$EI \frac{dy}{dx} = \frac{w_0}{L^2} \left(\frac{1}{15}x^5 - \frac{1}{6}Lx^4 + \frac{1}{6}L^3x^2 \right) + C_3$$

$$EIy = \frac{w_0}{L^2} \left(\frac{1}{90}x^6 - \frac{1}{30}Lx^5 + \frac{1}{18}L^3x^3 \right) + C_3x + C_4$$

$$[x = 0, y = 0] \quad 0 = 0 + 0 + 0 + 0 + C_4 \quad C_4 = 0$$

$$[x = L, y = 0] \quad 0 = \frac{w_0}{L^2} \left(\frac{1}{90}L^6 - \frac{1}{30}L^6 + \frac{1}{18}L^6 \right) + C_3L + 0 \quad C_3 = -\frac{1}{30}w_0L^3$$

(a) Elastic curve.

$$y = \frac{w_0}{EIL^2} \left(\frac{1}{90}x^6 - \frac{1}{30}Lx^5 + \frac{1}{18}L^3x^3 - \frac{1}{30}L^5x \right) \quad \blacktriangleleft$$

$$\frac{dy}{dx} = \frac{w_0}{EIL^2} \left(\frac{1}{15}x^5 - \frac{1}{6}Lx^4 + \frac{1}{6}L^3x^2 - \frac{1}{30}L^5 \right)$$

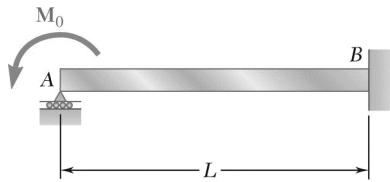
PROBLEM 9.18 (Continued)

(b) Slope at end A. Set $x = 0$ in $\frac{dy}{dx}$. $\left. \frac{dy}{dx} \right|_A = -\frac{1}{30} \frac{w_0 L^3}{EI}$

$$\theta_A = \frac{1}{30} \frac{w_0 L^3}{EI} \quad \swarrow \blacktriangleleft$$

(c) Deflection at midpoint. Set $x = \frac{L}{2}$ in y .

$$\begin{aligned} y_C &= \frac{w_0 L^4}{EI} \left\{ \left(\frac{1}{90} \right) \left(\frac{1}{2} \right)^6 - \left(\frac{1}{30} \right) \left(\frac{1}{2} \right)^5 + \frac{1}{18} \left(\frac{1}{2} \right)^3 - \frac{1}{30} \left(\frac{1}{2} \right) \right\} \\ &= \frac{w_0 L^4}{EI} \left\{ \frac{1}{5760} - \frac{1}{960} + \frac{1}{144} - \frac{1}{60} \right\} = -\frac{61}{5760} \frac{w_0 L^4}{EI} \quad y_C = \frac{61}{5760} \frac{w_0 L^4}{EI} \downarrow \blacktriangleleft \end{aligned}$$



PROBLEM 9.19

For the beam and loading shown, determine the reaction at the roller support.

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

$$\left[x = L, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

Boundary conditions are shown above.

Using free body AJ ,

$$+\circlearrowleft \Sigma M_J = 0: M_0 - R_A x + M = 0$$

$$M = R_A x - M_0$$

$$EI \frac{d^2 y}{dx^2} = R_A x - M_0$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - M_0 x + C_1$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad 0 = \frac{1}{2} R_A L^2 - M_0 L + C_1$$

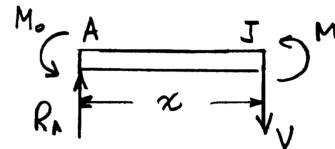
$$C_1 = M_0 L - \frac{1}{2} R_A L^2$$

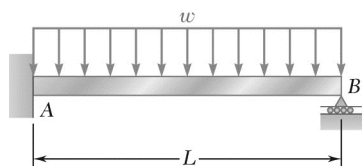
$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{2} M_0 x^2 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad C_2 = 0$$

$$[x = L, y = 0] \quad 0 = \frac{1}{6} R_A L^3 - \frac{1}{2} M_0 L^2 + \left(M_0 L - \frac{1}{2} R_A L^2 \right) L + 0$$

$$R_A = \frac{3}{2} \frac{M_0}{L} \uparrow \blacktriangleleft$$





PROBLEM 9.20

For the beam and loading shown, determine the reaction at the roller support.

$$[x = 0, y = 0] \quad [x = L, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

Boundary conditions are shown above.

Using free body KB ,

$$+\curvearrowright M_K = 0: R_B(L - x) - w(L - x)\left(\frac{L - x}{2}\right) - M = 0$$

$$M = R_B(L - x) - \frac{1}{2}w(L - x)^2$$

$$EI \frac{d^2y}{dx^2} = R_B(L - x) - \frac{1}{2}w(L - x)^2$$

$$EI \frac{dy}{dx} = -\frac{1}{2}R_B(L - x)^2 + \frac{1}{6}w(L - x)^3 + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: 0 = -\frac{1}{2}R_B L^2 + \frac{1}{6}wL^3 + C_1$$

$$C_1 = \frac{1}{2}R_B L^2 - \frac{1}{6}wL^3$$

$$EI_y = \frac{1}{6}R_B(L - x)^3 - \frac{1}{24}w(L - x)^4 + C_1x + C_2$$

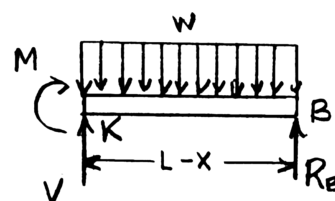
$$[x = 0, y = 0]: 0 = \frac{1}{6}R_B L^3 - \frac{1}{24}wL^4 + C_2$$

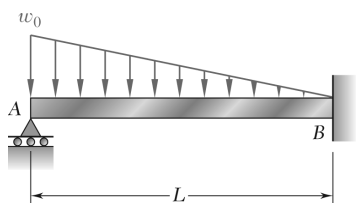
$$C_2 = -\frac{1}{6}R_B L^3 + \frac{1}{24}wL^4$$

$$[x = L, y = 0]: 0 = 0 - 0 + C_1 L + C_2$$

$$\frac{1}{2}R_B L^3 - \frac{1}{6}wL^4 - \frac{1}{6}R_B L^3 + \frac{1}{24}wL^4 = 0$$

$$R_B = \frac{3}{8}wL \uparrow \blacktriangleleft$$





PROBLEM 9.21

For the beam and loading shown, determine the reaction at the roller support.

$$[x = 0, y = 0] \quad [x = L, y = 0]$$

$$\left[x = L, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

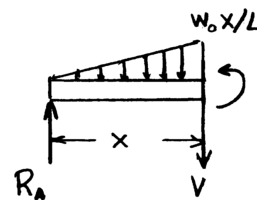
Boundary conditions are shown above.

$$w = \frac{w_0}{L}(L - x)$$

$$\frac{dV}{dx} = -w = -\frac{w_0}{L}(L - x)$$

$$\frac{dM}{dx} = V = -\frac{w_0}{L}\left(Lx - \frac{1}{2}x^2\right) + R_A$$

$$M = -\frac{w_0}{L}\left(\frac{1}{2}Lx^2 - \frac{1}{6}x^3\right) + R_Ax$$



$$EI \frac{d^2y}{dx^2} = -\frac{w_0}{L}\left(\frac{1}{2}Lx^2 - \frac{1}{6}x^3\right) + R_Ax$$

$$EI \frac{dy}{dx} = -\frac{w_0}{L}\left(\frac{1}{6}Lx^3 - \frac{1}{24}x^4\right) + \frac{1}{2}R_Ax^2 + C_1$$

$$EIy = -\frac{w_0}{L}\left(\frac{1}{24}Lx^4 - \frac{1}{120}x^5\right) + \frac{1}{6}R_Ax^3 + C_1x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 + 0 + 0 + C_2 \quad C_2 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad -\frac{w_0}{L}\left(\frac{1}{6}L^4 - \frac{1}{24}L^4\right) + \frac{1}{2}R_AL^2 + C_1 = 0$$

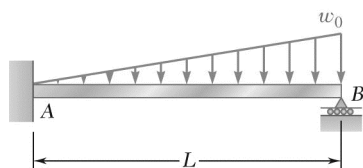
$$C_1 = \frac{1}{8}w_0L^3 - \frac{1}{2}R_AL^2$$

$$[x = L, y = 0] \quad -\frac{w_0}{L}\left(\frac{1}{24}L^5 - \frac{1}{120}L^5\right) + \frac{1}{6}R_AL^3 + \left(\frac{1}{8}w_0L^3 - \frac{1}{2}R_AL^2\right)L = 0$$

$$\left(\frac{1}{2} - \frac{1}{6}\right)R_A = \left(\frac{1}{8} - \frac{1}{24} + \frac{1}{120}\right)w_0L$$

$$\frac{1}{3}R_A = \frac{11}{120}w_0L$$

$$R_A = \frac{11}{40}w_0L \uparrow \blacktriangleleft$$



PROBLEM 9.22

For the beam and loading shown, determine the reaction at the roller support.

$$[x = 0, y = 0]$$

$$[x =$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

Boundary conditions are shown above.

Using free body JB ,

$$+\circlearrowleft \Sigma M_J = 0: -M + R_B(L - x) + \frac{1}{2}w_0(L - x)\frac{2}{3}(L - x)$$

$$+ \frac{1}{2} \frac{w_0 x}{L} (L - x) \frac{1}{3} (L - x) = 0$$

$$M = R_B(L - x) - \frac{w_0}{6L} [2L(L - x)^2 + x(L - x)^2]$$

$$= R_B(L - x) - \frac{w_0}{6L} [2L^3 - 4L^2x + 2Lx^2 + xL^2 - 2Lx^2 + x^3]$$

$$= R_B(L - x) - \frac{w_0}{6L} (x^3 - 3L^2x + 2L^3)$$

$$EI \frac{d^2y}{dx^2} = R_B(L - x) - \frac{w_0}{6L} (x^3 - 3L^2x + 2L^3)$$

$$EI \frac{dy}{dx} = R_B \left(Lx - \frac{1}{2}x^2 \right) - \frac{w_0}{6L} \left(\frac{1}{4}x^4 - \frac{3}{2}L^2x^2 + 2L^3x \right) + C_1$$

$$EIy = R_B \left(\frac{1}{2}Lx^2 - \frac{1}{6}x^3 \right) - \frac{w_0}{6L} \left(\frac{1}{20}x^5 - \frac{1}{2}L^2x^3 + L^3x^2 \right) + C_1x + C_2$$

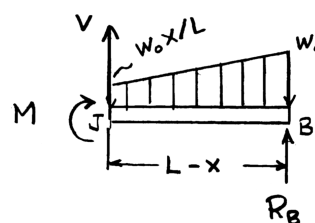
$$[x = 0, y = 0] \rightarrow C_2 = 0$$

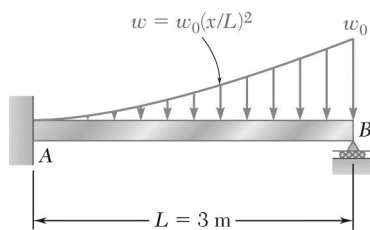
$$\left[x = 0, \frac{dy}{dx} = 0 \right] \rightarrow C_1 = 0$$

$$[x = L, y = 0] \quad 0 = R_B L^3 \left(\frac{1}{2} - \frac{1}{6} \right) - \frac{w_0 L^4}{6} \left(\frac{1}{20} - \frac{1}{2} + 1 \right)$$

$$\frac{1}{3} R_B = \left(\frac{1}{6} \right) \left(\frac{11}{20} \right) w_0 L$$

$$R_B = \frac{11}{40} w_0 L \uparrow \blacktriangleleft$$

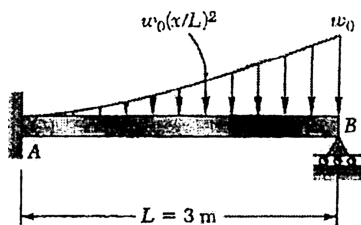




PROBLEM 9.23

For the beam shown, determine the reaction at the roller support when $w_0 = 15 \text{ kN/m}$.

SOLUTION



Reactions are statically indeterminate.

Boundary conditions are shown at left.

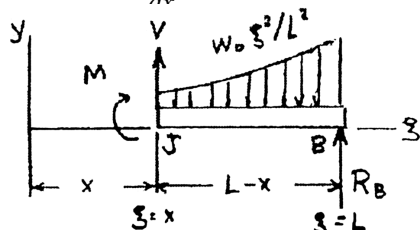
Using free body JB ,

$$+\circlearrowleft \sum M_J = 0 :$$

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

$$[x = 0, \frac{dy}{dx} = 0]$$



$$-M + \int_x^L \frac{w_0}{L^2} \xi^2 (\xi - x) d\xi + R_B(L - x) = 0$$

$$\begin{aligned} M &= \frac{w_0}{L^2} \int_x^L \xi^2 (\xi - x) d\xi - R_B(L - x) \\ &= \frac{w_0}{L^2} \left(\frac{1}{4} \xi^4 - \frac{1}{3} x \xi^3 \right) \bigg|_x^L - R_B(L - x) \\ &= \frac{w_0}{L^2} \left(\frac{1}{4} L^4 - \frac{1}{3} L^3 x + \frac{1}{12} x^4 \right) - R_B(L - x) \end{aligned}$$

$$EI \frac{d^2 y}{dx^2} = \frac{w_0}{L^2} \left(\frac{1}{4} L^4 - \frac{1}{3} L^3 x + \frac{1}{12} x^4 \right) - R_B(L - x)$$

$$EI \frac{dy}{dx} = \frac{w_0}{L^2} \left(\frac{1}{4} L^4 x - \frac{1}{6} L^3 x^2 + \frac{1}{60} x^5 \right) - R_B \left(Lx - \frac{1}{2} x^2 \right) + C_1$$

$$EI y = \frac{w_0}{L^2} \left(\frac{1}{8} L^4 x^2 - \frac{1}{18} L^3 x^3 + \frac{1}{360} x^6 \right) - R_B \left(\frac{1}{2} Lx^2 - \frac{1}{6} x^3 \right) + C_1 x + C_2$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 = 0 + 0 + C_1 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 = 0 + 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, y = 0] \quad \left(\frac{1}{8} - \frac{1}{18} + \frac{1}{360} \right) w_0 L^4 - \left(\frac{1}{2} - \frac{1}{6} \right) R_B L^3 = 0$$

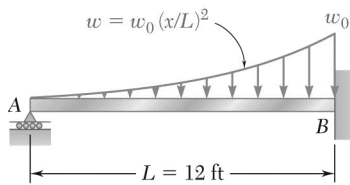
$$\frac{13}{180} w_0 L^4 - \frac{1}{3} R_B L^3 = 0 \quad R_B = \frac{13}{60} w_0 L$$

Data: $w_0 = 15 \text{ kN/m}$ $L = 3 \text{ m}$

$$R_B = \frac{13}{60} (15)(3) = 9.75 \text{ kN}$$

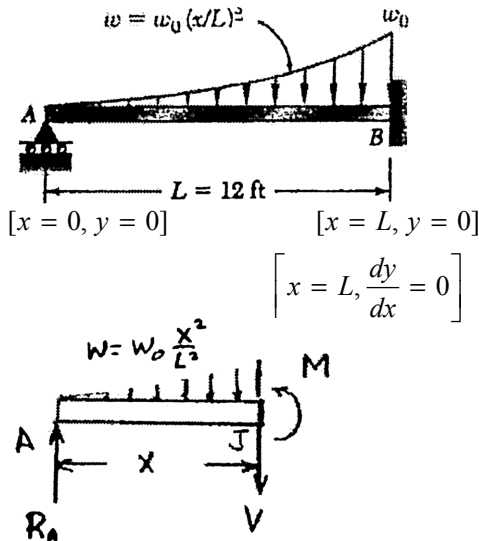
$$R_B = 9.75 \text{ kN} \uparrow \blacktriangleleft$$

PROBLEM 9.24



For the beam shown, determine the reaction at the roller support when $w_0 = 6$ kips/ft.

SOLUTION



Reactions are statically indeterminate.

Boundary conditions are shown at left.

$$w = w_0 \frac{x^2}{L^2}$$

$$\frac{dV}{dx} = -w = -\frac{w_0}{L^2} x^2$$

$$\frac{dM}{dx} = V = -\frac{w_0}{L^2} \frac{x^3}{3} + R_A$$

$$M = -\frac{w_0}{L^2} \frac{x^4}{12} + R_A x$$

$$EI \frac{d^2 y}{dx^2} = -\frac{w_0}{L^2} \frac{x^4}{12} + R_A x$$

$$EI \frac{dy}{dx} = -\frac{w_0}{L^2} \frac{x^5}{60} + \frac{1}{2} R_A x^2 + C_1$$

$$EI y = -\frac{w_0}{L^2} \frac{x^6}{360} + \frac{1}{6} R_A x^3 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 + 0 + 0 + C_2 \quad C_2 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad -\frac{1}{60} w_0 L^3 + \frac{1}{2} R_A L^2 + C_1 = 0 \quad C_1 = \frac{1}{60} w_0 L^3 - \frac{1}{2} R_A L^2$$

$$[x = L, y = 0] \quad -\frac{1}{360} w_0 L^4 + \frac{1}{6} R_A L^3 + \left(\frac{1}{60} w_0 L^4 - \frac{1}{2} R_A L^3 \right) L = 0$$

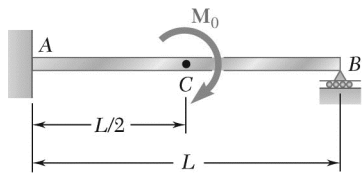
$$\left(\frac{1}{2} - \frac{1}{6} \right) R_A = \left(\frac{1}{60} - \frac{1}{360} \right) w_0 L$$

$$\frac{1}{3} R_A = \frac{1}{72} w_0 L \quad \Rightarrow \quad R_A = \frac{1}{18} w_0 L$$

Data: $w_0 = 6$ kips/ft, $L = 12$ ft

$$R_A = \frac{1}{18} (6)(12)$$

$$R_A = 4.00 \text{ kips } \uparrow \quad \blacktriangleleft$$



PROBLEM 9.25

Determine the reaction at the roller support and draw the bending moment diagram for the beam and loading shown.

$$[x = 0, y = 0]$$

$$[x =$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

$$+\uparrow \Sigma F_y = 0: \quad R_A + R_B = 0 \quad R_A = -R_B$$

$$+\curvearrowright \Sigma M_A = 0: \quad -M_A - M_0 + R_B L = 0 \quad M_A = R_B L - M_0$$

$$0 < x < \frac{L}{2}$$

$$M = R_B x + M_A = -M_0 + R_B L - R_B x$$

$$EI \frac{d^2 y}{dx^2} = -M_0 + R_B(L - x)$$

$$EI \frac{dy}{dx} = -M_0 x + R_B \left(Lx - \frac{1}{2} x^2 \right) + C_1$$

$$EI y = -\frac{1}{2} M_0 x^2 + R_B \left(\frac{1}{2} Lx^2 - \frac{1}{6} x^3 \right) + C_1 x + C_2$$

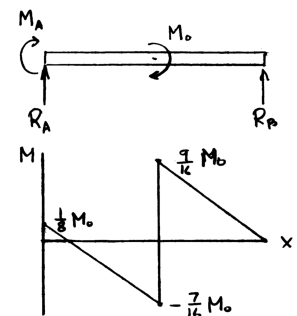
$$\frac{L}{2} < x < L$$

$$M = R_B(L - x)$$

$$EI \frac{d^2 y}{dx^2} = R_B(L - x)$$

$$EI \frac{dy}{dx} = R_B \left(Lx - \frac{1}{2} x^2 \right) + C_3$$

$$EI y = R_B \left(\frac{1}{2} Lx^2 - \frac{1}{6} x^3 \right) + C_3 x + C_4$$



PROBLEM 9.25 (Continued)

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right]$$

$$-M_0 \frac{L}{2} + R_B \left(\frac{1}{2} L^2 - \frac{1}{6} L^2 \right) = R_B \left(\frac{1}{2} L^2 - \frac{1}{6} L^2 \right) + C_3 \quad C_3 = -\frac{M_0 L}{2}$$

$$\left[x = \frac{L}{2}, y = y \right]$$

$$-\frac{1}{2} M_0 \left(\frac{L}{2} \right)^2 + R_B \left(\frac{1}{8} L^3 - \frac{1}{48} L^3 \right) = R_B \left(\frac{1}{8} L^3 - \frac{1}{48} L^3 \right) + C_3 \frac{L}{2} + C_4$$

$$C_4 = -\frac{1}{8} M_0 L^2 - \frac{1}{2} C_3 L$$

$$= \left(-\frac{1}{8} + \frac{1}{4} \right) M_0 L^2 = \frac{1}{8} M_0 L^2$$

$$[x = L, y = 0]$$

$$R_B \left(\frac{1}{2} L^3 - \frac{1}{6} L^3 \right) + \frac{M_0 L}{2} L + \frac{1}{8} M_0 L^2 = 0$$

$$\left(\frac{1}{2} - \frac{1}{6} \right) R_B L^3 = \left(\frac{1}{2} - \frac{1}{8} \right) M_0 L^2 \quad \frac{1}{3} R_B = \frac{3}{8} \frac{M_0}{L}$$

$$R_B = \frac{9}{8} \frac{M_0}{L} \uparrow \blacktriangleleft$$

$$M_A = \frac{9}{8} M_0 - M_0 = \frac{1}{8} M_0$$

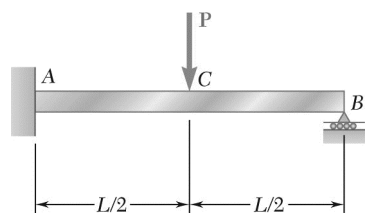
$$M_A = \frac{1}{8} M_0 \blacktriangleleft$$

$$M_{C^-} = -M_0 + \frac{9}{8} \frac{M_0}{L} \left(\frac{L}{2} \right) = -\frac{7}{16} M_0$$

$$M_{C^-} = -\frac{7}{16} M_0 \blacktriangleleft$$

$$M_{C^+} = R_B \left(L - \frac{L}{2} \right) = \frac{9}{8} \frac{M_0}{L} \left(\frac{L}{2} \right) = \frac{9}{16} M_0$$

$$M_{C^+} = \frac{9}{16} M_0 \blacktriangleleft$$



PROBLEM 9.26

Determine the reaction at the roller support and draw the bending moment diagram for the beam and loading shown.

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

$$+\uparrow \Sigma F_y = 0: R_A + R_B - P = 0 \quad R_A = P - R_B$$

$$+\curvearrowright \Sigma M_A = 0: -M_A + \frac{1}{2}PL - R_B L = 0$$

$$M_A = R_B L - \frac{1}{2}PL$$

$$0 < x < \frac{1}{2}L :$$

$$M = M_A + R_A x$$

$$EI \frac{d^2 y}{dx^2} = M_A + R_A x$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 + C_1$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 + C_1 x + C_2$$

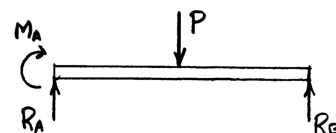
$$\frac{1}{2}L < x < L :$$

$$M = M_A + R_A x - P \left(x - \frac{1}{2}L \right)$$

$$EI \frac{d^2 y}{dx^2} = M = M_A + R_A x - P \left(x - \frac{1}{2}L \right)$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 - \frac{1}{2} P \left(x - \frac{1}{2}L \right)^2 + C_3$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 - \frac{1}{6} P \left(x - \frac{1}{2}L \right)^3 + C_3 x + C_4$$



PROBLEM 9.26 (Continued)

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right]$$

$$\frac{1}{2}M_A L + \frac{1}{8}R_A L^2 + 0 = \frac{1}{2}M_A L + \frac{1}{8}R_A L^2 - 0 + C_3 \quad C_3 = 0$$

$$\left[x = \frac{L}{2}, y = y \right]$$

$$\frac{1}{8}M_A L^2 + \frac{1}{48}R_A L^3 + 0 + 0 = \frac{1}{8}M_A L^2 + \frac{1}{48}R_A L^3 - 0 + 0 + C_4 \quad C_4 = 0$$

$$[x = L, y = 0]$$

$$\frac{1}{2}M_A L^2 + \frac{1}{6}R_A L^3 - \frac{1}{48}PL^3 + 0 + 0 = 0$$

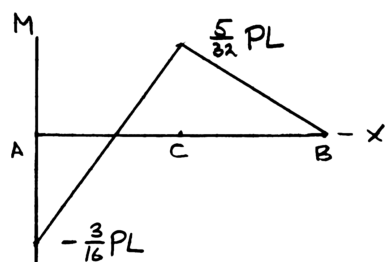
$$\frac{1}{2}\left(R_B L - \frac{1}{2}P\right)L^3 + \frac{1}{6}(P - R_B)L^3 - \frac{1}{48}PL^3 = 0 \quad R_B = \frac{5}{16}P \uparrow \blacktriangleleft$$

$$R_A = P - \frac{5}{16}P \quad R_A = \frac{7}{16}P \uparrow$$

$$M_A = \frac{5}{16}PL - \frac{1}{2}PL \quad M_A = -\frac{3}{16}PL \blacktriangleleft$$

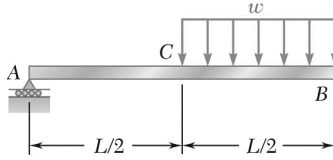
$$M_C = R_B\left(\frac{L}{2}\right) = \left(\frac{5}{16}P\right)\left(\frac{L}{2}\right) \quad M_C = \frac{5}{32}PL \blacktriangleleft$$

$$M_B = 0 \blacktriangleleft$$



Bending moment diagram

PROBLEM 9.27



Determine the reaction at the roller support and draw the bending moment diagram for the beam and loading shown.

SOLUTION

Reactions are statically indeterminate.

$$0 < x < \frac{L}{2}$$

$$EI \frac{d^2 y}{dx^2} = M = R_A x \quad (1)$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 + C_1 \quad (2)$$

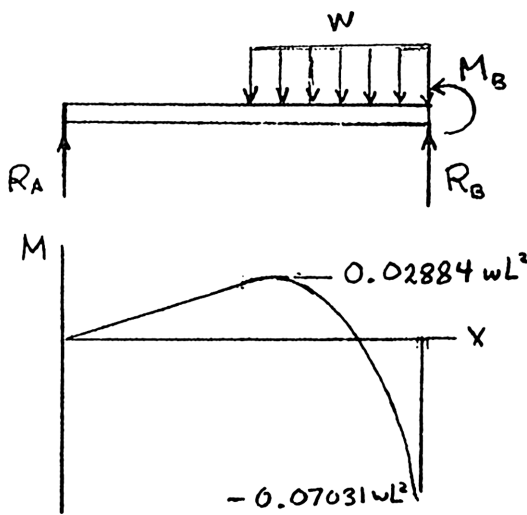
$$EI y = \frac{1}{6} R_A x^3 + C_1 x + C_2 \quad (3)$$

$$\frac{L}{2} < x < L$$

$$EI \frac{d^2 y}{dx^2} = M = R_A x - \frac{1}{2} w \left(x - \frac{L}{2} \right)^2 \quad (4)$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{6} w \left(x - \frac{L}{2} \right)^3 + C_3 \quad (5)$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{24} w \left(x - \frac{L}{2} \right)^4 + C_3 x + C_4 \quad (6)$$



$$[x = 0, y = 0] \quad 0 = 0 + 0 + C_2 \quad C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right] \quad \frac{1}{2} R_A \left(\frac{L}{2} \right)^2 + C_1 = \frac{1}{2} R_A \left(\frac{L}{2} \right)^2 + 0 + C_3 \quad C_1 = C_3$$

$$\left[x = \frac{L}{2}, y = y \right] \quad \frac{1}{6} R_A \left(\frac{L}{2} \right)^3 + C_1 \frac{L}{2} + C_2 = \frac{1}{6} R_A \left(\frac{L}{2} \right)^3 - 0 + C_3 \frac{L}{2} + C_4 \quad C_2 = C_4 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad \frac{1}{2} R_A L^2 - \frac{1}{6} w \left(\frac{L}{2} \right)^3 + C_3 = 0 \quad C_3 = \frac{1}{48} w L^3 - \frac{1}{2} R_A L^2$$

$$[x = L, y = 0] \quad \frac{1}{6} R_A L^2 - \frac{1}{24} w \left(\frac{L}{2} \right)^4 + \left(\frac{1}{48} w L^3 - \frac{1}{2} R_A L^2 \right) L + 0 = 0$$

PROBLEM 9.27 (Continued)

$$\left(\frac{1}{2} - \frac{1}{6}\right) R_A L^3 = \left(\frac{1}{48} - \frac{1}{384}\right) w L^4 \qquad \frac{1}{3} R_A = \frac{7}{384} w L \qquad R_A = \frac{7}{128} w L \uparrow \blacktriangleleft$$

From (1), with $x = \frac{L}{2}$, $M_C = R_A \left(\frac{L}{2}\right) = \frac{7}{256} w L^2 \qquad M_C = 0.02734 w L^2 \blacktriangleleft$

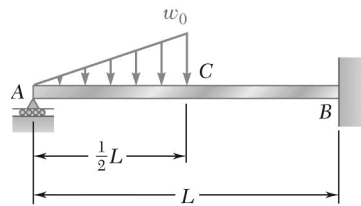
From (4), with $x = L$, $M_B = R_A L - \frac{1}{2} w \left(\frac{L}{2}\right)^2 = \left(\frac{7}{128} - \frac{1}{8}\right) w L - \frac{9}{128} w L^2$
 $M_B = -0.07031 w L \blacktriangleleft$

Location of maximum positive M :

$$\frac{L}{2} < x < L \qquad V_m = R_A - w \left(x_m - \frac{L}{2}\right) = 0 \qquad x_m - \frac{L}{2} = \frac{R_A}{w} = \frac{7}{128} L$$

$$x_m = \frac{L}{2} + \frac{7}{128} L = \frac{71}{128} L$$

From (4), with $x = x_m$, $M_m = R_A x_m - \frac{1}{2} w \left(x_m - \frac{L}{2}\right)^2$
 $= \left(\frac{7}{128} w L\right) \left(\frac{71}{128} L\right) - \frac{1}{2} w \left(\frac{7}{128} L\right)^2 \qquad M_m = 0.02884 w L^2 \blacktriangleleft$



PROBLEM 9.28

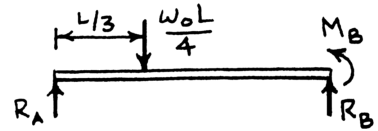
Determine the reaction at the roller support and draw the bending moment diagram for the beam and loading shown.

SOLUTION

Reactions are statically indeterminate.

$$+\uparrow \Sigma F_y = 0 : R_A + R_B - \frac{w_0 L}{4} = 0 \quad R_B = \frac{w_0 L}{4} - R_A$$

$$+\curvearrowright \Sigma M_B = 0 : -R_A L + \left(\frac{w_0 L}{4} \right) \left(\frac{2L}{3} \right) + M_B = 0 \quad M_B = R_A L - \frac{w_0 L^2}{6}$$



$$0 \leq x \leq \frac{L}{2} :$$

$$w = \frac{2w_0}{L}x$$

$$V = R_A - \frac{w_0}{L}x^2$$

$$M = R_A x - \frac{w_0}{3L}x^3$$

$$EI \frac{d^2 y}{dx^2} = R_A x - \frac{w_0}{3L}x^3$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{12} \frac{w_0}{L} x^4 + C_1$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{60} \frac{w_0}{L} x^5 + C_1 x + C_2$$

$$\frac{L}{2} \leq x \leq L :$$

$$M = R_A x - \frac{w_0 L}{4} \left(x - \frac{L}{3} \right)$$

$$EI \frac{d^2 y}{dx^2} = R_A x - w_0 L \left(\frac{1}{4} x - \frac{1}{12} L \right)$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - w_0 L \left(\frac{1}{8} x^2 - \frac{1}{12} L x \right) + C_3$$

$$EI y = \frac{1}{6} R_A x^3 - w_0 L \left(\frac{1}{24} x^3 - \frac{1}{24} L x^2 \right) + C_3 x + C_4$$

$$[x = 0, y = 0] : 0 = 0 - 0 + 0 + C_2 \quad \therefore C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right] : \cancel{\frac{1}{8} R_A L^2} - \frac{1}{192} w_0 L^3 + C_1 = \cancel{\frac{1}{8} R_A L^2} + \frac{1}{96} w_0 L^3 + C_3 \quad \therefore C_3 = C_1 - \frac{1}{64} w_0 L$$

$$\left[x = \frac{L}{2}, y = y \right] : \cancel{\frac{1}{48} R_A L^3} - \frac{1}{1920} w_0 L^4 + \cancel{\frac{1}{2} C_1 L} + 0$$

$$= \cancel{\frac{1}{48} R_A L^3} - w_0 L \left(\frac{1}{192} L^3 - \frac{1}{96} L^3 \right)$$

$$+ \left(\cancel{C_1} - \frac{1}{64} w_0 L^3 \right) \left(\frac{L}{2} \right) + C_4 \quad \therefore C_4 = \frac{1}{480} w_0 L^4$$

PROBLEM 9.28 (Continued)

$$\left[x = L, \frac{dy}{dx} = 0 \right] : \frac{1}{2} R_A L^2 - w_0 L \left(\frac{1}{8} L^2 - \frac{1}{12} L^2 \right) + C_3 = 0 \quad \therefore \quad C_3 = -\frac{1}{2} R_A L^2 + \frac{1}{24} w_0 L^3$$

$$[x = L, y = 0] : \frac{1}{6} R_A L^3 - w_0 L \left(\frac{1}{24} L^3 - \frac{1}{24} L^3 \right) + \left(-\frac{1}{2} R_A L^2 + \frac{1}{24} w_0 L^3 \right) (L) + \frac{1}{480} w_0 L^4 = 0$$

$$R_A = \frac{21}{160} w_0 L \uparrow \blacktriangleleft$$

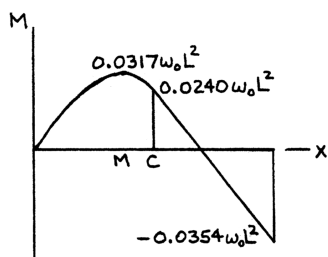
$$R_B = \frac{w_0 L}{4} - \frac{21}{160} w_0 L$$

$$R_B = \frac{19}{160} w_0 L \uparrow$$

$$M_B = \frac{21}{160} w_0 L^2 - \frac{w_0 L^2}{6}$$

$$M_B = -\frac{17}{480} w_0 L^2 = -0.0354 w_0 L^2 \blacktriangleleft$$

Over

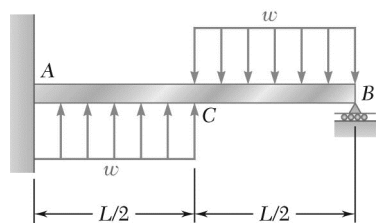


$$0 < x < \frac{L}{2}, \quad V = R_A - \frac{w_0}{L} x^2 = \frac{21}{160} w_0 L - \frac{w_0}{L} x^2$$

$$V = 0 \quad \text{at} \quad x = x_m = 0.36228L$$

$$M = \frac{21}{160} w_0 L x - \frac{w_0}{3L} x^3 \quad M_A = M(x = 0) = 0$$

$$M_C = M\left(x = \frac{L}{2}\right) = 0.0240 w_0 L^2 \quad M_m = M(x_m = 0.36228L) = 0.0317 w_0 L^2 \blacktriangleleft$$



PROBLEM 9.29

Determine the reaction at the roller support and the deflection at point C.

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

$$\left[x = \frac{L}{2}, y = y \right]$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right]$$

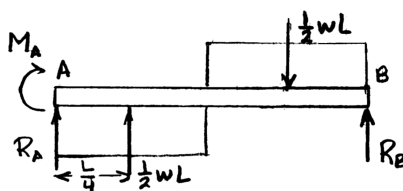
SOLUTION

Reactions are statically indeterminate.

$$+\uparrow \Sigma F_y = 0: R_A + \frac{1}{2}wL - \frac{1}{2}wL + R_B = 0 \quad R_A = -R_B$$

$$+\curvearrowright \Sigma M_A = 0: -M_A - \left(\frac{1}{2}wL \right) \frac{L}{2} + R_B L = 0$$

$$M_A = R_B L - \frac{1}{4}wL^2$$



From A to C:

$$0 < x \leq \frac{L}{2}$$

$$EI \frac{d^2 y}{dx^2} = M = M_A + R_A x + \frac{1}{2}wx^2$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2}R_A x^2 + \frac{1}{6}wx^3 + C_1$$

$$EI y = \frac{1}{2}M_A x^2 + \frac{1}{6}R_A x^3 + \frac{1}{24}wx^4 + C_1 x + C_2$$

PROBLEM 9.29 (Continued)

From C to B:

$$\frac{L}{2} \leq x < L$$

$$EI \frac{d^2 y}{dx^2} = M = M_A + R_A x + \frac{1}{2} wL \left(x - \frac{L}{4} \right) - \frac{1}{2} w \left(x - \frac{L}{2} \right)^2$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 + \frac{1}{4} wL \left(x - \frac{L}{4} \right)^2 - \frac{1}{6} w \left(x - \frac{L}{2} \right)^3 + C_3$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 + \frac{1}{12} wL \left(x - \frac{L}{4} \right)^3 - \frac{1}{24} w \left(x - \frac{L}{2} \right)^4 + C_3 x + C_4$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 + 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right]$$

$$\cancel{\frac{M_A L}{2}} - \cancel{\frac{1}{2} R_A \left(\frac{L}{2} \right)^2} + \frac{1}{6} w \left(\frac{L}{2} \right)^3 = \cancel{\frac{M_A L}{2}} + \cancel{\frac{1}{2} R_A \left(\frac{L}{2} \right)^2} + \frac{1}{4} wL \left(\frac{L}{4} \right)^2 - 0 + C_3$$

$$C_3 = \left(\frac{1}{48} - \frac{1}{64} \right) wL^3 = \frac{1}{192} wL^3$$

$$\left[x = \frac{L}{2}, y = y \right]$$

$$\begin{aligned} & \cancel{\frac{1}{2} M_A \left(\frac{L}{2} \right)^2} + \cancel{\frac{1}{6} R_A \left(\frac{L}{2} \right)^3} + \frac{1}{24} w \left(\frac{L}{2} \right)^4 \\ &= \cancel{\frac{1}{2} M_A \left(\frac{L}{2} \right)^2} + \cancel{\frac{1}{6} R_A \left(\frac{L}{2} \right)^3} + \frac{1}{12} wL \left(\frac{L}{4} \right)^3 \\ & \quad - 0 + \frac{1}{192} wL^3 \left(\frac{L}{2} \right) + C_4 \end{aligned}$$

PROBLEM 9.29 (Continued)

$$C_4 = \left(\frac{1}{384} - \frac{1}{768} - \frac{1}{384} \right) wL^4 = -\frac{1}{768} wL^4$$

$$[x = L, y = 0]$$

$$\begin{aligned} \frac{1}{2} M_A L^2 + \frac{1}{6} R_A L^3 + \frac{1}{12} wL \left(\frac{3L}{4} \right)^3 - \frac{1}{24} w \left(\frac{L}{2} \right)^4 \\ + \frac{1}{192} wL^3(L) - \frac{1}{768} wL^4 = 0 \end{aligned}$$

$$\frac{1}{2} \left(R_B L - \frac{1}{4} wL^2 \right) L^2 + \frac{1}{6} (-R_B) L^3 + \left(\frac{27}{768} - \frac{1}{384} + \frac{1}{192} - \frac{1}{768} \right) wL^4 = 0$$

$$\left(\frac{1}{2} - \frac{1}{6} \right) R_B L^3 = \left(\frac{1}{8} - \frac{7}{192} \right) wL^4 \quad \frac{1}{3} R_B = \frac{17}{192} wL \quad R_B = \frac{17}{64} wL \uparrow \blacktriangleleft$$

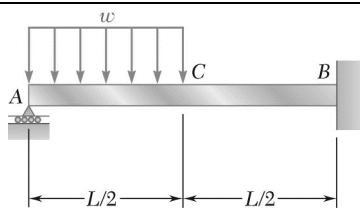
$$R_A = -R_B = -\frac{17}{64} wL$$

$$M_A = R_B L - \frac{1}{4} wL^2 = \left(\frac{17}{64} - \frac{1}{4} \right) wL^2 = \frac{1}{64} wL^2$$

Deflection at C.

$$\left(y \text{ at } x = \frac{L}{2} \right)$$

$$\begin{aligned} EI y_C &= \frac{1}{2} M_A \left(\frac{L}{2} \right)^2 + \frac{1}{6} R_A \left(\frac{L}{2} \right)^3 + \frac{1}{24} w \left(\frac{L}{2} \right)^4 \\ &= \frac{1}{2} \left(\frac{1}{64} wL^2 \right) \left(\frac{L}{2} \right)^2 + \frac{1}{6} \left(-\frac{17}{64} wL \right) \left(\frac{L}{2} \right)^3 + \frac{1}{24} w \left(\frac{L}{2} \right)^4 \\ &= \left(\frac{1}{512} - \frac{17}{3072} + \frac{1}{384} \right) wL^4 = -\frac{1}{1024} wL^4 \quad y_C = \frac{1}{1024} \frac{wL^4}{EI} \downarrow \blacktriangleleft \end{aligned}$$

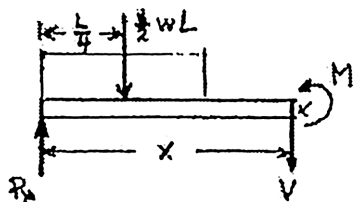


PROBLEM 9.30

Determine the reaction at the roller support and the deflection at point C.

SOLUTION

Reactions are statically indeterminate.



$$0 < x < \frac{L}{2}$$

$$EI \frac{d^2 y}{dx^2} = M = R_A x - \frac{1}{2} w x^2$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{6} w x^3 + C_1$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{24} w x^4 + C_1 x + C_2$$

$$\frac{L}{2} < x < L$$

(See free body diagram.)

$$+\circlearrowleft \sum M_K = 0: -R_A x + \frac{1}{2} w L \left(x - \frac{L}{4} \right) + M = 0$$

$$EI \frac{d^2 y}{dx^2} = M = R_A x - \frac{1}{2} w L \left(x - \frac{1}{4} L \right)$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{4} w L \left(x - \frac{L}{4} \right)^2 + C_3$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{12} w L \left(x - \frac{L}{4} \right)^3 + C_3 x + C_4$$

$$[x = 0, y = 0]: \quad 0 - 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = \frac{dy}{dx} \right]: \quad \cancel{\frac{1}{2} R_A \left(\frac{L}{2} \right)^2} - \frac{1}{6} w \left(\frac{L}{2} \right)^3 + C_1 = \cancel{\frac{1}{2} R_A \left(\frac{L}{2} \right)^2} - \frac{1}{4} w L \left(\frac{L}{4} \right)^2 + C_3$$

$$C_1 = C_3 + \frac{1}{48} w L^3 - \frac{1}{64} w L^3 = C_3 + \frac{1}{192} w L^3$$

$$\left[x = \frac{L}{2}, y = y \right]: \quad \cancel{\frac{1}{6} R_A \left(\frac{L}{2} \right)^3} - \frac{1}{24} w \left(\frac{L}{2} \right)^4 + \left(\cancel{C_1} + \frac{1}{192} w L^3 \right) \frac{L}{2} = \frac{1}{6} R_A \left(\frac{L}{2} \right)^3 - \frac{1}{12} w L \left(\frac{L}{4} \right)^3 + \cancel{C_3} \frac{L}{2} + C_4$$

$$C_4 = -\frac{1}{384} w L^4 + \frac{1}{384} w L^4 + \frac{1}{768} w L^4 = \frac{1}{768} w L^4$$

PROBLEM 9.30 (Continued)

$$\left[x = L, \frac{dy}{dx} = 0 \right]: \quad \frac{1}{2} R_A L^2 - \frac{1}{4} wL \left(\frac{3L}{4} \right)^2 + C_3 = 0 \quad C_3 = \frac{9}{64} wL^3 - \frac{1}{2} R_A L^2$$

$$[x = L, y = 0]: \quad \frac{1}{6} R_A L^3 - \frac{1}{12} wL \left(\frac{3L}{4} \right)^3 + \left(\frac{9}{64} wL^3 - \frac{1}{2} R_A L^2 \right) L + \frac{1}{768} wL^4 = 0$$

$$\left(\frac{1}{2} - \frac{1}{6} \right) R_A L^3 = \left(\frac{9}{64} - \frac{27}{768} + \frac{1}{768} \right) wL^4 \quad \frac{1}{3} R_A = \frac{41}{384} wL \quad R_A = \frac{41}{128} wL \uparrow \blacktriangleleft$$

$$C_3 = \frac{9}{64} wL^3 - \frac{1}{2} \frac{41}{128} wL^3 = -\frac{5}{256} wL^3$$

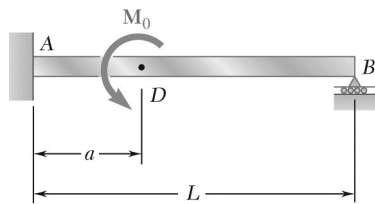
$$C_1 = -\frac{5}{256} wL^3 + \frac{1}{192} wL^3 = -\frac{11}{768} wL^3$$

Deflection at C. $\left(y \text{ at } x = \frac{L}{2} \right)$

$$\begin{aligned} y_C &= \frac{wL^4}{EI} \left\{ \frac{1}{6} \cdot \frac{41}{128} \cdot \left(\frac{1}{2} \right)^3 - \frac{1}{24} \cdot \left(\frac{1}{2} \right)^4 - \frac{11}{768} \cdot \frac{1}{2} + 0 \right\} \\ &= \left(\frac{41}{6144} - \frac{1}{384} - \frac{11}{1536} \right) \frac{wL^4}{EI} = -\frac{19}{6144} \frac{wL^4}{EI} \quad y_C = \frac{19}{6144} \frac{wL^4}{EI} \downarrow \blacktriangleleft \end{aligned}$$

or

$$\begin{aligned} y_C &= \frac{wL^4}{EI} \left\{ \frac{1}{6} \cdot \frac{41}{128} \left(\frac{1}{2} \right)^3 - \frac{1}{12} \cdot \left(\frac{1}{4} \right)^3 + \frac{5}{256} \cdot \frac{1}{2} + \frac{1}{768} \right\} \\ &= \left(\frac{41}{6144} - \frac{1}{\cancel{768}} - \frac{5}{512} + \frac{1}{\cancel{768}} \right) \frac{wL^4}{EI} = -\frac{19}{6144} \frac{wL^4}{EI} \end{aligned}$$



PROBLEM 9.31

Determine the reaction at the roller support and the deflection at point D if a is equal to $L/3$.

SOLUTION

$$+\uparrow \Sigma F_y = 0: R_A + R_B = 0 \quad R_A = -R_B$$

$$+\curvearrowright \Sigma M_A = 0: M_0 - M_A + R_B L = 0 \quad M_A = R_B L + M_0$$

$$0 \leq x \leq a:$$

$$M = M_A + R_A x$$

$$EI \frac{d^2 y}{dx^2} = M = M_A + R_A x$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 + C_1$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 + C_1 x + C_2$$

$$a \leq x \leq L:$$

$$M = M_A + R_A x - M_0$$

$$EI \frac{d^2 y}{dx^2} = M = M_A + R_A x - M_0$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 - M_0 x + C_3$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 - \frac{1}{2} M_0 x^2 + C_3 x + C_4$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: 0 + 0 + C_1 = 0 \quad \therefore C_1 = 0$$

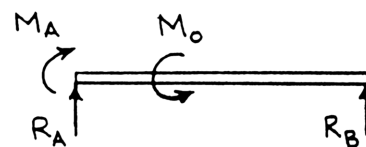
$$[x = 0, y = 0]: 0 + 0 + 0 + C_2 = 0 \quad \therefore C_2 = 0$$

$$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right]:$$

$$M_A a + \frac{1}{2} R_A a^2 = M_A a + \frac{1}{2} R_A a^2 - M_0 a + C_3 \quad \therefore C_3 = M_0 a$$

$$[x = a, y = y]:$$

$$\frac{1}{2} M_A a^2 + \frac{1}{6} R_A a^3 = \frac{1}{2} M_A a^2 + \frac{1}{6} R_A a^3 - \frac{1}{2} M_0 a^2 + (M_0 a)(a) + C_4 \quad \therefore C_4 = -\frac{1}{2} M_0 a^2$$



PROBLEM 9.31 (Continued)

$$[x = L, y = 0] :$$

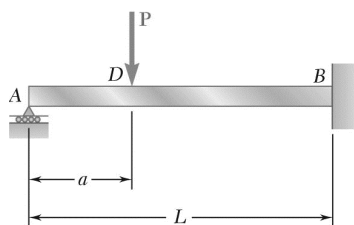
$$\frac{1}{2}M_A L^2 + \frac{1}{6}R_A L^3 - \frac{1}{2}M_0 L^2 + (M_0 a)(L) - \frac{1}{2}M_0 a^2 = 0$$

$$\frac{1}{2}(R_B L + M_0)L^2 + \frac{1}{6}(-R_B)L^3 - \frac{1}{2}M_0 L^2 + M_0 aL - \frac{1}{2}M_0 a^2 = 0$$

$$R_B = \frac{3M_0 a}{2L^3}(a - 2L) = \frac{3M_0}{2L^3}\left(\frac{L}{3}\right)\left(\frac{L}{3} - 2L\right) = -\frac{5M_0}{6L} \qquad R_B = \frac{5M_0}{6L} \downarrow \blacktriangleleft$$

Deflection at D. $\left(y \text{ at } x = a = \frac{L}{3}\right)$

$$\begin{aligned} y_D &= \frac{1}{EI} \left\{ \frac{1}{2}M_A x^2 + \frac{1}{6}R_A x^3 \right\} \\ &= \frac{1}{EI} \left\{ \frac{1}{2} \left(-\frac{5M_0}{6L}L + M_0 \right) \left(\frac{L}{3} \right)^2 + \frac{1}{6} \left(+\frac{5M_0}{6L} \right) \left(\frac{L}{3} \right)^3 \right\} \\ &= -\frac{7M_0 L^2}{486EI} \qquad y_D = \frac{7M_0 L^2}{486EI} \uparrow \blacktriangleleft \end{aligned}$$



PROBLEM 9.32

Determine the reaction at the roller support and the deflection at point D if a is equal to $L/3$.

SOLUTION

$$0 \leq x \leq a:$$

$$M = R_A x$$

$$EI \frac{d^2 y}{dx^2} = M = R_A x$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 + C_1$$

$$EI y = \frac{1}{6} R_A x^3 + C_1 x + C_2$$

$$a \leq x \leq L:$$

$$M = R_A x - P(x - a)$$

$$EI \frac{d^2 y}{dx^2} = M = R_A x - P(x - a)$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - \frac{1}{2} P(x - a)^2 + C_3$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{6} P(x - a)^3 + C_3 x + C_4$$

$$[x = 0, y = 0]: \quad 0 + 0 + C_2 = 0 \quad \therefore C_2 = 0$$

$$\left[x = a, \frac{dy}{dx} = \frac{dy}{dx} \right]:$$

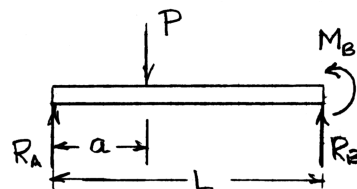
$$\frac{1}{2} R_A a^2 + C_1 = \frac{1}{2} R_A a^2 - 0 + C_3 \quad \therefore C_1 = C_3$$

$$[x = a, y = y]:$$

$$\frac{1}{6} R_A a^3 + C_1 a + 0 = \frac{1}{6} R_A a^3 - 0 + C_1 a + C_4 \quad \therefore C_4 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right]:$$

$$\frac{1}{2} R_A L^2 - \frac{1}{2} P(L - a)^2 + C_3 = 0 \quad \therefore C_3 = \frac{1}{2} P(L - a)^2 - \frac{1}{2} R_A L^2$$



PROBLEM 9.32 (Continued)

$$[x = L, y = 0] :$$

$$\frac{1}{6}R_A L^3 - \frac{1}{6}P(L-a)^3 + \left[\frac{1}{2}P(L-a)^2 - \frac{1}{2}R_A L^2 \right](L) + 0 = 0$$

$$R_A = \frac{P}{2L^3}(2L^3 - 3aL^2 + a^3) = \frac{P}{2L^3}\left(2L^3 - L^3 + \frac{L^3}{9}\right) \quad R_A = \frac{14}{27}P \uparrow \blacktriangleleft$$

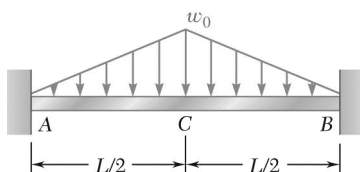
Deflection at D. $\left(y \text{ at } x = a = \frac{L}{3}\right)$

$$y_D = \frac{1}{EI} \left\{ \frac{1}{6}R_A \left(\frac{L}{3}\right)^3 + C_1 \left(\frac{L}{3}\right) \right\}$$

$$= \frac{1}{EI} \left\{ \frac{1}{6} \left(\frac{14}{27}P \right) \left(\frac{L}{3}\right)^3 + \left[\frac{1}{2}P \left(L - \frac{L}{3} \right)^2 - \frac{1}{2} \left(\frac{14}{27}P \right) L^2 \right] \left(\frac{L}{3}\right) \right\}$$

$$= -\frac{20}{2187} \frac{PL^3}{EI}$$

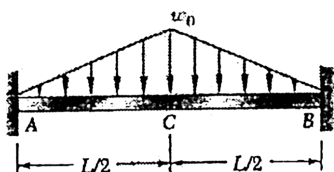
$$y_D = \frac{20}{2187} \frac{PL^3}{EI} \downarrow \blacktriangleleft$$



PROBLEM 9.33

Determine the reaction at A and draw the bending moment diagram for the beam and loading shown.

SOLUTION

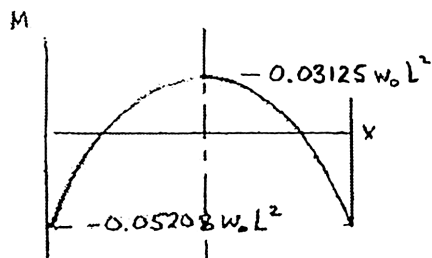


Reactions are statically indeterminate.

Because of symmetry, $\frac{dy}{dx} = 0$ and $V = 0$ at $x = \frac{L}{2}$.

$$\begin{aligned} & [x = 0, y = 0] \quad \left| \quad [x = L, y = 0] \right. \\ & \left[x = 0, \frac{dy}{dx} = 0 \right] \left| \left[x = L, \frac{dy}{dx} = 0 \right] \right. \\ & \left[x = \frac{L}{2}, V = 0 \right] \\ & \left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right] \end{aligned}$$

Use portion AC of beam. $\left(0 < x \leq \frac{L}{2} \right)$



$$\frac{dV}{dx} = -w = -2 \frac{w_0}{L} x$$

$$\frac{dM}{dx} = V = -\frac{w_0}{L} x^2 + R_A \quad (1)$$

$$EI \frac{d^2 y}{dx^2} = M = -\frac{1}{3} \frac{w_0}{L} x^3 + R_A x + M_A \quad (2)$$

$$EI \frac{dy}{dx} = -\frac{1}{12} \frac{w_0}{L} x^4 + \frac{1}{2} R_A x^2 + M_A x + C_1 \quad (3)$$

$$EI y = -\frac{1}{60} \frac{w_0}{L} x^5 + \frac{1}{6} R_A x^3 + \frac{1}{2} M_A x^2 + C_1 x + C_2 \quad (4)$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] : \quad 0 = 0 + 0 + 0 + C_1 \quad C_1 = 0$$

$$[x = 0, y = 0] : \quad 0 = 0 + 0 + 0 + 0 + C_2 \quad C_2 = 0$$

$$\left[x = \frac{L}{2}, V = 0 \right] : \quad -\frac{w_0}{L} \left(\frac{L}{2} \right)^2 + R_A = 0 \quad R_A = \frac{wL}{4} \quad \blacktriangleleft$$

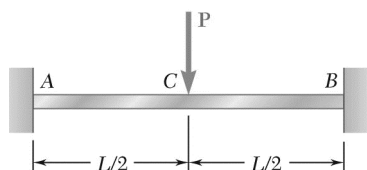
$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right] : \quad -\frac{1}{12} \frac{w_0}{L} \left(\frac{L}{2} \right)^4 + \frac{1}{2} \left(\frac{1}{4} w_0 L \right) \left(\frac{L}{2} \right)^2 + M_A \frac{L}{2} + 0 = 0$$

PROBLEM 9.33 (Continued)

$$M_A = -2\left(\frac{1}{32} - \frac{1}{192}\right)w_0L^2 = -\frac{5}{96}w_0L^2 \quad M_A = -0.05208w_0L^2 \quad \blacktriangleleft$$

From (2), with $x = \frac{L}{2}$,

$$\begin{aligned} M_C &= -\frac{1}{3}\frac{w_0}{L}\left(\frac{L}{2}\right)^3 + \left(\frac{1}{4}w_0L\right)\left(\frac{L}{2}\right) - \frac{5}{96}w_0L^2 \\ &= \left(-\frac{1}{24} + \frac{1}{8} - \frac{5}{96}\right)w_0L^2 = \frac{1}{32}w_0L^2 \quad M_C = 0.03125w_0L^2 \quad \blacktriangleleft \end{aligned}$$



PROBLEM 9.34

Determine the reaction at A and draw the bending moment diagram for the beam and loading shown.

$$[x = 0, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right]$$

SOLUTION

By symmetry, $R_A = R_B$ and $\frac{dy}{dx} = 0$ at $x = \frac{L}{2}$.

$$+\uparrow \Sigma F_y = 0 : R_A + R_B - P = 0$$

$$R_A = R_B = \frac{1}{2}P \quad \blacktriangleleft$$

Moment reaction is statically indeterminate.

$$0 < x < \frac{L}{2} :$$

$$M = M_A + R_A x = M_A + \frac{1}{2}Px$$

$$EI \frac{d^2 y}{dx^2} = M_A + \frac{1}{2}Px$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{4}Px^2 + C_1$$

$$y_B = \frac{P}{EI} \left\{ -0 + 0 + \frac{1}{12}a^3 \right\} = \frac{Pa^3}{12EI}$$

$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right] \quad M_A \frac{L}{2} + \frac{1}{4}P \left(\frac{L}{2} \right)^2 + 0 = 0$$

$$M_A = -\frac{1}{8}PL$$

$$M_A = \frac{1}{8}PL \quad \blacktriangleleft$$

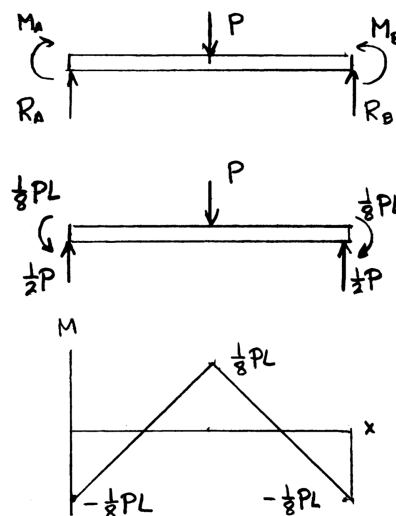
By symmetry,

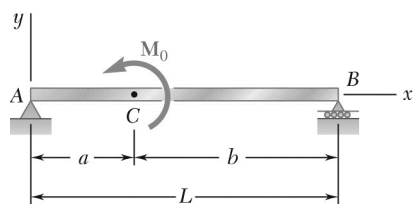
$$M_B = M_A$$

$$M_B = \frac{1}{8}PL \quad \blacktriangleleft$$

$$M_C = M_A + \frac{1}{2}P \frac{L}{2} = -\frac{1}{8}PL + \frac{1}{4}PL$$

$$M_C = \frac{1}{8}PL \quad \blacktriangleleft$$





PROBLEM 9.35

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the slope at end A, (c) the deflection of point C.

$$[x = 0, y = 0]$$

$$[x = L, y = 0]$$

SOLUTION

Reactions:

$$R_A = \frac{M_0}{L} \uparrow, \quad R_B = \frac{M_0}{L} \downarrow$$

$$0 < x < a \quad M = R_A x$$

$$a < x < L \quad M = R_A x - M_0$$

Using singularity functions,

$$EI \frac{d^2 y}{dx^2} = M = R_A x - M_0 \langle x - a \rangle^0$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - M_0 \langle x - a \rangle^1 + C_1$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{2} M_0 \langle x - a \rangle^2 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 - 0 + 0 + C_2 \quad C_2 = 0$$

$$[x = L, y = 0] \quad \frac{1}{6} R_A L^3 - \frac{1}{2} M_0 (L - a)^2 + C_1 L + 0 = 0$$

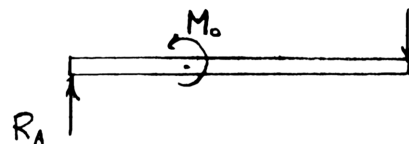
$$C_1 L = -\frac{1}{6} \frac{M_0}{L} L^3 + \frac{1}{2} M_0 b^2$$

$$C_1 = \frac{M_0}{6L} (3b^2 - L^2)$$

(a) Elastic curve.
$$y = \frac{1}{EI} \left\{ \frac{1}{6} \frac{M_0}{L} x^3 - \frac{1}{2} M_0 \langle x - a \rangle^2 + \frac{M_0}{6L} (3b^2 - L^2) x \right\}$$

$$y = \frac{M_0}{6EIL} \{ x^3 - 3L \langle x - a \rangle^2 + (3b^2 - L^2) x \} \quad \blacktriangleleft$$

$$\frac{dy}{dx} = \frac{M_0}{6EIL} \{ 3x^2 - 6L \langle x - a \rangle^1 + (3b^2 - L^2) \}$$



PROBLEM 9.35 (Continued)

(b) Slope at A. $\left(\frac{dy}{dx} \text{ at } x = 0 \right)$

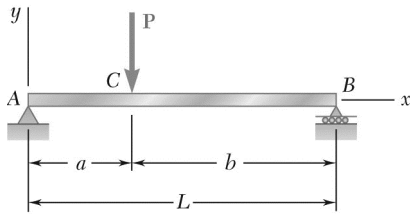
$$\theta_A = \frac{M_0}{6EI L} \{0 - 0 + 3Lb^2 - L^3\}$$

$$\theta_A = \frac{M_0}{6EI L} (3b^2 - L^2) \quad \nwarrow \blacktriangleleft$$

(c) Deflection at C. $(y \text{ at } x = a)$

$$\begin{aligned} y_C &= \frac{M_0}{6EI L} \{a^3 - 0 + (3b^2 - L^2)a\} \\ &= \frac{M_0 a}{6EI L} \{a^2 + 3b^2 - (a + b)^2\} \\ &= \frac{M_0 a}{6EI L} \{a^2 + 3b^2 - a^2 - 2ab - b^2\} \\ &= \frac{M_0 a}{6EI L} \{2b^2 - 2ab\} \end{aligned}$$

$$y_C = \frac{M_0 ab}{3EI L} (b - a) \quad \uparrow \blacktriangleleft$$



PROBLEM 9.36

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the slope at end A, (c) the deflection of point C.

$$\begin{aligned} [x = 0, M = 0] & \quad [x = L, M = 0] \\ [x = 0, y = 0] & \quad [x = L, y = 0] \end{aligned}$$

SOLUTION

$$+\circlearrowleft \Sigma M_B = 0: -R_A L + Pb = 0 \quad R_A = \frac{Pb}{L}$$

$$\frac{dM}{dx} = V = R_A - P\langle x - a \rangle^0 = \frac{Pb}{L} - P\langle x - a \rangle^0$$

$$M = \frac{Pb}{L}x - P\langle x - a \rangle^1 + \cancel{M_A}$$

$$EI \frac{d^2 y}{dx^2} = \frac{Pb}{L}x - P\langle x - a \rangle$$

$$EI \frac{dy}{dx} = \frac{Pb}{2L}x^2 - \frac{1}{2}P\langle x - a \rangle^2 + C_1$$

$$EI y = \frac{Pb}{6L}x^3 - \frac{1}{6}P\langle x - a \rangle^3 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad C_2 = 0$$

$$[x = L, y = 0] \quad \frac{Pb}{6L}L^3 - \frac{1}{6}P(L - a)^3 + C_1 L = 0$$

$$C_1 = -\frac{1}{6} \frac{P}{L} (bL^2 - b^3) = -\frac{1}{6} \frac{Pb}{L} (L^2 - b^2)$$

(a) Elastic curve.
$$y = \frac{P}{EI} \left\{ \frac{b}{6L}x^3 - \frac{1}{6}\langle x - a \rangle^3 - \frac{1}{6} \frac{b}{L} (L^2 - b^2)x \right\}$$

$$y = \frac{P}{6EIL} \{ bx^3 - L\langle x - a \rangle^3 - b(L^2 - b^2)x \} \quad \blacktriangleleft$$

(b) Slope at end A.

$$EI \left. \frac{dy}{dx} \right|_{x=0} = C_1 = -\frac{Pb}{6L} (L^2 - b^2)$$

$$\theta_A = -\frac{Pb}{6EIL} (L^2 - b^2)$$

$$\theta_A = \frac{Pb}{6EIL} (L^2 - b^2) \quad \nwarrow \blacktriangleleft$$

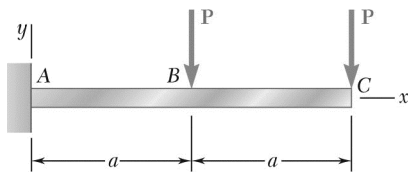
PROBLEM 9.36 (Continued)

(c) Deflection at C.

$$\begin{aligned}EIy_C &= \frac{Pb}{6L}a^3 + C_1a = \frac{Pba^3}{6L} - \frac{Pb}{6L}(L^2 - b^2)a \\ &= \frac{Pba}{6L}(a^2 - L^2 - b^2)\end{aligned}$$

$$\begin{aligned}y_C &= -\frac{Pab}{6EIL}(L^2 - a^2 - b^2) \\ &= -\frac{Pab}{6EIL}\{a^2 + 2ab + b^2 - a^2 - b^2\} \\ &= -\frac{Pa^2b^2}{3EIL}\end{aligned}$$

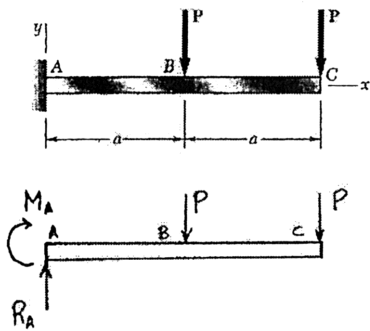
$$y_C = \frac{Pa^2b^2}{3EIL} \downarrow \blacktriangleleft$$



PROBLEM 9.37

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the slope at the free end, (c) the deflection of the free end.

SOLUTION



$$+\uparrow \Sigma F_y = 0: R_A - P - P = 0 \quad R_A = 2P$$

$$+\circlearrowleft \Sigma M_A = 0: -M_A - Pa - P(2a) = 0 \quad M_A = -3Pa$$

$$\frac{dM}{dx} = V = 2P - P\langle x - a \rangle^0$$

$$EI \frac{d^2y}{dx^2} = M = 2Px - P\langle x - a \rangle^1 - 3Pa$$

$$EI \frac{dy}{dx} = Px^2 - \frac{1}{2}P\langle x - a \rangle^2 - 3Pax + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: 0 - 0 - 0 + C_1 = 0 \quad C_1 = 0$$

$$EIy = \frac{1}{3}Px^3 - \frac{1}{6}P\langle x - a \rangle^3 - \frac{3}{2}Pax^2 + C_2$$

$$[x = 0, y = 0]: 0 - 0 - 0 + C_2 = 0 \quad C_2 = 0$$

$$EIy = \frac{1}{3}Px^3 - \frac{1}{6}P\langle x - a \rangle^3 - \frac{3}{2}Pax^2$$

(a) Elastic curve.

$$y = \frac{P}{EI} \left\{ \frac{1}{3}x^3 - \frac{1}{6}\langle x - a \rangle^3 - \frac{3}{2}ax^2 \right\} \quad \blacktriangleleft$$

$$\frac{dy}{dx} = \frac{P}{EI} \left\{ x^2 - \frac{1}{2}\langle x - a \rangle^2 - 3ax \right\}$$

(b) Slope at end C.

$$\left(\frac{dy}{dx} \text{ at } x = 2a \right)$$

$$\left. \frac{dy}{dx} \right|_A = \frac{P}{EI} \left\{ (2a)^2 - \frac{1}{2}a^2 - (3a)(2a) \right\} = -\frac{5Pa^2}{2EI}$$

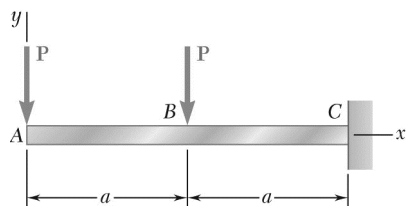
$$\theta_C = \frac{5Pa^2}{2EI} \quad \nwarrow \blacktriangleleft$$

(c) Deflection at end C.

(y at x = 2a)

$$y_A = \frac{P}{EI} \left\{ \frac{1}{3}(2a)^3 - \frac{1}{6}a^3 - \left(\frac{3}{2}a \right)(2a)^2 \right\} = -\frac{7Pa^3}{2EI}$$

$$y_C = \frac{7Pa^3}{2EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.38

For the beam and loading shown, determine (a) the equation curve, (b) the slope at the free end, (c) the deflection of the free end.

SOLUTION

$$\frac{dM}{dx} = V = -P - P\langle x - a \rangle^0$$

$$EI \frac{d^2y}{dx^2} = M = -Px - P\langle x - a \rangle^1$$

$$EI \frac{dy}{dx} = -\frac{1}{2}Px^2 - \frac{1}{2}P\langle x - a \rangle^2 + C_1$$

$$EIy = -\frac{1}{6}Px^3 - \frac{1}{6}P\langle x - a \rangle^3 + C_1x + C_2$$

$$\left[x = 2a, \frac{dy}{dx} = 0 \right] \quad -\frac{1}{2}P(2a)^2 - \frac{1}{2}P(a)^2 + C_1 = 0 \quad C_1 = \frac{5}{2}Pa^2$$

$$\left[x = 2a, y = 0 \right] \quad -\frac{1}{6}P(2a)^3 - \frac{1}{6}P(a)^3 + \frac{5}{2}Pa^2(2a) + C_2 = 0 \quad C_2 = -\frac{7}{2}Pa^3$$

$$EIy = -\frac{1}{6}Px^3 - \frac{1}{6}P\langle x - a \rangle^3 + \frac{5}{2}Pa^2x - \frac{7}{2}Pa^3$$

(a) Elastic curve.

$$y = \frac{P}{EI} \left\{ -\frac{1}{6}x^3 - \frac{1}{6}\langle x - a \rangle^3 + \frac{5}{2}a^2x - \frac{7}{2}a^3 \right\} \blacktriangleleft$$

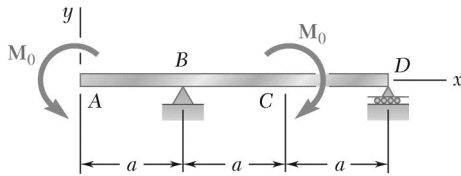
$$\frac{dy}{dx} = \frac{P}{EI} \left\{ -\frac{1}{2}x^2 - \frac{1}{2}\langle x - a \rangle^2 + \frac{5}{2}a^2 \right\}$$

(b) Slope at A.

$$\left(\frac{dy}{dx} \text{ at } x = 0 \right) \quad \frac{dy}{dx} \Big|_A = \frac{5Pa^2}{2EI} \quad \theta_A = \frac{5Pa^2}{2EI} \nearrow \blacktriangleleft$$

(c) Deflection at A.

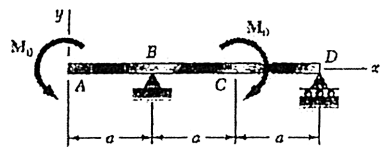
$$(y \text{ at } x = 0) \quad y_A = -\frac{7Pa^3}{2EI} \quad y_A = \frac{7Pa^3}{2EI} \downarrow \blacktriangleleft$$



PROBLEM 9.39

For the beam and loading shown, determine (a) the deflection end A, (b) the deflection at point C, (c) the slope at end D.

SOLUTION



Since loads self-equilibrate,

$$R_B = 0, \quad R_D = 0$$

$$(0 < x < 2a): \quad M = -M_0$$

$$[x = a, y = 0] \quad [x = 3a, y = 0] \quad (2a < x < 3a): \quad M = -M_0 + M_0 = 0$$

Using singularity functions,

$$EI \frac{d^2 y}{dx^2} = M = -M_0 + M_0 \langle x - 2a \rangle^0$$

$$EI \frac{dy}{dx} = -M_0 x + M_0 \langle x - 2a \rangle^1 + C_1$$

$$EI y = -\frac{1}{2} M_0 x^2 + \frac{1}{2} M_0 \langle x - 2a \rangle^2 + C_1 x + C_2$$

$$[x = 3a, y = 0] \quad -\frac{1}{2} M_0 (3a)^2 + \frac{1}{2} M_0 a^2 + C_1 (3a) + C_2 = 0 \quad 3aC_1 + C_2 = 4M_0 a^2$$

$$[x = a, y = 0] \quad -\frac{1}{2} M_0 a^2 + 0 + C_1 a + C_2 = 0 \quad aC_1 + C_2 = \frac{1}{2} M_0 a^2$$

Subtracting, $2aC_1 = \frac{7}{2} M_0 a^2 \quad C_1 = \frac{7}{4} M_0 a$

$$C_2 = \frac{1}{2} M_0 a^2 - aC_1 = -\frac{5}{4} M_0 a^2$$

$$y = \frac{M_0}{EI} \left\{ -\frac{1}{2} x^2 + \frac{1}{2} \langle x - 2a \rangle^2 + \frac{7}{4} ax - \frac{5}{4} a^2 \right\}$$

$$\frac{dy}{dx} = \frac{M_0}{EI} \left\{ -x + \langle x - a \rangle^1 + \frac{7}{4} a \right\}$$

(a) Deflection at A. (y at $x = 0$)

$$y_A = \frac{M_0 a^2}{EI} \left\{ -0 + 0 + 0 - \frac{5}{4} \right\} = -\frac{5}{4} \frac{M_0 a^2}{EI},$$

$$y_A = \frac{5}{4} \frac{M_0 a^2}{EI} \downarrow \blacktriangleleft$$

PROBLEM 9.39 (Continued)

(b) Deflection at C. (y at $x = 2a$)

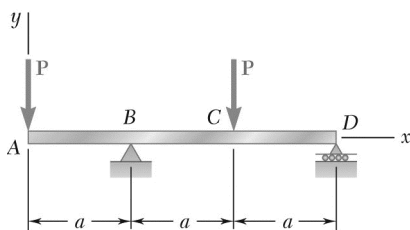
$$y_C = \frac{M_0 a^2}{EI} \left\{ -\frac{1}{2}(2)^2 + 0 + \frac{7}{4}(2) - \frac{5}{4} \right\} = \frac{1}{4} \frac{M_0 a^2}{EI}$$

$$y_C = \frac{1}{4} \frac{M_0 a^2}{EI} \uparrow \blacktriangleleft$$

(c) Slope at D. $\left(\frac{dy}{dx} \text{ at } x = 3a \right)$

$$\theta_D = \frac{M_0 a}{EI} \left\{ -3 + 1 + \frac{7}{4} \right\} = -\frac{1}{4} \frac{M_0 a}{EI},$$

$$\theta_D = \frac{1}{4} \frac{M_0 a}{EI} \searrow \blacktriangleleft$$

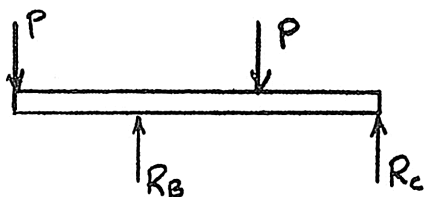


$$[x = a, y = 0] \quad [x = 3a, y = 0]$$

PROBLEM 9.40

For the beam and loading shown, determine (a) the deflection at end A, (b) the deflection at point C, (c) the slope at end D.

SOLUTION



Reactions: $R_B = 2P \uparrow, \quad R_D = 0$

$$(0 < x < a): \quad V = -P$$

$$(a < x < 2a): \quad V = -P + 2P$$

$$(2a < x < 3a): \quad V = -P + 2P - P$$

Using singularity functions,

$$\frac{dM}{dx} = V = -P + 2P\langle x - a \rangle^0 - P\langle x - 2a \rangle^0$$

$$M = -Px + 2P\langle x - a \rangle^1 - P\langle x - 2a \rangle^1 + M_A$$

But $M = 0$ at $x = 0$ $M_A = 0$

$$EI \frac{d^2 y}{dx^2} = M = -Px + 2P\langle x - a \rangle^1 - P\langle x - 2a \rangle^1 \quad (1)$$

$$EI \frac{dy}{dx} = -\frac{1}{2}Px^2 + P\langle x - a \rangle^2 - \frac{1}{2}P\langle x - 2a \rangle^2 + C_1 \quad (2)$$

$$EIy = -\frac{1}{6}Px^3 + \frac{1}{3}P\langle x - a \rangle^3 - \frac{1}{6}P\langle x - 2a \rangle^3 + C_1x + C_2 \quad (3)$$

$$[x = a, y = 0] \quad -\frac{1}{6}Pa^3 + 0 - 0 + C_1a + C_2 = 0 \quad aC_1 + C_2 = \frac{1}{6}Pa^3 \quad (4)$$

$$[x = 3a, y = 0] \quad -\frac{1}{6}P(3a)^3 + \frac{1}{3}P(2a)^3 - \frac{1}{6}Pa^3 + C_1(3a) + C_2 = 0 \quad 3aC_1 + C_2 = 2Pa^2 \quad (5)$$

$$\text{Eq (5) - Eq (4)} \quad 2C_1a + \frac{11}{6}Pa^2 \quad C_1 + \frac{11}{12}Pa^2$$

$$C_2 = \frac{1}{6}Pa^2 - aC_1 = -\frac{3}{4}Pa^3$$

$$y = \frac{P}{EI} \left\{ -\frac{1}{6}x^3 + \frac{1}{3}\langle x - a \rangle^3 - \frac{1}{6}\langle x - 2a \rangle^3 + \frac{11}{12}a^2x - \frac{3}{4}a^3 \right\}$$

$$\frac{dy}{dx} = \frac{P}{EI} \left\{ -\frac{1}{2}x^2 + \langle x - a \rangle^2 - \frac{1}{2}\langle x - 2a \rangle^2 + \frac{11}{12}a^2 \right\}$$

PROBLEM 9.40 (Continued)

(a) Deflection at A.

(y at $x = 0$)

$$y_A = \frac{Pa^3}{EI} \left\{ -0 + 0 - 0 + 0 - \frac{3}{4} \right\} = -\frac{3}{4} \frac{Pa^3}{EI}$$

$$y_A = \frac{3}{4} \frac{Pa^3}{EI} \downarrow \blacktriangleleft$$

(b) Deflection at C.

(y at $x = 2a$)

$$y_C = \frac{Pa^3}{EI} \left\{ -\frac{1}{6}(2)^3 + \frac{1}{3}(1)^3 - 0 + \frac{11}{12}(2) - \frac{3}{4} \right\}$$

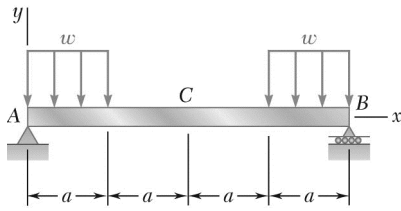
$$y_C = \frac{1}{12} \frac{Pa^3}{EI} \uparrow \blacktriangleleft$$

(c) Slope at D.

$\left(\frac{dy}{dx} \text{ at } x = 3a \right)$

$$\theta_D = \frac{Pa^2}{EI} \left\{ -\frac{1}{2}(3)^2 + (2)^2 - \frac{1}{2}(1)^2 + \frac{11}{12} \right\} = -\frac{1}{12} \frac{Pa^2}{EI}$$

$$\theta_D = \frac{1}{12} \frac{Pa^2}{EI} \nearrow \blacktriangleleft$$



PROBLEM 9.41

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the deflection at the midpoint C.

SOLUTION

By symmetry,

$$R_A = R_B = wa$$

$$w(x) = w - w\langle x - a \rangle^0 + w\langle x - 3a \rangle^0$$

$$\frac{dV}{dx} = -w(x) = -w + w\langle x - a \rangle^0 - w\langle x - 3a \rangle^0$$

$$\frac{dM}{dx} = V = R_A - wx + w\langle x - a \rangle^1 - w\langle x - 3a \rangle^1$$

$$M = M_A + R_A x - \frac{1}{2}wx^2 + \frac{1}{2}w\langle x - a \rangle^2 - \frac{1}{2}w\langle x - 3a \rangle^2 \quad (\text{with } M_A = 0)$$

$$EI \frac{d^2y}{dx^2} = M = wax - \frac{1}{2}wx^2 + \frac{1}{2}w\langle x - a \rangle^2 - \frac{1}{2}w\langle x - 3a \rangle^2$$

$$EI \frac{dy}{dx} = \frac{1}{2}wax^2 - \frac{1}{6}wx^3 + \frac{1}{6}w\langle x - a \rangle^3 - \frac{1}{6}w\langle x - 3a \rangle^3 + C_1$$

$$EIy = \frac{1}{6}wax^3 - \frac{1}{24}wx^4 + \frac{1}{24}w\langle x - a \rangle^4 - \frac{1}{24}w\langle x - 3a \rangle^4 + C_1x + C_2$$

$$[x = 0, y = 0] : \quad 0 - 0 + 0 - 0 + 0 + C_2 = 0 \quad \therefore \quad C_2 = 0$$

$$[x = 4a, y = 0] : \quad \frac{1}{6}wa(4a)^3 - \frac{1}{24}w(4a)^4 + \frac{1}{24}w(3a)^4 - \frac{1}{24}w(a)^4 + C_1(4a) = 0$$

$$\therefore \quad C_1 = -\frac{5}{6}wa^3$$

(a) Equation of elastic curve.

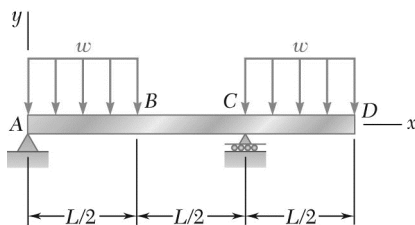
$$y = \frac{w}{EI} \left[\frac{1}{6}ax^3 - \frac{1}{24}x^4 + \frac{1}{24}\langle x - a \rangle^4 - \frac{1}{24}\langle x - 3a \rangle^4 - \frac{5}{6}a^3x \right] \quad \blacktriangleleft$$

(b) Deflection at C.

(y at $x = 2a$)

$$\begin{aligned} y_C &= \frac{wa^4}{EI} \left[\frac{1}{6}(2)^3 - \frac{1}{24}(2)^4 + \frac{1}{24}(1)^4 - 0 - \frac{5}{6}(2) \right] \\ &= -\frac{23}{24} \frac{wa^4}{EI} \end{aligned}$$

$$y_C = \frac{23}{24} \frac{wa^4}{EI} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.42

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the deflection at point B, (c) the deflection at point D.

SOLUTION

Use free body ABCD with the distributed loads replaced by equivalent concentrated loads.

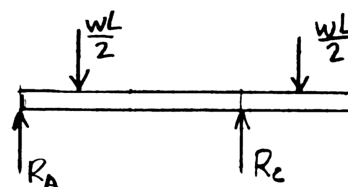
$$+\circlearrowleft \Sigma M_C = 0: -R_A L + \left(\frac{wL}{2}\right)\left(\frac{3L}{4}\right) - \left(\frac{wL}{2}\right)\left(\frac{L}{4}\right) = 0$$

$$R_A = \frac{1}{4}wL$$

$$+\circlearrowleft \Sigma M_A = 0: R_C L - \left(\frac{wL}{2}\right)\left(\frac{L}{4}\right) - \left(\frac{wL}{2}\right)\left(\frac{5L}{4}\right) = 0$$

$$R_C = \frac{3}{4}wL$$

$$\frac{dV}{dx} = -w = -w + w\left\langle x - \frac{L}{2} \right\rangle^0 - w\langle x - L \rangle^0$$



Integrating and adding terms to account for the reactions,

$$\frac{dM}{dx} = V = -wx + w\left\langle x - \frac{L}{2} \right\rangle^1 - w\langle x - L \rangle^1 + R_A + R_C\langle x - L \rangle^0$$

$$EI \frac{d^2y}{dx^2} = M = -\frac{1}{2}wx^2 + \frac{1}{2}w\left\langle x - \frac{L}{2} \right\rangle^2 - \frac{1}{2}w\langle x - L \rangle^2 + R_A x + R_C\langle x - L \rangle^1$$

$$EI \frac{dy}{dx} = -\frac{1}{6}wx^3 + \frac{1}{6}w\left\langle x - \frac{L}{2} \right\rangle^3 - \frac{1}{6}w\langle x - L \rangle^3 + \frac{1}{2}R_A x^2 + \frac{1}{2}R_C\langle x - L \rangle^2 + C_1$$

$$EIy = -\frac{1}{24}wx^4 + \frac{1}{24}w\left\langle x - \frac{L}{2} \right\rangle^4 - \frac{1}{24}w\langle x - L \rangle^4 + \frac{1}{6}R_A x^3 + \frac{1}{6}R_C\langle x - L \rangle^3 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad -0 + 0 - 0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = L, y = 0] \quad -\frac{1}{24}wL^4 + \frac{1}{24}w\left(\frac{L}{2}\right)^4 - 0 + \frac{1}{6}\left(\frac{wL}{4}\right)L^3 + 0 + C_1 L + 0 = 0 \quad C_1 = -\frac{1}{384}wL^3$$

$$EIy = -\frac{1}{24}wx^4 + \frac{1}{24}w\left\langle x - \frac{L}{2} \right\rangle^4 + \frac{1}{24}w\langle x - L \rangle^4 + \frac{1}{6}\left(\frac{wL}{4}\right)x^3 + \frac{1}{6}\left(\frac{3wL}{4}\right)\langle x - L \rangle^3 - \frac{1}{384}wL^3 x$$

PROBLEM 9.42 (Continued)

(a) Elastic curve.

$$y = \frac{w}{24EI} \left\{ -x^4 + \left\langle x - \frac{L}{2} \right\rangle^4 - \langle x - L \rangle^4 + Lx^3 + 3L\langle x - L \rangle^3 - \frac{1}{16}L^3x \right\} \blacktriangleleft$$

(b) Deflection at B.

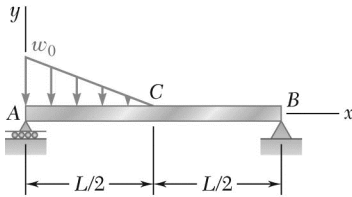
$$\left(y \text{ at } x = \frac{L}{2} \right)$$

$$y_B = \frac{w}{24EI} \left\{ -\left(\frac{L}{2}\right)^4 + 0 - 0 + (L)\left(\frac{L}{2}\right)^3 + 0 - \left(\frac{1}{16}L^3\right)\left(\frac{L}{2}\right) \right\} \qquad y_B = \frac{wL^4}{768EI} \uparrow \blacktriangleleft$$

(c) Deflection at D.

$$\left(y \text{ at } x = \frac{3L}{2} \right)$$

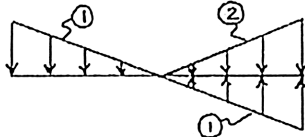
$$\begin{aligned} y_D &= \frac{w}{24EI} \left\{ -\left(\frac{3L}{2}\right)^4 + L^4 - \left(\frac{L}{2}\right)^4 + (L)\left(\frac{3L}{2}\right)^3 + (3L)\left(\frac{L}{2}\right)^3 - \left(\frac{1}{16}L\right)\left(\frac{3L}{2}\right) \right\} \\ &= -\frac{5wL^4}{256EI} \qquad y_D = \frac{5wL^4}{256EI} \downarrow \blacktriangleleft \end{aligned}$$



PROBLEM 9.43

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the deflection at the midpoint C.

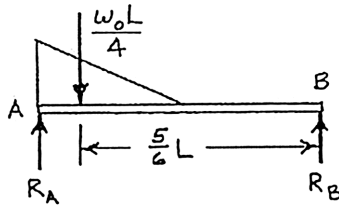
SOLUTION



Distributed loads: $k = \frac{2w_0}{L}$

$$(1) \quad w_1(x) = w_0 - kx \quad (2) \quad w_2(x) = k \left\langle x - \frac{L}{2} \right\rangle^1$$

$$+\circlearrowleft \Sigma M_B = 0: \left(\frac{w_0 L}{4} \right) \left(\frac{5}{6} L \right) - R_A L = 0 \quad R_A = \frac{5}{24} w_0 L \uparrow$$



$$w(x) = w_0 - kx + k \left\langle x - \frac{L}{2} \right\rangle^1 = w_0 - \frac{2w_0}{L} x + \frac{2w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^1$$

$$\frac{dV}{dx} = -w = -w_0 + \frac{2w_0}{L} x - \frac{2w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^1$$

$$\frac{dM}{dx} = V = \frac{5}{24} w_0 L - w_0 x + \frac{w_0}{L} x^2 - \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^2$$

$$EI \frac{d^2 y}{dx^2} = M = \frac{5}{24} w_0 L x - \frac{1}{2} w_0 x^2 + \frac{1}{3} \frac{w_0}{L} x^3 - \frac{1}{3} \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^3$$

$$EI \frac{dy}{dx} = \frac{5}{48} w_0 L x^2 - \frac{1}{6} w_0 x^3 + \frac{1}{12} \frac{w_0}{L} x^4 - \frac{1}{12} \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^4 + C_1$$

$$EI y = \frac{5}{144} w_0 L x^3 - \frac{1}{24} w_0 x^4 + \frac{1}{60} \frac{w_0}{L} x^5 - \frac{1}{60} \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^5 + C_1 x + C_2$$

$$[x = 0, y = 0]:$$

$$C_2 = 0$$

$$[x = L, y = 0]: \frac{5}{144} w_0 L^4 - \frac{1}{24} w_0 L^4 + \frac{1}{60} w_0 L^4 - \frac{1}{60} \frac{w_0}{L} \left(\frac{L}{2} \right)^5 + C_1 L = 0 \quad C_1 = -\frac{53}{5760} w_0 L^3$$

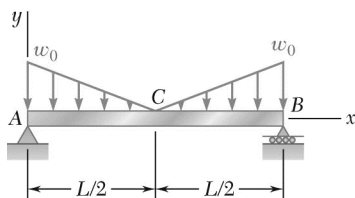
(a) Equation of elastic curve.

$$y = w_0 \left[96x^5 - 96 \left\langle x - \frac{L}{2} \right\rangle^5 - 240Lx^4 + 200L^2x^3 - 53L^4x \right] / 5760 EI \quad \blacktriangleleft$$

(b) Deflection at C. $\left(y \text{ at } x = \frac{L}{2} \right)$

$$y_C = \frac{w_0 L^4}{5760 EI} \left(\frac{96}{32} - 0 - \frac{240}{16} + \frac{200}{8} - \frac{53}{2} \right) = -\frac{3w_0 L^4}{1280 EI}$$

$$y_C = \frac{3w_0 L^4}{1280 EI} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.44

For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the deflection at the midpoint C.

SOLUTION

Distributed loads: $k_1 = \frac{2w_0}{L}$ $k_2 = \frac{4w_0}{L}$

$$(1) \quad w_1(x) = w_0 - k_1 x$$

$$(2) \quad w_2(x) = k_2 \left\langle x - \frac{L}{2} \right\rangle^1$$

$$+\circlearrowleft \Sigma M_B = 0: \left(\frac{w_0 L}{4} \right) \left(\frac{5}{6} L \right) + \left(\frac{w_0 L}{4} \right) \left(\frac{L}{6} \right) + R_A L = 0$$

$$R_A = \frac{w_0 L}{4} \uparrow$$

$$w(x) = w_0 - k_1 x + k_2 \left\langle x - \frac{L}{2} \right\rangle^1$$

$$= w_0 - \frac{2w_0}{L} x + \frac{4w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^1$$

$$\frac{dV}{dx} = -w = -w_0 + \frac{2w_0}{L} x - \frac{4w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^1$$

$$\frac{dM}{dx} = V = \frac{w_0 L}{4} - w_0 x + \frac{w_0}{L} x^2 - \frac{2w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^2$$

$$EI \frac{d^2 y}{dx^2} = M = \frac{1}{4} w_0 L x - \frac{1}{2} w_0 x^2 + \frac{1}{3} \frac{w_0}{L} x^3 - \frac{2}{3} \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^3$$

$$EI \frac{dy}{dx} = \frac{1}{8} w_0 L x^2 - \frac{1}{6} w_0 x^3 + \frac{1}{12} \frac{w_0}{L} x^4 - \frac{1}{6} \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^4 + C_1$$

$$EI y = \frac{1}{24} w_0 L x^3 - \frac{1}{24} w_0 x^4 + \frac{1}{60} \frac{w_0}{L} x^5 - \frac{1}{30} \frac{w_0}{L} \left\langle x - \frac{L}{2} \right\rangle^5 + C_1 x + C_2$$

$$[x = 0, y = 0]: \quad C_2 = 0$$

$$[x = L, y = 0]: \quad \frac{1}{24} w_0 L^4 - \frac{1}{24} w_0 L^4 + \frac{1}{60} w_0 L^4 - \frac{1}{30} \frac{w_0}{L} \left(\frac{L}{2} \right)^5 + C_1 L = 0$$

$$C_1 = -\frac{1}{64} w_0 L^3$$

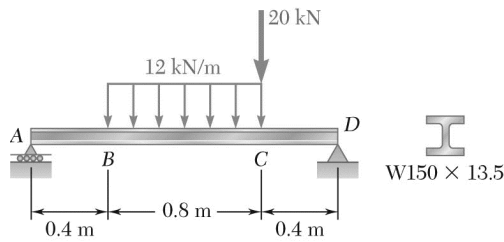
PROBLEM 9.44 (Continued)

(a) Equation of elastic curve.

$$y = w_0 \left[16x^5 - 32 \left\langle x - \frac{L}{2} \right\rangle^5 - 40Lx^4 + 40L^2x^3 - 15L^4x \right] / 960 EI \quad \blacktriangleleft$$

(b) Deflection at C. $\left(y \text{ at } x = \frac{L}{2} \right)$

$$y_C = \frac{w_0 L^4}{960 EI} \left(\frac{1}{2} - 0 - \frac{5}{2} + 5 - \frac{15}{2} \right) = -\frac{3w_0 L^4}{640 EI} \quad y_C = \frac{3w_0 L^4}{640 EI} \downarrow \quad \blacktriangleleft$$



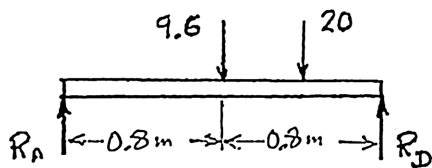
PROBLEM 9.45

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at point C. Use $E = 200$ GPa.



SOLUTION

Units: Forces in kN, lengths in m



$$+\circlearrowleft M_D = 0:$$

$$-1.6 R_A + (9.6)(0.8) + (20)(0.4) = 0$$

$$R_A = 9.8 \text{ kN}$$

$$w(x) = 12\langle x - 0.4 \rangle^0 - 12\langle x - 1.2 \rangle^0 \text{ kN/m}$$

$$\frac{dV}{dx} = -w(x) = -12\langle x - 0.4 \rangle^0 + 12\langle x - 1.2 \rangle^0 \text{ kN/m}$$

$$\frac{dM}{dx} = V = 9.8 - 12\langle x - 0.4 \rangle^1 + 12\langle x - 1.2 \rangle^1 - 20\langle x - 1.2 \rangle^0 \text{ kN}$$

$$EI \frac{d^2y}{dx^2} = M = 9.8x - 6\langle x - 0.4 \rangle^2 + 6\langle x - 1.2 \rangle^2 - 20\langle x - 1.2 \rangle^1 \text{ kN} \cdot \text{m}$$

$$EI \frac{dy}{dx} = 4.9x^2 - 2\langle x - 0.4 \rangle^3 + 2\langle x - 1.2 \rangle^3 - 10\langle x - 1.2 \rangle^2 + C_1 \text{ kN} \cdot \text{m}^2$$

$$EI y = 1.63333x^3 - \frac{1}{2}\langle x - 0.4 \rangle^4 + \frac{1}{2}\langle x - 1.2 \rangle^4 - \frac{10}{3}\langle x - 1.2 \rangle^3 + C_1 x + C_2 \text{ kN} \cdot \text{m}^3$$

$$[x = 0, y = 0]: 0 - 0 + 0 - 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = 1.6, y = 0]: (1.63333)(1.6)^3 - \frac{1}{2}(1.2)^4 + \frac{1}{2}(0.4)^4 - \frac{10}{3}(0.4)^3 + C_1(1.6) + 0 = 0$$

$$C_1 = -3.4080 \text{ kN} \cdot \text{m}^2$$

Data: $E = 200 \times 10^9 \text{ Pa}$, $I = 6.83 \times 10^6 \text{ mm}^4 = 6.83 \times 10^{-6} \text{ m}^4$

$$EI = (200 \times 10^9)(6.83 \times 10^{-6}) = 1.366 \times 10^6 \text{ N} \cdot \text{m}^2 = 1366 \text{ kN} \cdot \text{m}^2$$

(a) Slope at A. $\left(\frac{dy}{dx} \text{ at } x = 0 \right)$

$$EI \frac{dy}{dx} = 0 - 0 + 0 - 0 - 3.4080 \text{ kN} \cdot \text{m}^2$$

$$\theta_A = -\frac{3.4080}{1366} = -2.49 \times 10^{-3} \text{ rad}$$

$$\theta_A = 2.49 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

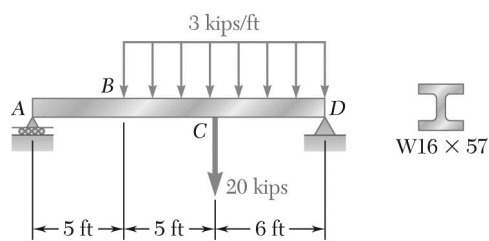
PROBLEM 9.45 (Continued)

(b) Deflection at C. (y at $x = 1.2$ m)

$$\begin{aligned}EIy_C &= (1.63333)(1.2)^3 - \frac{1}{2}(0.8)^4 + 0 - 0 - (3.4080)(1.2) + 0 \\&= -1.4720 \text{ kN} \cdot \text{m}^3\end{aligned}$$

$$y_C = -\frac{1.4720}{1366} = -1.078 \times 10^{-3} \text{ m}$$

$$y_C = 1.078 \text{ mm} \downarrow \blacktriangleleft$$

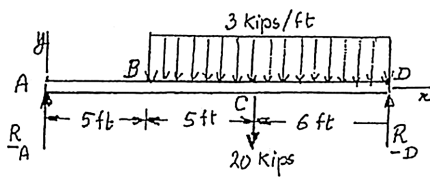


PROBLEM 9.46

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at the point C. Use $E = 29 \times 10^6$ psi.



SOLUTION



Units: Forces in kips, lengths in feet.

$$E = 29 \times 10^3 \text{ ksi}, \quad I = 758 \text{ in}^4$$

$$EI = (29 \times 10^3)(758) = 21.982 \times 10^6 \text{ kip} \cdot \text{in}^2 \\ = 152,650 \text{ kip} \cdot \text{ft}^2$$

$$+\circlearrowleft \Sigma M_D = 0: \quad (20)(6) + (3 \times 11)(5.5) - R_A(16) = 0 \quad R_A = 18.844 \text{ kips} \uparrow$$

Express the loading as a singularity function.

$$w(x) = 3\langle x - 5 \rangle^0$$

$$V(x) = R_A - \int w \, dx - 20\langle x - 10 \rangle^0 = 18.844 - 3\langle x - 5 \rangle^1 - 20\langle x - 10 \rangle^0$$

$$EI \frac{d^2 y}{dx^2} = M(x) = 18.844x - 1.5\langle x - 5 \rangle^2 - 20\langle x - 10 \rangle^1$$

$$EI \frac{dy}{dx} = 9.422x^2 - 0.5\langle x - 5 \rangle^3 - 10\langle x - 10 \rangle^2 + C_1$$

$$EI y = 3.141x^3 - 0.125\langle x - 5 \rangle^4 - 3.333\langle x - 10 \rangle^3 + C_1 x + C_2$$

Boundary conditions:

$$[x = 0, y = 0]: \quad C_2 = 0$$

$$[x = 16, y = 0]: \quad 3.141(16)^3 - (0.125)(11)^4 - (3.333)(6)^3 + 16 C_1 = 0 \quad C_1 = -644.7 \text{ kip} \cdot \text{ft}^2$$

$$EI \frac{dy}{dx} = 9.442x^2 - 0.5\langle x - 5 \rangle^3 - 10\langle x - 10 \rangle^2 - 644.7$$

$$EI y = 3.141x^3 - 0.125\langle x - 5 \rangle^4 - 3.333\langle x - 10 \rangle^3 - 644.7x$$

(a) Slope at A. $\left(\frac{dy}{dx} \text{ at } x = 0 \right)$

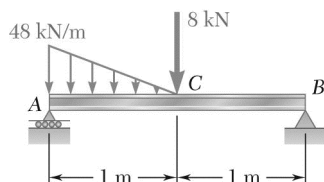
$$152,650 \theta_A = 0 + 0 + 0 - 644.7 \quad \theta_A = -4.22 \times 10^{-3} \text{ rad} \quad \theta_A = 4.22 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at C. $(y \text{ at } x = 10 \text{ ft})$

$$152,650 y_C = (3.141)(10)^3 - (0.125)(5)^4 + 0 - (644.7)(10) = -3384$$

$$y_C = -0.02217 \text{ ft}$$

$$y_C = 0.266 \text{ in.} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.47

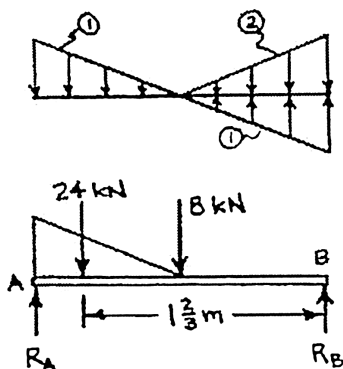
For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at the midpoint C. Use $E = 200$ GPa.

SOLUTION

Distributed loads: (1) $w_1(x) = w_0 - kx$ (2) $w_2 = k\langle x - 1 \rangle^1$

$$w_0 = 48 \text{ kN/m}, \quad k = 48 \text{ kN/m}^2$$

$$+\circlearrowleft \Sigma M_B = 0: -2R_A + (24)\left(1\frac{2}{3}\right) + (8)(1) = 0 \quad R_A = 24 \text{ kN} \uparrow$$



$$w(x) = w_0 - kx + k\langle x - 1 \rangle^1 = 48 - 48x + 48\langle x - 1 \rangle^1$$

$$\frac{dV}{dx} = -w = -48 + 48x - 48\langle x - 1 \rangle^1 \text{ kN/m}$$

$$\frac{dM}{dx} = V = 24 - 48x + 24x^2 - 24\langle x - 1 \rangle^2 - 8\langle x - 1 \rangle^0 \text{ kN}$$

$$EI \frac{d^2y}{dx^2} = M = 24x - 24x^2 + 8x^3 - 8\langle x - 1 \rangle^3 - 8\langle x - 1 \rangle^1 \text{ kN} \cdot \text{m}$$

$$EI \frac{dy}{dx} = 12x^2 - 8x^3 + 2x^4 - 2\langle x - 1 \rangle^4 - 4\langle x - 1 \rangle^2 + C_1 \text{ kN} \cdot \text{m}^2$$

$$EIy = 4x^3 - 2x^4 + \frac{2}{5}x^5 - \frac{2}{5}\langle x - 1 \rangle^5 - \frac{4}{3}\langle x - 1 \rangle^3 + C_1x + C_2 \text{ kN} \cdot \text{m}^3$$

$$[x = 0, y = 0]: 0 - 0 + 0 - 0 - 0 + 0 + C_2 = 0 \quad \therefore C_2 = 0$$

$$[x = 2, y = 0]: 4(2)^3 - 2(2)^4 + \frac{2}{5}(2)^5 - \frac{2}{5}(1)^5 - \frac{4}{3}(1)^3 + C_1(2) = 0 \quad \therefore C_1 = -\frac{83}{15} \text{ kN} \cdot \text{m}^2$$

Data: $E = 200(10^6) \text{ kN/m}^2 \quad I = 5.12(10^6) \text{ mm}^4 = 5.12(10^{-6}) \text{ m}^4$

$$EI = (200 \times 10^6)(5.12 \times 10^{-6}) = 1024 \text{ kN} \cdot \text{m}^2$$

(a) Slope at A. $\left(\frac{dy}{dx} \text{ at } x = 0 \right)$

$$EI\theta_A = 0 - 0 + 0 - 0 - 0 - \frac{83}{15} \text{ kN} \cdot \text{m}^2$$

$$\theta_A = -\frac{83/15}{1024} = -5.4036 \times 10^{-3} \text{ rad}$$

$$\theta_A = -5.40 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

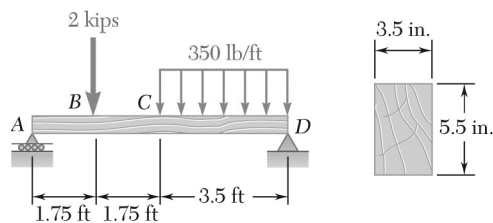
PROBLEM 9.47 (Continued)

(b) Deflection at C. (y at $x = 1$ m)

$$EIy_C = 4(1)^3 - 2(1)^4 + \frac{2}{5}(1)^4 - 0 - 0 - \frac{83}{15}(1) = -3.1333 \text{ kN} \cdot \text{m}^3$$

$$y_C = -\frac{3.1333}{1024} = -3.0599 \times 10^{-3} \text{ m}$$

$$y_C = 3.06 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.48

For the timber beam and loading shown, determine (a) the slope at end A, (b) the deflection at the midpoint C. Use $E = 1.6 \times 10^6$ psi.

SOLUTION

Units: Forces in kips; lengths in ft.

$$+\circlearrowleft \Sigma M_D = 0: -7R_A + (2)(5.25) + (1.225)(1.75) = 0$$

$$R_A = 1.80625 \text{ kips}$$

$$w(x) = 0.350\langle x - 3.5 \rangle^0$$

$$\frac{dV}{dx} = -w = -0.35\langle x - 3.5 \rangle^0$$

$$\frac{dM}{dx} = V = 1.80625 - 2\langle x - 1.75 \rangle^0 - 0.35\langle x - 3.5 \rangle^1 \quad R_A \quad R_D$$

$$EI \frac{d^2y}{dx^2} = M = 1.80625x - 2\langle x - 1.75 \rangle^1 - 0.175\langle x - 3.5 \rangle^2 \quad \text{kip} \cdot \text{ft}$$

$$EI \frac{dy}{dx} = 0.903125x^2 - 1\langle x - 1.75 \rangle^2 - 0.05833\langle x - 3.5 \rangle^3 + C_1 \quad \text{kip} \cdot \text{ft}^2$$

$$EIy = 0.301042x^3 - \frac{1}{3}\langle x - 1.75 \rangle^3 - 0.014583\langle x - 3.5 \rangle^4 + C_1x + C_2 \quad \text{kip} \cdot \text{ft}^3$$

$$[x = 0, y = 0] \quad C_2 = 0$$

$$[x = 7, y = 0] \quad (0.301042)(7)^3 - \frac{1}{3}(5.25)^3 - 0.014583(3.5)^4 + C_1(7) + 0 = 0$$

$$C_1 = -7.54779 \text{ kip} \cdot \text{ft}^2$$

Data: $E = 1.6 \times 10^6 \text{ psi} = 1.6 \times 10^3 \text{ ksi}$

$$I = \frac{1}{12}(3.5)(5.5)^3 = 48.526 \text{ in}^3$$

$$EI = (1.6 \times 10^3)(48.526) = 77.6417 \text{ kip} \cdot \text{in}^2 = 539.18 \text{ kip} \cdot \text{ft}^2$$

(a) Slope at A. $\left(\frac{dy}{dx} \text{ at } x = 0 \right)$

$$EI \frac{dy}{dx} = 0 - 0 - 0 - 7.54779 \text{ kip} \cdot \text{ft}^2$$

$$\theta_A = -\frac{7.54779}{539.18} = -14.00 \times 10^{-3} \text{ rad}$$

$$\theta_A = 14.00 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

PROBLEM 9.48 (Continued)

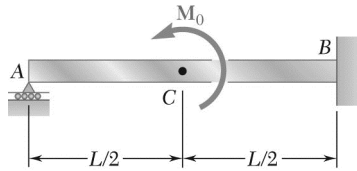
(b) Deflection at C.

(y at $x = 3.5$ ft)

$$\begin{aligned}EIy_C &= (0.301042)(3.5)^3 - \frac{1}{3}(1.75)^3 - 0 - (7.54779)(3.5) + 0 \\&= -15.297 \text{ kip} \cdot \text{ft}^3\end{aligned}$$

$$y_C = -\frac{15.297}{539.18} = -28.37 \times 10^{-3} \text{ ft}$$

$$y_C = 0.340 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.49

For the beam and loading shown, determine (a) the reaction at the roller support, (b) the deflection at point C.

$$[x = 0, y = 0] \quad \left[x = L, \frac{dy}{dx} = 0 \right]$$

$$[x = L, y = 0]$$

SOLUTION

For $0 \leq x \leq \frac{L}{2}, \quad M = R_A x$

For $\frac{L}{2} \leq x \leq L, \quad M = R_A x - M_0$

Then

$$EI \frac{d^2 y}{dx^2} = M = R_A x - M_0 \left\langle x - \frac{L}{2} \right\rangle^0$$

$$EI \frac{dy}{dx} = \frac{1}{2} R_A x^2 - M_0 \left\langle x - \frac{L}{2} \right\rangle^1 + C_1$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{1}{2} M_0 \left\langle x - \frac{L}{2} \right\rangle^2 + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 - 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad \frac{1}{2} R_A L^2 - M_0 \left(\frac{L}{2} \right) + C_1 = 0 \quad C_1 = \frac{1}{2} (M_0 L - R_A L^2)$$

$$[x = L, y = 0] \quad \frac{1}{6} R_A L^3 - \frac{1}{2} M_0 \left(\frac{L}{2} \right)^2 + \frac{1}{2} (M_0 L - R_A L^2) L + 0 = 0$$

$$-\frac{1}{3} R_A L^3 + \frac{3}{8} M_0 L^2 = 0$$

(a) Reaction at A. $M_A = 0, \quad R_A = \frac{9M_0}{8L} \uparrow \blacktriangleleft$

$$C_1 = \frac{1}{2} \left[M_0 L - \left(\frac{9M_0}{8L} \right) (L^2) \right] = -\frac{1}{16} M_0 L$$

$$EI y = \frac{1}{6} \left(\frac{9M_0}{8L} \right) x^3 - \frac{1}{2} M_0 \left\langle x - \frac{L}{2} \right\rangle^2 - \frac{1}{16} M_0 L x + 0$$

PROBLEM 9.49 (Continued)

Elastic curve.

$$y = \frac{M_0}{EIL} \left\{ \frac{9}{8}x^3 - \frac{1}{2}L \left\langle x - \frac{L}{2} \right\rangle^2 - \frac{1}{16}L^2x \right\}$$

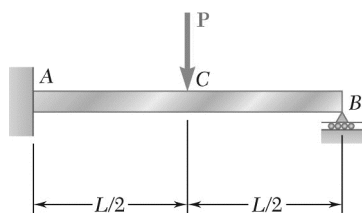
(b) Deflection at point C.

$$\left(y \text{ at } x = \frac{L}{2} \right)$$

$$y_C = \frac{M_0}{EIL} \left\{ \left(\frac{1}{6} \right) \left(\frac{9}{8} \right) \left(\frac{L}{2} \right)^3 - 0 - \left(\frac{1}{16}L^2 \right) \left(\frac{L}{2} \right) \right\}$$

$$= -\frac{M_0L^2}{128EI}$$

$$y_C = \frac{M_0L^2}{128EI} \downarrow \blacktriangleleft$$



PROBLEM 9.50

For the beam and loading shown, determine (a) the reaction at the roller support, (b) the deflection at point C.

$$[x = 0, y = 0] \quad [x = L, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

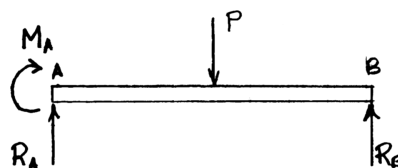
SOLUTION

$$+\uparrow \Sigma F_y = 0: R_A + R_B - P = 0 \quad R_A = P - R_B$$

$$+\curvearrowright \Sigma M_A = 0: -M_A - P\frac{L}{2} + R_B L = 0 \quad M_A = R_B L - \frac{1}{2}PL$$

Reactions are statically indeterminate.

$$\begin{aligned} \frac{dM}{dx} &= V = R_A - P\left\langle x - \frac{L}{2} \right\rangle^0 \\ EI \frac{d^2y}{dx^2} &= M = M_A + R_A x - P\left\langle x - \frac{L}{2} \right\rangle^1 \\ EI \frac{dy}{dx} &= M_A x + \frac{1}{2}R_A x^2 - \frac{1}{2}P\left\langle x - \frac{L}{2} \right\rangle^2 + C_1 \\ EI y &= \frac{1}{2}M_A x^2 + \frac{1}{6}R_A x^3 - \frac{1}{6}P\left\langle x - \frac{L}{2} \right\rangle^3 + C_1 x + C_2 \end{aligned}$$



$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 + 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = L, y = 0] \quad \frac{1}{2}M_A L^2 + \frac{1}{6}R_A L^3 - \frac{1}{6}P\left(\frac{L}{2}\right)^3 + 0 + 0 = 0$$

$$\frac{1}{2}\left(R_B L - \frac{1}{2}PL\right)L^2 + \frac{1}{6}(P - R_B)L^3 - \frac{1}{48}PL^3 = 0$$

$$\left(\frac{1}{2} - \frac{1}{6}\right)R_B L^3 = \left(\frac{1}{4} - \frac{1}{6} + \frac{1}{48}\right)PL^3 \quad \frac{1}{3}R_B = \frac{5}{48}P \quad (a) \quad R_B = \frac{5}{16}P \uparrow \blacktriangleleft$$

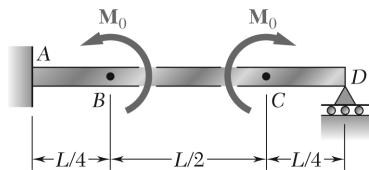
$$R_A = P - \frac{5}{16}P = \frac{11}{16}P$$

$$M_A = \frac{5}{16}PL - \frac{1}{2}PL = -\frac{3}{16}PL$$

PROBLEM 9.50 (Continued)

(b) Deflection at C. $\left(y \text{ at } x = \frac{L}{2} \right)$

$$\begin{aligned} y_C &= \frac{1}{EI} \left\{ \frac{1}{2} M_A \left(\frac{L}{2} \right)^2 + \frac{1}{6} R_A \left(\frac{L}{2} \right)^3 + 0 + 0 + 0 \right\} \\ &= \frac{PL^3}{EI} \left\{ \left(\frac{1}{2} \right) \left(-\frac{3}{16} \right) \left(\frac{1}{4} \right) + \left(\frac{1}{6} \right) \left(\frac{11}{16} \right) \left(\frac{1}{8} \right) \right\} = -\frac{7}{168} \frac{PL^3}{EI} \end{aligned} \quad y_C = \frac{7}{168} \frac{PL^3}{EI} \downarrow \blacktriangleleft$$



PROBLEM 9.51

For the beam and loading shown, determine (a) the reaction at the roller support, (b) the deflection at point B.

SOLUTION

$$+\uparrow \Sigma F_y = 0: R_A + R_D = 0 \quad R_A = -R_D$$

$$+\circlearrowleft \Sigma M_A = 0: -M_A + M_0 - M_0 + RL = 0$$

$$M_A = R_D L$$

$$M(x) = M_A + R_A x - M_0 \left\langle x - \frac{L}{4} \right\rangle^0 + M_0 \left\langle x - \frac{3L}{4} \right\rangle^0$$

$$EI \frac{d^2 y}{dx^2} = R_D L - R_D x - M_0 \left\langle x - \frac{L}{4} \right\rangle^0 + M_0 \left\langle x - \frac{3L}{4} \right\rangle^0$$

$$EI \frac{dy}{dx} = R_D Lx - \frac{1}{2} R_D x^2 - M_0 \left\langle x - \frac{L}{4} \right\rangle^1 + M_0 \left\langle x - \frac{3L}{4} \right\rangle^1 + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 - 0 - 0 + 0 + C_1 = 0$$

$$C_1 = 0$$

$$EI y = \frac{1}{2} R_D L x^2 - \frac{1}{6} R_D x^3 - \frac{1}{2} M_0 \left\langle x - \frac{L}{4} \right\rangle^1 + \frac{1}{2} M_0 \left\langle x - \frac{3L}{4} \right\rangle^1 + C_2$$

$$[x = 0, y = 0] \quad 0 - 0 - 0 + 0 + C_2 = 0$$

$$C_2 = 0$$

$$[x = L, y = 0] \quad \frac{1}{2} R_D L^3 - \frac{1}{6} R_D L^3 - \frac{1}{2} M_0 \left(\frac{3L}{4} \right)^2 + \frac{1}{2} M_0 \left(\frac{L}{4} \right)^2 = 0$$

(a) Reaction at D.

$$R_D = \frac{3M_0}{4L} \uparrow \blacktriangleleft$$

$$EI y = \frac{1}{2} \left(\frac{3M_0}{4L} \right) L x^2 - \frac{1}{6} \left(\frac{3M_0}{4L} \right) x^3 - \frac{1}{2} M_0 \left\langle x - \frac{L}{4} \right\rangle^2 + \frac{1}{2} M_0 \left\langle x - \frac{3L}{4} \right\rangle^2$$

Elastic curve.

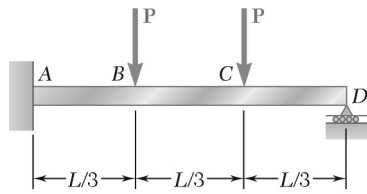
$$y = \frac{M_0}{EIL} \left\{ \frac{3}{8} L x^2 - \frac{1}{8} x^3 - \frac{1}{2} L \left\langle x - \frac{L}{4} \right\rangle^2 + \frac{1}{2} L \left\langle x - \frac{3L}{4} \right\rangle^2 \right\}$$

(b) Deflection at point B.

$$\left(y \text{ at } x = \frac{L}{4} \right)$$

$$y_B = \frac{M_0}{EIL} \left\{ \left(\frac{3}{8} L \right) \left(\frac{L}{4} \right)^2 - \left(\frac{1}{8} \right) \left(\frac{L}{4} \right)^3 - 0 + 0 \right\}$$

$$y_B = \frac{11M_0 L^2}{512EI} \uparrow \blacktriangleleft$$



PROBLEM 9.52

For the beam and loading shown, determine (a) the reaction at the roller support, (b) the deflection at point B.

SOLUTION

$$+\uparrow \Sigma F_y = 0: \quad R_A - P - P + R_D = 0 \quad R_A = 2P - R_D$$

$$+\circlearrowleft \Sigma M_A = 0: \quad -M_A - \frac{PL}{3} - \frac{2PL}{3} + R_DL = 0$$

$$M_A = R_DL - PL$$

$$\frac{dM}{dx} = V = R_A - P \left\langle x - \frac{L}{3} \right\rangle^0 - P \left\langle x - \frac{2L}{3} \right\rangle^0$$

$$EI \frac{d^2y}{dx^2} = M = M_A + R_Ax - P \left\langle x - \frac{L}{3} \right\rangle^1 - P \left\langle x - \frac{2L}{3} \right\rangle^1$$

$$EI \frac{dy}{dx} = M_Ax + \frac{1}{2}R_Ax^2 - \frac{1}{2}P \left\langle x - \frac{L}{3} \right\rangle^2 - \frac{1}{2}P \left\langle x - \frac{2L}{3} \right\rangle^2 + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 - 0 - 0 + C_1 = 0 \quad C_1 = 0$$

$$EIy = \frac{1}{2}M_Ax^2 + \frac{1}{6}R_Ax^3 - \frac{1}{6}P \left\langle x - \frac{L}{3} \right\rangle^3 - \frac{1}{6}P \left\langle x - \frac{2L}{3} \right\rangle^3 + C_2$$

$$[x = 0, y = 0] \quad 0 + 0 - 0 - 0 + C_2 = 0 \quad C_2 = 0$$

$$EIy = \frac{1}{2}M_Ax^2 + \frac{1}{6}R_Ax^3 - \frac{1}{6}P \left\langle x - \frac{L}{3} \right\rangle^3 - \frac{1}{6}P \left\langle x - \frac{2L}{3} \right\rangle^3$$

$$[x = L, y = 0] \quad \frac{1}{2}(R_DL - PL)L^2 + \frac{1}{6}(2P - R_D)L^3 - \frac{1}{6}P \left(\frac{2L}{3} \right)^3 - \frac{1}{6}P \left(\frac{L}{3} \right)^3 = 0$$

$$\frac{1}{3}R_DL^3 - \frac{2}{9}PL^3 = 0$$

(a) Reaction at D.

$$R_D = \frac{2}{3}P \uparrow \blacktriangleleft$$

$$M_A = \frac{2}{3}PL - PL = -\frac{1}{3}PL \quad R_A = 2P - \frac{2}{3}P = \frac{4}{3}P$$

$$EIy = \frac{1}{2} \left(-\frac{1}{3}PL \right) x^2 + \frac{1}{6} \left(\frac{4}{3}P \right) x^3 - \frac{1}{6}P \left\langle x - \frac{L}{3} \right\rangle^3 - \frac{1}{6}P \left\langle x - \frac{2L}{3} \right\rangle^3$$

PROBLEM 9.52 (Continued)

Elastic curve.

$$y = \frac{P}{EI} \left\{ -\frac{1}{6} Lx^2 + \frac{2}{9} x^3 - \frac{1}{6} \left\langle x - \frac{L}{3} \right\rangle^3 - \frac{1}{6} \left\langle x - \frac{2L}{3} \right\rangle^3 \right\}$$

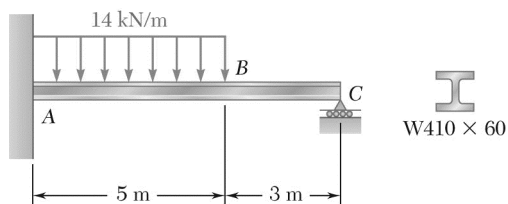
(b) Deflection at B.

$$\left(y \text{ at } x = \frac{L}{3} \right)$$

$$y_B = \frac{P}{EI} \left\{ -\left(\frac{1}{6} L \right) \left(\frac{L}{3} \right)^2 + \frac{2}{9} \left(\frac{L}{3} \right)^3 - 0 - 0 \right\}$$

$$= -\frac{5 PL^3}{486 EI}$$

$$y_B = \frac{5 PL^3}{486 EI} \downarrow \blacktriangleleft$$



PROBLEM 9.53

For the beam and loading shown, determine (a) the reaction at point C, (b) the deflection at point B. Use $E = 200$ GPa.

$$[x = 0, y = 0] \quad [x = 8, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Units: Forces in kN; lengths in m.

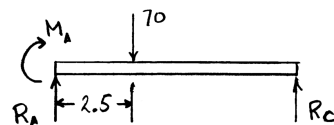
$$+\uparrow \Sigma F_y = 0: R_A - 70 + R_C = 0$$

$$R_A = 70 - R_C \quad \text{kN}$$

$$+\curvearrowright M_A = 0: -M_A - (70)(2.5) + 8R_C = 0$$

$$M_A = 8R_C - 175 \quad \text{kN} \cdot \text{m}$$

Reactions are statically indeterminate.



$$w(x) = 14 - 14\langle x - 5 \rangle^0 \quad \text{kN/m}$$

$$\frac{dV}{dx} = -w = -14 + 14\langle x - 5 \rangle^0 \quad \text{kN/m}$$

$$\frac{dM}{dx} = V = R_A - 14x + 14\langle x - 5 \rangle^1 \quad \text{kN}$$

$$EI \frac{d^2y}{dx^2} = M = M_A + R_A x - 7x^2 + 7\langle x - 5 \rangle^2 \quad \text{kN} \cdot \text{m}$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 - \frac{7}{3} x^3 + \frac{7}{3} \langle x - 5 \rangle^3 + C_1 \quad \text{kN} \cdot \text{m}^2$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 - \frac{7}{12} x^4 + \frac{7}{12} \langle x - 5 \rangle^4 + C_1 x + C_2 \quad \text{kN} \cdot \text{m}^3$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 + 0 + 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = 8, y = 0] \quad \frac{1}{2} M_A (8)^2 + \frac{1}{6} R_A (8)^3 - \frac{7}{12} (8)^4 + \frac{7}{12} (3)^4 + 0 + 0 = 0$$

PROBLEM 9.53 (Continued)

$$32(8R_C - 175) + \frac{512}{6}(70 - R_C) - \frac{28105}{12} = 0$$

$$170.667 R_C = 5600 - \frac{35840}{6} + \frac{28105}{12} = 1968.75 \quad R_C = 11.536 \text{ kN} \uparrow$$

(a) Reaction at C.

$$R_C = 11.546 \text{ kN} \uparrow \blacktriangleleft$$

$$M_A = (8)(11.536) - 175 = -82.715 \text{ kN} \cdot \text{m}$$

$$R_A = 70 - 11.536 = 58.464 \text{ kN}$$

Data:

$$E = 200 \times 10^9 \text{ Pa} \quad I = 216 \times 10^6 \text{ mm}^4 = 216 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(216 \times 10^{-6}) = 43.2 \times 10^6 \text{ N} \cdot \text{m}^2 = 43200 \text{ kN} \cdot \text{m}^2$$

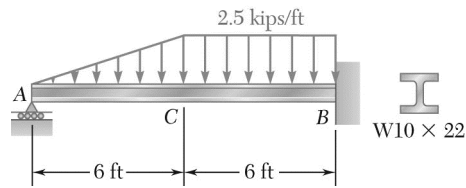
(b) Deflection at B.

(y at $x = 5 \text{ m}$)

$$EI y_B = \frac{1}{2}(-82.715)(5)^2 + \frac{1}{6}(58.464)(5)^3 - \frac{7}{12}(5)^4 = -180.52 \text{ kN} \cdot \text{m}^3$$

$$y_B = -\frac{180.52}{43200} = -4.18 \times 10^{-3} \text{ m}$$

$$y_B = 4.18 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.54

For the beam and loading shown, determine (a) the reaction at point A, (b) the deflection at point C. Use $E = 29 \times 10^6$ psi.

SOLUTION

$$k = \frac{2.5 \text{ kips/ft}}{6 \text{ ft}} = \frac{5}{12} \text{ kips/ft}^2$$

$$w(x) = \frac{5}{12}x - \frac{5}{12}\langle x - 6 \rangle^1 \text{ kip/ft}$$

$$\frac{dV}{dx} = -w(x) = -\frac{5}{12}x + \frac{5}{12}\langle x - 6 \rangle^1 \text{ kip/ft}$$

$$\frac{dM}{dx} = V = R_A - \frac{5}{24}x^2 + \frac{5}{24}\langle x - 6 \rangle^2 \text{ kip}$$

$$EI \frac{d^2y}{dx^2} = M = R_A x - \frac{5}{72}x^3 + \frac{5}{72}\langle x - 6 \rangle^3 \text{ kip} \cdot \text{ft}$$

$$EI \frac{dy}{dx} = \frac{1}{2}R_A x^2 - \frac{5}{288}x^4 + \frac{5}{288}\langle x - 6 \rangle^4 + C_1 \text{ kip} \cdot \text{ft}^2$$

$$EI y = \frac{1}{6}R_A x^3 - \frac{1}{288}x^5 + \frac{1}{288}\langle x - 6 \rangle^5 + C_1 x + C_2 \text{ kip} \cdot \text{ft}^3$$

$$[x = 0, y = 0] \quad 0 - 0 + 0 + 0 + C_2 = 0 \quad \therefore C_2 = 0$$

$$\left[x = 12 \text{ ft}, \frac{dy}{dx} = 0 \right] \quad \frac{1}{2}R_A(12)^2 - \frac{5}{288}(12)^4 + \frac{5}{288}(6)^4 + C_1 = 0$$

$$\therefore C_1 = 337.5 - 72R_A \text{ kip} \cdot \text{ft}^2$$

$$[x = 12 \text{ ft}, y = 0] \quad \frac{1}{6}R_A(12)^3 - \frac{1}{288}(12)^5 + \frac{1}{288}(6)^5 + (337.5 - 72R_A)(12) = 0$$

$$(864 - 288)R_A = 3213$$

$$R_A = 5.5781 \text{ kips}$$

(a) Reaction at A.

$$R_A = 5.58 \text{ kips} \uparrow \blacktriangleleft$$

$$C_1 = 337.5 - 72(5.5781) = -64.123 \text{ kip} \cdot \text{ft}^2$$

Data:

$$E = 29 \times 10^3 \text{ ksi} \quad I = 118 \text{ in}^4$$

$$EI = (29 \times 10^3)(118) = 3.422 \times 10^6 \text{ kip} \cdot \text{in}^2 = 23764 \text{ kip} \cdot \text{ft}^2$$

(b) Deflection at C.

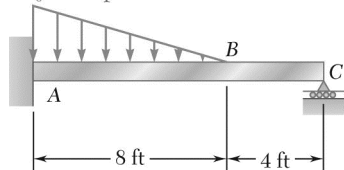
(y at $x = 6 \text{ ft}$)

$$\begin{aligned} EI y_C &= \frac{1}{6}(5.5781)(6)^3 - \frac{1}{288}(6)^5 + 0 - 64.123(6) \\ &= -210.93 \text{ kip} \cdot \text{ft}^3 \end{aligned}$$

$$y_C = -\frac{210.93}{23764} = -8.8760 \times 10^{-3} \text{ ft}$$

$$y_C = 0.1065 \text{ in.} \downarrow \blacktriangleleft$$

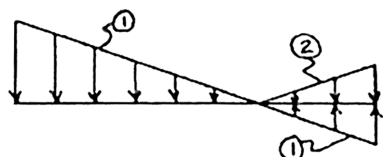
$$w_0 = 9 \text{ kips/ft}$$



PROBLEM 9.55

For the beam and loading shown, determine (a) the reaction at point C, (b) the deflection at point B. Use $E = 29 \times 10^6$ psi.

SOLUTION



Distributed loads:

$$k = \frac{9}{8}$$

$$(1) \quad w_1(x) = w_0 - kx \quad (2) \quad w_2(x) = k\langle x - 8 \rangle^1$$

$$w_0 = 9 \text{ kips/ft}, \quad k = \frac{9}{8} \text{ kips/ft}^2$$

$$+\uparrow \sum F_y = 0: R_A - 36 + R_C = 0 \quad R_A = (36 - R_C) \text{ kips}$$

$$+\curvearrowright \sum M_A = 0: 12R_C - M_A - \left(\frac{8}{3}\right)(36) = 0$$

$$M_A = (12R_C - 96) \text{ kip} \cdot \text{ft}$$

$$w(x) = w_0 - kx + k\langle x - 8 \rangle^1 = 9 - \frac{9}{8}x + \frac{9}{8}\langle x - 8 \rangle^1 \text{ kips/ft}$$

$$\frac{dV}{dx} = -w = -9 + \frac{9}{8}x - \frac{9}{8}\langle x - 8 \rangle^1 \text{ kips/ft}$$

$$\frac{dM}{dx} = V = R_A - 9x + \frac{9}{16}x^2 - \frac{9}{16}\langle x - 8 \rangle^2 \text{ kips}$$

$$EI \frac{d^2y}{dx^2} = M = M_A + R_Ax - \frac{9}{2}x^2 + \frac{3}{16}x^3 - \frac{3}{16}\langle x - 8 \rangle^3 \text{ kip} \cdot \text{ft}$$

$$EI \frac{dy}{dx} = M_Ax + \frac{1}{2}R_Ax^2 - \frac{3}{2}x^3 + \frac{3}{64}x^4 - \frac{3}{64}\langle x - 8 \rangle^4 + C_1 \text{ kip} \cdot \text{ft}^2$$

$$EIy = \frac{1}{2}M_Ax^2 + \frac{1}{6}R_Ax^3 - \frac{3}{8}x^4 + \frac{3}{320}x^5 - \frac{3}{320}\langle x - 8 \rangle^5 + C_1x + C_2 \text{ kip} \cdot \text{ft}^3$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]: C_1 = 0 \quad \left[x = 0, y = 0 \right]: C_2 = 0$$

$$\left[x = 12, y = 0 \right]: \frac{1}{2}M_A(12)^2 + \frac{1}{6}R_A(12)^3 - \frac{3}{8}(12)^4 + \frac{3}{320}(12)^5 - \frac{3}{320}(4)^5 + 0 + 0 = 0$$

$$72(12R_C - 96) + 288(36 - R_C) - 5452.8 = 0$$

$$R_C = 3.4667$$

$$R_C = 3.47 \text{ kips} \uparrow \blacktriangleleft$$

PROBLEM 9.55 (Continued)

(a) Reaction at C:

$$R_C = 3.47 \text{ kips } \uparrow \blacktriangleleft$$

$$M_A = 12(3.4667) - 96 = -54.400 \text{ kip} \cdot \text{ft} \quad R_A = 36 - 3.4667 = 32.533 \text{ kips}$$

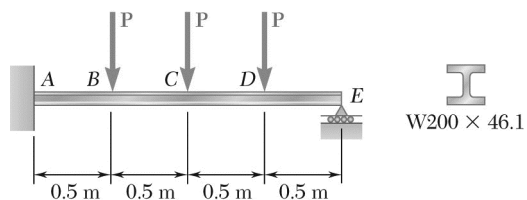
Data: $E = 29 \times 10^3 \text{ ksi} \quad I = 307 \text{ in}^4 \quad EI = (29 \times 10^3)(307) = 8.903 \times 10^6 \text{ kip} \cdot \text{in}^2$
 $= 61,826 \text{ kip} \cdot \text{ft}^2$

(b) Deflection at B.

(y at x = 8 ft)

$$EIy_B = \frac{1}{2}(-54.400)(8)^2 + \frac{1}{6}(32.533)(8)^3 - \frac{3}{8}(8)^4 + \frac{3}{320}(8)^5 - 0 = -193.451 \text{ kip} \cdot \text{ft}^3$$

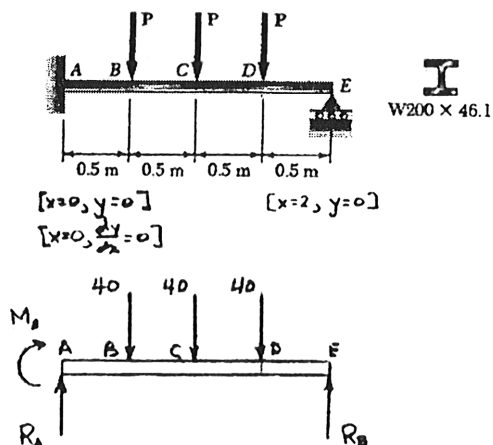
$$y_B = \frac{-193.451}{61,826} = -3.1290 \times 10^{-3} \text{ ft} \quad y_B = 0.0376 \text{ in. } \downarrow \blacktriangleleft$$



PROBLEM 9.56

For the beam shown and knowing that $P = 40$ kN, determine (a) the reaction at point E, (b) the deflection at point C. Use $E = 200$ GPa.

SOLUTION



Units: Forces in kN; lengths in m.

$$+\uparrow \sum F_y = 0: R_A - 40 - 40 - 40 + R_E = 0$$

$$R_A = 120 - R_E \quad \text{kN}$$

$$+\curvearrowright \sum M_A = 0: -M_A - 20 - 40 - 60 + 2R_E = 0$$

$$M_A = 2R_E - 120 \quad \text{kN} \cdot \text{m}$$

Reactions are statically indeterminate.

$$\frac{dM}{dx} = V = R_A - 40\langle x - 0.5 \rangle^0 - 40\langle x - 1 \rangle^0 - 40\langle x - 1.5 \rangle^0$$

$$EI \frac{d^2 y}{dx^2} = M = M_A + R_A x - 40\langle x - 0.5 \rangle^1 - 40\langle x - 1 \rangle^1 - 40\langle x - 1.5 \rangle^1$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 - 20\langle x - 0.5 \rangle^2 - 20\langle x - 1 \rangle^2 - 20\langle x - 1.5 \rangle^2 + C_1$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 - \frac{20}{3} \langle x - 0.5 \rangle^3 - \frac{20}{3} \langle x - 1 \rangle^3 - \frac{20}{3} \langle x - 1.5 \rangle^3 + C_1 x + C_2$$

$$[x = 0, \frac{dy}{dx} = 0] \quad 0 + 0 + 0 + 0 + 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = 2, y = 0] \quad \frac{1}{2} M_A (2)^2 + \frac{1}{6} R_A (2)^3 - \frac{20}{3} (1.5)^3 - \frac{20}{3} (1)^3 - \frac{20}{3} (0.5)^3 + 0 + 0 = 0$$

(a) Reaction at E.

$$\frac{1}{2} (2R_E - 120)(2)^2 + \frac{1}{6} (120 - R_E)(2)^3 = 30$$

$$2.66667 R_E = 30 + 240 - 160 = 110$$

$$R_E = 41.25 \text{ kN } \uparrow \quad \blacktriangleleft$$

$$M_A = (2)(41.25) - 120 = -37.5 \text{ kN} \cdot \text{m}$$

$$R_A = 120 - 41.25 = 78.25 \text{ kN}$$

PROBLEM 9.56 (Continued)

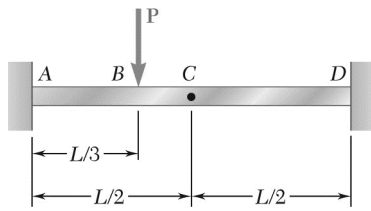
Data: $E = 200 \times 10^9 \text{ Pa}, \quad I = 45.8 \times 10^6 \text{ mm}^4 = 45.8 \times 10^{-6} \text{ m}^4$
 $EI = (200 \times 10^9)(45.8 \times 10^{-6}) = 9.16 \times 10^6 \text{ N} \cdot \text{m}^2 = 9160 \text{ kN} \cdot \text{m}^2$

(b) Deflection at C. (y at x = 1 m)

$$EIy_C = \frac{1}{2}(-37.5)(1)^2 + \frac{1}{6}(78.75)(1)^3 - \frac{20}{3}(0.5)^3 - 0 - 0 + 0 + 0$$
$$= -6.4583 \text{ kN} \cdot \text{m}^3$$

$$y_C = -\frac{6.4583}{9160} = -0.705 \times 10^{-3} \text{ m}$$

$$y_C = 0.705 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.57

For the beam and loading shown, determine (a) the reaction at point A, (b) the deflection at midpoint C.

SOLUTION

$$\frac{dM}{dx} = V = R_A - P \left\langle x - \frac{L}{3} \right\rangle^0$$

$$EI \frac{d^2y}{dx^2} = M = M_A + R_A x - P \left\langle x - \frac{L}{3} \right\rangle^1$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 - \frac{1}{2} P \left\langle x - \frac{L}{3} \right\rangle^2 + C_1$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 - \frac{1}{6} P \left\langle x - \frac{L}{3} \right\rangle^3 + C_1 x + C_2$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 - 0 + C_1 = 0 \quad \therefore C_1 = 0$$

$$\left[x = 0, y = 0 \right] \quad 0 + 0 - 0 + C_2 = 0 \quad \therefore C_2 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad M_A L + \frac{1}{2} R_A L^2 - \frac{1}{2} P \left(\frac{2L}{3} \right)^2 = 0 \quad (1)$$

$$\left[x = L, y = 0 \right] \quad \frac{1}{2} M_A L^2 + \frac{1}{6} R_A L^3 - \frac{1}{6} P \left(\frac{2L}{3} \right)^3 = 0 \quad (2)$$

(a) Solving Eqs. (1) and (2) simultaneously,

$$R_A = \frac{20}{27} P \quad M_A = -\frac{4}{27} PL$$

$$R_A = \frac{20}{27} P \uparrow \blacktriangleleft$$

$$M_A = \frac{4}{27} PL \curvearrowright \blacktriangleleft$$

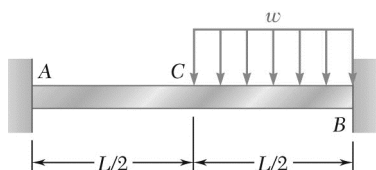
Elastic curve.

$$y = \frac{P}{EI} \left[-\frac{2}{27} Lx^2 + \frac{10}{81} x^3 - \frac{1}{6} \left\langle x - \frac{L}{3} \right\rangle^3 \right]$$

(b) Deflection at midpoint C. $\left(y \text{ at } x = \frac{L}{2} \right)$

$$y_C = \frac{P}{EI} \left[-\frac{2}{27} L \left(\frac{L}{2} \right)^2 + \frac{10}{81} \left(\frac{L}{2} \right)^3 - \frac{1}{6} \left(\frac{L}{6} \right)^3 \right] = -\frac{5PL^3}{1296EI}$$

$$y_C = \frac{5PL^3}{1296EI} \downarrow \blacktriangleleft$$



PROBLEM 9.58

For the beam and loading shown, determine (a) the reaction at point A, (b) the deflection at midpoint C.

SOLUTION

$$w(x) = w \left\langle x - \frac{L}{2} \right\rangle^0$$

$$\frac{dV}{dx} = -w(x) = -w \left\langle x - \frac{L}{2} \right\rangle^0$$

$$\frac{dM}{dx} = V = R_A - w \left\langle x - \frac{L}{2} \right\rangle^1$$

$$EI \frac{d^2y}{dx^2} = M = M_A + R_A x - \frac{1}{2} w \left\langle x - \frac{L}{2} \right\rangle^2$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{2} R_A x^2 - \frac{1}{6} w \left\langle x - \frac{L}{2} \right\rangle^3 + C_1$$

$$EI y = \frac{1}{2} M_A x^2 + \frac{1}{6} R_A x^3 - \frac{1}{24} w \left\langle x - \frac{L}{2} \right\rangle^4 + C_1 x + C_2$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 - 0 + C_1 = 0 \quad C_1 = 0$$

$$[x = 0, y = 0] \quad 0 + 0 - 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad M_A L + \frac{1}{2} R_A L^2 - \frac{1}{6} w \left(\frac{L}{2} \right)^3 = 0 \quad (1)$$

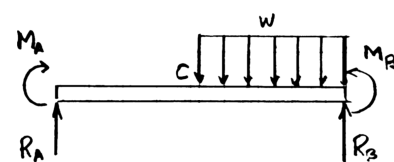
$$[x = L, y = 0] \quad \frac{1}{2} M_A L^2 + \frac{1}{6} R_A L^2 - \frac{1}{24} w \left(\frac{L}{2} \right)^4 = 0 \quad (2)$$

Solving Eqs. (1) and (2) simultaneously,

$$(a) \quad R_A = \frac{3wL}{32} \quad M_A = -\frac{5wL^2}{192} \quad R_A = \frac{3wL}{32} \uparrow \blacktriangleleft$$

$$M_A = \frac{5wL^2}{192} \curvearrowright \blacktriangleleft$$

$$EI y = -\frac{5}{384} w L^2 x^2 + \frac{3}{192} w L x^3 - \frac{1}{24} w \left\langle x - \frac{L}{2} \right\rangle^4$$



PROBLEM 9.58 (Continued)

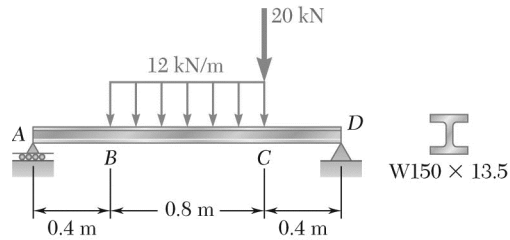
Elastic curve.

$$y = \frac{w}{EI} \left\{ -\frac{5}{384} L^2 x^2 + \frac{3}{192} L x^3 - \frac{1}{24} \left\langle x - \frac{L}{2} \right\rangle^4 \right\}$$

(b) Deflection at midpoint C. $\left(y \text{ at } x = \frac{L}{2} \right)$

$$\begin{aligned} y_C &= \frac{w}{EI} \left\{ \left(-\frac{5}{384} L^2 \right) \left(\frac{L}{2} \right)^2 + \left(\frac{3}{192} L \right) \left(\frac{L}{2} \right)^3 - 0 \right\} \\ &= -\frac{wL^4}{768EI} \end{aligned}$$

$$y_B = \frac{wL^4}{768EI} \downarrow \blacktriangleleft$$



PROBLEM 9.59

For the beam and loading of Prob. 9.45, determine the magnitude and location of the largest downward deflection.

SOLUTION

See solution to Prob. 9.45 for the derivation of the equations used in the following:

$$EI = EI = 1366 \text{ kN} \cdot \text{m}^2$$

$$EI \frac{dy}{dx} = 4.9x^2 - 2\langle x - 0.4 \rangle^3 + 2\langle x - 1.2 \rangle^3 - 10\langle x - 1.2 \rangle^2 - 3.4080 \text{ kN} \cdot \text{m}^2$$

$$EIy = 1.63333x^3 - \frac{1}{2}\langle x - 0.4 \rangle^4 + \frac{1}{2}\langle x - 1.2 \rangle^4 - \frac{10}{3}\langle x - 1.2 \rangle^3 - 3.4080x \text{ kN} \cdot \text{m}^3$$

To find the location of maximum $|y|$, set $\frac{dy}{dx} = 0$. Assume $0.4 < x < 1.2$.

$$4.9x^2 - 2(x - 0.4)^3 - 3.4080 = f(x) = 0$$

Solve by iteration: $x = 0.8 \quad 0.858 \quad 0.857 \quad 0.8570$

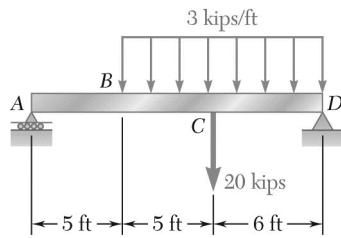
$$x_m = 0.8570 \text{ m} \quad \blacktriangleleft$$

$$df/dx = 6.88 \quad 7.123 \quad 7.145$$

$$\begin{aligned} EIy_m &= (1.63333)(0.8570)^3 - \frac{1}{2}(0.8570 - 0.4)^4 - (3.4080)(0.8570) \\ &= -1.9144 \text{ kN} \cdot \text{m} \end{aligned}$$

$$y_m = -\frac{1.9144}{1366} = -1.401 \times 10^{-3} \text{ m}$$

$$y_m = 1.401 \text{ mm} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.60

For the beam and loading of Prob. 9.46, determine the magnitude and location of the largest downward deflection.

SOLUTION

See solution to Prob. 9.46 for the derivation of following:

$$EI = 152,650 \text{ kip} \cdot \text{ft}^2$$

$$EI \frac{dy}{dx} = 9.422x^2 - 0.5\langle x - 5 \rangle^3 - 10\langle x - 10 \rangle^2 - 644.7 \quad \text{kip} \cdot \text{ft}^2$$

$$EIy = 3.141x^3 - 0.125\langle x - 5 \rangle^4 - 3.333\langle x - 10 \rangle^3 - 644.7x \quad \text{kip} \cdot \text{ft}^3$$

To find the location of maximum $|y|$, set $\frac{dy}{dx} = 0$.

Assume $5 \leq x < 10$.

$$EI \frac{dy}{dx} = 9.422x^2 - 0.5(x - 5)^3 - 644.7 = f(x) = 0 \quad (1)$$

$$\frac{df}{dx} = 18.844x - 1.5(x - 5)^2$$

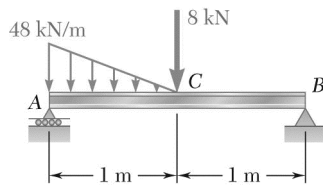
Solve Eq. (1) by iteration:

$$x_{i+1} = x_i - \frac{f}{df/dx}$$

$x = 9$	8.406	8.397	$x_m = 8.40 \text{ ft} \blacktriangleleft$
$f = 86.48$	1.310	0.0415	
$df/dx = 145.6$	141.0		

$$EIy_m = (3.141)(8.397)^3 - (0.125)(8.397 - 5)^4 - (644.7)(8.397) = -3570.5 \text{ kip} \cdot \text{ft}^3$$

$$152,650 y_m = -3570.5 \quad y_m = -0.02339 \text{ ft} \quad y_m = 0.281 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.61

For the beam and loading of Prob. 9.47, determine the magnitude and location of the largest downward deflection.

SOLUTION

See solution to Prob. 9.47 for the derivation of the equations used in the following:

$$EI = 1024 \text{ kN} \cdot \text{m}^2$$

$$EI \frac{dy}{dx} = 12x^2 - 8x^3 + 2x^4 - 2\langle x - 1 \rangle^4 - 4\langle x - 1 \rangle^2 - \frac{83}{15} \text{ kN} \cdot \text{m}^2$$

$$EI y = 4x^3 - 2x^4 + \frac{2}{5}x^5 - \frac{2}{5}\langle x - 1 \rangle^5 - \frac{4}{3}\langle x - 1 \rangle^3 - \frac{83}{15}x \text{ kN} \cdot \text{m}^3$$

To find location of maximum $|y|$, set $\frac{dy}{dx} = 0$. Assume $0 < x < 1 \text{ m}$.

$$EI \frac{dy}{dx} = 12x^2 - 8x^3 + 2x^4 - \frac{83}{15} = 0$$

Solving:

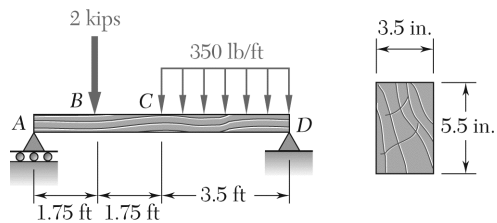
$$x = 0.94166 \text{ m}$$

$$x_m = 0.942 \text{ m} \quad \blacktriangleleft$$

$$\begin{aligned} EI y_m &= 4(0.94166)^3 - 2(0.94166)^4 + \frac{2}{5}(0.94166)^5 - \frac{83}{15}(0.94166) \\ &= -3.1469 \text{ kN} \cdot \text{m}^3 \end{aligned}$$

$$y_m = -\frac{3.1469}{1024} = -3.0731 \times 10^{-3} \text{ m}$$

$$y_m = 3.07 \text{ mm} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.62

For the beam and loading of Prob. 9.48, determine the magnitude and location of the largest downward deflection.

SOLUTION

See solution to Prob. 9.48 for the derivation of the equations used in the following:

$$EI = 539.18 \text{ kip} \cdot \text{ft}^2$$

$$EI \frac{dy}{dx} = 0.903125x^2 - 1(x - 1.75)^2 - 0.05833(x - 3.5)^2 - 7.54779 \text{ kip} \cdot \text{ft}^2$$

$$EIy = 0.301042x^3 - \frac{1}{3}(x - 1.75)^3 - 0.014583(x - 3.5)^4 - 7.54779x \text{ kip} \cdot \text{ft}^3$$

To find the location of maximum |y|, set $\frac{dy}{dx} = 0$. Assume $1.75 < x_m < 3.5$.

$$0.903125x_m^2 - 1(x_m - 1.75)^2 - 7.54779 = 0$$

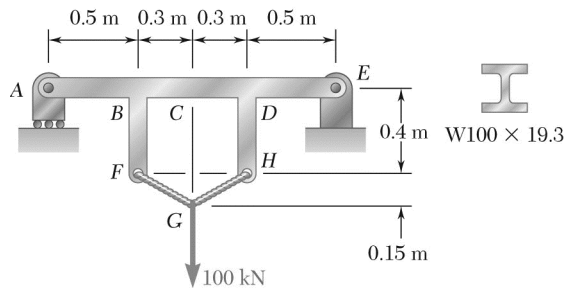
$$0.096875x_m^2 - 8.5x_m + 10.61029 = 0$$

$$x_m = \frac{3.5 - \sqrt{(3.5)^2 - (4)(0.096875)(10.61029)}}{(2)(0.096875)} \quad x_m = 3.34 \text{ ft} \quad \blacktriangleleft$$

$$EIy_m = (0.301042)(3.340)^3 - \frac{1}{3}(3.340 - 1.75)^3 - (7.54779)(3.340)$$

$$= -15.3328 \text{ kip} \cdot \text{ft}^3$$

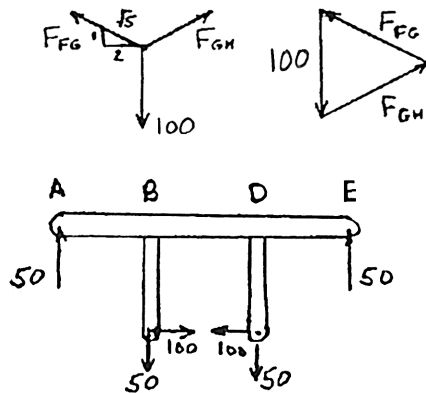
$$y_m = -\frac{15.3328}{539.18} = -12.44 \times 10^{-3} \text{ ft} \quad y_m = 0.341 \text{ in.} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.63

The rigid bars BF and DH are welded to the rolled-steel beam AE as shown. Determine for the loading shown (a) the deflection at point B , (b) the deflection at midpoint C of the beam. Use $E = 200$ GPa.

SOLUTION



Use joint G as a free body,

By symmetry, $F_{GH} = F_{FG}$

$$+\uparrow \sum F_y = 0: 2F_{GHy} - 100 = 0 \quad F_{GHy} = 50 \text{ kN}$$

$$F_{GHx} = 2 F_{GHy} = 100 \text{ kN.}$$

Forces in kN; lengths in m.

$$V = 50 - 50\langle x - 0.5 \rangle^0 - 50\langle x - 1.1 \rangle^0 \quad \text{kN}$$

$$M = 50x - 50\langle x - 0.5 \rangle^1 - 50\langle x - 1.1 \rangle^0 + 40\langle x - 0.5 \rangle^0 - 40\langle x - 1.1 \rangle^0 \quad \text{kN} \cdot \text{m}$$

$$EI \frac{dy}{dx} = 25x^2 - 25\langle x - 0.5 \rangle^2 - 25\langle x - 1.1 \rangle^2 - 40\langle x - 0.5 \rangle^1 + 40\langle x - 1.1 \rangle^1 + C_1 \quad \text{kN} \cdot \text{m}^2$$

$$EIy = \frac{25}{3}x^3 - \frac{25}{3}\langle x - 0.5 \rangle^3 - \frac{25}{3}\langle x - 1.1 \rangle^3 - 20\langle x - 0.5 \rangle^2 + 20\langle x - 1.1 \rangle^2 + C_1x + C_2 \quad \text{kN} \cdot \text{m}^3$$

$$[x = 0, y = 0] \quad C_2 = 0$$

$$[x = 1.6, y = 0]$$

$$\left(\frac{25}{3}\right)(1.6)^3 - \left(\frac{25}{3}\right)(1.1)^3 - \left(\frac{25}{3}\right)(0.5)^3 - (20)(1.1)^2 + (20)(0.5)^2 + C_1(1.6) + 0 = 0$$

$$C_1 = -1.75 \text{ kN} \cdot \text{m}^3$$

For EIy_B , $x = 0.5 \text{ m}$

$$EIy_B = \left(\frac{25}{3}\right)(0.5)^3 - 0 - 0 + 0 - 0 - (1.75)(0.5) = 0.1667 \text{ kN} \cdot \text{m}^3$$

For EIy_C , $x = 0.8 \text{ m}$

$$EIy_C = \left(\frac{25}{3}\right)(0.8)^3 - \left(\frac{25}{3}\right)(0.3)^3 - 0 - (20)(0.3)^2 - 0 - (1.75)(0.8) + 0$$

$$= -0.8417 \text{ kN} \cdot \text{m}^3$$

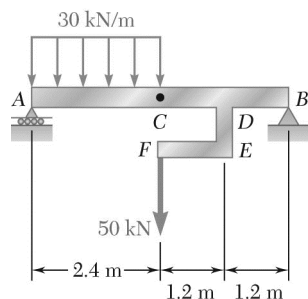
PROBLEM 9.63 (Continued)

For W 100 × 19.3 rolled-steel shape, $I = 4.70 \times 10^6 \text{ mm}^4 = 4.70 \times 10^{-6} \text{ m}^4$

$$EI = (200 \times 10^9)(4.70 \times 10^{-6}) = 940 \times 10^3 \text{ N} \cdot \text{m}^2 = 940 \text{ kN} \cdot \text{m}^2$$

$$(a) \quad y_B = \frac{0.1667}{940} = 0.177 \times 10^{-3} \text{ m} \qquad y_B = 0.177 \text{ mm} \uparrow \blacktriangleleft$$

$$(b) \quad y_C = \frac{0.8417}{940} = 0.895 \times 10^{-3} \text{ m} \qquad y_C = 0.895 \text{ mm} \uparrow \blacktriangleleft$$



PROBLEM 9.64

The rigid bar DEF is welded at point D to the rolled-steel beam AB . For the loading shown, determine (a) the slope at point A , (b) the deflection at midpoint C of the beam. Use $E = 200$ GPa.

SOLUTION

Units: Forces in kN; lengths in meters.

$$\begin{aligned} +\curvearrowright M_B = 0: & -4.8R_A + (30)(2.4)(3.6) \\ & + (50)(2.4) = 0 \end{aligned}$$

$$R_A = 79 \text{ kN} \uparrow$$

$$I = 212 \times 10^6 \text{ mm}^4 = 212 \times 10^{-6} \text{ m}^4$$

$$\begin{aligned} EI &= (200 \times 10^9)(212 \times 10^{-6}) = 42.4 \times 10^6 \text{ N} \cdot \text{m}^2 \\ &= 42400 \text{ kN} \cdot \text{m}^2 \end{aligned}$$

$$w(x) = 30 - 30\langle x - 2.4 \rangle^0$$

$$\frac{dV}{dx} = -w = -30 + 30\langle x - 2.4 \rangle^0 \quad \text{kN/m}$$

$$\frac{dM}{dx} = V = 79 - 30x + 30\langle x - 2.4 \rangle^1 - 50\langle x - 3.6 \rangle^0 \quad \text{kN}$$

$$EI \frac{d^2y}{dx^2} = M = 79x - 15x^2 + 15\langle x - 2.4 \rangle^2 - 50\langle x - 3.6 \rangle^1 - 60\langle x - 3.6 \rangle^0 \quad \text{kN} \cdot \text{m}$$

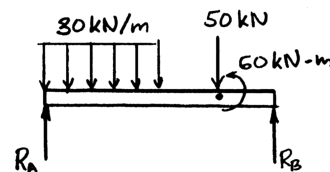
$$EI \frac{dy}{dx} = \frac{79}{2}x^2 - 5x^3 + 5\langle x - 2.4 \rangle^3 - 25\langle x - 3.6 \rangle^2 - 60\langle x - 3.6 \rangle^1 + C_1 \quad \text{kN} \cdot \text{m}^2$$

$$EIy = \frac{79}{6}x^3 - \frac{5}{4}x^4 + \frac{5}{4}\langle x - 2.4 \rangle^4 - \frac{25}{3}\langle x - 3.6 \rangle^3 - 30\langle x - 3.6 \rangle^2 + C_1x + C_2 \quad \text{kN} \cdot \text{m}^3$$

$$[x = 0, y = 0] \quad 0 - 0 + 0 - 0 - 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$\begin{aligned} [x = 4.8, y = 0] \quad & \left(\frac{79}{6}\right)(4.8)^3 - \left(\frac{5}{4}\right)(4.8)^4 + \left(\frac{5}{4}\right)(2.4)^4 \\ & - \left(\frac{25}{3}\right)(1.2)^3 - (30)(1.2)^2 + 4.8C_1 = 0 \end{aligned}$$

$$C_1 = -161.76 \text{ kN} \cdot \text{m}^2$$



PROBLEM 9.64 (Continued)

(a) Slope at point A. $\left(\frac{dy}{dx} \text{ at } x = 0\right)$

$$EI\left(\frac{dy}{dx}\right)_A = 0 - 0 + 0 - 0 - 0 - 161.76$$

$$= -161.76 \text{ kN} \cdot \text{m}^2$$

$$\left(\frac{dy}{dx}\right)_A = \frac{-161.76}{42400} = -3.82 \times 10^{-3}$$

$$\theta_A = 3.82 \times 10^{-3} \text{ rad.} \quad \nwarrow \swarrow \blacktriangleleft$$

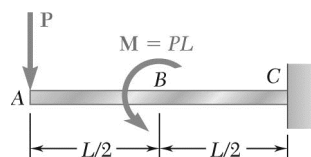
(b) Deflection at midpoint C. (y at $x = 2.4$)

$$EIy_C = \left(\frac{79}{6}\right)(2.4)^3 - \left(\frac{5}{4}\right)(2.4)^4 + 0 - 0 - 0 - (161.76)(2.4) + 0$$

$$= -247.68 \text{ kN} \cdot \text{m}^3$$

$$y_C = \frac{-247.68}{42400} = -5.84 \times 10^{-3} \text{ m}$$

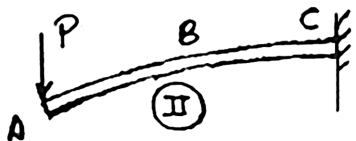
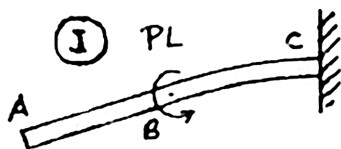
$$y_C = 5.84 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.65

For the cantilever beam and loading shown, determine the slope and deflection at the free end.

SOLUTION



Loading I: Counterclockwise couple PL at B .

Case 3 of Appendix D applied to portion BC .

$$\theta'_B = -\frac{(PL)(L/2)}{EI} = -\frac{1}{2} \frac{PL^2}{EI}$$

$$y'_B = \frac{(PL)(L/2)^2}{2EI} = \frac{1}{8} \frac{PL^3}{EI}$$

AB remains straight.

$$\theta'_A = \theta'_B = \frac{1}{2} \frac{PL^2}{EI}$$

$$y'_A = y'_B - \left(\frac{L}{2}\right)\theta'_B = -\frac{1}{8} \frac{PL^3}{EI} - \frac{1}{4} \frac{PL^3}{EI} = -\frac{3}{8} \frac{PL^3}{EI}$$

Loading II: Case 1 of Appendix D.

$$\theta''_A = \frac{PL^2}{2EI}, \quad y''_A = \frac{PL^3}{3EI}$$

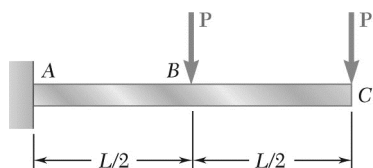
By superposition,

$$\theta_A = \theta'_A + \theta''_A = \frac{1}{2} \frac{PL^2}{EI} + \frac{1}{2} \frac{PL^2}{EI} = \frac{PL^2}{EI}$$

$$\theta_A = \frac{PL^2}{EI} \nearrow \blacktriangleleft$$

$$y_A = y'_A + y''_A = -\frac{3}{8} \frac{PL^3}{EI} - \frac{1}{3} \frac{PL^3}{EI} = -\frac{17}{24} \frac{PL^3}{EI}$$

$$y_A = \frac{17}{24} \frac{PL^3}{EI} \downarrow \blacktriangleleft$$



PROBLEM 9.66

For the cantilever beam and loading shown, determine the slope and deflection at the free end.

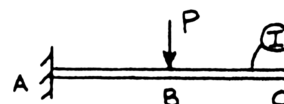
SOLUTION

Loading I: P downward at B .

Case 1 of Appendix D applied to portion AB .

$$\theta'_B = -\frac{P(L/2)^2}{2EI} = -\frac{1}{8} \frac{PL^2}{EI}$$

$$y'_B = -\frac{P(L/2)^3}{3EI} = -\frac{1}{24} \frac{PL^3}{EI}$$



BC remains straight.

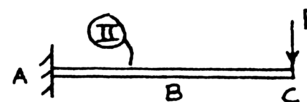
$$\theta'_C = \theta'_B = -\frac{1}{8} \frac{PL^2}{EI}$$

$$\begin{aligned} y'_C &= y'_B - \left(\frac{L}{2}\right) \theta'_B = -\frac{1}{24} \frac{PL^3}{EI} - \frac{1}{16} \frac{PL^3}{EI} \\ &= -\frac{5}{48} \frac{PL^3}{EI} \end{aligned}$$

Loading II: P downward at C .

Case 1 of Appendix D.

$$\theta''_C = -\frac{PL^2}{2EI} \quad y''_C = -\frac{PL^3}{3EI}$$



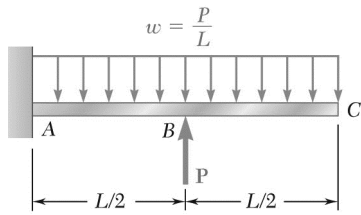
By superposition,

$$\theta_C = \theta'_C + \theta''_C = -\frac{1}{8} \frac{PL^2}{EI} - \frac{1}{2} \frac{PL^2}{EI} = -\frac{5}{8} \frac{PL^2}{EI}$$

$$\frac{5PL^2}{8EI} \swarrow \blacktriangleleft$$

$$y_C = y'_C + y''_C = -\frac{5}{48} \frac{PL^3}{EI} - \frac{1}{3} \frac{PL^3}{EI} = -\frac{21}{48} \frac{PL^3}{EI}$$

$$\frac{7PL^3}{16EI} \downarrow \blacktriangleleft$$



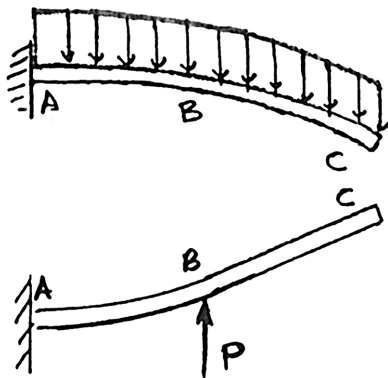
PROBLEM 9.67

For the cantilever beam and loading shown, determine the slope and deflection at the end.

SOLUTION

Loading I: Uniformly distributed downward loading with $w = P/L$.

Case 2 of Appendix D.



$$\theta'_C = -\frac{(P/L)L^3}{6EI} = -\frac{1}{6} \frac{PL^2}{EI}$$

$$y'_C = -\frac{(P/L)L^4}{8EI} = -\frac{1}{8} \frac{PL^3}{EI}$$

Loading II: Upward concentrated load at P at point B .

Case 1 of Appendix D applied to portion AB .

$$\theta''_B = \frac{P(L/2)^2}{2EI} = \frac{1}{8} \frac{PL^2}{EI}$$

$$y''_B = \frac{P(L/2)^3}{3EI} = \frac{1}{24} \frac{PL^3}{EI}$$

Portion BC remains straight.

$$\theta''_C = \theta''_B = \frac{1}{8} \frac{PL^2}{EI}$$

$$y''_C = y''_B + \frac{L}{2} \theta''_B = \frac{1}{24} \frac{PL^3}{EI} + \frac{1}{16} \frac{PL^3}{EI} = \frac{5}{48} \frac{PL^3}{EI}$$

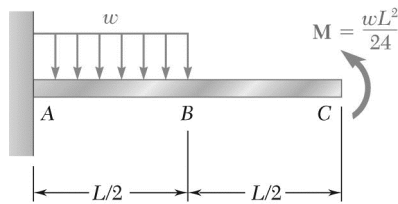
By superposition,

$$\theta_C = \theta'_C + \theta''_C = -\frac{1}{6} \frac{PL^2}{EI} + \frac{1}{8} \frac{PL^2}{EI} = -\frac{1}{24} \frac{PL^2}{EI}$$

$$\theta_C = \frac{PL^2}{24EI} \swarrow \blacktriangleleft$$

$$y_C = y'_C + y''_C = -\frac{1}{8} \frac{PL^3}{EI} + \frac{5}{48} \frac{PL^3}{EI} = -\frac{1}{48} \frac{PL^3}{EI}$$

$$y_C = \frac{PL^3}{48EI} \downarrow \blacktriangleleft$$



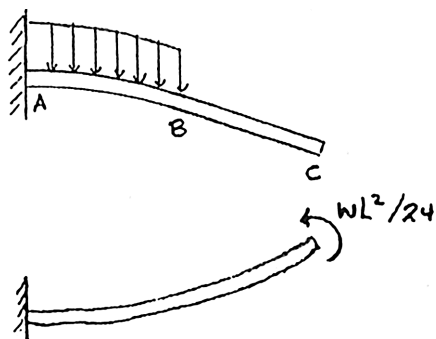
PROBLEM 9.68

For the cantilever beam and loading shown, determine the slope and deflection at the free end.

SOLUTION

Loading I: Downward distributed load w applied to portion AB .

Case 2 of Appendix *D* applied to portion AB .



$$\theta'_B = -\frac{w(L/2)^3}{6EI} = -\frac{1}{48} \frac{wL^2}{EI}$$

$$y'_B = -\frac{w(L/2)^4}{8EI} = -\frac{1}{128} \frac{wL^4}{EI}$$

Portion BC remains straight.

$$\theta'_C = \theta'_B = -\frac{1}{48} \frac{wL^3}{EI}$$

$$y'_C = y'_B + \left(\frac{L}{2}\right)\theta'_B = -\frac{1}{128} \frac{wL^4}{EI} - \frac{1}{96} \frac{wL^4}{EI} = -\frac{7}{384} \frac{wL^4}{EI}$$

Loading II: Counterclockwise couple $\frac{wL^2}{24}$ applied at C .

Case 3 of Appendix *D*.

$$\theta''_C = \frac{(wL^2/24)L}{EI} = \frac{1}{24} \frac{wL^3}{EI}$$

$$y''_C = \frac{(wL^2/24)L^2}{2EI} = \frac{1}{48} \frac{wL^4}{EI}$$

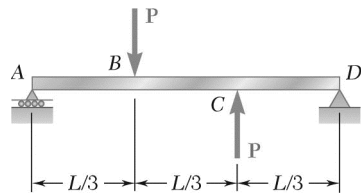
By superposition,

$$\theta_C = \theta'_C + \theta''_C = -\frac{1}{48} \frac{wL^3}{EI} + \frac{1}{24} \frac{wL^3}{EI}$$

$$y_C = y'_C + y''_C = -\frac{7}{384} \frac{wL^4}{EI} + \frac{1}{48} \frac{wL^4}{EI}$$

$$\theta_C = \frac{1}{48} \frac{wL^3}{EI} \nearrow \blacktriangleleft$$

$$y_C = \frac{1}{384} \frac{wL^4}{EI} \uparrow \blacktriangleleft$$



PROBLEM 9.69

For the beam and loading shown, determine (a) the deflection at C, (b) the slope at end A.

SOLUTION

Loading I: Downward load P at B .

Use Case 5 of Appendix D with

$$P = P, \quad a = \frac{L}{3}, \quad b = \frac{2L}{3}, \quad L = L, \quad x = \frac{2L}{3}$$

For $x < a$, given elastic curve is $y = \frac{Pb}{EIL} [x^3 - (L^2 - b^2)x]$

To obtain elastic curve for $x > a$, replace x by $L - x$ and interchange a and b to get

$$y = \frac{Pa}{6EIL} [(L - x)^3 - (L^2 - a^2)(L - x)] \text{ with } x = \frac{2L}{3} \text{ at point C.}$$

$$y_C = \frac{P(L/3)}{6EIL} \left[\left(\frac{L}{3} \right)^3 - \left(L^2 - \left(\frac{L}{3} \right)^2 \right) \left(\frac{L}{3} \right) \right] = -\frac{7}{486} \frac{PL^3}{EI}$$

$$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL} = -\frac{P(2L/3)[L^2 - (2L/3)^2]}{6EIL} = -\frac{5}{81} \frac{PL^2}{EI}$$

Loading II: Upward load at C. Use Case 5 of Appendix D with

$$P = -P, \quad a = \frac{2L}{3}, \quad b = \frac{L}{3}, \quad L = L, \quad x = a = \frac{2L}{3}$$

$$y_C = -\frac{(-P)(2L/3)^2(L/3)^2}{3EIL} = \frac{4}{243} \frac{PL^3}{EI}$$

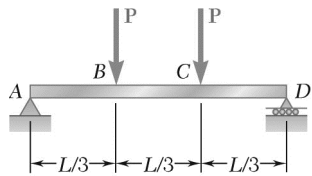
$$\theta_A = -\frac{(-P)(L/3)(L^2 - (L/3)^2)}{6EIL} = \frac{4}{81} \frac{PL^2}{EI}$$

(a) Deflection at C. $y_C = -\frac{7}{486} \frac{PL^3}{EI} + \frac{4}{243} \frac{PL^3}{EI}$

$$y_C = \frac{1}{486} \frac{PL^3}{EI} \uparrow \blacktriangleleft$$

(b) Slope at A. $\theta_A = -\frac{5}{81} \frac{PL^2}{EI} + \frac{4}{81} \frac{PL^2}{EI}$

$$\theta_A = \frac{1}{81} \frac{PL^2}{EI} \curvearrowright \blacktriangleleft$$



PROBLEM 9.70

For the beam and loading shown, determine (a) the deflection at C, (b) the slope at end A.

SOLUTION

Loading I: Load at B. Case 5 of Appendix D.

$$a = \frac{L}{3}, \quad b = \frac{2L}{3}, \quad x = \frac{2L}{3}$$

For $x > a$, replace x by $L - x$ and interchange a and b .

$$y = \frac{Pa}{6EIL}[(L - x)^3 - (L^2 - a^2)(L - x)]$$

$$y_C = \frac{P(L/3)}{6EIL} \left[\left(L - \frac{2L}{3} \right)^3 - \left(L^2 - \frac{L^2}{9} \right) \left(L - \frac{2L}{3} \right) \right] = -\frac{7}{486} \frac{PL^3}{EI}$$

$$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL} = -\frac{P(2L/3)(L^2 - 4L^2/9)}{6EIL} = -\frac{5}{81} \frac{PL^2}{EI}$$

Loading II: Load at C. Case 5 of Appendix D.

$$a = \frac{2L}{3}, \quad b = \frac{L}{3}, \quad x = \frac{2L}{3}$$

$$y_C = \frac{Pb}{6EIL} [x^3 - (L^2 - b^2)x] = \frac{P(L/3)}{6EIL} \left[\frac{8L^3}{27} - \left(L^2 - \frac{L^2}{9} \right) \left(\frac{2L}{3} \right) \right]$$

$$= -\frac{8}{486} \frac{PL^3}{EI}$$

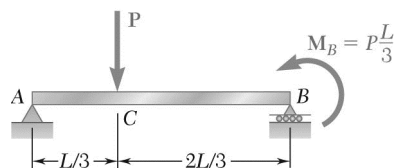
$$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL} = -\frac{P(L/3)(L^2 - L^2/9)}{6EIL} = -\frac{4}{81} \frac{PL^2}{EI}$$

(a) Deflection at C. $y_C = -\frac{7}{486} \frac{PL^3}{EI} - \frac{8}{486} \frac{PL^3}{EI} = -\frac{15}{486} \frac{PL^3}{EI}$

$$y_C = \frac{5}{162} \frac{PL^3}{EI} \downarrow \blacktriangleleft$$

(b) Slope at A. $\theta_A = -\frac{5}{81} \frac{PL^2}{EI} - \frac{4}{81} \frac{PL^2}{EI} = -\frac{1}{9} \frac{PL^2}{EI}$

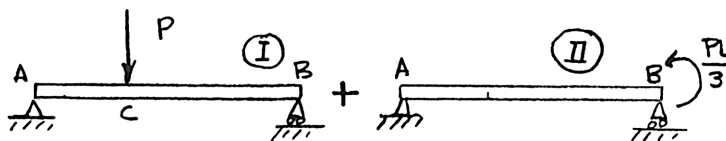
$$\theta_A = \frac{1}{9} \frac{PL^2}{EI} \nearrow \blacktriangleleft$$



PROBLEM 9.71

For the beam and loading shown, determine (a) the deflection at C, (b) the slope at end A.

SOLUTION



Loading I: Case 5. $a = \frac{L}{3}$, $b = \frac{2L}{3}$, $P = P$, $x = a$

$$y_C = -\frac{Pa^2b^2}{6EIL} = -\frac{P}{6EIL} \left(\frac{L}{3}\right)^2 \left(\frac{2L}{3}\right)^2 = -\frac{4}{243} \frac{PL^3}{EI}$$

$$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL} = -\frac{P}{6EIL} \left(\frac{2L}{3}\right) \left[L^2 - \left(\frac{2L}{3}\right)^2\right] = -\frac{5}{81} \frac{PL^2}{EI}$$

Loading II: Case 7. $M = -\frac{PL}{3}$, $x = \frac{L}{3}$

$$y_C = -\frac{M}{6EIL} (x^3 - L^2x) = +\frac{PL/3}{6EIL} \left\{ \left(\frac{L}{3}\right)^3 - L^2 \left(\frac{L}{3}\right) \right\} = -\frac{4}{243} \frac{PL^3}{EI}$$

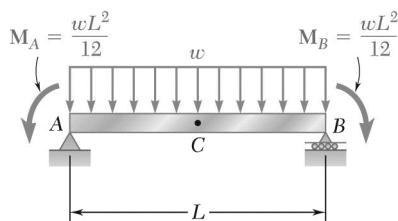
$$\theta_A = +\frac{ML}{6EI} = \frac{-(PL/3)L}{6EI} = -\frac{1}{18} \frac{PL^2}{EI}$$

(a) Deflection at C. $y_C = -\frac{4}{243} \frac{PL^3}{EI} - \frac{4}{243} \frac{PL^3}{EI} = -\frac{8}{243} \frac{PL^3}{EI}$

$$y_C = \frac{8}{243} \frac{PL^3}{EI} \downarrow \blacktriangleleft$$

(b) Slope at A. $\theta_A = -\frac{5}{81} \frac{PL^2}{EI} - \frac{1}{18} \frac{PL^2}{EI} = -\frac{19}{162} \frac{PL^2}{EI}$

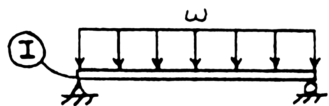
$$\theta_A = \frac{19}{162} \frac{PL^2}{EI} \nwarrow \blacktriangleleft$$



PROBLEM 9.72

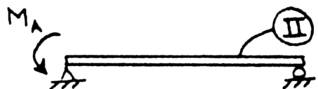
For the beam and loading shown, determine (a) the deflection at C, (b) the slope at end A.

SOLUTION



Loading I: Case 6 in Appendix D.

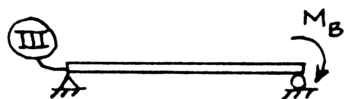
$$y_C = -\frac{5wL^4}{384EI} \quad \theta_A = -\frac{wL^3}{24EI}$$



Loading II: Case 7 in Appendix D.

$$y_C = -\frac{M_A}{6EI} \left[\left(\frac{L}{2} \right)^3 - L^2 \left(\frac{L}{2} \right) \right] = \frac{1}{16} \frac{M_A L^2}{EI} \quad \theta_A = \frac{M_A L}{3EI}$$

with $M_A = \frac{wL^2}{12} \quad y_C = \frac{1}{192} \frac{wL^4}{EI} \quad \theta_A = \frac{1}{36} \frac{wL^3}{EI}$



Loading III: Case 7 in Appendix D.

$$y_C = \frac{1}{16} \frac{M_B L^2}{EI} \quad (\text{using Loading II result})$$

$$\theta_A = \frac{M_B L}{6EI}$$

with $M_B = \frac{wL^2}{12} \quad y_C = \frac{1}{192} \frac{wL^4}{EI} \quad \theta_A = \frac{1}{72} \frac{wL^3}{EI}$

(a) Deflection at C.

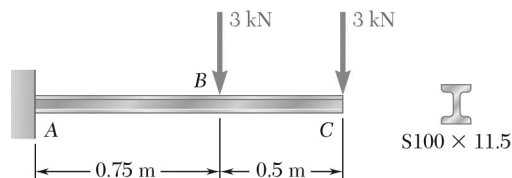
$$y_C = -\frac{5}{384} \frac{wL^4}{EI} + \frac{1}{192} \frac{wL^4}{EI} + \frac{1}{192} \frac{wL^4}{EI} = -\frac{1}{384} \frac{wL^4}{EI}$$

$$y_C = \frac{1}{384} \frac{wL^4}{EI} \downarrow \blacktriangleleft$$

(b) Slope at A.

$$\theta_A = -\frac{1}{24} \frac{wL^3}{EI} + \frac{1}{36} \frac{wL^3}{EI} + \frac{1}{72} \frac{wL^3}{EI} = 0$$

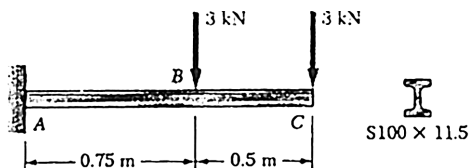
$$\theta_A = 0 \blacktriangleleft$$



PROBLEM 9.73

For the cantilever beam and loading shown, determine the slope and deflection at end C. Use $E = 200 \text{ GPa}$.

SOLUTION



Units: Forces in kN; lengths in m.

Loading I: Concentrated load at B

Case 1 of Appendix D applied to portion AB.

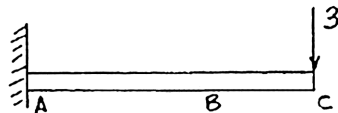
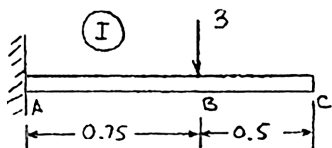
$$\theta'_B = -\frac{PL^2}{2EI} = -\frac{(3)(0.75)^2}{2EI} = -\frac{0.84375}{EI}$$

$$y'_B = -\frac{PL^3}{3EI} = -\frac{(3)(0.75)^3}{3EI} = -\frac{0.421875}{EI}$$

Portion BC remains straight.

$$\theta'_C = \theta'_B = -\frac{0.84375}{EI}$$

$$y'_C = y'_B - (0.5)\theta'_B = -\frac{0.84375}{EI}$$



Loading II: Concentrated load at C. Case 1 of Appendix D.

$$\theta''_A = -\frac{PL^2}{2EI} = -\frac{(3)(1.25)^2}{2EI} = -\frac{2.34375}{EI}$$

$$y''_A = -\frac{PL^3}{3EI} = -\frac{(3)(1.25)^3}{3EI} = -\frac{1.953125}{EI}$$

By superposition,

$$\theta_A = \theta'_A + \theta''_A = -\frac{3.1875}{EI}$$

$$y_A = y'_A + y''_A = -\frac{2.796875}{EI}$$

Data:

$$E = 200 \times 10^9 \text{ Pa}, \quad I = 2.52 \times 10^6 \text{ mm}^4 = 2.52 \times 10^{-6} \text{ m}^4$$

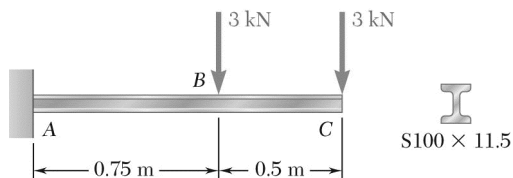
$$EI = (200 \times 10^4)(2.52 \times 10^{-6}) = 504 \times 10^3 \text{ N} \cdot \text{m}^2 = 504 \text{ kN} \cdot \text{m}^2$$

Slope at C.

$$\theta_C = -\frac{3.1875}{504} = -6.32 \times 10^{-3} \text{ rad} \quad \theta_C = 6.32 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

Deflection at C.

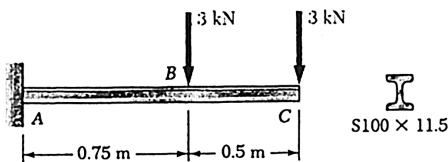
$$y_C = -\frac{2.796875}{504} = -5.55 \times 10^{-3} \text{ m} \quad y_C = 5.55 \text{ mm} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.74

For the cantilever beam and loading shown, determine the slope and deflection at point B . Use $E = 200$ GPa.

SOLUTION



Units: Forces in kN; lengths in m.

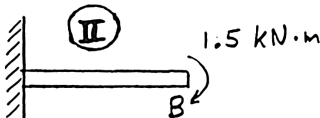
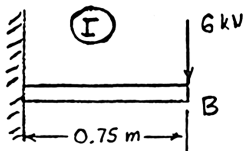
The slope and deflection at B depend only on the deformation of portion AB .

Reduce the force at C to an equivalent force-couple system at B and add the force already at B to obtain the loadings I and II shown.

Loading I: Case 1 of Appendix D.

$$\theta'_B = -\frac{PL^2}{2EI} = -\frac{(6)(0.75)^2}{2EI} = -\frac{1.6875}{EI}$$

$$y'_B = -\frac{PL^3}{3EI} = -\frac{(6)(0.75)^3}{3EI} = -\frac{0.84375}{EI}$$



Loading II: Case 3 of Appendix D.

$$\theta''_B = -\frac{ML}{EI} = -\frac{(1.5)(0.75)}{EI} = -\frac{1.125}{EI}$$

$$y''_B = -\frac{ML^2}{2EI} = -\frac{(1.5)(0.75)^2}{2EI} = -\frac{0.421875}{EI}$$

By superposition,

$$\theta_B = \theta'_B + \theta''_B = -\frac{2.8125}{EI}$$

$$y_B = y'_B + y''_B = -\frac{1.265625}{EI}$$

Data:

$$E = 200 \times 10^9 \text{ Pa}, \quad I = 2.52 \times 10^6 \text{ mm}^4 = 2.52 \times 10^{-6} \text{ m}^4$$

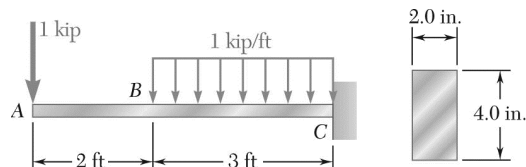
$$EI = (200 \times 10^9)(2.52 \times 10^{-6}) = 504 \times 10^3 \text{ N} \cdot \text{m}^2 = 504 \text{ kN} \cdot \text{m}^2$$

Slope at B.

$$\theta_B = -\frac{2.8125}{504} = -5.58 \times 10^{-3} \text{ rad} \qquad \theta_B = 5.58 \times 10^{-3} \text{ rad} \swarrow \blacktriangleleft$$

Deflection at B.

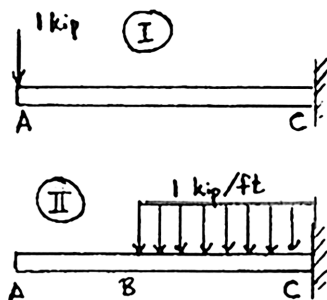
$$y_B = -\frac{1.265625}{504} = -2.51 \times 10^{-3} \text{ m} \qquad y_B = 2.51 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.75

For the cantilever beam and loading shown, determine the slope and deflection at end A . Use $E = 29 \times 10^6$ psi.

SOLUTION



Units: Forces in kips; lengths in ft.

Loading I: Concentrated load at A .

Case 1 of Appendix D .

$$\theta'_A = \frac{PL^2}{2EI} = \frac{(1)(5)^2}{2EI} = \frac{12.5}{EI}$$

$$y'_A = -\frac{PL^3}{3EI} = -\frac{(1)(5)^3}{3EI} = -\frac{41.667}{EI}$$

Loading II: Uniformly distributed load over portion BC .

Case 2 of Appendix D applied to portion BC .

$$\theta''_B = \frac{wL^3}{6EI} = \frac{(1)(3)^3}{6EI} = \frac{4.5}{EI}$$

$$y''_B = -\frac{wL^4}{8EI} = -\frac{(1)(3)^4}{8EI} = -\frac{10.125}{EI}$$

Portion AB remains straight. $\theta''_A = \theta''_B = \frac{4.5}{EI}$

$$y''_A = y''_B - a\theta''_B = -\frac{10.125}{EI} - (2)\left(\frac{4.5}{EI}\right) = -\frac{19.125}{EI}$$

By superposition,

$$\theta_A = \theta'_A + \theta''_A = \frac{12.5}{EI} + \frac{4.5}{EI} = \frac{17}{EI}$$

$$y_A = y'_A + y''_A = -\frac{41.667}{EI} - \frac{19.125}{EI} = -\frac{60.792}{EI}$$

Data:

$$E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi}$$

$$I = \frac{1}{12}(2.0)(4.0)^3 = 10.667 \text{ in}^4$$

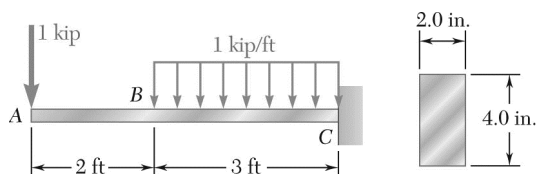
$$EI = (29 \times 10^3)(10.667) = 309.33 \times 10^3 \text{ kip} \cdot \text{in}^2 = 2148 \text{ kip} \cdot \text{ft}^2$$

Slope at A .

$$\theta_A = \frac{17}{2148} \qquad \theta_A = 7.91 \times 10^{-3} \text{ rad} \nearrow \blacktriangleleft$$

Deflection at A .

$$y_A = -\frac{60.792}{2148} = -28.30 \times 10^{-3} \text{ ft} \qquad y_A = 0.340 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.76

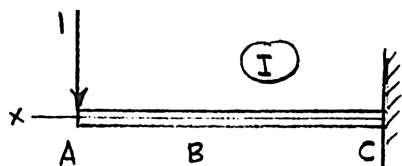
For the cantilever beam and loading shown, determine the slope and deflection at point *B*. Use $E = 29 \times 10^6$ psi.

SOLUTION

Units: Forces in kips; lengths in ft.

Loading I: Concentrated load at *A*.

Case 1 of Appendix *D*.



$$y = \frac{P}{6EI} [x^3 - 3Lx^2]$$

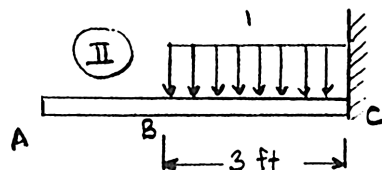
$$\frac{dy}{dx} = \frac{P}{6EI} [3x^2 - 6Lx]$$

with $P = 1$ kip, $L = 5$ ft, $x = 3$ ft

$$y'_B = \frac{1}{6EI} [(3)^3 - (3)(5)(3)^2] = -\frac{18}{EI}$$

$$\frac{dy}{dx}\bigg|_B = \frac{1}{6EI} [(3)(3)^2 - (6)(5)(3)] = -\frac{10.5}{EI}$$

Adjusting the sign, $\theta'_B = \frac{10.5}{EI}$



Loading II: Uniformly distributed load over portion *BC*.

Case 2 of Appendix *D* applied to portion *BC*.

$$y''_B = -\frac{wL^4}{8EI} = -\frac{(1)(3)^4}{8EI} = -\frac{10.125}{EI} \quad \theta''_B = \frac{wL^3}{6EI} = \frac{(1)(3)^3}{6EI} = \frac{4.5}{EI}$$

By superposition,

$$\theta_B = \theta'_B + \theta''_B = \frac{10.5}{EI} + \frac{4.5}{EI} = \frac{15}{EI} \quad y_B = y'_B + y''_B = -\frac{18}{EI} - \frac{10.125}{EI} = -\frac{28.125}{EI}$$

Data:

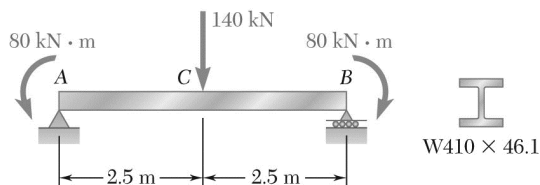
$$E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi} \quad I = \frac{1}{12}(2.0)(4.0)^3 = 10.667 \text{ in}^4$$

$$EI = (29 \times 10^3)(10.667) = 309.33 \times 10^3 \text{ kip} \cdot \text{in}^2 = 2148 \text{ kip} \cdot \text{ft}^2$$

PROBLEM 9.76 (Continued)

Slope at B . $\theta_B = \frac{15}{2148} = 6.98 \times 10^{-3}$ $\theta_B = 6.98 \times 10^{-3} \text{ rad}$ ↗ ◀

Deflection at B . $y_B = -\frac{28.125}{2148} = -13.09 \times 10^{-3} \text{ ft}$ $y_B = 0.1571 \text{ in.}$ ↓ ◀



PROBLEM 9.77

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at point C. Use $E = 200$ GPa.

SOLUTION

Units: Forces in kN; lengths in m.

Loading I: Moment at B.

Case 7 of Appendix D. $M = 80$ kN·m, $L = 5.0$ m, $x = 2.5$ m

$$\theta_A = \frac{ML}{6EI} = \frac{(80)(5.0)}{6EI} = \frac{66.667}{EI}$$

$$y_C = -\frac{M}{6EIL}(x^3 - L^2x) = -\frac{80}{6EI(5.0)}[2.5^3 - (5.0)^2(2.5)] = \frac{125}{EI}$$

Loading II: Moment at A. Case 7 of Appendix D.

$M = 80$ kN·m, $L = 5.0$ m, $x = 2.5$ m

$$\theta_A = \frac{ML}{3EI} = \frac{(80)(5.0)}{3EI} = \frac{133.333}{EI}$$

$$y_C = \frac{125}{EI} \quad (\text{Same as loading I.})$$

Loading III: 140 kN concentrated load at C. $P = 140$ kN

$$\theta_A = -\frac{PL^2}{16EI} = -\frac{(140)(5.0)^2}{16EI} = -\frac{218.75}{EI}$$

$$y_C = -\frac{PL^3}{48EI} = -\frac{(140)(5.0)^3}{48EI} = -\frac{364.583}{EI}$$

Data: $E = 200 \times 10^9$ Pa, $I = 156 \times 10^6$ mm⁴ = 156×10^{-6} m⁴

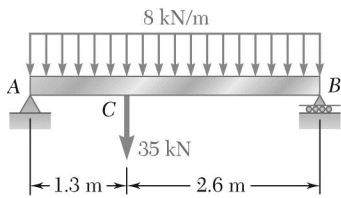
$$EI = (200 \times 10^9)(156 \times 10^{-6}) = 31.2 \times 10^6 \text{ N} \cdot \text{m}^2 = 31200 \text{ kN} \cdot \text{m}^2$$

(a) Slope at A. $\theta_A = \frac{66.667 + 133.333 - 218.75}{31200} = -0.601 \times 10^{-3} \text{ rad}$

$$\theta_A = 0.601 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at C. $y_C = \frac{125 + 125 - 364.583}{31200} = -3.67 \times 10^{-3} \text{ m}$

$$y_C = 3.67 \text{ mm} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.78

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at point C. Use $E = 200$ GPa.

SOLUTION

Units: Forces in kN; lengths in m.

Loading I: 8 kN/m uniformly distributed.

Case 6: $w = 8$ kN/m, $L = 3.9$ m, $x = 1.3$ m

$$\theta_A = -\frac{WL^3}{24EI} = -\frac{(8)(3.9)^3}{24EI} = -\frac{19.773}{EI}$$

$$y_C = -\frac{w}{24EI}[x^4 - 2Lx^3 + L^3x] = -\frac{8}{24EI}[(1.3)^4 - (2)(3.9)(1.3)^3 + (3.9)^3(1.3)]$$

$$= -\frac{20.945}{EI}$$

Loading II: 35 kN concentrated load at C. Case 5 of Appendix D.

$P = 35$ kN, $L = 3.9$ m, $a = 1.3$ m, $b = 2.6$ m, $x = a = 1.3$ m

$$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL} = -\frac{(35)(2.6)(3.9^2 - 2.6^2)}{6EI(3.9)} = -\frac{32.861}{EI}$$

$$y_C = -\frac{Pa^2b^2}{3EIL} = -\frac{(35)(1.3)^2(2.6)^2}{3EI(3.9)} = -\frac{34.176}{EI}$$

Data: $E = 200 \times 10^9$, $I = 102 \times 10^6 \text{ mm}^4 = 102 \times 10^{-6} \text{ m}^4$

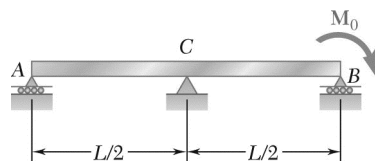
$EI = (200 \times 10^9)(102 \times 10^{-6}) = 20.4 \times 10^6 \text{ N} \cdot \text{m}^2 = 20,400 \text{ kN} \cdot \text{m}^2$

(a) Slope at A. $\theta_A = -\frac{19.773 + 32.861}{20,400} = -2.58 \times 10^{-3} \text{ rad}$

$$\theta_A = 2.58 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at C. $y_C = -\frac{20.934 + 34.176}{20,400} = -2.70 \times 10^{-3} \text{ m}$

$$y_C = 2.70 \text{ mm} \quad \downarrow \blacktriangleleft$$

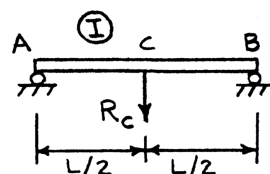


PROBLEM 9.79

For the uniform beam shown, determine the reaction at each of the three supports.

SOLUTION

Consider R_C as redundant and replace loading system by I and II.

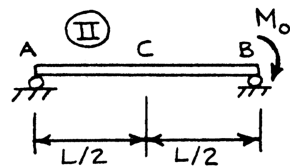


Loading I: (Case 4 of Appendix D.) $y'_C = -\frac{R_C L^3}{48EI}$

Loading II: (Case 7 of Appendix D.)

$$y''_C = -\frac{M_0}{6EI} \left[\left(\frac{L}{2} \right)^3 - L^2 \left(\frac{L}{2} \right) \right]$$

$$= \frac{M_0 L^2}{16EI}$$



Superposition and constraint:

$$y_C = y'_C + y''_C = 0$$

$$-\frac{R_C L^3}{48EI} + \frac{M_0 L^2}{16EI} = 0$$

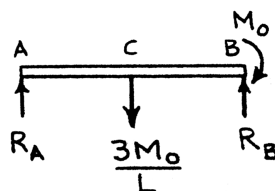
$$R_C = \frac{3M_0}{L} \downarrow \blacktriangleleft$$

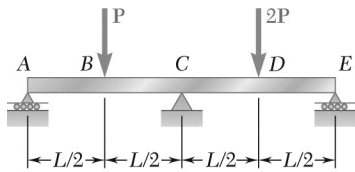
$$+\circlearrowleft \Sigma M_B = 0: -R_A L + \left(\frac{3M_0}{L} \right) \left(\frac{L}{2} \right) - M_0 = 0$$

$$R_A = \frac{M_0}{2L} \uparrow \blacktriangleleft$$

$$+\uparrow \Sigma F_Y = 0: \frac{M_0}{2L} - \frac{3M_0}{L} + R_B = 0$$

$$R_B = \frac{5M_0}{2L} \uparrow \blacktriangleleft$$

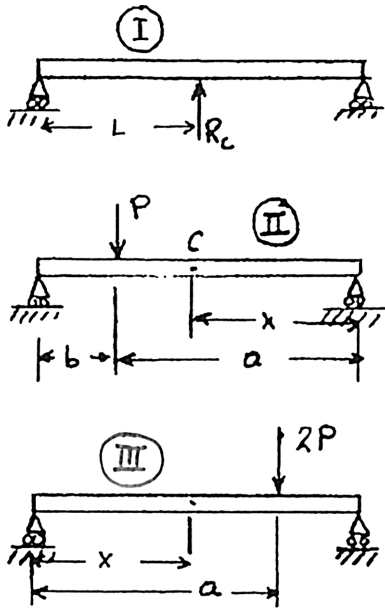




PROBLEM 9.80

For the uniform beam shown, determine the reaction at each of the three supports.

SOLUTION



Beam is indeterminate to first degree. Consider R_C to be the redundant reaction, and replace the loading by loadings I, II, and III.

Loading I: Case 4 of Appendix D.

$$(y_C)_I = \frac{R_C(2L)^3}{48EI} = \frac{1}{6} \frac{R_C L^2}{EI}$$

Loading II: Case 5 of Appendix D.

$$\begin{aligned} (y_C)_{II} &= \frac{Pb}{6EI(2L)} [x^3 - \{(2L)^2 - b^2\}x] \\ &= \frac{P(L/2)}{12EI} \left[L^3 - \left\{ 4L^2 - \left(\frac{L}{2} \right)^2 \right\} L \right] = -\frac{11}{48} \frac{PL^3}{EI} \end{aligned}$$

Loading III: Case 5 of Appendix D.

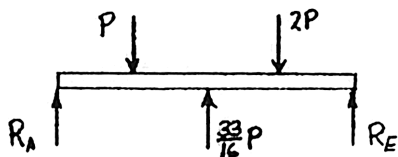
$$(y_C)_{III} = 2(y_C)_{II} = -\frac{11}{24} \frac{PL^3}{EI}$$

Superposition and constraint:

$$y_C = (y_C)_I + (y_C)_{II} + (y_C)_{III} = 0$$

$$\frac{1}{6} \frac{R_C L^3}{EI} - \frac{11}{48} \frac{PL^3}{EI} - \frac{11}{24} \frac{PL^3}{EI} = \frac{1}{6} \frac{R_C L^3}{EI} - \frac{11}{16} \frac{PL^3}{EI} = 0$$

$$R_C = \frac{33}{16} P \uparrow \blacktriangleleft$$



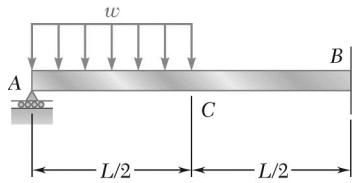
Statics: $+\circlearrowleft \sum M_E = 0$:

$$-R_A(2L) + P\left(\frac{3L}{2}\right) - \left(\frac{33}{16}P\right)L + (2P)\left(\frac{L}{2}\right) = 0$$

$$R_A = \frac{7}{32} P \uparrow \blacktriangleleft$$

$$+\uparrow \sum F_y = 0: \frac{7}{32} P - P + \frac{33}{16} P - 2P + R_E = 0$$

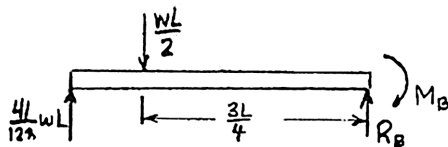
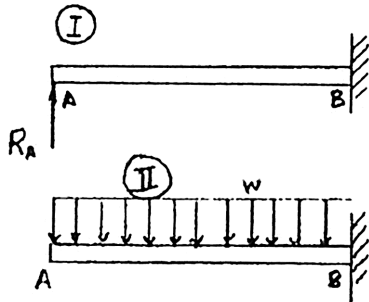
$$R_E = \frac{23}{32} P \uparrow \blacktriangleleft$$



PROBLEM 9.81

For the uniform beam shown, determine (a) the reaction at A, (b) the reaction at B.

SOLUTION



Beam is indeterminate to first degree. Consider R_A as redundant and replace the given loading by loadings I, II, and III.

Loading I: Case 1 of Appendix D.

$$(y_A)_I = \frac{R_A L^3}{3EI}$$

Loading II: Case 2 of Appendix D.

$$(y_A)_{II} = -\frac{wL^4}{8EI}$$

Loading III: Case 2 of Appendix D (portion CB).

$$(\theta_C)_{III} = -\frac{w(L/2)^3}{6EI} = -\frac{1}{48} \frac{wL^3}{EI}$$

$$(y_C)_{III} = \frac{w(L/2)^4}{8EI} = \frac{1}{128} \frac{wL^4}{EI}$$

Portion AC remains straight.

$$(y_A)_{III} = (y_C)_{III} + \frac{L}{2}(\theta_C)_{III} = \frac{7}{384} \frac{wL^4}{EI}$$

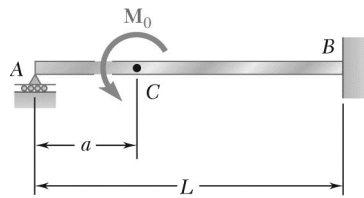
Superposition and constraint: $y_A = (y_A)_I + (y_A)_{II} + (y_A)_{III} = 0$

$$(a) \quad \frac{1}{3} \frac{R_A L^3}{EI} - \frac{1}{8} \frac{wL^4}{EI} + \frac{7}{384} \frac{wL^4}{EI} = \frac{1}{3} \frac{R_A L^3}{EI} - \frac{41}{384} \frac{wL^4}{EI} = 0 \quad R_A = \frac{41}{128} wL \uparrow \blacktriangleleft$$

Statics:

$$(b) \quad +\uparrow \Sigma F_y = 0 : \quad \frac{41}{128} wL - \frac{1}{2} wL + R_B = 0 \quad R_B = \frac{23}{128} wL \uparrow \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_B = 0 : \quad -\left(\frac{41}{128} wL\right)L - \left(\frac{1}{2} wL\right)\left(\frac{3L}{4}\right) - M_B = 0 \quad M_B = \frac{7}{128} wL^2 \curvearrowright \blacktriangleleft$$



PROBLEM 9.82

For the uniform beam shown, determine (a) the reaction at A, (b) the reaction at B.

SOLUTION

Consider R_A as redundant and replace loading system by I and II.

Loading I: (Case 1 of Appendix D.) $y'_A = \frac{R_A L^3}{3EI}$

Loading II: Portion BC. (Case 3 of Appendix D.)

$$y''_C = -\frac{M_0(L-a)^2}{2EI} \quad \theta''_C = \frac{M_0(L-a)}{EI}$$

Portion AC is straight.

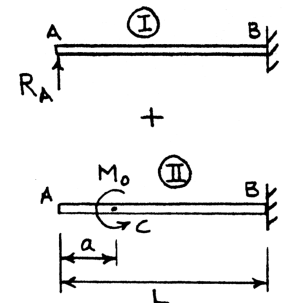
$$\begin{aligned} y''_A &= y''_C - (a)\theta''_C \\ &= -\frac{M_0(L-a)^2}{2EI} - \frac{aM_0(L-a)}{EI} \end{aligned}$$

(a) Superposition and constraint: $y_A = y'_A + y''_A = 0$

$$\frac{R_A L^3}{3EI} - \frac{M_0(L-a)^2}{2EI} - \frac{aM_0(L-a)}{EI} = 0$$

$$\frac{2}{3}R_A L^3 - M_0(L-a)(L-a+2a) = 0$$

$$\frac{2}{3}R_A L^3 = M_0(L^2 - a^2)$$



$$R_A = \frac{3M_0}{2L^3}(L^2 - a^2) \uparrow \blacktriangleleft$$

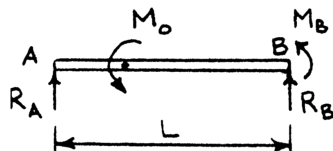
(b) $+\uparrow \Sigma F_y = 0: R_A + R_B = 0 \quad R_B = -R_A$

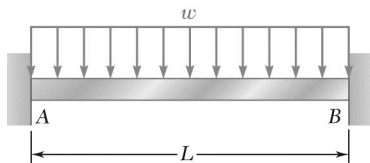
$$R_B = \frac{3M_0}{2L^3}(L^2 - a^2) \downarrow \blacktriangleleft$$

$+\curvearrowright \Sigma M_B = 0: M_B + M_0 - R_A L = 0 \quad M_B + M_0 - \frac{3}{2} \frac{M_0}{L^2}(L^2 - a^2) = 0$

$$M_B = \frac{3M_0}{2L^2} \left(L^2 - a^2 - \frac{2}{3} L^2 \right)$$

$$M_B = \frac{M_0}{2L^2}(L^2 - 3a^2) \curvearrowright \blacktriangleleft$$





PROBLEM 9.83

For the beam shown, determine the reaction at B .

SOLUTION

Beam is second degree indeterminate. Choose R_B and M_B as redundant reactions.

Loading I: Case 1 of Appendix D.

$$(y_B)_I = \frac{R_B L^3}{3EI} \quad (\theta_B)_I = \frac{R_B L^2}{2EI}$$

Loading II: Case 3 of Appendix D.

$$(y_B)_{II} = -\frac{M_B L^2}{2EI} \quad (\theta_B)_{II} = -\frac{M_B L}{EI}$$

Loading III: Case 2 of Appendix D.

$$(y_B)_{III} = -\frac{w L^4}{8EI} \quad (\theta_B)_{III} = -\frac{w L^3}{6EI}$$

Superposition and constraint:

$$y_B = (y_B)_I + (y_B)_{II} + (y_B)_{III} = 0$$

$$\frac{L^3}{3EI} R_B - \frac{L^2}{2EI} M_B - \frac{w L^4}{8EI} = 0 \quad (1)$$

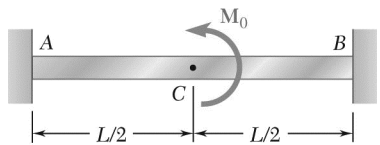
$$\theta_B = (\theta_B)_I + (\theta_B)_{II} + (\theta_B)_{III} = 0$$

$$\frac{L^2}{2EI} R_B - \frac{L}{EI} M_B - \frac{w L^3}{6EI} = 0 \quad (2)$$

Solving Eqs. (1) and (2) simultaneously,

$$R_B = \frac{1}{2} w L \uparrow \blacktriangleleft$$

$$M_B = \frac{1}{12} w L^2 \curvearrowright \blacktriangleleft$$



PROBLEM 9.84

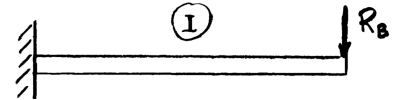
For the beam shown, determine the reaction at B.

SOLUTION

Beam is second degree indeterminate. Choose R_B and M_B as redundant reactions.

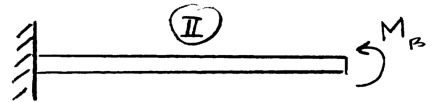
Loading I: Case 1 of Appendix D.

$$(y_B)_I = -\frac{R_B L^3}{3EI}, \quad (\theta_B)_I = -\frac{R_B L^2}{2EI}$$



Loading II: Case 3 of Appendix D.

$$(y_B)_{II} = \frac{M_B L^2}{2EI}, \quad (\theta_B)_{II} = \frac{M_B L}{EI}$$



Loading III: Case 3 applied to portion AC.

$$(y_C)_{III} = \frac{M_0 (L/2)^2}{2EI} = \frac{M_0 L^2}{8EI}$$

$$(\theta_C)_{III} = \frac{M_0 (L/2)}{EI} = \frac{M_0 L}{2EI}$$



Portion CB remains straight.

$$(y_B)_{III} = (y_C)_{III} + \frac{L}{2} (\theta_C)_{III} = \frac{3 M_0 L^2}{8 EI}$$

$$(\theta_B)_{III} = (\theta_C)_{III} = \frac{1 M_0 L}{2 EI}$$

Superposition and constraint:

$$y_B = (y_B)_I + (y_B)_{II} + (y_B)_{III} = 0$$

$$-\frac{L^3}{3EI} R_B + \frac{L^2}{2EI} M_B + \frac{3 M_0 L^2}{8 EI} = 0 \quad (1)$$

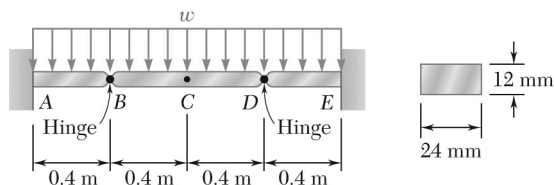
$$\theta_B = (\theta_B)_I + (\theta_B)_{II} + (\theta_B)_{III} = 0$$

$$-\frac{L^2}{2EI} R_B + \frac{L}{EI} M_B + \frac{1 M_0 L}{2 EI} = 0 \quad (2)$$

Solving Eqs. (1) and (2) simultaneously,

$$R_B = \frac{3 M_0}{2 L} \downarrow \blacktriangleleft$$

$$M_B = \frac{1}{4} M_0 \curvearrowright \blacktriangleleft$$



PROBLEM 9.85

A central beam BD is joined at hinges to two cantilever beams AB and DE . All beams have the cross section shown. For the loading shown, determine the largest w so that the deflection at C does not exceed 3 mm. Use $E = 200$ GPa.

SOLUTION

Let $a = 0.4$ m.

Cantilever beams AB and CD .

Cases 1 and 2 of Appendix D.
$$y_B = y_D = -\frac{(wa)a^3}{3EI} - \frac{wa^4}{8EI} = -\frac{11}{24} \frac{wa^4}{EI}$$

Beam BCD , with $L = 0.8$ m, assuming that points B and D do not move.

Case 6 of Appendix D.

$$y'_C = -\frac{5wL^4}{384EI}$$

Additional deflection due to movement of points B and D .

$$y''_C = y_B = y_D = -\frac{11}{24} \frac{wa^4}{EI}$$

Total deflection at C .

$$y_C = y'_C + y''_C$$

$$y_C = -\frac{w}{EI} \left\{ \frac{5L^4}{384} + \frac{11a^4}{24} \right\}$$

Data:

$$E = 200 \times 10^9 \text{ Pa},$$

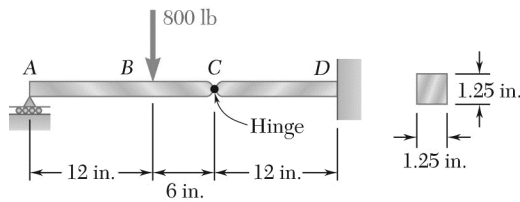
$$I = \frac{1}{12} (24)(12)^3 = 3.456 \times 10^{-3} \text{ mm}^4 = 3.456 \times 10^{-9} \text{ m}^4$$

$$EI = (200 \times 10^9)(3.456 \times 10^{-9}) = 691.2 \text{ N} \cdot \text{m}^2$$

$$y_C = -3 \times 10^{-3} \text{ m}$$

$$-3 \times 10^{-3} = -\frac{w}{691.2} \left\{ \frac{(5)(0.8)^4}{384} + \frac{(11)(0.4)^4}{24} \right\} = -24.69 \times 10^{-6} w$$

$$w = 121.5 \text{ N/m} \quad \blacktriangleleft$$



PROBLEM 9.86

The two beams shown have the same cross section and are joined by a hinge at C. For the loading shown, determine (a) the slope at point A, (b) the deflection at point B. Use $E = 29 \times 10^6$ psi.

SOLUTION

Using free body ABC,

$$+\circlearrowleft \Sigma M_A = 0: 18R_C - (12)(800) = 0 \quad R_C = 533.33 \text{ lb}$$

$$E = 29 \times 10^6 \text{ psi}$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(1.25)(1.25)^3 = 0.20345 \text{ in}^4$$

$$EI = (29 \times 10^6)(0.20345) = 5.900 \times 10^6 \text{ lb} \cdot \text{in}^2$$

Using cantilever beam CD with load R_C ,

Case 1 of Appendix D.

$$y_C = -\frac{R_C L_{CD}^3}{3EI} = -\frac{(533.33)(12)^3}{(3)(5.900 \times 10^6)} = -52.067 \times 10^{-3} \text{ in.}$$

Calculation of θ'_A and y'_B assuming that point C does not move.

Case 5 of Appendix D.

$$P = 800 \text{ lb}, \quad L = 18 \text{ in.}, \quad a = 12 \text{ in.}, \quad b = 6 \text{ in.}$$

$$\theta'_A = -\frac{Pb(L^2 - b^2)}{6EIL} = -\frac{(800)(6)(18^2 - 6^2)}{(6)(5.900 \times 10^6)(18)} = -2.1695 \times 10^{-3} \text{ rad}$$

$$y'_B = -\frac{Pb^2a^2}{3EIL} = -\frac{(800)(6)^2(12)^2}{(3)(5.900 \times 10^6)(18)} = -13.017 \times 10^{-3} \text{ in.}$$

Additional slope and deflection due to movement of point C.

$$\theta''_A = \frac{y_C}{L_{AC}} = -\frac{52.067 \times 10^{-3}}{18} = -2.8926 \times 10^{-3} \text{ rad}$$

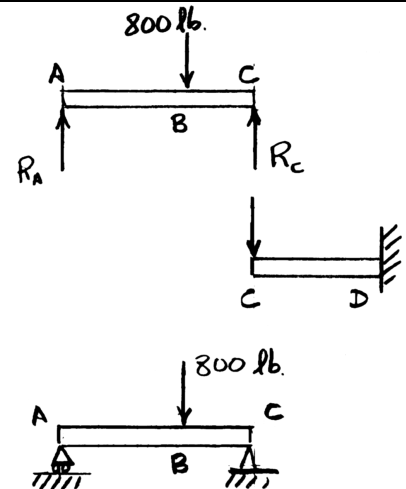
$$y''_B = \frac{a}{L} y_C = -\frac{(12)(52.067 \times 10^{-3})}{18} = -34.711 \times 10^{-3} \text{ in.}$$

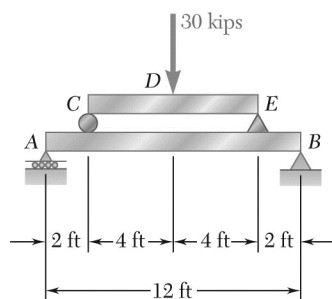
$$(a) \quad \text{Slope at A.} \quad \theta_A = \theta'_A + \theta''_A = -2.1695 \times 10^{-3} - 2.8926 \times 10^{-3}$$

$$\theta_A = 5.06 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

$$(b) \quad \text{Deflection at B.} \quad y_B = y'_B + y''_B = -13.017 \times 10^{-3} - 34.711 \times 10^{-3}$$

$$y_B = 0.0477 \text{ in.} \quad \downarrow \blacktriangleleft$$





PROBLEM 9.87

Beam CE rests on beam AB as shown. Knowing that a $W10 \times 30$ rolled-steel shape is used for each beam, determine for the loading shown the deflection at point D . Use $E = 29 \times 10^6$ psi.

SOLUTION

For $W10 \times 30$, $I = 170 \text{ in}^4$

$$EI = (29 \times 10^6)(170) = 4.93 \times 10^9 \text{ lb} \cdot \text{in}^2 = 34,236 \text{ kip} \cdot \text{ft}^2$$

Beam AB : 15 kip downward loads at C and E .

Refer to Case 5 of Appendix D.

Loading I: $(y_C)_1 = -\frac{Pa^2b^2}{3EIL}$

with $a = 2 \text{ ft}$, $b = 10 \text{ ft}$, $L = 12 \text{ ft}$

$$(y_C)_1 = -\frac{(15)(2)^2(10)^2}{(3)(34236)(12)} = -4.8682 \times 10^{-3} \text{ ft}$$

Loading II: $(y_C)_2 = \frac{Pb[x^2 - (L^2 - b^2)x]}{6EIL}$

with $b = 2 \text{ ft}$, $x = 2 \text{ ft}$, $L = 12 \text{ ft}$

$$(y_C)_2 = \frac{(15)(2)[2^3 - (12^2 - 2^2)(2)]}{(6)(34,236)(12)} = -3.3104 \times 10^{-3} \text{ ft}$$

$$y_C = (y_C)_1 + (y_C)_2 = -8.1786 \times 10^{-3} \text{ ft}$$

By symmetry, $y_E = y_C$

Beam CDE : 30 kip downward loads at D .

Refer to Case 4 of Appendix D.

$$y_{D/C} = -\frac{PL^3}{48EI}$$

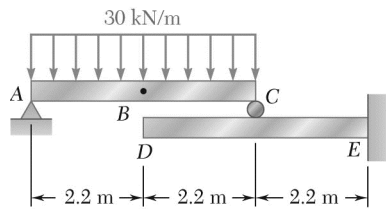
with $P = 30 \text{ kips}$ and $L = 8 \text{ ft}$

$$y_{D/C} = -\frac{(30)(8)^3}{(48)(34,236)} = -9.3469 \times 10^{-3} \text{ ft}$$

Total deflection at D .

$$y_D = y_C + y_{D/C} = 17.5255 \times 10^{-3} \text{ ft}$$

$$y_D = 0.210 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.88

Beam AC rests on the cantilever beam DE as shown. Knowing that a $W410 \times 38.8$ rolled-steel shape is used for each beam, determine for the loading shown (a) the deflection at point B , (b) the deflection at point D . Use $E = 200$ GPa.

SOLUTION

Units: Forces in kN; lengths in m.

Using free body ABC ,

$$+\circlearrowleft \sum M_A = 0: 4.4 R_C - (4.4)(30)(2.2) = 0$$

$$R_C = 66.0 \text{ kN}$$

$$E = 200 \times 10^9 \text{ Pa}$$

$$I = 125 \times 10^6 \text{ mm}^4 = 125 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(125 \times 10^{-6}) = 25.0 \times 10^6 \text{ N} \cdot \text{m}^2 \\ = 25,000 \text{ kN} \cdot \text{m}^2$$

For slope and deflection at C , use Case 1 of Appendix D applied to portion CE of beam DCE .

$$\theta_C = \frac{R_C L^2}{2EI} = \frac{(66.0)(2.2)^2}{(2)(25,000)} = 6.3888 \times 10^{-3} \text{ rad}$$

$$y_C = -\frac{R_C L^3}{3EI} = \frac{(66.0)(2.2)^3}{(3)(25,000)} = -9.3702 \times 10^{-3} \text{ m}$$

Deflection at B , assuming that point C does not move.

$$\text{Use Case 6 of Appendix D.} \quad (y_B)_1 = -\frac{5WL^4}{384EI} = -\frac{(5)(30)(4.4)^4}{(384)(25,000)} = -5.8564 \times 10^{-3}$$

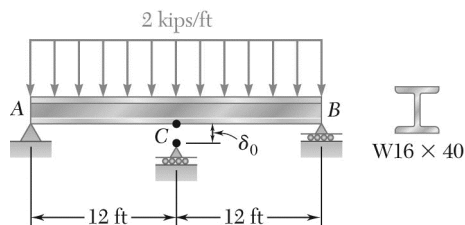
$$\text{Additional deflection at } B \text{ due to movement of point } C: \quad (y_B)_2 = \frac{1}{2} y_C = -4.6851 \times 10^{-3} \text{ m}$$

$$(a) \quad \text{Total deflection at } B. \quad y_B = (y_B)_1 + (y_B)_2 = -10.54 \times 10^{-3} \text{ m} \quad y_B = 10.54 \text{ mm} \downarrow \blacktriangleleft$$

Portion DC of beam DCB remains straight.

$$(b) \quad \text{Deflection at } D. \quad y_D = y_C - a\theta_C = -9.3702 \times 10^{-3} - (2.2)(6.3888 \times 10^{-3}) = -23.4 \times 10^{-3} \text{ m}$$

$$y_D = 23.4 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.89

Before the 2-kip/ft load is applied, a gap, $\delta_0 = 0.8$ in., exists between the W16×40 beam and the support at C. Knowing that $E = 29 \times 10^6$ psi, determine the reaction at each support after the uniformly distributed load is applied.

SOLUTION

Data:

$$\delta_0 = 0.8 \text{ in.} = 66.667 \times 10^{-3} \text{ ft}$$

$$E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi}$$

$$I = 518 \text{ in}^4$$

$$EI = 15.022 \times 10^6 \text{ kip} \cdot \text{in}^2$$

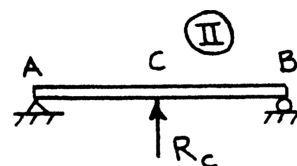
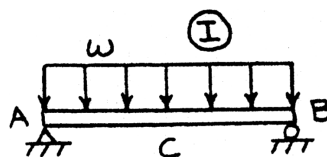
$$= 104.319 \times 10^3 \text{ kip} \cdot \text{ft}^2$$

Loading I: Case 6 of Appendix D.

$$y'_C = -\frac{5wL^4}{384EI}$$

$$= -\frac{5(2)(24)^4}{384(104.319 \times 10^3)}$$

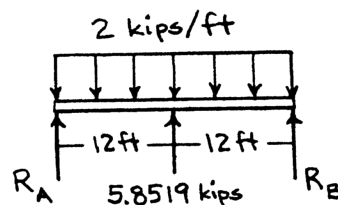
$$= -82.823 \times 10^{-3} \text{ ft}$$



Loading II: Case 4 of Appendix D.

$$y''_C = \frac{R_C L^3}{48EI} = \frac{R_C (24)^3}{48(104.319 \times 10^3)}$$

$$= 2.7608 \times 10^{-3} R_C$$



Deflection at C.

$$y_C = y'_C + y''_C = -\delta_0$$

$$-82.823 \times 10^{-3} + 2.7608 \times 10^{-3} R_C = -66.667 \times 10^{-3}$$

$$R_C = 5.8519 \text{ kips}$$

$$R_C = 5.85 \text{ kips} \uparrow \blacktriangleleft$$

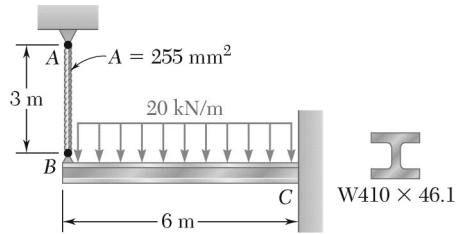
$$+\circlearrowleft \Sigma M_B = 0: (2)(24)(12) - R_A(24) - (5.8519)(12) = 0$$

$$R_A = 21.074 \text{ kips}$$

$$R_A = 21.1 \text{ kips} \uparrow \blacktriangleleft$$

$$+\uparrow \Sigma F_y = 0: 21.074 - 2(24) + 5.8519 + R_B = 0$$

$$R_B = 21.1 \text{ kips} \uparrow \blacktriangleleft$$



PROBLEM 9.90

The cantilever beam BC is attached to the steel cable AB as shown. Knowing that the cable is initially taut, determine the tension in the cable caused by the distributed load shown. Use $E = 200$ GPa.

SOLUTION

Let P be the tension developed in member AB and δ_B be the elongation of that member.

Cable AB :

$$A = 255 \text{ mm}^2 = 255 \times 10^{-6} \text{ m}^2$$

$$\begin{aligned} \delta_B &= \frac{PL}{EA} = \frac{(P)(3)}{(200 \times 10^9)(255 \times 10^{-6})} \\ &= 58.82 \times 10^{-9} P \end{aligned}$$

Beam BC :

$$\begin{aligned} I &= 156 \times 10^6 \text{ mm}^4 = 156 \times 10^{-6} \text{ m}^4 \\ EI &= (200 \times 10^9)(156 \times 10^{-6}) \\ &= 31.2 \times 10^6 \text{ N} \cdot \text{m}^2 \end{aligned}$$

Loading I: 20 kN/m downward.

Refer to Case 2 of Appendix D.

$$\begin{aligned} (y_B)_1 &= -\frac{wL^4}{8EI} = -\frac{(20 \times 10^3)(6)^4}{(8)(31.2 \times 10^6)} \\ &= -103.846 \times 10^{-3} \text{ m} \end{aligned}$$

Loading II: Upward force P at Point B .

Refer to Case 1 of Appendix D.

$$(y_B)_2 = \frac{PL^3}{3EI} = \frac{P(6)^3}{(3)(31.2 \times 10^6)} = 2.3077 \times 10^{-6} P$$

By superposition, $y_B = (y_B)_1 + (y_B)_2$

Also, matching the deflection at B ,

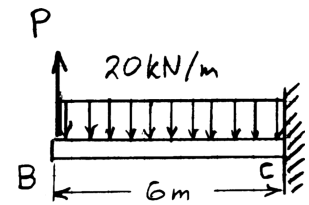
$$y_B = -\delta_B$$

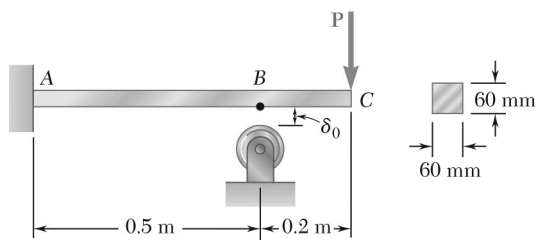
$$-103.846 \times 10^{-3} + 2.3077 \times 10^{-6} P = -58.82 \times 10^{-9} P$$

$$2.3666 \times 10^{-6} P = 103.846 \times 10^{-3}$$

$$P = 43.9 \times 10^3 \text{ N}$$

$$P = 43.9 \text{ kN} \quad \blacktriangleleft$$





PROBLEM 9.91

Before the load P was applied, a gap, $\delta_0 = 0.5 \text{ mm}$, existed between the cantilever beam AC and the support at B . Knowing that $E = 200 \text{ GPa}$, determine the magnitude of P for which the deflection at C is 1 mm .

SOLUTION

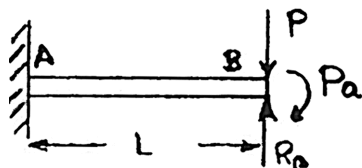
Let length $AB = L = 0.5 \text{ m}$

length $BC = a = 0.2 \text{ m}$

Consider portion AB of beam ABC .

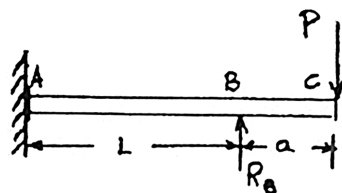
The loading becomes forces P and R_B at B plus the couple Pa .

The deflection at B is δ_0 . Using Cases 1 and 3 of Appendix D,



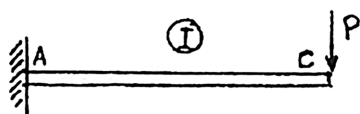
$$\delta_0 = \frac{(P - R_B)L^3}{3EI} + \frac{PaL^2}{2EI}$$

$$\left(\frac{L^3}{3} + \frac{L^2a}{2}\right)P - \frac{L^3}{3}R_B = EI\delta_0 \quad (1)$$



The deflection at C depends on the deformation of beam ABC subjected to loads P and R_B . For loading I, using Case 1 of Appendix D,

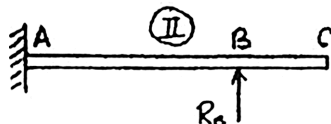
$$(\delta_C)_I = \frac{P(L+a)^3}{3EI}$$



For loading II, using Case 1 of Appendix D,

$$y_B = \frac{R_B L^3}{3EI} \quad \theta_B = \frac{R_B L^2}{2EI}$$

Portion BC remains straight.



$$y_C = y_B + a\theta_B = \left(\frac{L^3}{3} + \frac{L^2a}{2}\right)\frac{R_B}{EI}$$

By superposition, the downward deflection at C is

$$\delta_C = \frac{P(L+a)^3}{3EI} - \left(\frac{L^3}{3} + \frac{L^2a}{2}\right)\frac{R_B}{EI}$$

$$\frac{(L+a)^3}{3}P - \left(\frac{L^3}{3} + \frac{L^2a}{2}\right)R_B = EI\delta_C \quad (2)$$

PROBLEM 9.91 (Continued)

Data: $E = 200 \times 10^9 \text{ Pa}$ $I = \frac{1}{12}(60)(60)^3 = 1.08 \times 10^6 \text{ mm}^4 = 1.08 \times 10^{-6} \text{ m}^4$

$$EI = 216 \times 10^3 \text{ N} \cdot \text{m}^2$$

$$\delta_0 = 0.5 \times 10^{-3} \text{ m} \quad \delta_C = 1.0 \times 10^{-3} \text{ m}$$

Using the data, eqs (1) and (2) become

$$0.06667 P - 0.04167 R_B = 108 \quad (1)'$$

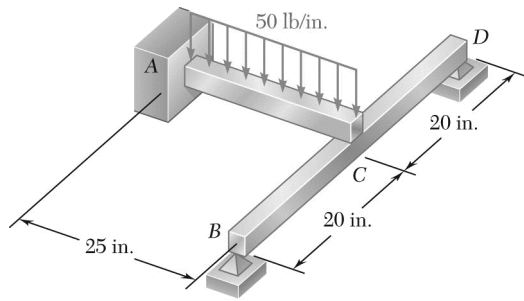
$$0.11433 P - 0.06667 R_B = 216 \quad (2)'$$

Solving simultaneously,

$$P = 5.63 \times 10^3 \text{ N}$$

$$P = 5.63 \text{ kN} \downarrow \blacktriangleleft$$

$$R_B = 6.42 \times 10^3 \text{ N}$$



PROBLEM 9.92

For the loading shown, knowing that beams AC and BD have the same flexural rigidity, determine the reaction at B .

SOLUTION

Consider the two beams shown at the right.

Let R_C be the contact force between beams AC and BCD .

Applying Cases 1 and 2 of Appendix D to cantilever beam AC ,

$$y_C = \frac{R_C a^3}{3EI} - \frac{wa^4}{8EI}$$

Applying Case 4 of Appendix D to simply supported beam BCD ,

$$y_C = -\frac{R_C L^3}{48EI}$$

Equating expressions for y_C ,

$$\frac{R_C a^3}{3EI} - \frac{wa^4}{8EI} = -\frac{R_C L^3}{48EI}$$

$$(16a^3 + L^3)R_C = 6wa^4$$

$$R_C = \frac{6wa}{16 + L^3/a^3}$$

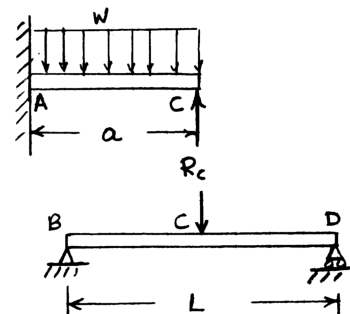
Data: $w = 50 \text{ lb/in.}$, $a = 25 \text{ in.}$, $L = 20 + 20 = 40 \text{ in.}$

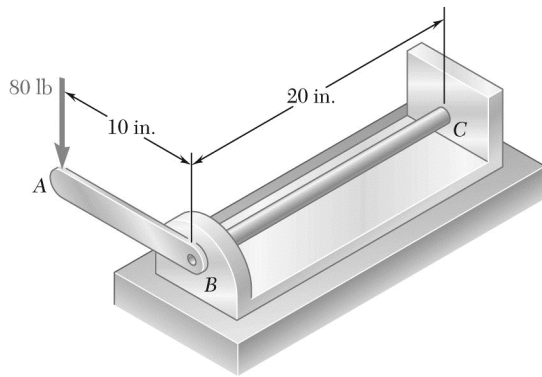
$$R_C = \frac{(6)(50)(25)}{16 + (40/25)^3} = 373.21 \text{ lb}$$

Using beam BCD as a free body,

$$+\circlearrowleft \Sigma M_D = 0: -R_B L + R_C \frac{L}{2} = 0$$

$$R_B = \frac{1}{2} R_C = 186.6 \text{ lb } \uparrow \blacktriangleleft$$





PROBLEM 9.93

A $\frac{7}{8}$ -in.-diameter rod BC is attached to the lever AB and to the fixed support at C . Lever AB has a uniform cross section $\frac{3}{8}$ in. thick and 1 in. deep. For the loading shown, determine the deflection of point A . Use $E = 29 \times 10^6$ psi and $G = 11.2 \times 10^6$ psi.

SOLUTION

Deformation of rod BC . (Torsion)

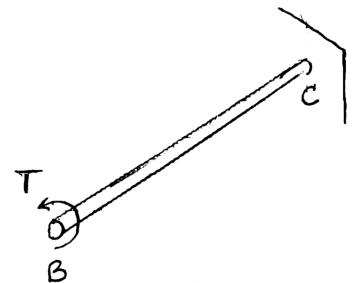
$$c = \frac{1}{2}d = \frac{1}{2}\left(\frac{7}{8}\right) = 0.4375 \text{ in.}$$

$$J = \frac{\pi}{2}C^4 = 57.548 \times 10^{-3} \text{ in}^4$$

$$T = Pa = (80)(10) = 800 \text{ lb} \cdot \text{in}$$

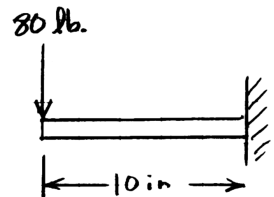
$$L = 20 \text{ in.}$$

$$\begin{aligned}\phi_B &= \frac{TL}{GJ} = \frac{(800)(20)}{(11.2 \times 10^6)(57.548 \times 10^{-3})} \\ &= 24.824 \times 10^{-3} \text{ rad}\end{aligned}$$



Deflection of point A assuming lever AB to be rigid.

$$\begin{aligned}(y_A)_1 &= a\phi_B = (10)(24.824 \times 10^{-3}) \\ &= 0.24824 \text{ in.} \downarrow\end{aligned}$$



Additional deflection due to bending of lever AB .

Refer to Case 1 of Appendix D.

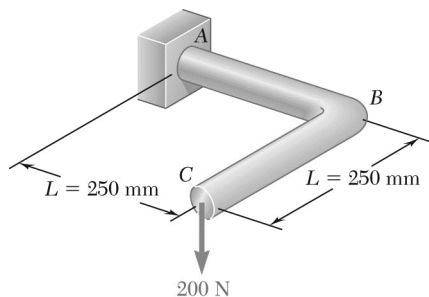
$$I = \frac{1}{12}\left(\frac{3}{8}\right)(1)^3 = 31.25 \times 10^{-3} \text{ in}^4$$

$$\begin{aligned}(y_A)_2 &= \frac{PL^3}{3EI} = \frac{(80)(10)^3}{(3)(29 \times 10^6)(31.25 \times 10^{-3})} \\ &= 0.02943 \text{ in.} \downarrow\end{aligned}$$

Total deflection at point A .

$$y_A = (y_A)_1 + (y_A)_2$$

$$y_A = 0.278 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.94

A 16-mm-diameter rod has been bent into the shape shown. Determine the deflection of end C after the 200-N force is applied. Use $E = 200$ GPa and $G = 80$ GPa.

SOLUTION

Let $200 \text{ N} = P$.

Consider torsion of rod AB .

$$\phi_B = \frac{TL}{JG} = \frac{(PL)L}{JG} = \frac{PL^2}{JG}$$

$$y'_C = -L\phi_B = -\frac{PL^3}{JG}$$

Consider bending of AB . (Case 1 of Appendix D.)

$$y''_C = y_B = -\frac{PL^3}{3EI}$$

Consider bending of BC . (Case 1 of Appendix D.)

$$y'''_C = -\frac{PL^3}{3EI}$$

Superposition:

$$y_C = y'_C + y''_C + y'''_C$$

$$= -\frac{PL^3}{JG} - \frac{PL^3}{3EI} - \frac{PL^3}{3EI} = -\frac{PL^3}{EI} \left(\frac{EI}{JG} + \frac{2}{3} \right)$$

Data:

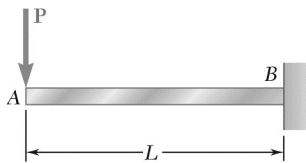
$$G = 80(10^9) \text{ Pa} \quad J = \frac{1}{2}\pi(0.008)^4 = 6.4340(10^{-9}) \text{ m}^4$$

$$E = 200(10^9) \text{ Pa} \quad I = \frac{1}{2}J = 3.2170(10^{-9}) \text{ m}^4$$

$$EI = 643.40 \text{ N} \cdot \text{m}^2 \quad JG = 514.72 \text{ N} \cdot \text{m}^2$$

$$y_C = -\frac{(200)(0.25)^3}{643.40} \left(\frac{643.40}{514.72} + \frac{2}{3} \right) = -9.3093(10^{-3}) \text{ m}$$

$$y_C = 9.31 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.95

For the uniform cantilever beam and loading shown, determine (a) the slope at the free end, (b) the deflection at the free end.

SOLUTION

Place reference tangent at B .

Draw M/EI diagram.

$$A = \frac{1}{2} \left(-\frac{PL}{EI} \right) L = -\frac{PL^2}{2EI}$$

$$\bar{x} = \frac{2}{3}L$$

$$\theta_{B/A} = A = -\frac{PL^2}{2EI}$$

$$t_{A/B} = A\bar{x} = \left(-\frac{PL^2}{2EI} \right) \left(\frac{2}{3}L \right) = -\frac{PL^3}{3EI}$$

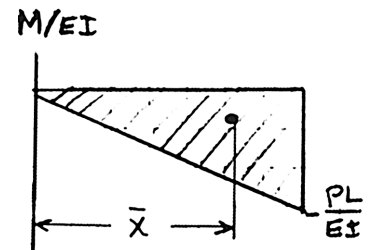
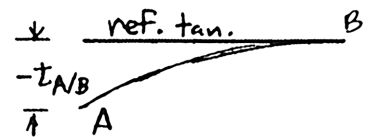
(a) Slope at end A .

$$\theta_B = \theta_A + A$$

$$0 = \theta_A - \frac{PL^2}{2EI}$$

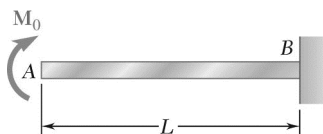
(b) Deflection at A .

$$y_A = t_{A/B} = -\frac{PL^3}{3EI}$$



$$\theta_A = \frac{PL^2}{2EI} \nearrow \blacktriangleleft$$

$$y_A = \frac{PL^3}{3EI} \downarrow \blacktriangleleft$$



PROBLEM 9.96

For the uniform cantilever beam and loading shown, determine (a) the slope at the free end, (b) the deflection at the free end.

SOLUTION

Place reference tangent at B .

Draw M/EI diagram.

$$A = \left(\frac{M_0}{EI} \right) L = \frac{M_0 L}{EI}$$

$$\bar{x} = \frac{1}{2} L$$

$$\theta_{B/A} = A = \frac{M_0 L}{EI}$$

$$t_{B/A} = A\bar{x} = \left(\frac{M_0 L}{EI} \right) \left(\frac{1}{2} L \right) = \frac{M_0 L^2}{2EI}$$

(a) Slope at end A .

$$\theta_B = \theta_A + A$$

$$0 = \theta_A + \frac{M_0 L}{EI}$$

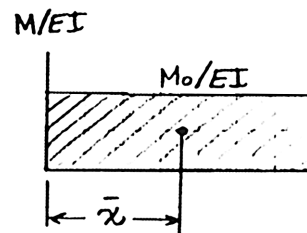
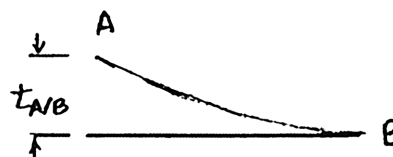
$$\theta_A = -\frac{M_0 L}{EI}$$

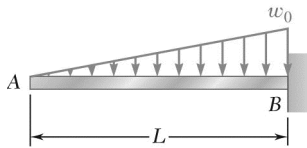
$$\theta_A = \frac{M_0 L}{EI} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at A .

$$y_A = t_{A/B} = \frac{M_0 L^2}{2EI}$$

$$y_A = \frac{M_0 L^2}{2EI} \quad \uparrow \blacktriangleleft$$





PROBLEM 9.97

For the uniform cantilever beam and loading shown, determine (a) the slope at the free end, (b) the deflection at the free end.

SOLUTION

Place reference tangent at B.

$$\theta_B = 0$$

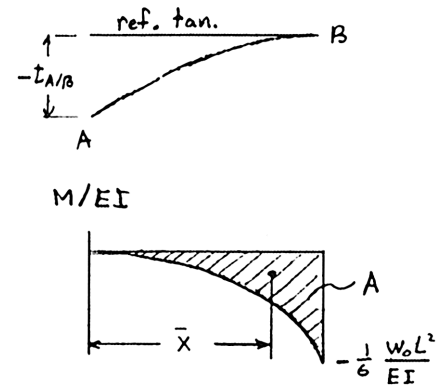
$$+\circlearrowleft \Sigma M_B = 0: \left(\frac{1}{2} w_0 L \right) \frac{L}{3} + M_B = 0$$

$$M_B = -\frac{1}{6} w_0 L^2$$

Draw M/EI curve as cubic parabola.

$$A = -\frac{1}{4} \left(\frac{1}{6} \frac{w_0 L^2}{EI} \right) L = -\frac{1}{24} \frac{w_0 L^3}{EI}$$

$$\bar{x} = L - \frac{1}{5} L = \frac{4}{5} L$$



By first moment-area theorem,

$$\theta_{B/A} = A = -\frac{1}{24} \frac{w_0 L^3}{EI}$$

$$\theta_B = \theta_A + \theta_{B/A}$$

$$\theta_A = \theta_B - \theta_{B/A} = 0 + \frac{1}{24} \frac{w_0 L^3}{EI} = \frac{1}{24} \frac{w_0 L^3}{EI}$$

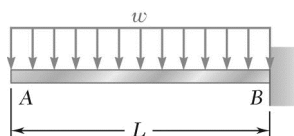
By second moment-area theorem,

$$t_{A/B} = \bar{x} A = \left(\frac{4}{5} L \right) \left(-\frac{1}{24} \frac{w_0 L^3}{EI} \right) = -\frac{1}{30} \frac{w_0 L^4}{EI}$$

$$y_A = t_{A/B} = -\frac{1}{30} \frac{w_0 L^4}{EI}$$

$$(a) \quad \theta_A = \frac{w_0 L^3}{24EI} \nearrow \blacktriangleleft$$

$$(b) \quad y_A = \frac{w_0 L^4}{30EI} \downarrow \blacktriangleleft$$



PROBLEM 9.98

For the uniform cantilever beam and loading shown, determine (a) the slope at the free end, (b) the deflection at the free end.

SOLUTION

Place reference tangent at B.

$$\theta_B = 0$$

Draw M/EI curve as parabola.

$$A = -\frac{1}{3} \left(\frac{wL^2}{2EI} \right) L = -\frac{1}{6} \frac{wL^3}{EI}$$

$$\bar{x} = L - \frac{1}{4}L = \frac{3}{4}L$$

By first moment-area theorem,

$$\theta_{B/A} = A = -\frac{1}{6} \frac{wL^3}{EI}$$

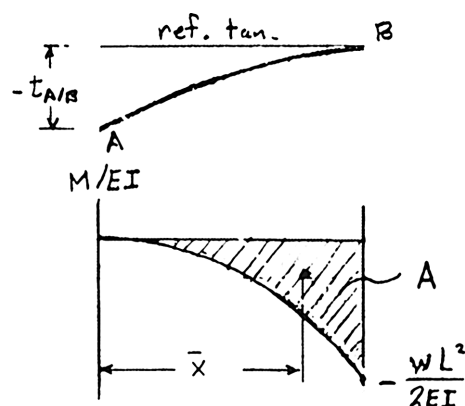
$$\theta_B = \theta_A + \theta_{B/A}$$

$$\theta_A = \theta_B - \theta_{B/A} = 0 + \frac{1}{6} \frac{wL^3}{EI} = \frac{1}{6} \frac{wL^3}{EI}$$

By second moment-area theorem,

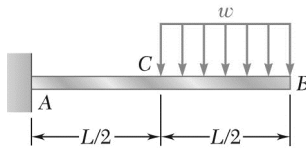
$$t_{A/B} = \bar{x}A = \left(\frac{3}{4}L \right) \left(-\frac{1}{6} \frac{wL^3}{EI} \right) = -\frac{1}{8} \frac{wL^4}{EI}$$

$$y_A = t_{A/B} = -\frac{1}{8} \frac{wL^4}{EI}$$



$$(a) \quad \theta_A = \frac{wL^3}{6EI} \nearrow \blacktriangleleft$$

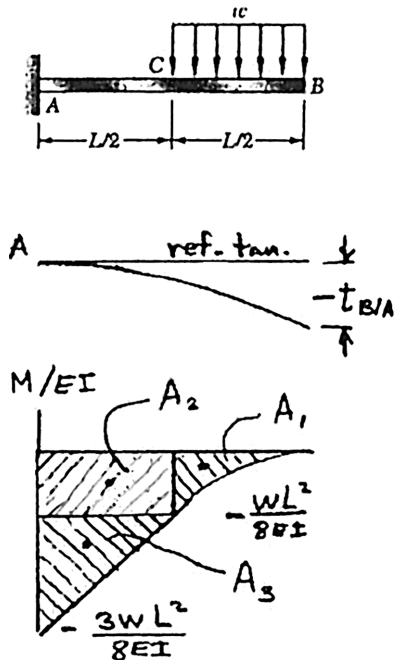
$$(b) \quad y_A = \frac{wL^4}{8EI} \downarrow \blacktriangleleft$$



PROBLEM 9.99

For the uniform cantilever beam and loading shown, determine the slope and deflection at (a) point B, (b) point C.

SOLUTION



Place reference tangent at A.

Draw $\frac{M}{EI}$ diagram.

$$A_1 = \frac{1}{3} \left(-\frac{wL^2}{8EI} \right) \left(\frac{L}{2} \right) = -\frac{wL^3}{48EI}$$

$$A_2 = \left(-\frac{wL^2}{8EI} \right) \left(\frac{L}{2} \right) = -\frac{wL^3}{16EI}$$

$$A_3 = \frac{1}{2} \left(-\frac{2wL^2}{8EI} \right) \left(\frac{L}{2} \right) = -\frac{wL^3}{16EI}$$

$$\theta_A = 0, \quad y_A = 0$$

(a) Slope at B.

$$\begin{aligned} \theta_B &= \theta_A + A_1 + A_2 + A_3 \\ &= -\frac{7wL^3}{48EI} \end{aligned}$$

$$\theta_B = \frac{7wL^3}{48EI} \quad \swarrow \blacktriangleleft$$

Deflection at B. $y_B = t_{B/A} = A_1 \left(\frac{3}{4} \cdot \frac{L}{2} \right) + A_2 \left(\frac{L}{2} + \frac{1}{2} \cdot \frac{L}{2} \right) + A_3 \left(\frac{L}{2} + \frac{2}{3} \cdot \frac{L}{2} \right)$

$$= -\frac{wL^4}{128EI} - \frac{3wL^4}{64EI} - \frac{5wL^4}{96EI} = -\frac{41wL^4}{384EI}$$

$$y_B = \frac{41wL^4}{384EI} \quad \downarrow \blacktriangleleft$$

(b) Slope at C.

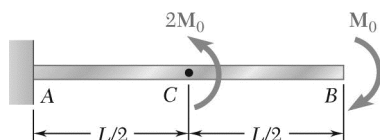
$$\theta_C = \theta_A + A_2 + A_3 = -\frac{wL^3}{8EI}$$

$$\theta_C = \frac{wL^3}{8EI} \quad \swarrow \blacktriangleleft$$

Deflection at C.

$$\begin{aligned} y_C = t_{C/A} &= A_2 \left(\frac{1}{2} \cdot \frac{L}{2} \right) + A_3 \left(\frac{R}{3} \cdot \frac{L}{2} \right) \\ &= -\frac{7wL^4}{192EI} \end{aligned}$$

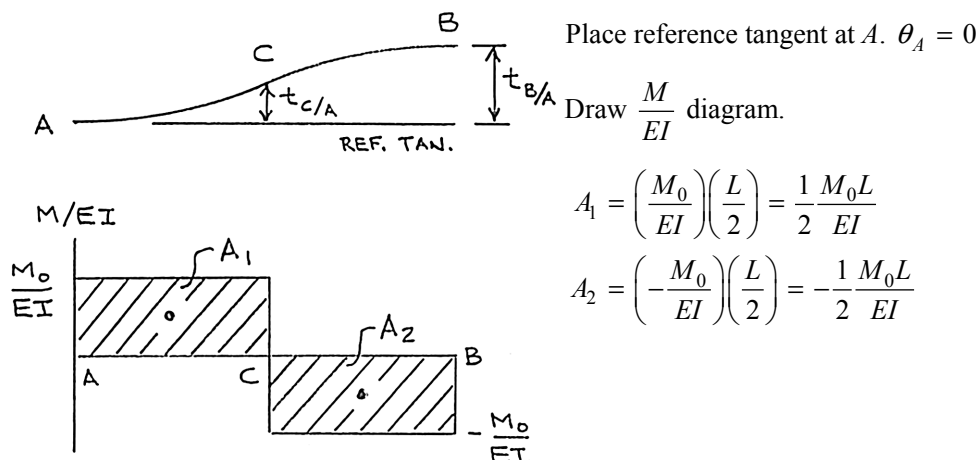
$$y_C = \frac{7wL^4}{192EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.100

For the uniform cantilever beam and loading shown, determine the slope and deflection at (a) point B, (b) point C.

SOLUTION



(a) Slope at B.

$$\theta_{B/A} = A_1 + A_2 = \frac{1}{2} \frac{M_0 L}{EI} - \frac{1}{2} \frac{M_0 L}{EI} = 0$$

$$\theta_B = \theta_A + \theta_{B/A} = 0$$

$$\theta_B = 0 \quad \blacktriangleleft$$

Deflection at B.

$$y_B = t_{B/A} = A_1 \left(\frac{L}{2} + \frac{1}{2} \cdot \frac{L}{2} \right) + A_2 \left(\frac{1}{2} \cdot \frac{L}{2} \right)$$

$$= \frac{3}{8} \frac{M_0 L^2}{EI} - \frac{1}{8} \frac{M_0 L^2}{EI} = \frac{1}{4} \frac{M_0 L^2}{EI}$$

$$y_B = \frac{1}{4} \frac{M_0 L^2}{EI} \uparrow \quad \blacktriangleleft$$

(b) Slope at C.

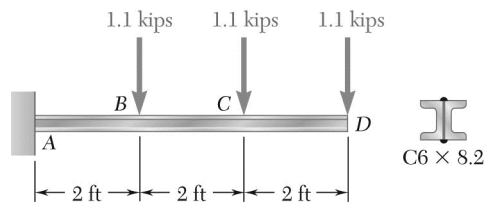
$$\theta_{C/A} = A_1 = \frac{1}{2} \frac{M_0 L}{EI} \quad \theta_C = \theta_A + \theta_{C/A}$$

$$\theta_C = \frac{1}{2} \frac{M_0 L}{EI} \quad \blacktriangleleft$$

Deflection at C.

$$y_C = t_{C/A} = A_1 \left(\frac{1}{2} \cdot \frac{L}{2} \right) = \frac{1}{8} \frac{M_0 L^2}{EI}$$

$$y_C = \frac{1}{8} \frac{M_0 L^2}{EI} \uparrow \quad \blacktriangleleft$$



PROBLEM 9.101

Two C6×8.2 channels are welded back to back and loaded as shown. Knowing that $E = 29 \times 10^6$ psi, determine (a) the slope at point D, (b) the deflection at point D.

SOLUTION

Units: Forces in kips; lengths in ft.

$$E = 29 \times 10^6 \text{ psi} = 29 \times 10^3 \text{ ksi} \quad I = (2)(13.1) = 26.2 \text{ in}^4$$

$$EI = (29 \times 10^3)(26.2) = 759.8 \times 10^3 \text{ kip} \cdot \text{in}^2 = 5276 \text{ kip} \cdot \text{ft}^2$$

Draw M/EI diagram by parts.

$$\frac{M_1}{EI} = \frac{(1.1)(6)}{EI} = -\frac{6.6}{EI} \text{ ft}^{-1}$$

$$A_1 = \frac{1}{2} \left(\frac{6.6}{EI} \right) (6) = -\frac{19.8}{EI} \quad \bar{x}_1 = \frac{1}{3} (6) = 2 \text{ ft}$$

$$\frac{M_2}{EI} = -\frac{(1.1)(4)}{EI} = -\frac{4.4}{EI} \text{ ft}^{-1}$$

$$A_2 = \frac{1}{2} \left(-\frac{4.4}{EI} \right) (4) = -\frac{8.8}{EI} \quad \bar{x}_2 = \frac{1}{3} (4) = \frac{4}{3} \text{ ft}$$

$$\frac{M_3}{EI} = -\frac{(1.1)(2)}{EI} = -\frac{2.2}{EI} \text{ ft}^{-1}$$

$$A_3 = \frac{1}{2} \left(-\frac{2.2}{EI} \right) (2) = -\frac{2.2}{EI} \quad \bar{x}_3 = \frac{1}{3} (2) = \frac{2}{3} \text{ ft}$$

Place reference tangent at A.

$$\theta_A = 0$$

(a) Slope at D. $\theta_{D/A} = A_1 + A_2 + A_3 = -\frac{30.8}{EI} = -\frac{30.8}{5276} = -5.84 \times 10^{-3} \text{ rad}$

$$\theta_D = \theta_A + \theta_{D/A}$$

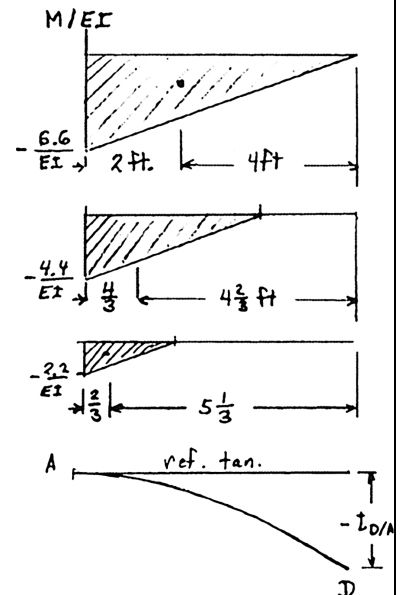
$$\theta_D = 5.89 \times 10^{-3} \text{ rad.} \quad \swarrow \blacktriangleleft$$

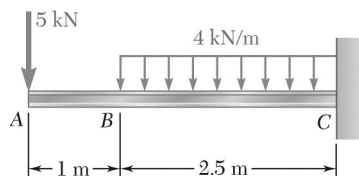
(b) Deflection at D.
$$t_{D/A} = \left(-\frac{19.8}{EI} \right) (4) + \left(-\frac{8.8}{EI} \right) \left(4\frac{2}{3} \right) + \left(-\frac{2.2}{EI} \right) \left(5\frac{1}{3} \right)$$

$$= -\frac{132.0}{EI} = -\frac{132.0}{5276} = 25.02 \times 10^{-3} \text{ ft}$$

$$y_D = t_{D/A} = 25.02 \times 10^{-3} \text{ ft}$$

$$y_D = 0.300 \text{ in.} \quad \downarrow \blacktriangleleft$$



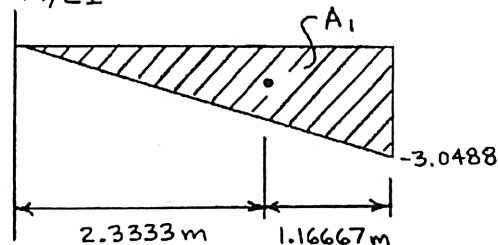


PROBLEM 9.102

For the cantilever beam and loading shown, determine (a) the slope at point A, (b) the deflection at point A. Use $E = 200 \text{ GPa}$.

SOLUTION

10^{-3} M/EI



Units: Forces in kN; lengths in m.

$$E = 200 \times 10^9 \text{ Pa}$$

$$I = 28.7 \times 10^6 \text{ mm}^4 = 28.7 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(28.7 \times 10^{-6})$$

$$= 5.74 \times 10^6 \text{ N} \cdot \text{m}^2$$

$$= 5740 \text{ kN} \cdot \text{m}^2$$

Draw M/EI diagram by parts:

$$\frac{M_1}{EI} = -\frac{(5)(3.5)}{5740} = -3.0488 \times 10^{-3} \text{ m}^{-1}$$

$$A_1 = \frac{1}{2}(-3.0488 \times 10^{-3})(3.5) = -5.3354 \times 10^{-3}$$

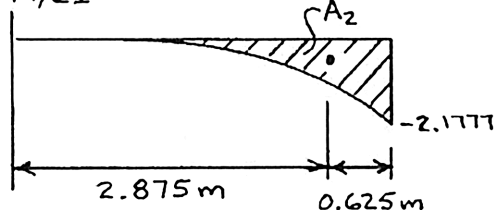
$$\bar{x}_1 = \frac{1}{3}(3.5) = 1.16667 \text{ m}$$

$$\frac{M_2}{EI} = -\frac{(4)(2.5)^2}{(2)(5740)} = -2.1777 \times 10^{-3} \text{ m}^{-1}$$

$$A_2 = \frac{1}{3}(-2.1777 \times 10^{-3})(2.5) = -1.81475 \times 10^{-3}$$

$$\bar{x}_2 = \frac{1}{4}(2.5) = 0.625 \text{ m}$$

10^{-3} M/EI



Place reference tangent at C.

$$\theta_C = 0$$

$$\theta_{C/A} = A_1 + A_2 = -7.1502 \times 10^{-3}$$

(a) Slope at A.

$$\theta_A = \theta_C - \theta_{C/A} = 7.1502 \times 10^{-3}$$

$$\theta_A = 7.15 \times 10^{-3} \text{ rad} \swarrow \blacktriangleleft$$

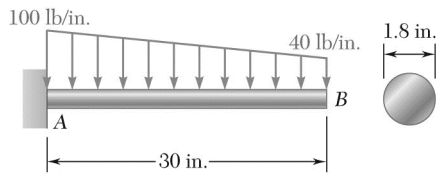
$$t_{A/C} = (2.3333)(-5.3354 \times 10^{-3}) + (2.875)(-1.81475 \times 10^{-3})$$

$$= -17.6665 \times 10^{-3} \text{ m}$$

(b) Deflection at A.

$$y_A = t_{AC} = -17.67 \times 10^{-3} \text{ m}$$

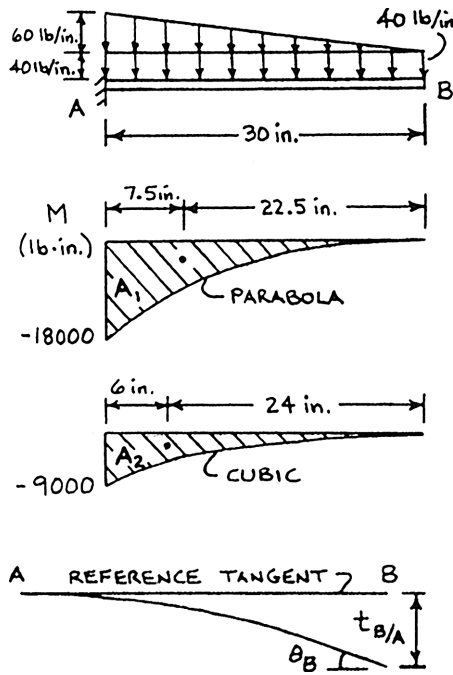
$$y_A = 17.67 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.103

For the cantilever beam and loading shown, determine (a) the slope at point B, (b) the deflection at point B. Use $E = 29 \times 10^6$ psi.

SOLUTION



$$I = \frac{\pi}{4} \left(\frac{1.8}{2} \right)^4 = 0.51530 \text{ in}^4$$

$$EI = (29 \times 10^6)(0.51530) = 14.9437 \times 10^6 \text{ lb} \cdot \text{in}^2$$

$$M_A = -\frac{wL^2}{2} = -\frac{(40)(30)^2}{2} = -18000 \text{ lb} \cdot \text{in}$$

$$A_1 = \frac{1}{3}(-18000)(30) = -180 \times 10^3 \text{ lb} \cdot \text{in}^2$$

$$M_A = -\frac{wL^2}{6} = -\frac{(60)(30)^2}{6} = -9000 \text{ lb} \cdot \text{in}$$

$$A_2 = \frac{1}{4}(-9000)(30) = -67.5 \times 10^3 \text{ lb} \cdot \text{in}^2$$

(a) Slope at B.

$$EI\theta_{B/A} = A_1 + A_2 = -180 \times 10^3 - 67.5 \times 10^3$$

$$= -247.5 \times 10^3 \text{ lb} \cdot \text{in}^2$$

$$\theta_B = \theta_{B/A} = \frac{-247.5 \times 10^3}{14.9437 \times 10^6} = -16.5622 \times 10^{-3} \text{ rad}$$

$$\theta_B = 16.56 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at B.

$$EIy_B = EI t_{B/A}$$

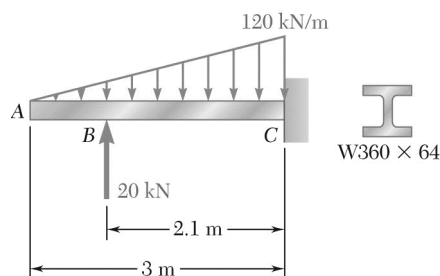
$$= (-180 \times 10^3)(22.5) + (-67.5 \times 10^3)(24)$$

$$= -5.67 \times 10^6 \text{ lb} \cdot \text{in}^3$$

$$y_B = \frac{-5.67 \times 10^6}{14.9437 \times 10^6}$$

$$= -0.37942 \text{ in.}$$

$$y_B = 0.379 \text{ in.} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.104

For the cantilever beam and loading shown, determine (a) the slope at point A, (b) the deflection at point A. Use $E = 200$ GPa.

SOLUTION

Units: Forces in kN; lengths in meters.

$$I = 178 \times 10^6 \text{ mm}^4 = 178 \times 10^{-6}$$

$$EI = (200 \times 10^9)(178 \times 10^{-6}) = 35600 \text{ kN} \cdot \text{m}^2$$

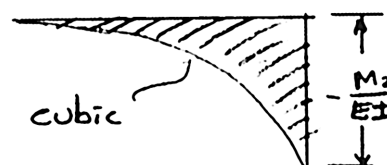
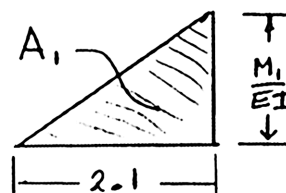
Draw $\frac{M}{EI}$ diagram by parts.

$$\frac{M_1}{EI} = \frac{(20)(2.1)}{35600} = 1.17978 \times 10^{-3} \text{ m}^{-1}$$

$$A_1 = \left(\frac{1}{2}\right)(2.1)(1.17978 \times 10^{-3}) = 1.23876 \times 10^{-3}$$

$$M_2 = -\frac{\left(\frac{1}{2}\right)(120)(3)(1)}{35600} = -5.0562 \times 10^{-3} \text{ m}^{-1}$$

$$A_2 = \left(\frac{1}{4}\right)(3)(-5.0562 \times 10^{-3}) = -3.7921 \times 10^{-3}$$



Place reference tangent at C.

$$\theta_C = 0$$

(a) Slope at A.

$$\theta_A = -\theta_{C/A} = -A_1 - A_2$$

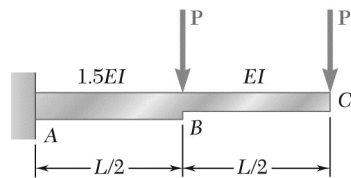
$$\theta_A = 2.55 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at A.

$$y_A = t_{A/C}$$

$$y_C = A_1\left(3 - 0.7\right) + A_2\left(3 - \frac{3}{5}\right) = -6.25 \times 10^{-3} \text{ m}$$

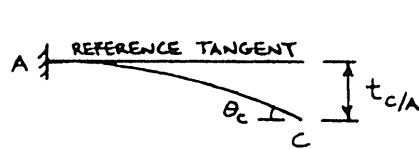
$$y_C = 6.25 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.105

For the cantilever beam and loading shown, determine (a) the slope at point C, (b) the deflection at point C.

SOLUTION



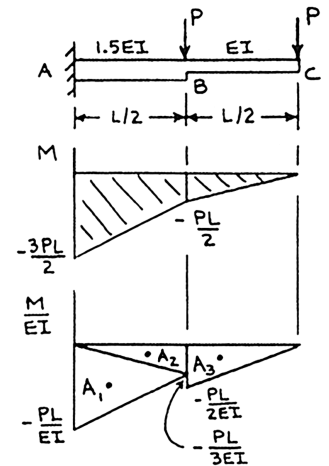
$$A_1 = \frac{1}{2} \left(-\frac{PL}{EI} \right) \left(\frac{L}{2} \right) = -\frac{PL^2}{4EI}$$

$$A_2 = \frac{1}{2} \left(-\frac{PL}{3EI} \right) \left(\frac{L}{2} \right) = -\frac{PL^2}{12EI}$$

$$A_3 = \frac{1}{2} \left(-\frac{PL}{2EI} \right) \left(\frac{L}{2} \right) = -\frac{PL^2}{8EI}$$

(a) Slope at C.

$$\begin{aligned} \theta_C &= A_1 + A_2 + A_3 \\ &= -\frac{PL^2}{EI} \left(\frac{1}{4} + \frac{1}{12} + \frac{1}{8} \right) = -\frac{11PL^2}{24EI} \end{aligned}$$

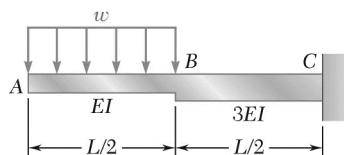


$$\theta_C = \frac{11PL^2}{24EI} \quad \swarrow \blacktriangleleft$$

(b) Deflection at C.

$$\begin{aligned} y_C = t_{C/A} &= A_1 \left(\frac{5}{6}L \right) + A_2 \left(\frac{2}{3}L \right) + A_3 \left(\frac{1}{3}L \right) \\ &= \left(-\frac{PL^2}{4EI} \right) \left(\frac{5}{6}L \right) + \left(-\frac{PL^2}{12EI} \right) \left(\frac{2}{3}L \right) + \left(-\frac{PL^2}{8EI} \right) \left(\frac{1}{3}L \right) = -\frac{22PL^3}{72EI} \end{aligned}$$

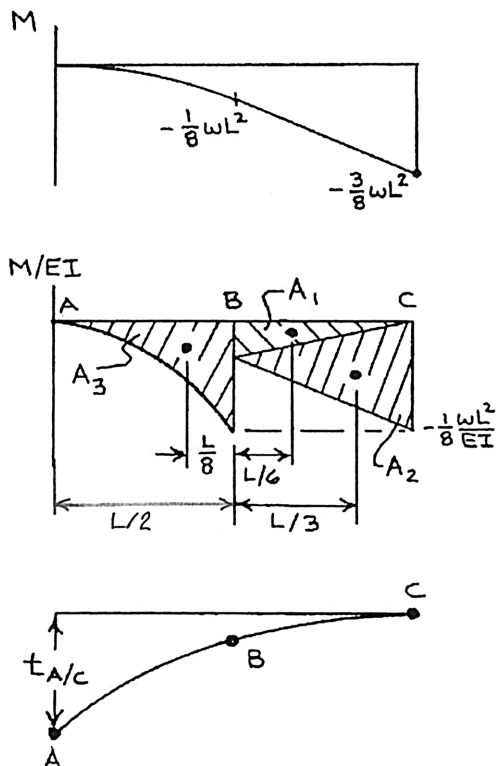
$$y_C = \frac{11PL^3}{36EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.106

For the cantilever beam and loading shown, determine (a) the slope at point A, (b) the deflection at point A.

SOLUTION



Draw the $\frac{M}{EI}$ diagram using the M diagram.

$$A_1 = \frac{1}{2} \left(-\frac{1}{24} \frac{wL^2}{EI} \right) \left(\frac{L}{2} \right) = -\frac{1}{96} \frac{wL^3}{EI}$$

$$A_2 = \frac{1}{2} \left(-\frac{1}{8} \frac{wL^2}{EI} \right) \left(\frac{L}{2} \right) = -\frac{1}{32} \frac{wL^3}{EI}$$

$$A_3 = \frac{1}{3} \left(-\frac{1}{8} \frac{wL^2}{EI} \right) \left(\frac{L}{2} \right) = -\frac{1}{48} \frac{wL^3}{EI}$$

Place reference tangent at C.

(a) Slope at A. $\theta_A = -\theta_{C/A}$

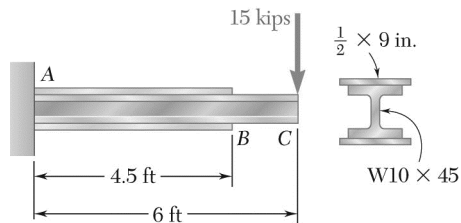
$$\theta_A = -A_1 - A_2 - A_3 = \frac{1}{16} \frac{wL^3}{EI}$$

$$\theta_A = \frac{1}{16} \frac{wL^3}{EI} \nearrow \blacktriangleleft$$

(b) Deflection at A. $y_A = t_{A/C}$

$$\begin{aligned} y_A &= A_1 \left(\frac{2}{3} L \right) + A_2 \left(\frac{5}{6} L \right) + A_3 \left(\frac{3}{8} L \right) \\ &= -\frac{47}{1152} \frac{wL^4}{EI} \end{aligned}$$

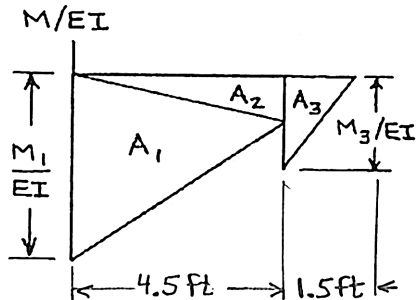
$$y_A = \frac{47}{1152} \frac{wL^4}{EI} \downarrow \blacktriangleleft$$



PROBLEM 9.107

Two cover plates are welded to the rolled-steel beam as shown. Using $E = 29 \times 10^6$ psi, determine (a) the slope at end C, (b) the deflection at end C.

SOLUTION



Portion BC: $I = 248 \text{ in}^4$

$$EI = (29 \times 10^6)(248) = 7.192 \times 10^9 \text{ lb} \cdot \text{in}^2 = 49,944 \text{ kip} \cdot \text{ft}^2$$

Portion AB:

	$A(\text{in}^2)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
Top plate	4.5	5.3	126.405	0.09375
W12 x 45				248
Bot. plate	4.5	5.3	126.405	0.09375
Σ			252.81	248.19

$$I = 252.81 + 248.19 = 501.00 \text{ in}^4 \quad EI = (29 \times 10^6)(501) = 14.529 \times 10^9 \text{ lb} \cdot \text{in}^2 = 100,896 \text{ kip} \cdot \text{ft}^2$$

Draw $\frac{M}{EI}$ diagram: $\frac{M_1}{EI} = -\frac{(15)(6)}{100,896} = -892.01 \times 10^{-6} \text{ ft}^{-1}$

$$\frac{M_3}{EI} = -\frac{(15)(1.5)}{49,944} = -450.50 \times 10^{-6} \text{ ft}^{-1}$$

$$A_1 = \frac{1}{2}(4.5)(-892.01 \times 10^{-6}) = -2.0070 \times 10^{-3}$$

$$A_2 = \frac{1.5}{6} A_1 = -0.50175 \times 10^{-3}$$

$$A_3 = \frac{1}{2}(1.5)(-450.50 \times 10^{-6}) = -0.33788 \times 10^{-3}$$

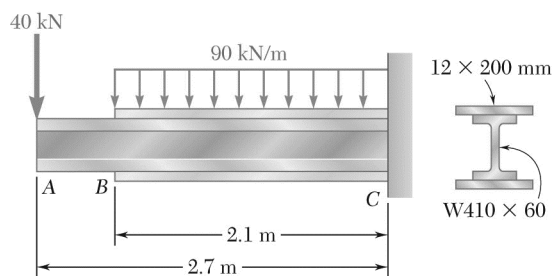
Place reference tangent at A.

(a) Slope at C. $\theta_C = \theta_{C/A} = A_1 + A_2 + A_3 \quad \theta_C = 2.85 \times 10^{-3} \text{ rad} \curvearrowright \blacktriangleleft$

(b) Deflection at C. $y_C = t_{C/A}$

$$y_C = (4.5)(A_1) + (3)(A_2) + (1)(A_3) = -10.8746 \times 10^{-3} \text{ ft}$$

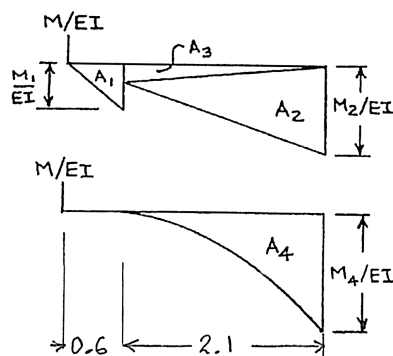
$$y_C = 0.1305 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.108

Two cover plates are welded to the rolled-steel beam as shown. Using $E = 200$ GPa, determine (a) the slope at end A , (b) the deflection at end A .

SOLUTION



Portion AB : $I = 216 \times 10^6 \text{ mm}^4$

$$EI = (200 \times 10^6 \text{ kPa})(216 \times 10^{-6} \text{ m}^4) = 43,200 \text{ kN} \cdot \text{m}^2$$

Portion BC :

	$A(\text{mm}^2)$	$d(\text{mm})$	$Ad^2(\text{mm}^4)$	$\bar{I}(\text{mm}^4)$
Top plate	2400	209	104.834×10^6	28,800
W410 x 60				216×10^6
Bot. plate	2400	209	104.834×10^6	28,800
Σ			209.67×10^6	216.06×10^6

$$I = 209.67 \times 10^6 + 216.06 \times 10^6 = 425.73 \times 10^6 \text{ mm}^4$$

$$EI = (200 \times 10^6 \text{ kPa})(425.73 \times 10^{-6} \text{ m}^4) = 85,146 \text{ kN} \cdot \text{m}^2$$

Draw $\frac{M}{EI}$ diagram:

$$\frac{M_1}{EI} = -\frac{(40)(0.6)}{43200} = -0.55556 \times 10^{-3} \text{ m}^{-1}$$

$$\frac{M_2}{EI} = -\frac{(40)(2.7)}{85146} = -1.26841 \times 10^{-3} \text{ m}^{-1}$$

$$\frac{M_4}{EI} = -\frac{(90)(2.1)(1.05)}{85146} = -2.3307 \times 10^{-3} \text{ m}^{-1}$$

$$A_1 = \frac{1}{2}(0.6)(-0.55556 \times 10^{-3}) = -0.166668 \times 10^{-3}$$

$$A_2 = \frac{1}{2}(2.1)(-1.26841 \times 10^{-3}) = -1.33183 \times 10^{-3}$$

$$A_3 = \frac{0.6}{2.7}A_2 = -0.29596 \times 10^{-3}$$

$$A_4 = \frac{1}{3}(2.1)(-2.3307 \times 10^{-3}) = -1.63149 \times 10^{-3}$$

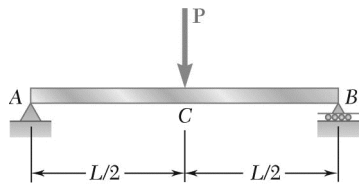
PROBLEM 9.108 (Continued)

Place reference tangent at C . $\theta_C = 0$

(a) Slope at A . $\theta_A = \theta_C - \theta_{A/C} = 0 - (A_1 + A_2 + A_3 + A_4) \quad \theta_A = 3.43 \times 10^{-3} \text{ rad } \nearrow \blacktriangleleft$

(b) Deflection at A . $y_A = t_{A/C}$
 $y_A = (0.4)(A_1) + (2)(A_2) + (1.3)(A_3) + (2.175)(A_4) = -6.66 \times 10^{-3} \text{ m}$

$y_A = 6.66 \text{ mm } \downarrow \blacktriangleleft$



PROBLEM 9.109

For the prismatic beam and loading shown, determine (a) the slope at end A, (b) the deflection at the center C of the beam.

SOLUTION

Symmetric beam and loading.

Place reference tangent at C.

Reactions:

$$R_A = R_B = \frac{1}{2}P$$

Bending moment at C.

$$M_C = \frac{1}{4}PL$$

$$A = \frac{1}{2} \left(\frac{1}{4} \frac{PL}{EI} \right) \left(\frac{L}{2} \right) = \frac{1}{16} \frac{PL^2}{EI}$$

(a) Slope at A.

$$\theta_A = \theta_C - \theta_{C/A}$$

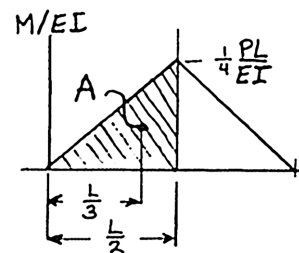
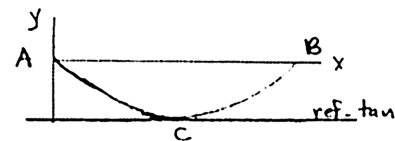
$$\theta_A = 0 - \frac{1}{16} \frac{PL^2}{EI}$$

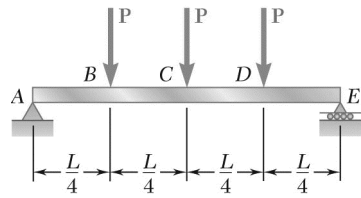
$$\theta_A = -\frac{1}{16} \frac{PL^2}{EI} \swarrow \blacktriangleleft$$

(b) Deflection at C.

$$y_C = -t_{A/C} = -A \left(\frac{L}{3} \right) = - \left(\frac{1}{16} \frac{PL^2}{EI} \right) \left(\frac{L}{3} \right)$$

$$y_C = \frac{1}{48} \frac{PL^3}{EI} \downarrow \blacktriangleleft$$

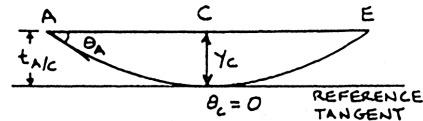
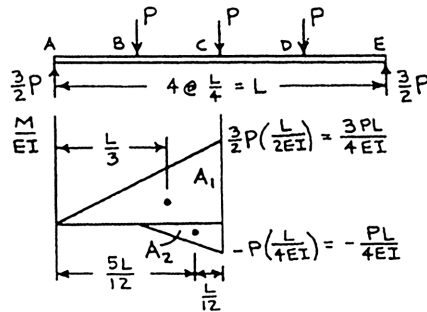




PROBLEM 9.110

For the prismatic beam and loading shown, determine (a) the slope at end A, (b) the deflection at the center C of the beam.

SOLUTION



$$A_1 = \frac{1}{2} \left(\frac{3PL}{4EI} \right) \left(\frac{L}{2} \right) = \frac{3PL^2}{16EI}$$

$$A_2 = \frac{1}{2} \left(-\frac{PL}{4EI} \right) \left(\frac{L}{4} \right) = -\frac{PL^2}{32EI}$$

(a) Slope at A. $\theta_C = \theta_A + \theta_{C/A}; \quad \theta_A = 0 - \theta_{C/A}$

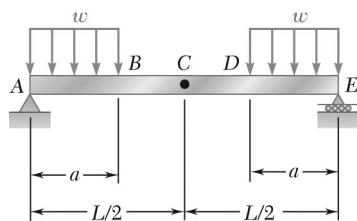
$$\begin{aligned} \theta_A &= -\theta_{C/A} = -(A_1 + A_2) \\ &= -\left[\frac{3PL^2}{16EI} - \frac{PL^2}{32EI} \right] \\ &= -\frac{5PL^2}{32EI} \end{aligned}$$

$$\theta_A = \frac{5PL^2}{32EI} \quad \swarrow \blacktriangleleft$$

(b) Deflection at C.

$$\begin{aligned} t_{A/C} &= A_1 \left(\frac{L}{3} \right) + A_2 \left(\frac{5L}{12} \right) \\ &= \left(\frac{3PL^2}{16EI} \right) \left(\frac{L}{3} \right) + \left(-\frac{PL^2}{32EI} \right) \left(\frac{5L}{12} \right) \\ &= \frac{19PL^3}{384EI} \end{aligned}$$

$$y_C = -t_{A/C} = -\frac{19PL^3}{384EI} \quad y_C = \frac{19PL^3}{384EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.111

For the prismatic beam and loading shown, determine (a) the slope at end A, (b) the deflection at the center C of the beam.

SOLUTION

Symmetric beam and loading.

Place reference tangent at C.

Reactions:

$$\theta_C = 0$$

$$R_A = R_E = wa$$

Bending moment.

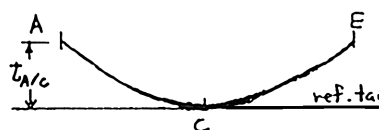
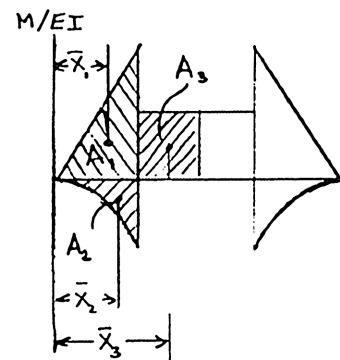
Over AB:

$$M = wax - \frac{1}{2}wa^2$$

Over BD:

$$M = \frac{1}{2}wa^2$$

Draw M/EI diagram by parts.



$$\frac{M_1}{EI} = \frac{wa^2}{EI}$$

$$\frac{M_2}{EI} = -\frac{1}{2} \frac{wa^2}{EI}$$

$$\frac{M_3}{EI} = \frac{1}{2} \frac{wa^2}{EI}$$

$$A_1 = \frac{1}{2} \frac{M_1}{EI} a = \frac{1}{2} \frac{wa^3}{EI}$$

$$A_2 = -\frac{1}{3} \frac{M_2}{EI} a = -\frac{1}{6} \frac{wa^3}{EI}$$

$$A_3 = \frac{M_3}{EI} \left(\frac{L}{2} - a \right) = \frac{1}{4} \frac{wa^2}{EI} (L - 2a)$$

(a) Slope at A.

$$\theta_A = \theta_C - \theta_{C/A} = 0 - (A_1 + A_2 + A_3)$$

$$= -\frac{1}{2} \frac{wa^3}{EI} + \frac{1}{6} \frac{wa^3}{EI} - \frac{1}{4} \frac{wa^2}{EI} (L - 2a)$$

$$= -\frac{wa^2}{EI} \left(\frac{1}{4}L - \frac{1}{6}a \right)$$

$$= -\frac{1}{12} \frac{wa^2}{EI} (3L - 2a)$$

$$\theta_A = \frac{wa^2}{12EI} (3L - 2a) \quad \nwarrow \blacktriangleleft$$

PROBLEM 9.111 (Continued)(b) Deflection at C.

$$y_C = -t_{C/A}$$

$$\bar{x}_1 = \frac{2}{3}a,$$

$$\bar{x}_2 = \frac{3}{4}a,$$

$$\bar{x}_3 = a + \frac{1}{2}\left(\frac{L}{2} - a\right) = \frac{1}{4}(L + 2a)$$

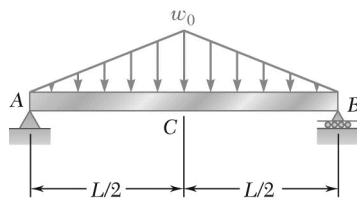
$$y_C = -t_{C/A} = -A_1\bar{x}_1 - A_2\bar{x}_2 - A_3\bar{x}_3$$

$$= -\left(\frac{1}{2}\frac{wa^3}{EI}\right)\left(\frac{2}{3}a\right) + \left(\frac{1}{6}\frac{wa^3}{EI}\right)\left(\frac{3}{4}a\right) - \frac{1}{4}\left(\frac{wa^2}{EI}\right)(L - 2a)\frac{1}{4}(L + 2a)$$

$$= -\frac{1}{3}\frac{wa^3}{EI} + \frac{1}{8}\frac{wa^3}{EI} - \frac{1}{16}\frac{wa^2}{EI}(L^2 - 4a^2)$$

$$= -\frac{wa^2}{EI}\left(\frac{1}{16}L^2 - \frac{1}{24}a^2\right) = -\frac{1}{48}\frac{wa^2}{EI}(3L^2 - 2a^2)$$

$$y_A = \frac{wa^2}{48EI}(3L^2 - 2a^2) \downarrow \blacktriangleleft$$



PROBLEM 9.112

For the prismatic beam and loading shown, determine (a) the slope at end A, (b) the deflection at the center C of the beam.

SOLUTION

Symmetric beam and loading.

Place reference tangent at C. $\theta_C = 0$

Reactions: $R_A = R_B = \frac{w_0 L}{4}$

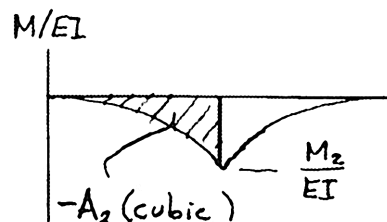
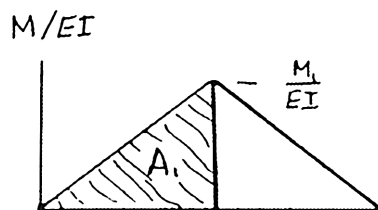
Draw $\frac{M}{EI}$ diagram by parts.

$$\frac{M_1}{EI} = \frac{R_A L}{2} = \frac{w_0 L^2}{8EI}$$

$$A_1 = \frac{1}{2} \left(\frac{L}{2} \right) \left(\frac{M_1}{EI} \right) = \frac{w_0 L^3}{32EI}$$

$$\frac{M_2}{EI} = \frac{1}{EI} \cdot \frac{1}{2} \left(-\frac{w_0 L}{2} \right) \left(\frac{1}{3} \cdot \frac{L}{2} \right) = -\frac{w_0 L^2}{24EI}$$

$$A_2 = \frac{1}{4} \left(\frac{L}{2} \right) \left(-\frac{w_0 L^2}{24EI} \right) = -\frac{w_0 L^3}{192EI}$$



(a) Slope at A.

$$\theta_A = -\theta_{C/A}$$

$$\theta_A = -A_1 - A_2 = \left(-\frac{1}{32} + \frac{1}{192} \right) \frac{w_0 L^3}{EI}$$

$$\theta_A = \frac{5w_0 L^3}{192EI} \quad \swarrow \blacktriangleleft$$

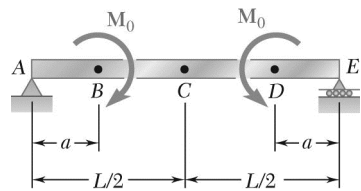
(b) Deflection at C.

$$y_C = t_{A/C}$$

$$t_{A/C} = A_1 \left[\left(\frac{2}{3} \right) \left(\frac{L}{2} \right) \right] + A_2 \left[\left(\frac{4}{5} \right) \left(\frac{L}{2} \right) \right] = \left[\left(\frac{1}{3} \right) \left(\frac{1}{32} \right) - \left(\frac{2}{5} \right) \left(\frac{1}{192} \right) \right] \frac{w_0 L^4}{EI}$$

$$= \frac{w_0 L^4}{120EI}$$

$$y_C = \frac{w_0 L^4}{120EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.113

For the prismatic beam and loading shown, determine (a) the slope at end A, (b) the deflection at the center C of the beam.

SOLUTION

Symmetric beam and loading.

Place reference tangent at C. $\theta_C = 0$.

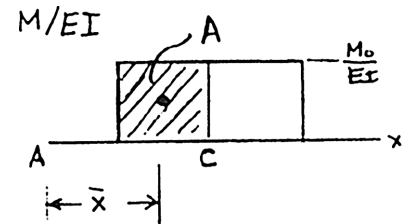
Draw $\frac{M}{EI}$ diagram.

(a) Slope at A.

$$\theta_A = 0$$

$$A = \frac{M_0}{EI} \left(\frac{L}{2} - a \right) = \frac{1}{2} \frac{M_0}{EI} (L - 2a)$$

$$\theta_A = \theta_C - \theta_{C/A} = 0 - A = -\frac{1}{2} \frac{M_0}{EI} (L - 2a)$$

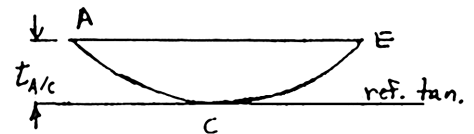


$$\theta_A = \frac{1}{2} \frac{M_0}{EI} (L - 2a) \quad \nwarrow \blacktriangleleft$$

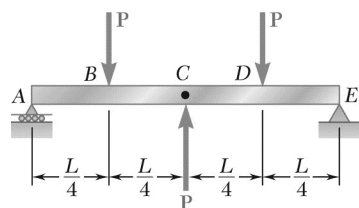
(b) Deflection at C.

$$\bar{x} = a + \frac{1}{2} \left(\frac{L}{2} - a \right) = \frac{1}{4} (L + 2a)$$

$$\begin{aligned} y_C &= -t_{C/A} = A\bar{x} \\ &= -\frac{1}{2} \frac{M_0}{EI} (L - 2a) \frac{1}{4} (L + 2a) \\ &= -\frac{1}{8} \frac{M_0}{EI} (L^2 - 4a^2) \end{aligned}$$



$$y_C = \frac{1}{8} \frac{M_0}{EI} (L^2 - 4a^2) \quad \downarrow \blacktriangleleft$$



PROBLEM 9.114

For the prismatic beam and loading shown, determine (a) the slope at end A , (b) the deflection at the center C of the beam.

SOLUTION

Symmetric beam and loading.

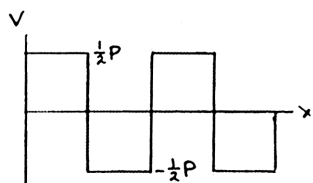
Place reference tangent at C .

$$\theta_C = 0$$

Reactions:

$$R_A = R_E = \frac{1}{2}P$$

Draw V (shear) and M/EI diagrams.



$$A_1 = A_2 = \frac{1}{2} \left(\frac{1}{8} \frac{PL}{EI} \right) \frac{L}{4} = \frac{1}{64} \frac{PL^2}{EI}$$

(a) Slope at A .

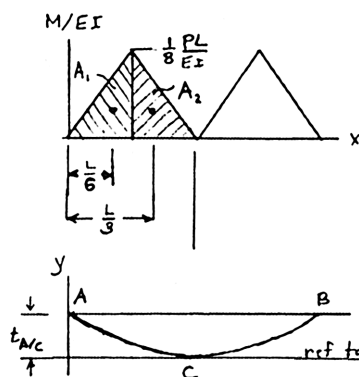
$$\begin{aligned} \theta_A &= \theta_C - \theta_{A/C} = 0 - A_1 - A_2 \\ &= -\frac{1}{32} \frac{PL^2}{EI} \end{aligned}$$

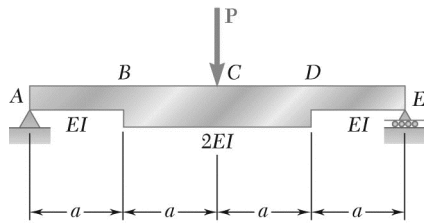
$$\theta_A = \frac{PL^2}{32EI} \swarrow \blacktriangleleft$$

(b) Deflection at C .

$$\begin{aligned} y_C &= -t_{A/C} = - \left(A_1 \frac{L}{6} + A_2 \frac{L}{3} \right) \\ &= - \left(\frac{1}{64} \frac{PL^3}{EI} \cdot \frac{L}{6} + \frac{1}{64} \frac{PL^3}{EI} \cdot \frac{L}{3} \right) \\ &= -\frac{1}{128} \frac{PL^3}{EI} \end{aligned}$$

$$y_C = \frac{PL^3}{128EI} \downarrow \blacktriangleleft$$





PROBLEM 9.115

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at the center C of the beam.

SOLUTION

Symmetric beam and loading.

$$R_A = R_E = \frac{1}{2}P$$

$$M_{\max} = \left(\frac{1}{2}P\right)(2a) = Pa$$

Draw M and M/EI diagrams.

$$A_1 = \frac{1}{2} \left(\frac{Pa}{2EI} \right) a = \frac{1}{4} \frac{Pa^2}{EI}$$

$$A_2 = \frac{1}{2} \left(\frac{Pa}{4EI} \right) a = \frac{1}{8} \frac{Pa^2}{EI}$$

$$A_3 = \frac{1}{2} \left(\frac{Pa}{2EI} \right) a = \frac{1}{4} \frac{Pa^2}{EI}$$

Place reference tangent at C.

$$\theta_C = 0$$

(a) Slope at A.

$$\theta_A = \theta_C - \theta_{C/A} = 0 - (A_1 + A_2 + A_3)$$

$$= -\frac{5}{8} \frac{Pa^2}{EI}$$

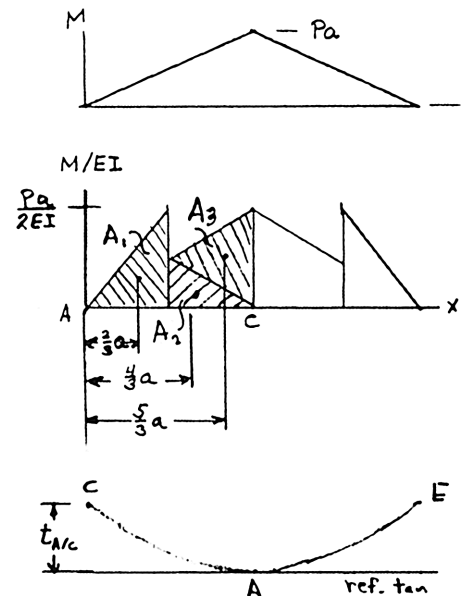
$$\theta_A = \frac{5Pa^2}{8EI} \quad \swarrow \blacktriangleleft$$

(b) Deflection at C.

$$|y_C| = t_{A/C} = A_1 \left(\frac{2}{3}a \right) + A_2 \left(\frac{4}{3}a \right) + A_3 \left(\frac{5}{3}a \right)$$

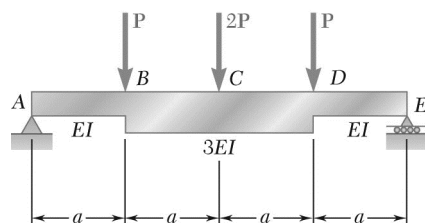
$$= \frac{1}{6} \frac{Pa^3}{EI} + \frac{1}{6} \frac{Pa^3}{EI} + \frac{5}{12} \frac{Pa^3}{EI} = \frac{3}{4} \frac{Pa^3}{EI}$$

$$y_C = \frac{3Pa^3}{4EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.116

For the beam and loading shown, determine (a) the slope at end A , (b) the deflection at the center C of the beam.

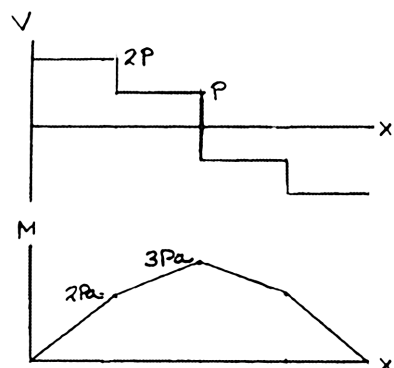


SOLUTION

Symmetric beam and loading.

$$R_A = R_E = 2P.$$

Draw V , M , and M/EI diagrams.



$$A_1 = \frac{1}{2} \left(\frac{2Pa}{EI} \right) a = \frac{Pa^2}{EI}$$

$$A_2 = \frac{1}{2} \left(\frac{2}{3} \frac{Pa}{EI} \right) a = \frac{1}{3} \frac{Pa^2}{EI}$$

$$A_3 = \frac{1}{2} \left(\frac{Pa}{EI} \right) a = \frac{1}{2} \frac{Pa^2}{EI}$$

Place reference tangent at C .

$$\theta_C = 0$$

(a) Slope at A .

$$\theta_A = \theta_C - \theta_{C/A} = 0 - (A_1 + A_2 + A_3)$$

$$= -\frac{11}{6} \frac{Pa^2}{EI}$$

$$\theta_A = \frac{11Pa^2}{6EI} \quad \swarrow \blacktriangleleft$$

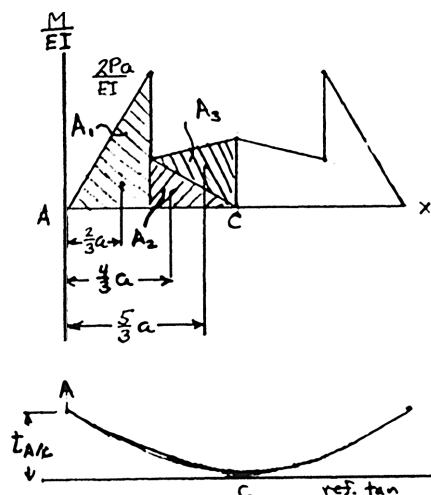
(b) Deflection at C .

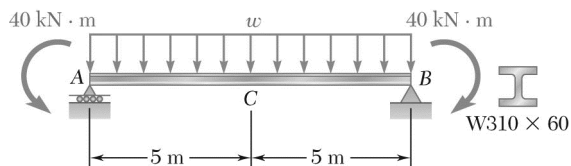
$$|y_C| = t_{A/C}$$

$$= A_1 \left(\frac{2}{3} a \right) + A_2 \left(\frac{4}{3} a \right) + A_3 \left(\frac{5}{3} a \right)$$

$$= \frac{35}{18} \frac{Pa^3}{EI}$$

$$y_C = \frac{35Pa^3}{18EI} \quad \downarrow \blacktriangleleft$$

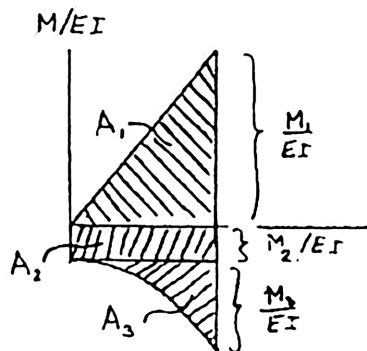
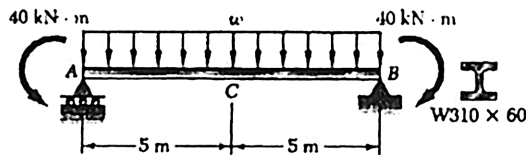




PROBLEM 9.117

For the beam and loading shown and knowing that $w = 8 \text{ kN/m}$, determine (a) the slope at end A, (b) the deflection at midpoint C. Use $E = 200 \text{ GPa}$.

SOLUTION



$$E = 200 \times 10^9 \text{ Pa}$$

$$I = 128 \times 10^6 \text{ mm}^4 = 128 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(128 \times 10^{-6}) = 25.6 \times 10^6 \text{ N} \cdot \text{m}^2 \\ = 25,600 \text{ kN} \cdot \text{m}^2$$

Symmetrical beam and loading.

$$R_A = R_B = \frac{1}{2}(8)(10) = 40 \text{ kN}$$

Bending moment:

$$M = 40x - 40 - \frac{1}{2}(8)x^2$$

At $x = 5$,

$$M = 200 - 40 - 100$$

Draw $\frac{M}{EI}$ diagram by parts.

$$\frac{M_1}{EI} = \frac{200}{25,600} = 7.8125 \times 10^{-3} \text{ m}^{-1}$$

$$\frac{M_2}{EI} = \frac{-40}{25,600} = -1.5625 \times 10^{-3} \text{ m}^{-1}$$

$$\frac{M_3}{EI} = \frac{-100}{25,600} = -3.9063 \times 10^{-3} \text{ m}^{-1}$$

$$A_1 = \frac{1}{2}(7.8125 \times 10^{-3})(5) = 19.5313 \times 10^{-3}$$

$$\bar{x}_1 = \left(\frac{2}{3}\right)(5) = 3.3333 \text{ m}$$

$$A_2 = -(1.5625)(5) = -7.8125 \times 10^{-3}$$

$$\bar{x}_2 = \left(\frac{1}{2}\right)(5) = 2.5 \text{ m}$$

$$A_3 = -\frac{1}{3}(3.9063)(5) = -6.5105 \times 10^{-3}$$

$$\bar{x}_3 = \left(\frac{3}{4}\right)(5) = 3.75 \text{ m}$$

Place reference tangent at C.

$$\theta_C = 0$$

PROBLEM 9.117 (Continued)

(a) Slope at A.

$$\theta_A = \theta_C - \theta_{C/A} = 0 - (A_1 + A_2 + A_3)$$

$$\theta_A = -(19.5313 \times 10^{-3} - 7.8125 \times 10^{-3} - 6.5105 \times 10^{-3}) = -5.21 \times 10^{-3}$$

$$\theta_A = 5.21 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

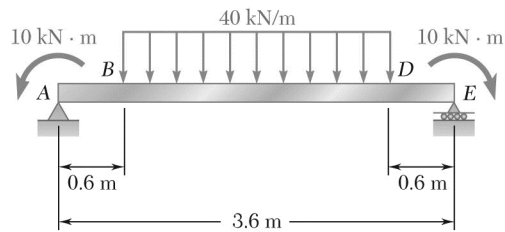
(b) Deflection at C.

$$|y_C| = t_{A/C}$$

$$= (19.5313 \times 10^{-3})(3.3333) - (7.8125 \times 10^{-3})(2.5) - (6.5105 \times 10^{-3})(3.75)$$

$$= 21.2 \times 10^{-3} \text{ m}$$

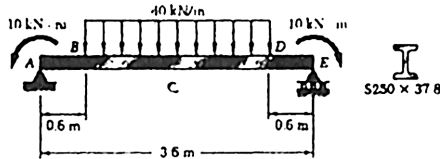
$$y_C = 21.2 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.118

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at the midpoint of the beam. Use $E = 200$ GPa.

SOLUTION



Use units of kN and m.

For S250 x 37.8

$$I = 51.2 \times 10^6 \text{ mm}^4 = 51.2 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(51.2 \times 10^{-6})$$

$$= 10.24 \times 10^6 \text{ N} \cdot \text{m}^2 = 10,240 \text{ kN} \cdot \text{m}^2$$

Place reference tangent at midpoint C.

Reactions: $R_A = R_E = \frac{1}{2}(40)(3.6 - 1.2) = 48 \text{ kN} \uparrow$

Draw bending moment diagram of left half of beam by parts.

$$M_1 = (48)(1.8) = 86.4 \text{ kN} \cdot \text{m}$$

$$A_1 = \frac{1}{2}(1.8)(86.4) = 77.76 \text{ kN} \cdot \text{m}^2$$

$$A_2 = (1.8)(-10) = -18 \text{ kN} \cdot \text{m}^2$$

$$M_3 = \frac{1}{2}(40)(1.8 - 0.6)^2 = -28.8 \text{ kN} \cdot \text{m}$$

$$A_3 = \frac{1}{3}(1.2)(-28.8) = -11.52 \text{ kN} \cdot \text{m}^2$$

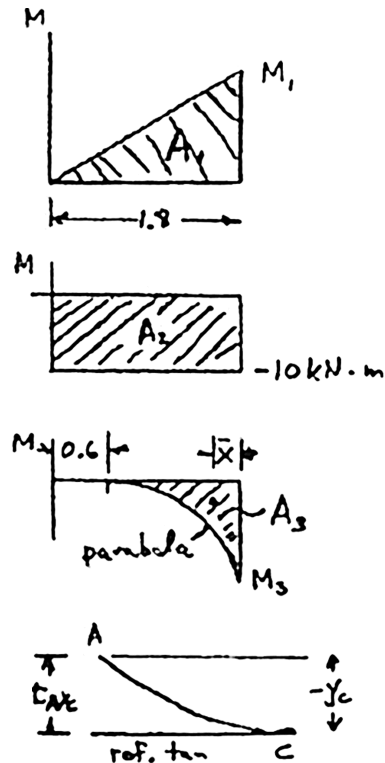
$$\bar{x} = \frac{1}{4}(1.2) = 0.30 \text{ m}$$

(a) Slope at end A. $\theta_A = -\theta_{A/C}$

$$\theta_A = \frac{1}{EI} \{-A_1 - A_2 - A_3\} = \frac{-77.76 + 18 + 11.52}{10,240}$$

$$= -4.71 \times 10^{-3} \text{ rad}$$

$$\theta_A = 4.71 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$



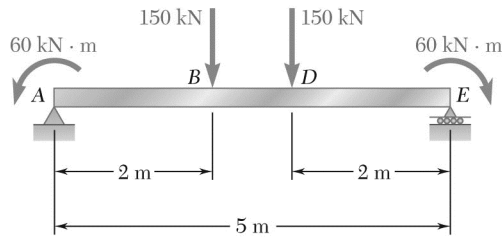
PROBLEM 9.118 (Continued)

(b) Deflection at midpoint C. $y_C = -t_{A/C}$

$$\begin{aligned} t_{A/C} &= \frac{1}{EI} \{1.2A_1 + 0.9A_2 + (1.8 - 0.3)A_3\} \\ &= \frac{(1.2)(77.76) - (0.9)(18) - (1.5)(11.52)}{10,240} = 5.84 \times 10^{-3} \text{ m} \end{aligned}$$

$$y_C = -5.84 \times 10^{-3} \text{ m}$$

$$y_C = 5.84 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.119

For the beam and loading shown, determine (a) the slope at end A , (b) the deflection at the midpoint of the beam. Use $E = 200$ GPa.

SOLUTION

Use units of kN and m.

For $W460 \times 74$,

$$\begin{aligned} I &= 333 \times 10^6 \text{ mm}^4 = 333 \times 10^{-6} \text{ m}^4 \\ EI &= (200 \times 10^9)(333 \times 10^{-6}) \\ &= 66.6 \times 10^6 \text{ N} \cdot \text{m}^2 = 66600 \text{ kN} \cdot \text{m}^2 \end{aligned}$$

Symmetric beam and loading. Place reference tangent at midpoint C where $\theta_C = 0$.

Reactions: $R_A = R_E = 150 \text{ kN} \uparrow$

Draw bending moment diagram of left half of beam by parts.

$$M_1 = (2)(150) = 300 \text{ kN} \cdot \text{m}$$

$$A_1 = \left(\frac{1}{2}\right)(2)(300) = 300 \text{ kN} \cdot \text{m}^2$$

$$A_2 = (0.5)(300) = 150 \text{ kN} \cdot \text{m}^2$$

$$M_3 = -60 \text{ kN} \cdot \text{m}$$

$$A_3 = (2.5)(-60) = -150 \text{ kN} \cdot \text{m}^2$$

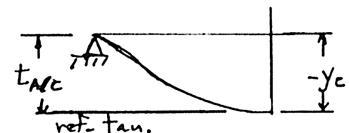
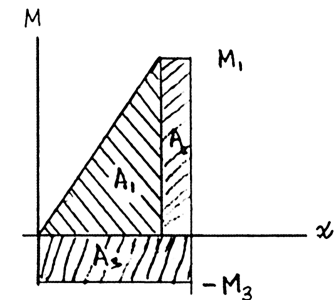
(a) Slope at end A .

$$\theta_A = -\theta_{C/A}$$

$$\begin{aligned} \theta_A &= \frac{1}{EI} \{-A_1 - A_2 - A_3\} \\ &= \frac{-300 - 150 + 150}{66600} \end{aligned}$$

$$= -4.50 \times 10^{-3} \text{ rad}$$

$$\theta_A = 4.50 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$



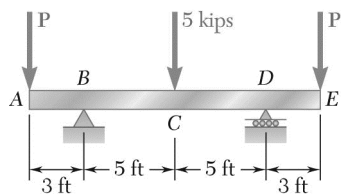
(b) Deflection at midpoint C .

$$y_C = -t_{A/C}$$

$$\begin{aligned} t_{A/C} &= \frac{1}{EI} \left\{ \left(\frac{2}{3} \cdot 2\right) A_1 + \left(2 + \frac{0.5}{2}\right) A_2 + \left(\frac{2.5}{2}\right) A_3 \right\} \\ &= \frac{400 + 337.5 - 187.5}{66600} = 8.26 \times 10^{-3} \text{ m} \end{aligned}$$

$$y_C = -8.26 \times 10^{-3} \text{ m}$$

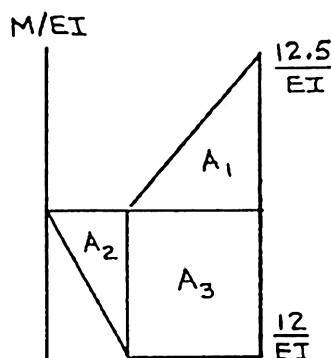
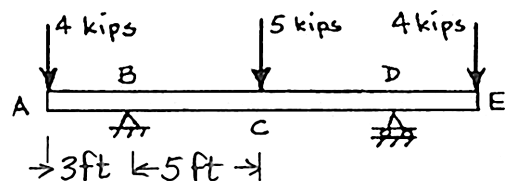
$$y_C = 8.26 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.120

Knowing that $P = 4$ kips, determine (a) the slope at end A , (b) the deflection at midpoint C of the beam. Use $E = 29 \times 10^6$ psi.

SOLUTION



$$E = 29 \times 10^3 \text{ ksi} \quad I = 39.6 \text{ in}^4$$

$$EI = (29 \times 10^3)(39.6) = 1.1484 \times 10^6 \text{ kip} \cdot \text{in}^2 \\ = 7975 \text{ kip} \cdot \text{ft}^2$$

Symmetric beam and loading:

$$R_A = R_B = P + 2.5 = 4 + 2.5 = 6.5 \text{ kips}$$

Bending moment:

$$\text{Over } AB: M = -Px = -4x$$

$$\text{Over } BC: M = -4x + 6.5(x - 3) \\ = 2.5(x - 3) - 12$$

Draw $\frac{M}{EI}$ diagram by parts.

$$A_1 = \frac{1}{2} \left(\frac{12.5}{EI} \right) (5) = \frac{31.25}{EI}$$

$$A_2 = -\frac{1}{2} \left(\frac{12}{EI} \right) (3) = -\frac{18}{EI}$$

$$A_3 = -\left(\frac{12}{EI} \right) (5) = -\frac{60}{EI}$$

Place reference tangent at C . $\theta_C = 0$

(a) Slope at A . $\theta_A = \theta_C - \theta_{C/A} = 0 - (A_1 + A_2 + A_3)$

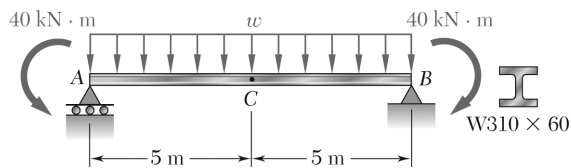
$$\theta_A = -\left(\frac{31.25}{EI} - \frac{18}{EI} - \frac{60}{EI} \right) = \frac{46.75}{EI} = \frac{46.75}{7975} \quad \theta_A = 5.86 \times 10^{-3} \text{ rad} \nearrow \blacktriangleleft$$

(b) Deflection at C . $y_C = -t_{B/C}$

$$= -(A_1 \bar{x}_1 + A_3 \bar{x}_3)$$

$$= -\left[\left(\frac{31.25}{EI} \right) \left(\frac{2}{3}(5) \right) - \left(\frac{60}{EI} \right) \left(\frac{1}{2}(5) \right) \right] = \frac{45.833}{EI} = \frac{45.833}{7975} = 5.7471 \times 10^{-3} \text{ ft}$$

$$y_C = 0.0690 \text{ in.} \uparrow \blacktriangleleft$$



PROBLEM 9.121

For the beam and loading of Prob. 9.117, determine the value of w for which the deflection is zero at the midpoint C of the beam. Use $E = 200$ GPa.

SOLUTION

Symmetric beam and loading:

$$R_A = R_B = 5w \quad (w \text{ in kN/m})$$

Bending moment in kN·m:

$$M = 5wx - 40 - \frac{1}{2}wx^2$$

At $x = 5$ m,

$$M = 25w - 40 - 12.5w$$

Draw M/EI diagram by parts.

$$A_1 = \frac{1}{2} \left(\frac{25w}{EI} \right) (5) = \frac{62.5w}{EI}$$

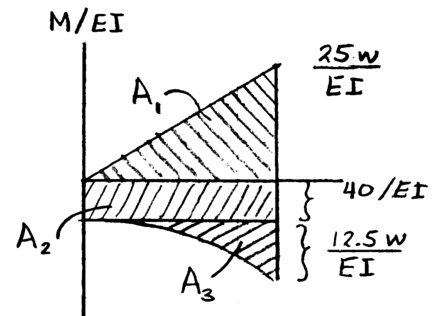
$$A_2 = -\frac{(40)(5)}{EI} = -\frac{200}{EI}$$

$$A_3 = -\frac{1}{3} \left(\frac{12.5w}{EI} \right) (5) = -\frac{20.833w}{EI}$$

$$\bar{x}_1 = \frac{2}{3} (5) = 3.3333 \text{ m}$$

$$\bar{x}_2 = \frac{1}{2} (5) = 2.5 \text{ m}$$

$$\bar{x}_3 = \frac{3}{4} (5) = 3.75 \text{ m}$$



Place reference tangent at C .

Deflection at C is zero.

$$t_{A/C} = y_A - y_C = 0$$

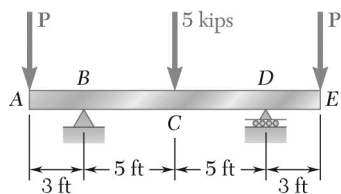
$$A_1 \bar{x}_1 + A_2 \bar{x}_2 + A_3 \bar{x}_3 = 0$$

$$\left(\frac{62.5w}{EI} \right) (3.3333) - \left(\frac{200}{EI} \right) (2.5) - \left(\frac{20.833w}{EI} \right) (3.75) = 0$$

$$\frac{130.21w}{EI} - \frac{500}{EI} = 0$$

$$w = \frac{500}{130.21} = 3.84 \text{ kN/m}$$

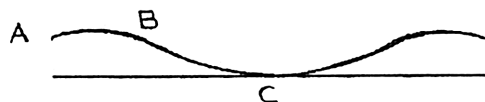
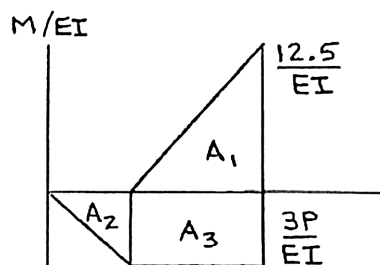
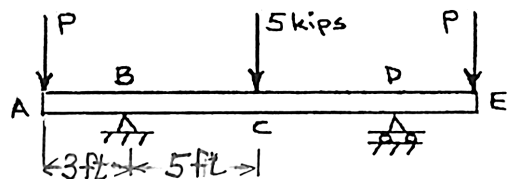
$$w = 3.84 \text{ kN/m} \quad \blacktriangleleft$$



PROBLEM 9.122

For the beam and loading of Prob 9.120, determine the magnitude of the forces P for which the deflection is zero at end A of the beam. Use $E = 29 \times 10^6$ psi.

SOLUTION



Symmetric beam and loading:

$$R_A = R_B = P + 2.5 \quad (P \text{ in kips})$$

Bending moment:

$$\text{Over } AB: \quad M = -Px$$

$$\begin{aligned} \text{Over } BC: \quad M &= -Px + (P + 2.5)(x - 3) \\ &= 2.5(x - 3) - P(3) \end{aligned}$$

$$\text{At } x = 8 \text{ ft}, \quad M = 12.5 - P(3)$$

Draw $\frac{M}{EI}$ diagram by parts.

$$A_1 = \frac{1}{2} \left(\frac{12.5}{EI} \right) (5) = \frac{31.25}{EI}$$

$$A_2 = -\frac{1}{2} \left(\frac{3P}{EI} \right) (3) = -\frac{4.5P}{EI}$$

$$A_3 = -\left(\frac{3P}{EI} \right) (5) = -\frac{15P}{EI}$$

Place reference tangent at C .

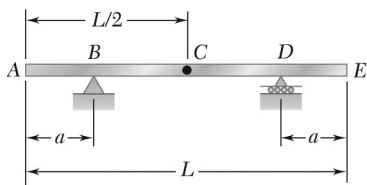
$$y_A = y_B = 0 \quad y_A - y_B = 0 \quad t_{A/C} - t_{B/C} = 0$$

$$\left[A_1 \left(3 + \frac{2}{3}(5) \right) + A_3 \left(3 + \frac{1}{2}(5) \right) + A_2 \left(\frac{2}{3}(3) \right) \right] - \left[A_1 \left(\frac{2}{3}(5) \right) + A_3 \left(\frac{1}{2}(5) \right) \right] = 0$$

$$A_1(3) + A_3(3) + A_2(2) = 0$$

$$\frac{93.75}{EI} - \frac{45P}{EI} - \frac{9P}{EI} = 0$$

$$P = 1.736 \text{ kips} \quad \blacktriangleleft$$



PROBLEM 9.123*

A uniform rod AE is to be supported at two points B and D . Determine the distance a for which the slope at ends A and E is zero.

SOLUTION

Let w = weight per unit length of rod.

Symmetric beam and loading:

$$R_B = R_D = \frac{1}{2}wL$$

Bending moment:

Over AB :

$$M = -\frac{1}{2}wx^2$$

Over BCD :

$$M = -\frac{1}{2}wx^2 + \frac{1}{2}wL(x - a)$$

Draw M/EI diagram by parts.

$$\frac{M_1}{EI} = \frac{1}{2} \frac{wL(\frac{L}{2} - a)}{EI} = \frac{1}{4} \frac{wL(L - 2a)}{EI}$$

$$\frac{M_2}{EI} = \frac{1}{2} \frac{w(\frac{L}{2})^2}{EI} = -\frac{1}{8} \frac{wL^2}{EI}$$

$$A_1 = \frac{1}{2} \frac{M_1}{EI} \left(\frac{L}{2} - a \right) = \frac{1}{16} \frac{wL(L - 2a)^2}{EI}$$

$$A_2 = \frac{1}{3} \left(\frac{M_2}{EI} \right) \frac{L}{2} = -\frac{1}{48} \frac{wL^3}{EI}$$

Place reference tangent at C .

$$\theta_C = 0$$

$$\theta_A = \theta_C - \theta_{C/A} = 0 - (A_1 + A_2) = 0$$

$$-\frac{1}{16} \frac{wL(L - 2a)^2}{EI} + \frac{1}{48} \frac{wL^3}{EI} = 0$$

Let $u = \frac{a}{L}$ and divide by $\frac{wL^3}{48EI}$.

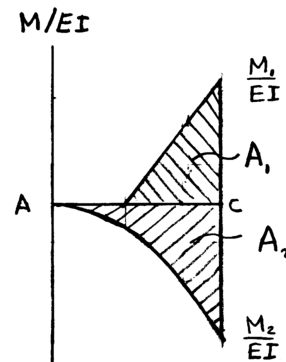
$$1 - 3(1 - 2u)^2 = 0$$

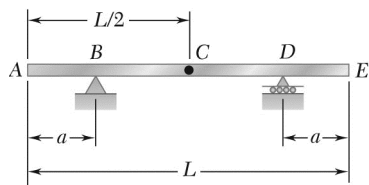
$$1 - 2u = \frac{\sqrt{3}}{3}$$

$$u = \frac{1}{2} \left(1 - \frac{\sqrt{3}}{3} \right) = 0.21132$$

$$\frac{a}{L} = 0.211$$

$$a = 0.211L \quad \blacktriangleleft$$





PROBLEM 9.124*

A uniform rod AE is to be supported at two points B and D . Determine the distance a from the ends of the rod to the points of support, if the downward deflections of points A , C , and E are to be equal.

SOLUTION

Let w = weight per unit length of rod.

Symmetric beam and loading:

$$R_B = R_D = \frac{1}{2} wL$$

Bending moment:

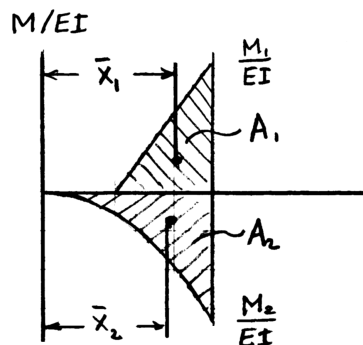
Over AB :

$$M = -\frac{1}{2} wx^2$$

Over BCD :

$$M = -\frac{1}{2} wx^2 + \frac{1}{2} wL(x - a)$$

Draw M/EI diagram by parts.



$$\frac{M_1}{EI} = \frac{1}{2} \frac{wL(\frac{L}{2} - a)}{EI} = \frac{1}{4} \frac{wL(L - 2a)}{EI}$$

$$\frac{M_2}{EI} = -\frac{1}{2} \frac{w(\frac{L}{2})^2}{EI} = -\frac{1}{8} \frac{wL^2}{EI}$$

$$A_1 = \frac{1}{2} \frac{M_1}{EI} \left(\frac{L}{2} - a \right) = \frac{1}{16} \frac{wL(L - 2a)^2}{EI}$$

$$A_2 = \frac{1}{3} \left(\frac{M_2}{EI} \right) \left(\frac{L}{2} \right) = -\frac{1}{48} \frac{wL^3}{EI}$$

$$\bar{x}_1 = a + \frac{2}{3} \left(\frac{L}{2} - a \right) = \frac{1}{3} (L + a)$$

$$\bar{x}_2 = \frac{L}{2} - \frac{1}{4} \left(\frac{L}{2} \right) = \frac{3}{8} L$$

A C

Place reference tangent at C .

$$y_A - y_C = t_{A/C} = 0$$

$$A_1 \bar{x}_1 + A_2 \bar{x}_2 = 0$$

$$\frac{1}{16} \frac{wL(L - 2a)^2}{EI} \frac{1}{3} (L + a) - \frac{1}{48} \frac{wL^3}{EI} \frac{3}{8} L = 0$$

PROBLEM 9.124* (Continued)

Let $u = \frac{a}{L}$. Divide by $\frac{wL^4}{48EI}$.

$$(1 - 2u)^2(1 + u) - \frac{3}{8} = 0$$

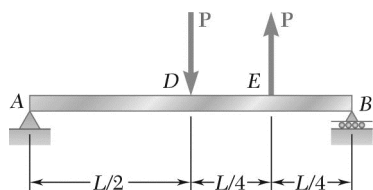
$$4u^3 - 3u + \frac{5}{8} = 0$$

Solving for u ,

$$u = 0.22315$$

$$\frac{a}{L} = 0.223$$

$$a = 0.223 L \quad \blacktriangleleft$$



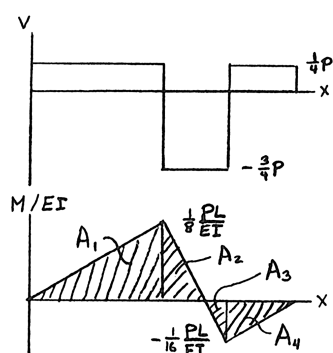
PROBLEM 9.125

For the prismatic beam and loading shown, determine (a) the deflection at point D, (b) the slope at end A.

SOLUTION

$$+\circlearrowleft \sum M_B = 0 : \quad -R_A L + \frac{PL}{2} - \frac{PL}{4} = 0 \quad R_A = \frac{1}{4}P \uparrow$$

$$+\circlearrowleft \sum M_A = 0 : \quad -\frac{PL}{2} + P\frac{3L}{4} + R_B L = 0 \quad R_B = \frac{1}{4}P \downarrow$$



Draw V (shear) diagram and $\frac{M}{EI}$ diagram.

$$A_1 = \frac{1}{2} \left(\frac{1}{8} \frac{PL}{EI} \right) \left(\frac{L}{2} \right) = \frac{1}{32} \frac{PL^2}{EI}$$

$$A_2 = \frac{1}{2} \left(\frac{1}{8} \frac{PL}{EI} \right) \left(\frac{L}{6} \right) = \frac{1}{96} \frac{PL^2}{EI}$$

$$A_3 = \frac{1}{2} \left(-\frac{1}{16} \frac{PL}{EI} \right) \left(\frac{L}{12} \right) = -\frac{1}{384} \frac{PL^2}{EI}$$

$$A_4 = \frac{1}{2} \left(-\frac{1}{16} \frac{PL}{EI} \right) \left(\frac{L}{4} \right) = -\frac{1}{128} \frac{PL^2}{EI}$$

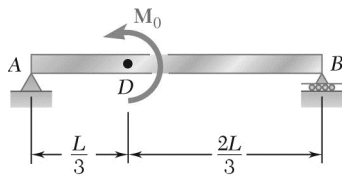
Place reference tangent at A.

$$\begin{aligned} t_{B/A} &= \left(\frac{1}{32} \frac{PL^2}{EI} \right) \left(\frac{2L}{3} \right) + \left(\frac{1}{96} \frac{PL^2}{EI} \right) \left(\frac{L}{2} - \frac{1}{3} \cdot \frac{L}{6} \right) \\ &\quad + \left(-\frac{1}{384} \frac{PL^2}{EI} \right) \left(\frac{L}{4} + \frac{1}{3} \cdot \frac{L}{12} \right) + \left(-\frac{1}{128} \frac{PL^2}{EI} \right) \left(\frac{2}{3} \cdot \frac{L}{4} \right) \\ &= \frac{1}{48} \frac{PL^3}{EI} + \frac{1}{216} \frac{PL^3}{EI} - \frac{5}{6912} \frac{PL^3}{EI} - \frac{1}{768} \frac{PL^3}{EI} = \frac{3}{128} \frac{PL^3}{EI} \end{aligned}$$

$$t_{D/A} = \left(\frac{1}{32} \frac{PL^2}{EI} \right) \left(\frac{1}{3} \cdot \frac{L}{2} \right) = \frac{1}{192} \frac{PL^3}{EI}$$

(a) Deflection at D. $y_D = t_{D/A} - \frac{x_D}{L} t_{B/A} = \frac{1}{192} \frac{PL^3}{EI} - \frac{1}{2} \left(\frac{3}{128} \frac{PL^3}{EI} \right) \quad y_D = -\frac{5}{768} \frac{PL^3}{EI} \blacktriangleleft$

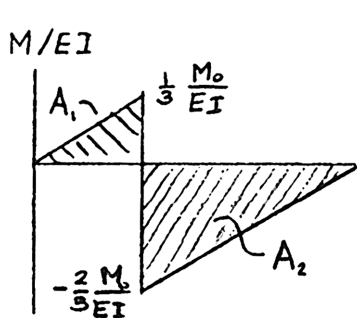
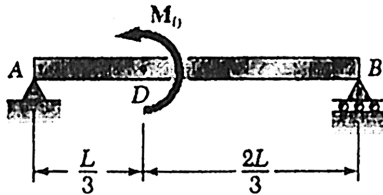
(b) Slope at A. $\theta_A = -\frac{t_{B/A}}{L} \quad \theta_A = -\frac{3}{128} \frac{PL^2}{EI} \blacktriangleleft$



PROBLEM 9.126

For the prismatic beam and loading shown, determine (a) the deflection at point D, (b) the slope at end A.

SOLUTION



Reactions: $R_A = \frac{M_0}{L} \uparrow$, $R_B = \frac{M_0}{L} \downarrow$

Draw $\frac{M}{EI}$ diagram.

$$A_1 = \frac{1}{2} \left(\frac{1}{3} \frac{M_0}{EI} \right) \frac{L}{3} = \frac{1}{18} \frac{M_0 L^2}{EI}$$

$$A_2 = -\frac{1}{2} \left(\frac{2}{3} \frac{M_0}{EI} \right) \frac{2L}{3} = -\frac{2}{9} \frac{M_0 L^2}{EI}$$

Place reference tangent at A.

$$\begin{aligned} t_{B/A} &= A_1 \left(\frac{L}{9} + \frac{2L}{3} \right) + A_2 \left(\frac{2}{3} \cdot \frac{2L}{3} \right) \\ &= \frac{7}{162} \frac{M_0 L^2}{EI} - \frac{8}{81} \frac{M_0 L^2}{EI} = -\frac{1}{18} \frac{M_0 L^2}{EI} \end{aligned}$$

$$t_{D/A} = A_1 \frac{L}{9} = \frac{1}{162} \frac{M_0 L^2}{EI}$$

(a) Deflection at D.

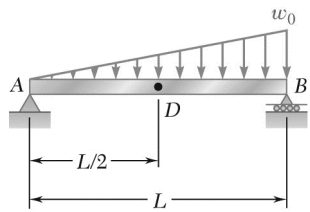
$$\begin{aligned} y_D &= t_{D/A} - \frac{x_D}{L} t_{B/A} \\ &= \frac{1}{162} \frac{M_0 L^2}{EI} - \frac{1}{3} \left(-\frac{1}{18} \frac{M_0 L^2}{EI} \right) = \frac{2}{81} \frac{M_0 L^2}{EI} \end{aligned}$$

$$y_D = \frac{2}{81} \frac{M_0 L^2}{EI} \uparrow \blacktriangleleft$$

(b) Slope at end A.

$$\theta_A = -\frac{t_{B/A}}{L} = \frac{1}{18} \frac{M_0 L}{EI}$$

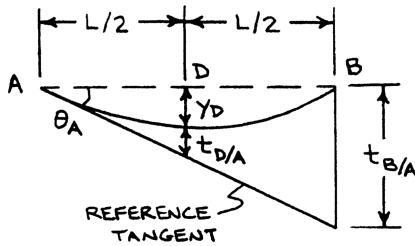
$$\theta_A = \frac{1}{18} \frac{M_0 L}{EI} \blacktriangleleft$$



PROBLEM 9.127

For the prismatic beam and loading shown, determine (a) the deflection at point D, (b) the slope at end A.

SOLUTION



$$t_{B/A} = \frac{1}{2} \left(\frac{1}{6} \frac{w_0 L^2}{EI} \right) (L) \left(\frac{L}{3} \right) + \frac{1}{4} \left(-\frac{1}{6} \frac{w_0 L^2}{EI} \right) (L) \left(\frac{L}{5} \right)$$

$$= \frac{7w_0 L^4}{360EI}$$

$$t_{D/A} = \frac{1}{2} \left(\frac{1}{12} \frac{w_0 L^2}{EI} \right) \left(\frac{L}{2} \right) \left(\frac{L}{6} \right) + \frac{1}{4} \left(-\frac{1}{48} \frac{w_0 L^2}{EI} \right) \left(\frac{L}{2} \right) \left(\frac{L}{10} \right)$$

$$= \frac{37w_0 L^4}{11520EI}$$

(a) Deflection at D.

$$y_D = \frac{1}{2} t_{B/A} - t_{D/A}$$

$$= \frac{1}{2} \left(\frac{7w_0 L^4}{360EI} \right) - \frac{37w_0 L^4}{11520EI}$$

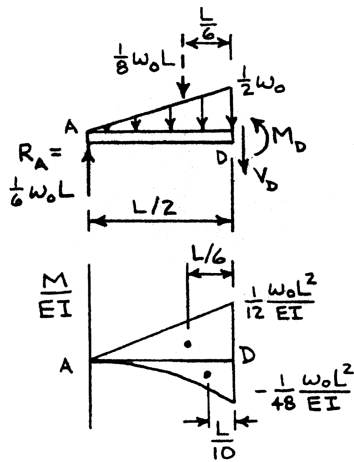
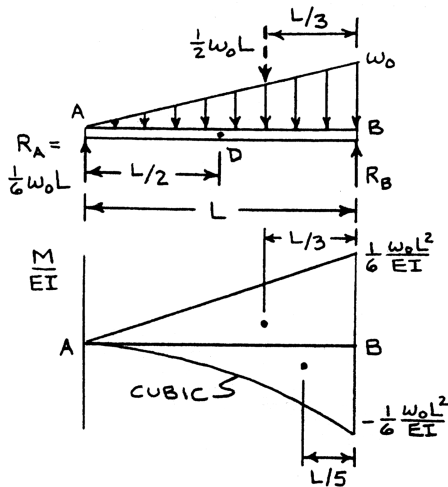
$$= \frac{75w_0 L^4}{11520EI}$$

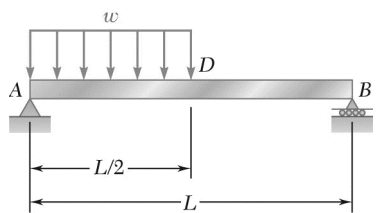
$$y_D = \frac{5w_0 L^4}{768EI} \downarrow \blacktriangleleft$$

(b) Slope at A.

$$\theta_A = -\frac{t_{B/A}}{L} = -\frac{7w_0 L^3}{360EI}$$

$$\theta_A = \frac{7w_0 L^3}{360EI} \nearrow \blacktriangleleft$$





PROBLEM 9.128

For the prismatic beam and loading shown, determine (a) the deflection at point D, (b) the slope at end A.

SOLUTION

$$+\circlearrowleft \Sigma M_A = 0: R_B L - \frac{wL}{2} \left(\frac{L}{4} \right) = 0$$

$$R_B = \frac{1}{8} wL \uparrow$$

Draw M/EI diagram by parts.

$$\frac{M_1}{EI} = \frac{R_B L}{EI} = \frac{wL^2}{8EI}$$

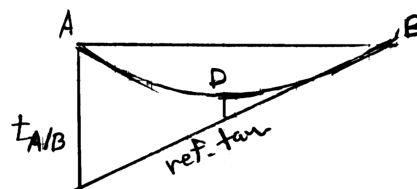
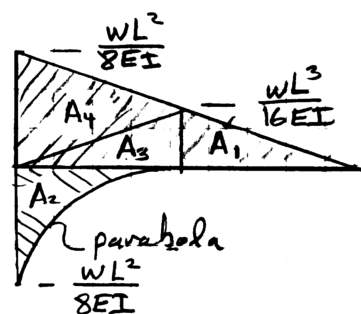
$$\frac{M_2}{EI} = -\frac{wL^3}{8EI}$$

$$A_1 = \left(\frac{1}{2} \right) \left(\frac{L}{2} \right) \frac{wL^2}{16EI} = \frac{wL^3}{64EI}$$

$$A_2 = \left(\frac{1}{3} \right) \left(\frac{L}{2} \right) \frac{wL^2}{8EI} = -\frac{wL^3}{48EI}$$

$$A_3 = \left(\frac{1}{2} \right) \left(\frac{L}{2} \right) \frac{wL^2}{16EI} = \frac{wL^3}{64EI}$$

$$A_4 = \left(\frac{1}{2} \right) \left(\frac{L}{2} \right) \frac{wL^2}{8EI} = \frac{wL^3}{32EI}$$



(a) Deflection at D.

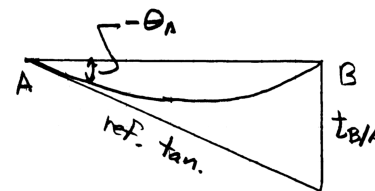
Place reference tangent at B.

$$y_D = t_{D/B} - \frac{L}{2} t_{A/B}$$

$$t_{D/A} = \left(\frac{1}{3} \cdot \frac{L}{2} \right) A_1 = \frac{wL^4}{384EI}$$

$$t_{B/A} = \frac{L}{3} (A_1 + A_3 + A_4) + \left(\frac{1}{4} \cdot \frac{L}{2} \right) A_2 = \frac{7wL^4}{384EI}$$

$$y_D = \frac{wL^4}{384EI} - \frac{1}{2} \cdot \frac{7wL^4}{384EI} = -\frac{5wL^4}{768EI}$$

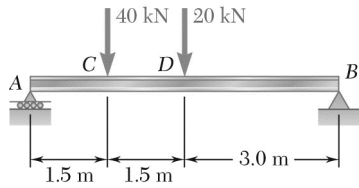


$$y_D = \frac{5wL^4}{768EI} \downarrow \blacktriangleleft$$

PROBLEM 9.128 (Continued)

(b) Slope at A. Place reference tangent at A.

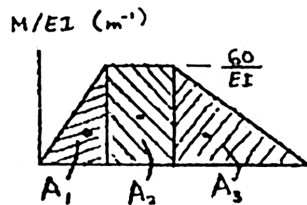
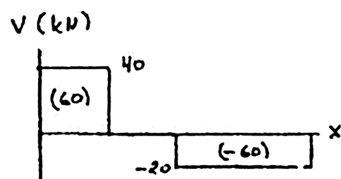
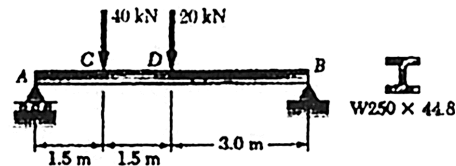
$$\begin{aligned}\theta_A &= -\frac{1}{L}t_{B/A} \\ &= -\left(\frac{1}{L}\right)\left\{\left(\frac{2L}{3}\right)(A_1 + A_3 + A_4) + \left(L - \frac{1}{4} \cdot \frac{L}{2}\right)A_2\right\} \\ &= -\frac{3wL^3}{128EI}\end{aligned}\quad \theta_A = \frac{3wL^3}{128EI} \swarrow \blacktriangleleft$$



PROBLEM 9.129

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at point D. Use $E = 200$ GPa.

SOLUTION



$$E = 200 \times 10^9 \text{ Pa}$$

$$I = 70.8 \times 10^6 \text{ mm}^4 = 70.8 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(70.8 \times 10^{-6}) = 19.16 \times 10^6 \text{ N} \cdot \text{m}^2 \\ = 14,160 \text{ kN} \cdot \text{m}^2$$

$$+\circlearrowleft \Sigma M_B = 0: -6R_A + (4.5)(40) + (3)(20) = 0$$

$$R_A = 40 \text{ kN}$$

Draw shear and $\frac{M}{EI}$ diagrams.

$$A_1 = \frac{1}{2} \left(\frac{60}{EI} \right) (1.5) = \frac{45}{EI}$$

$$A_2 = \left(\frac{60}{EI} \right) (1.5) = \frac{90}{EI}$$

$$A_3 = \frac{1}{2} \left(\frac{60}{EI} \right) (3) = \frac{90}{EI}$$

Place reference tangent at A.

$$t_{B/A} = A_1(4.5 + 0.5) + A_2(3 + 0.75) + A_3(2.0)$$

$$= \frac{742.5}{EI} \text{ m}$$

$$t_{D/A} = A_1(1.5 + 0.5) + A_2(0.75)$$

$$= \frac{157.5}{EI} \text{ m}$$

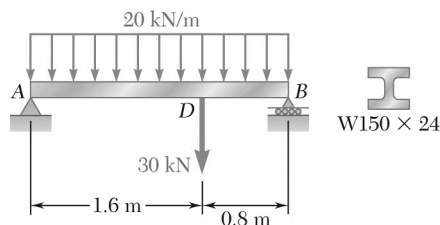
$$(a) \quad \text{Slope at A.} \quad \theta_A = -\frac{t_{B/A}}{L} = -\frac{742.5}{6EI} = -\frac{123.75}{EI} = -\frac{123.75}{14,160} \\ = -8.74 \times 10^{-3}$$

$$\theta_A = 8.74 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at D.

$$y_D = t_{D/A} - \frac{x_D}{L} t_{B/A} = \frac{157.5}{EI} - \left(\frac{3}{6} \right) \left(\frac{742.5}{EI} \right) = -\frac{213.75}{EI} \\ = -\frac{213.75}{14,160} = -15.10 \times 10^{-3} \text{ m}$$

$$y_D = 15.10 \text{ mm} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.130

For the beam and loading shown, determine (a) the slope at end A , (b) the deflection at point D . Use $E = 200$ GPa.



SOLUTION

Units: Forces in kN; lengths in meters.

For W150×24,

$$I = 13.4 \times 10^6 \text{ mm}^4 \\ = 13.4 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(13.4 \times 10^{-6}) = 2.68 \times 10^6 \text{ N} \cdot \text{m}^2 \\ = 2680 \text{ kN} \cdot \text{m}^2$$

$$+\circlearrowleft \sum M_B = 0: -2.4R_A + (0.8)(30) + (1.2)(2.4)(20) = 0 \\ R_A = 34 \text{ kN} \uparrow$$

Draw bending moment diagram by parts.

$$M_1 = (1.6)(34) = 54.4 \text{ kN} \cdot \text{m}$$

$$M_2 = (2.4)(34) = 81.6 \text{ kN} \cdot \text{m}$$

$$M_3 = -\frac{1}{2}(20)(1.6)^2 = -25.6 \text{ kN} \cdot \text{m}$$

$$M_4 = -\frac{1}{2}(20)(2.4)^2 = -57.6 \text{ kN} \cdot \text{m}$$

$$M_5 = -(0.8)(30) = -24 \text{ kN} \cdot \text{m}$$

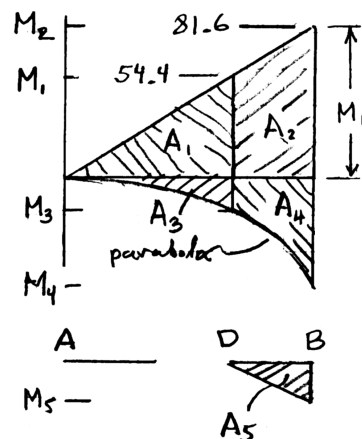
$$A_1 = \frac{1}{2}(1.6)(54.4) = 43.52 \text{ kN} \cdot \text{m}^2$$

$$A_1 + A_2 = \frac{1}{2}(2.4)(81.6) = 97.92 \text{ kN} \cdot \text{m}^2$$

$$A_3 = \frac{1}{3}(1.6)(-25.6) = -13.6533 \text{ kN} \cdot \text{m}^2$$

$$A_3 + A_4 = \frac{1}{3}(2.4)(-57.6) = -46.08 \text{ kN} \cdot \text{m}^2$$

$$A_5 = \frac{1}{2}(0.8)(-24) = -9.6 \text{ kN} \cdot \text{m}^2$$



PROBLEM 9.130 (Continued)

(a) Slope at A. Place reference tangent at A.

$$\theta_A = -\frac{1}{L}t_{B/A}$$

$$t_{B/A} = \frac{1}{EI} \left\{ (A_1 + A_2) \left(\frac{1}{3} \right) (2.4) + (A_3 + A_4) \left(\frac{1}{4} \right) (2.4) + A_5 \left(\frac{1}{3} \right) (0.8) \right\}$$

$$= \frac{48.128}{2680} = 17.9582 \times 10^{-3} \text{ m}$$

$$\theta_A = -\frac{17.9582 \times 10^{-3}}{2.4} = -7.48258 \times 10^{-3}$$

$$\theta_A = 7.48 \times 10^{-3} \text{ rad.} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at point D.

$$y_D = t_{D/A} + \theta_A x_D$$

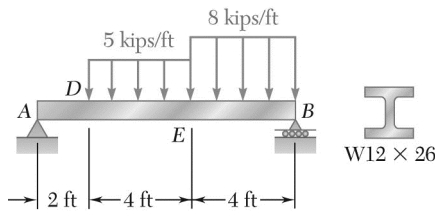
$$t_{D/A} = \frac{1}{EI} \left\{ A_1 \left(\frac{1}{3} \right) (1.6) + A_2 \left(\frac{1}{4} \right) (1.6) \right\}$$

$$= \frac{17.7493}{2680} = 6.62289 \times 10^{-3} \text{ m}$$

$$y_D = 6.62289 \times 10^{-3} + (-7.48258 \times 10^{-3})(1.6)$$

$$= -5.3492 \times 10^{-3} \text{ m}$$

$$y_D = 5.35 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.131

For the beam and loading shown, determine (a) the slope at point A, (b) the deflection at point E. Use $E = 29 \times 10^6$ psi.

SOLUTION

Units: Forces in kips; lengths in ft.

For W12 x 26, $I = 204 \text{ in}^4$

$$EI = (29 \times 10^6)(204) = 5.916 \times 10^9 \text{ lb} \cdot \text{in}^2 \\ = 41083 \text{ kip} \cdot \text{ft}^2$$

$$+\circlearrowleft \Sigma M_B = 0: -10R_A + (6)(4)(5) + (2)(4)(8) = 0 \quad R_A = 18.4 \text{ kips} \uparrow$$

Consider loading as 5 kips/ft from D to B plus 3 kips/ft from E to B. Draw bending moment diagram by parts.

$$M_1 = 10R_A = 184 \text{ kip} \cdot \text{ft}$$

$$M_2 = 6R_A = 110.4 \text{ kip} \cdot \text{ft}$$

$$M_3 = -\frac{1}{2}(5)(8)^2 = -160 \text{ kip} \cdot \text{ft}$$

$$M_4 = -\frac{1}{2}(5)(4)^2 = -40 \text{ kip} \cdot \text{ft}$$

$$M_5 = -\frac{1}{2}(3)(4)^2 = -24 \text{ kip} \cdot \text{ft}$$

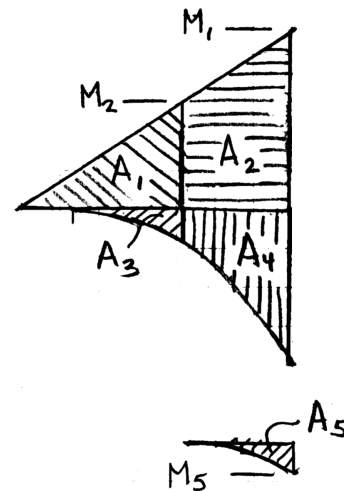
$$A_1 + A_2 = \frac{1}{2}(10)(184) = 920 \text{ kip} \cdot \text{ft}^2$$

$$A_1 = \frac{1}{2}(6)(110.4) = 331.2 \text{ kip} \cdot \text{ft}^2$$

$$A_3 + A_4 = \frac{1}{3}(8)(-160) = -426.667 \text{ kip} \cdot \text{ft}^2$$

$$A_3 = \frac{1}{3}(4)(-40) = -53.333 \text{ kip} \cdot \text{ft}^2$$

$$A_5 = \frac{1}{3}(4)(-24) = -32 \text{ kip} \cdot \text{ft}^2$$



PROBLEM 9.131 (Continued)

(a) Slope at A. $y_B = y_A + \theta_A L + t_{B/A} \quad y_A = y_B = 0$

$$\theta_A = -t_{B/A}/L$$

$$t_{B/A} = \frac{1}{EI} \left\{ (A_1 + A_2) \left(\frac{1}{3} \right) (10) + (A_3 + A_4) \left(\frac{1}{4} \right) (8) + (A_5) \left(\frac{1}{4} \right) (4) \right\}$$

$$= \frac{2181.33}{41083} = 53.096 \times 10^{-3} \text{ ft}$$

$$\theta_A = -\frac{53.096 \times 10^{-3}}{10} = -5.3096 \times 10^{-3}$$

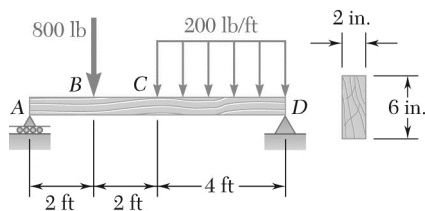
$$\theta_A = 5.31 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at E. $y_E = x_E \theta_A + t_{E/A}$

$$t_{E/A} = \frac{1}{EI} \left\{ (A_1) \left(\frac{1}{3} \right) (6) + (A_3) \left(\frac{1}{4} \right) (4) \right\} = \frac{609.067}{41083} = 14.8253 \times 10^{-3} \text{ ft}$$

$$y_E = (6)(-5.3096 \times 10^{-3}) + 14.8253 \times 10^{-3} = -17.0323 \times 10^{-3} \text{ ft}$$

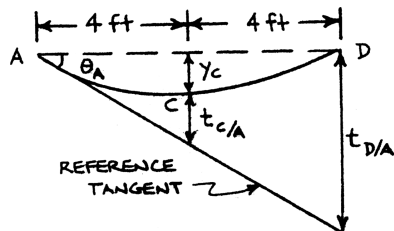
$$y_E = 0.204 \text{ in.} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.132

For the timber beam and loading shown, determine (a) the slope at point A, (b) the deflection at point C. Use $E = 1.7 \times 10^6$ psi.

SOLUTION



$$I = \frac{1}{12}(2)(6)^3 = 36 \text{ in}^4$$

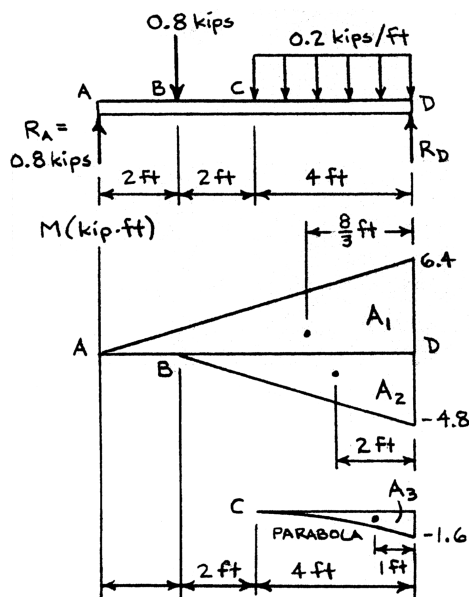
$$EI = (1.7 \times 10^3 \text{ ksi})(36 \text{ in}^4) = 61.2 \times 10^3 \text{ kip} \cdot \text{in}^2$$

$$A_1 = \frac{1}{2}(6.4)(8) = 25.6 \text{ kip} \cdot \text{ft}^2$$

$$A_2 = \frac{1}{2}(-4.8)(6) = -14.4 \text{ kip} \cdot \text{ft}^2$$

$$A_3 = \frac{1}{3}(-1.6)(4) = -2.1333 \text{ kip} \cdot \text{ft}^2$$

$$\begin{aligned} EIt_{D/A} &= A_1\left(\frac{8}{3} \text{ ft}\right) + A_2(2 \text{ ft}) + A_3(1 \text{ ft}) \\ &= (25.6)\left(\frac{8}{3}\right) + (-14.4)(2) + (-2.1333)(1) \\ &= 37.333 \text{ kip} \cdot \text{ft}^3 = 64512 \text{ kip} \cdot \text{in}^3 \\ t_{D/A} &= \frac{64512}{61.2 \times 10^3} = 1.05412 \text{ in.} \end{aligned}$$



PROBLEM 9.132 (Continued)

$$A_4 = \frac{1}{2}(3.2)(4) = 6.4 \text{ kip} \cdot \text{ft}^2$$

$$A_5 = \frac{1}{2}(-1.6)(2) = -1.6 \text{ kip} \cdot \text{ft}^2$$

$$EI t_{C/A} = A_4 \left(\frac{4}{3} \text{ ft} \right) + A_5 \left(\frac{2}{3} \text{ ft} \right) = (6.4) \left(\frac{4}{3} \right) + (-1.6) \left(\frac{2}{3} \right)$$

$$= 7.4667 \text{ kip} \cdot \text{ft}^3 = 12902.4 \text{ kip} \cdot \text{in}^3$$

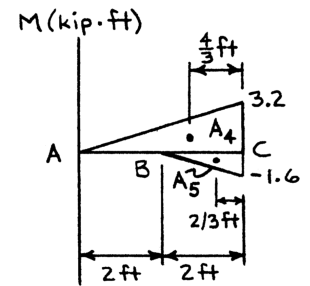
$$t_{C/A} = \frac{12902.4}{61.2 \times 10^3} = 0.21082 \text{ in.}$$

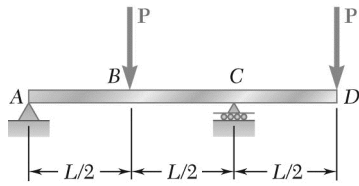
$$(a) \quad \theta_A = -\frac{t_{D/A}}{L} = -\frac{1.05412}{96} = -0.0109804 \text{ rad}$$

$$\theta_A = 10.98 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

$$(b) \quad y_C = t_{C/A} - \frac{1}{2} t_{D/A} = 0.21082 \text{ in.} - \frac{1}{2}(1.05412 \text{ in.})$$

$$y_C = 0.316 \text{ in.} \quad \downarrow \blacktriangleleft$$

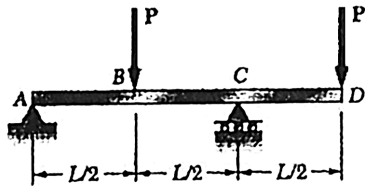




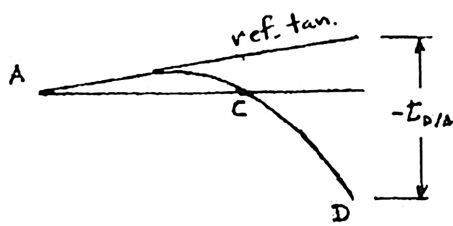
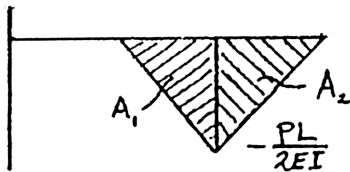
PROBLEM 9.133

For the beam and loading shown, determine (a) the slope at point A, (b) the deflection at point D.

SOLUTION



M/EI



$$+\circlearrowleft \sum M_C = 0 : -R_A L + P \frac{L}{2} - P \frac{L}{2} = 0 \quad R_A = 0$$

Draw $\frac{M}{EI}$ diagram.

$$A_1 = -\frac{1}{2} \left(\frac{PL}{2EI} \right) \left(\frac{L}{2} \right) = -\frac{1}{8} \frac{PL^2}{EI}$$

$$A_2 = -\frac{1}{2} \left(\frac{PL}{2EI} \right) \left(\frac{L}{2} \right) = -\frac{1}{8} \frac{PL^2}{EI}$$

Place reference tangent at A.

$$t_{C/A} = A_1 \cdot \left(\frac{1}{3} \cdot \frac{L}{2} \right) = -\frac{1}{48} \frac{PL^3}{EI}$$

(a) Slope at A.

$$\theta_A = -\frac{t_{C/A}}{L}$$

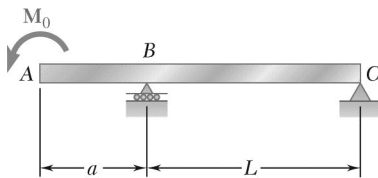
$$\theta_A = \frac{1}{48} \frac{PL^2}{EI} \quad \blacktriangleleft$$

(b) Deflection at D.

$$t_{D/A} = A_1 \left(\frac{L}{2} + \frac{L}{6} \right) + A_2 \left(\frac{2}{3} \cdot \frac{L}{2} \right) = -\frac{1}{8} \frac{PL^3}{EI}$$

$$y_D = t_{D/A} - \frac{x_D}{L} t_{C/A} = -\frac{1}{8} \frac{PL^3}{EI} - \left(\frac{3}{2} \right) \left(-\frac{1}{48} \frac{PL^3}{EI} \right)$$

$$y_D = -\frac{3}{32} \frac{PL^3}{EI} \quad \blacktriangleleft$$



PROBLEM 9.134

For the beam and loading shown, determine (a) the slope at point A, (b) the deflection at point A.

SOLUTION

$$A_1 = -\frac{M_0 a}{EI}$$

$$A_2 = -\frac{M_0 L}{2EI}$$

$$t_{C/B} = A_2 \left(\frac{2L}{3} \right)$$

$$= \left(-\frac{M_0 L}{2EI} \right) \left(\frac{2L}{3} \right)$$

$$= -\frac{M_0 L^2}{3EI}$$

(a) Slope at A.

$$\theta_B = \frac{t_{C/B}}{L} = \frac{M_0 L}{3EI}$$

$$\theta_B = \theta_A + \theta_{B/A} = \theta_A + A_1$$

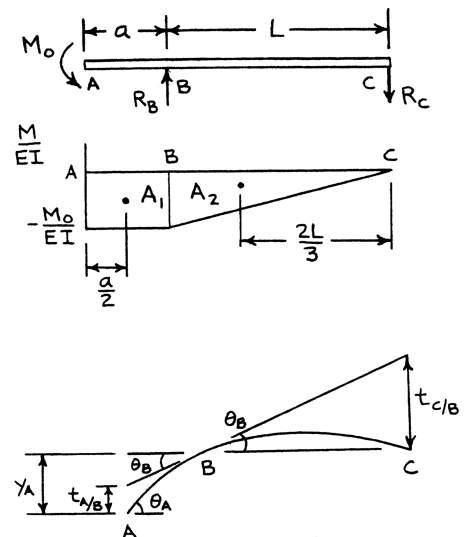
$$\frac{M_0 L}{3EI} = \theta_A - \frac{M_0 a}{EI}$$

(b) Deflection at A.

$$t_{A/B} = A_1 \left(\frac{a}{2} \right) = -\frac{M_0 a^2}{2EI}$$

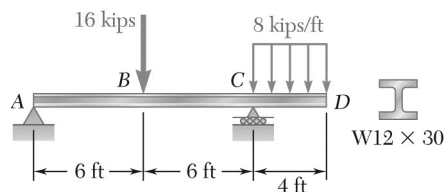
$$y_A = \frac{a}{L} t_{C/B} + t_{A/B}$$

$$= \frac{a}{L} \left(-\frac{M_0 L^2}{3EI} \right) - \frac{M_0 a^2}{2EI}$$



$$\theta_A = \frac{M_0}{3EI} (L + 3a) \quad \nearrow \blacktriangleleft$$

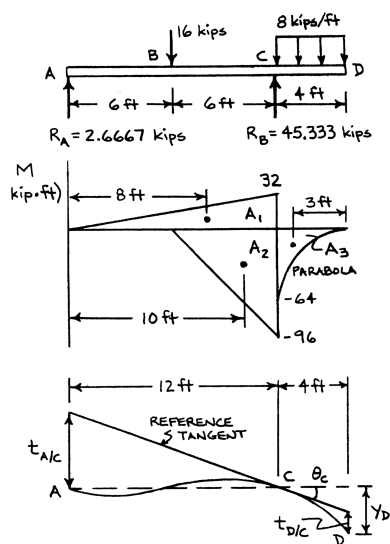
$$y_A = \frac{M_0 a}{6EI} (2L + 3a) \quad \downarrow \blacktriangleleft$$



PROBLEM 9.135

For the beam and loading shown, determine (a) the slope at point C, (b) the deflection at point D. Use $E = 29 \times 10^6$ psi.

SOLUTION



Free Body AD : $+\circlearrowleft \Sigma M_C = 0: (16)(6) - (32)(2) - 12R_A = 0$
 $R_A = 2.6667 \text{ kips } \uparrow$

$+\uparrow \Sigma F_y = 0: 2.6667 - 16 + R_B - 32 = 0$
 $R_B = 45.333 \text{ kips } \uparrow$

For $W12 \times 30$,

$I = 238 \text{ in}^4$

$EI = (29 \times 10^3 \text{ ksi})(238 \text{ in}^4)$
 $= 6.902 \times 10^6 \text{ kip} \cdot \text{in}^2 = 47931 \text{ kip} \cdot \text{ft}^2$

(a) Slope at C.

$A_1 = \frac{1}{2}(32)(12) = 192 \text{ kip} \cdot \text{ft}^2$

$A_2 = \frac{1}{2}(-96)(6) = -288 \text{ kip} \cdot \text{ft}^2$

$EIt_{A/C} = A_1(8 \text{ ft}) + A_2(10 \text{ ft})$
 $= (192)(8) + (-288)(10) = -1344 \text{ kip} \cdot \text{ft}^3$

$t_{A/C} = -\frac{1344}{47931} = -28.040 \times 10^{-3} \text{ ft} = -0.33648 \text{ in.}$

$\theta_C = \frac{t_{A/C}}{L} = -\frac{28.040 \times 10^{-3} \text{ ft}}{12 \text{ ft}}$

$\theta_C = 2.34 \times 10^{-3} \text{ rad } \searrow \blacktriangleleft$

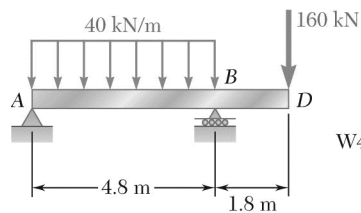
(b) Deflection at D.

$EIt_{D/C} = A_1(3 \text{ ft}) = \frac{1}{3}(-64)(4)(3) = -256 \text{ kip} \cdot \text{ft}^3$

$t_{D/C} = -\frac{256}{47931} = -5.3410 \times 10^{-3} \text{ ft} = -0.064092 \text{ in.}$

$y_D = t_{D/C} + \frac{4}{12}t_{A/C} = -0.064092 + \frac{1}{3}(-0.33648)$
 $= -0.176252 \text{ in.}$

$y_D = 0.1763 \text{ in. } \downarrow \blacktriangleleft$



PROBLEM 9.136

For the beam and loading shown, determine (a) the slope at point B, (b) the deflection at point D. Use $E = 200$ GPa.

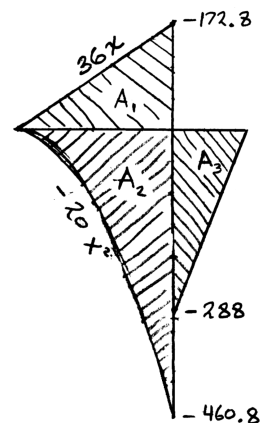
SOLUTION

Units: Forces in kN; lengths in meters.

$$\begin{aligned}
 I &= 462 \times 10^6 \text{ mm}^4 = 462 \times 10^{-6} \text{ m}^4 \\
 EI &= (200 \times 10^9)(462 \times 10^{-6}) \\
 &= 92.4 \times 10^6 \text{ N} \cdot \text{m}^2 = 92400 \text{ kN} \cdot \text{m}^2 \\
 + \sum M_B &= 0: -4.8 R_A + (40)(4.8)(2.4) - (160)(1.8) = 0 \\
 R_A &= 36 \text{ kN}
 \end{aligned}$$

Draw bending moment diagram by parts.

$$\begin{aligned}
 A_1 &= \frac{1}{2}(4.8)(172.8) = 414.72 \text{ kN} \cdot \text{m}^2 \\
 A_2 &= \frac{1}{3}(4.8)(-460.8) = -737.28 \text{ kN} \cdot \text{m}^2 \\
 A_3 &= \frac{1}{2}(1.8)(-288) = -259.2 \text{ kN} \cdot \text{m}^2
 \end{aligned}$$



Place reference tangent at B.

(a) Slope at B.

$$y_A = y_B - L\theta_B + t_{A/B}$$

$$\begin{aligned}
 \theta_B &= \frac{t_{B/A}}{L} = \frac{1}{EIL} \left\{ A_1 \left(\frac{2}{3} \right) (4.8) + A_2 \left(\frac{3}{4} \right) (4.8) \right\} \\
 &= \frac{-1327.104}{(92400)(4.8)} = -2.9922 \times 10^{-3}
 \end{aligned}$$

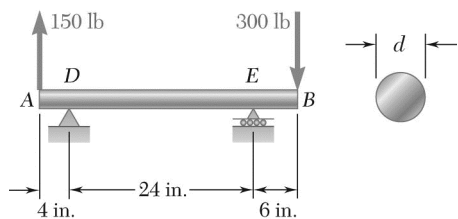


$$\theta_B = 2.99 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at D.

$$\begin{aligned}
 y_D &= y_B + a\theta_B + t_{D/B} \\
 &= 0 + (1.8)(-2.9922 \times 10^{-3}) - \frac{1}{EI} \left\{ A_3 \left(\frac{2}{3} \right) (1.8) \right\} \\
 &= -5.3860 \times 10^{-3} - \frac{311.04}{92400} \\
 &= -8.75 \times 10^{-3} \text{ m}
 \end{aligned}$$

$$y_D = 8.75 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.137

Knowing that the beam AB is made of a solid steel rod of diameter $d = 0.75$ in., determine for the loading shown (a) the slope at point D , (b) the deflection at point A . Use $E = 29 \times 10^6$ psi.

SOLUTION

Units: Forces in lb; lengths in inches.

$$c = \frac{1}{2} = \frac{1}{2}(0.75) = 0.375 \text{ in.}$$

$$I = \frac{\pi}{4}c^4 = \frac{\pi}{4}(0.375)^4 = 0.0155316 \text{ in}^4$$

$$EI = (29 \times 10^6)(0.0155316) = 450.4 \times 10^3 \text{ lb} \cdot \text{in}^2$$

Draw $\frac{M}{EI}$ diagram by parts by considering the bending moment diagram due to each of the applied loads.

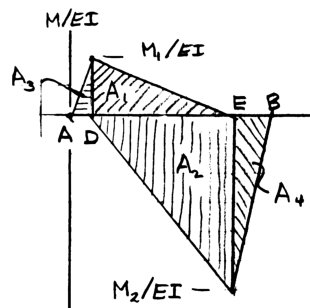
$$\frac{M_1}{EI} = \frac{(150)(4)}{450.4 \times 10^3} = 1.33215 \times 10^{-3} \text{ in}^{-1}$$

$$\frac{M_2}{EI} = -\frac{(300)(6)}{450.4 \times 10^3} = -3.99645 \times 10^{-3} \text{ in}^{-1}$$

$$A_1 = \frac{1}{2}(24)(1.33215 \times 10^{-3}) = 15.9858 \times 10^{-3}$$

$$A_2 = \frac{1}{2}(24)(-3.99645 \times 10^{-3}) = -47.9574 \times 10^{-3}$$

$$A_3 = \frac{1}{2}(4)(1.33215 \times 10^{-3}) = 2.6643 \times 10^{-3}$$



Place reference tangent at D .

(a) Slope at point D .

$$y_E = y_D + L\theta_D + t_{E/D} \quad \theta_D = -t_{E/D}/L$$

$$t_{E/A} = 16A_1 + 8A_2 = -127.8864 \times 10^{-3} \text{ in.}$$

$$\theta_D = \frac{-127.8864 \times 10^{-3}}{24} = 5.3286 \times 10^{-3}$$

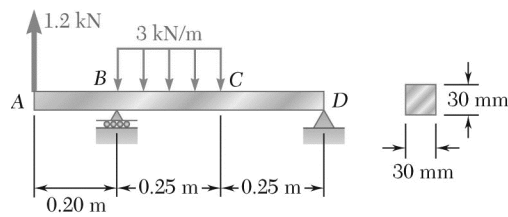
$$\theta_D = 5.33 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

(b) Deflection at A .

$$y_A = y_D - a\theta_D + t_{A/D} = t_{A/D} - a\theta_D$$

$$y_A = A_3 \left(\frac{2}{3} \right) (4) - (4)(5.3286 \times 10^{-3}) = -14.21 \times 10^{-3} \text{ in.}$$

$$y_A = 0.01421 \text{ in.} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.138

Knowing that the beam AD is made of a solid steel bar, determine the (a) slope at point B , (b) the deflection at point A . Use $E = 200$ GPa.

SOLUTION

$$E = 200 \times 10^9 \text{ Pa} \quad I = \frac{1}{12}(30)(30)^3 = 67.5 \times 10^3 \text{ mm}^4 = 67.5 \times 10^{-9} \text{ m}^4$$

$$EI = (200 \times 10^9)(67.5 \times 10^{-9}) = 13500 \text{ N} \cdot \text{m}^2 = 13.5 \text{ kN} \cdot \text{m}^2$$

$$+\circlearrowleft \Sigma M_B = 0: -(0.2)(1.2) - (3)(0.25)(0.125) + 5R_D = 0 \quad R_D = 0.6675 \text{ kN}$$

Draw $\frac{M}{EI}$ diagram by parts.

$$M_1 = (0.6675)(0.5) = 0.33375 \text{ kN} \cdot \text{m}$$

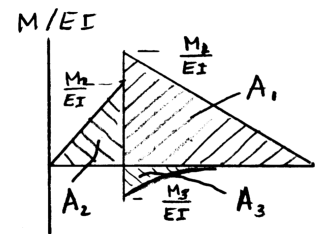
$$M_2 = (1.2)(0.2) = 0.240 \text{ kN} \cdot \text{m}$$

$$M_3 = -\frac{1}{2}(3)(0.25)^2 = -0.09375 \text{ kN} \cdot \text{m}$$

$$A_1 = \frac{1}{2}(0.33375)(0.5)/EI = 0.0834375/EI$$

$$A_2 = \frac{1}{2}(0.240)(0.2)/EI = 0.024/EI$$

$$A_3 = \frac{1}{3}(-0.09375)(0.25)/EI = -0.0078125/EI$$



Place reference tangent at B .

$$t_{D/B} = A_1 \left(\frac{2}{3} \cdot 0.5 \right) + A_3 \left(\frac{3}{4} \cdot (0.25) + 0.25 \right) = 0.024395/EI$$

$$(a) \quad \text{Slope at } B. \quad \theta_B = -\frac{t_{D/B}}{L} = -\frac{0.024395}{0.5EI} = -\frac{0.048789}{EI}$$

$$= -3.6140 \times 10^{-3} \text{ rad}$$

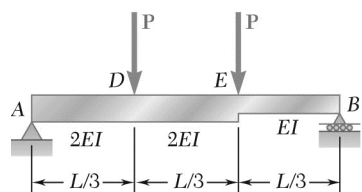
$$\theta_B = 3.61 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

$$t_{A/B} = A_2 \left(\frac{2}{3}(0.20) \right) = 0.0032/EI = 0.23704 \times 10^{-3} \text{ m}$$

$$(b) \quad \text{Deflection at } A. \quad y_A = t_{A/B} - L_{AB}\theta_B$$

$$= 0.23704 \times 10^{-3} - (0.2)(-3.6140 \times 10^{-3}) = 0.960 \times 10^{-3} \text{ m}$$

$$y_A = 0.960 \text{ mm} \quad \uparrow \blacktriangleleft$$



PROBLEM 9.139

For the beam and loading shown, determine the deflection (a) at point D, (b) at point E.

SOLUTION

$$A_1 = \frac{1}{2} \left(\frac{PL}{6EI} \right) \left(\frac{L}{3} \right) = \frac{PL^2}{36EI}$$

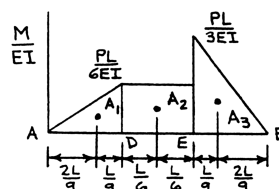
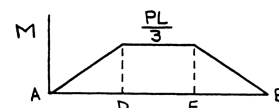
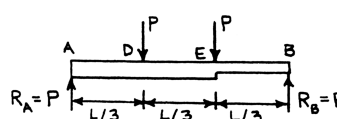
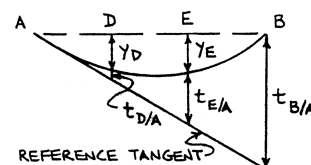
$$A_2 = \left(\frac{PL}{6EI} \right) \left(\frac{L}{3} \right) = \frac{PL^2}{18EI}$$

$$A_3 = \frac{1}{2} \left(\frac{PL}{3EI} \right) \left(\frac{L}{3} \right) = \frac{PL^2}{18EI}$$

$$t_{D/A} = A_1 \left(\frac{L}{9} \right) = \left(\frac{PL^2}{36EI} \right) \left(\frac{L}{9} \right) = \frac{PL^3}{324EI}$$

$$\begin{aligned} t_{E/A} &= A_1 \left(\frac{L}{9} + \frac{L}{3} \right) + A_2 \left(\frac{L}{6} \right) \\ &\quad + \left(\frac{PL^3}{36EI} \right) \left(\frac{4L}{9} \right) + \left(\frac{PL^2}{18EI} \right) \left(\frac{L}{6} \right) \\ &= \frac{7PL^3}{324EI} \end{aligned}$$

$$\begin{aligned} t_{B/A} &= A_1 \left(\frac{7L}{9} \right) + A_2 \left(\frac{L}{2} \right) + A_3 \left(\frac{2L}{9} \right) \\ &= \left(\frac{PL^3}{36EI} \right) \left(\frac{7L}{9} \right) + \left(\frac{PL^2}{18EI} \right) \left(\frac{L}{2} \right) + \left(\frac{PL^2}{18EI} \right) \left(\frac{2L}{9} \right) \\ &= \frac{5PL^3}{81EI} \end{aligned}$$



(a) Deflection at D.

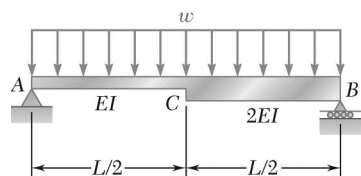
$$y_D = \frac{1}{3} t_{B/A} - t_{D/A} = \frac{1}{3} \left(\frac{5PL^3}{81EI} \right) - \frac{PL^3}{324EI} = \frac{17PL^3}{972EI}$$

$$y_D = \frac{17PL^3}{972EI} \downarrow \blacktriangleleft$$

(b) Deflection at E.

$$y_E = \frac{2}{3} t_{B/A} - t_{E/A} = \frac{2}{3} \left(\frac{5PL^3}{81EI} \right) - \frac{7PL^3}{324EI} = \frac{19PL^3}{972EI}$$

$$y_E = \frac{19PL^3}{972EI} \downarrow \blacktriangleleft$$



PROBLEM 9.140

For the beam and loading shown, determine (a) the slope at end A, (b) the slope at end B, (c) the deflection at the midpoint C.

SOLUTION

Reactions:

$$R_A = R_B = \frac{1}{2} wL$$

Draw bending moment and M/EI diagrams by parts as shown.

$$A_1 = \frac{1}{2} \cdot \frac{L}{2} \cdot \frac{wL^2}{4EI} = \frac{wL^3}{16EI}$$

$$A_2 = -\frac{1}{3} \cdot \frac{L}{2} \cdot \frac{wL^2}{8EI} = -\frac{wL^3}{48EI}$$

$$A_3 = \frac{1}{2} \cdot \frac{L}{2} \cdot \frac{wL^2}{8EI} = -\frac{wL^3}{32EI}$$

$$A_4 = -\frac{1}{3} \cdot \frac{L}{2} \cdot \frac{wL^2}{16EI} = -\frac{wL^3}{96EI}$$

Place reference tangent at A.

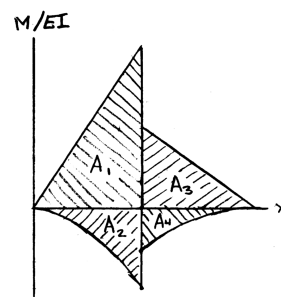
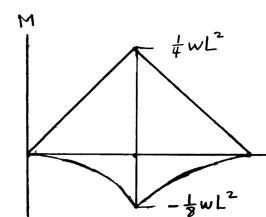
(a) Slope at end A.

$$y_B = y_A + L\theta_A + t_{B/A}$$

$$\theta_A = -t_{B/A}/L$$

$$\begin{aligned} t_{B/A} &= \left(\frac{L}{2} + \frac{L}{6}\right) A_1 + \left(\frac{L}{2} + \frac{L}{8}\right) A_2 + \frac{L}{3} A_3 + \frac{3L}{8} A_4 \\ &= \frac{wL^4}{EI} \left(\frac{1}{24} - \frac{5}{384} + \frac{1}{96} - \frac{1}{256} \right) = \frac{9wL^4}{256EI} \end{aligned}$$

$$\theta_A = -\frac{9wL^4}{256EI} \cdot \frac{1}{L} = -\frac{9wL^3}{256EI}$$



$$\theta_A = \frac{9wL^3}{256EI} \quad \swarrow \blacktriangleleft$$

(b) Slope at end B.

$$\theta_B = \theta_A + \theta_{B/A} = -\frac{9wL^3}{256EI} + A_1 + A_2 + A_3 + A_4$$

$$\theta_B = \frac{7wL^3}{256EI}$$

$$\theta_B = \frac{7wL^3}{256EI} \quad \nearrow \blacktriangleleft$$

PROBLEM 9.140 (Continued)

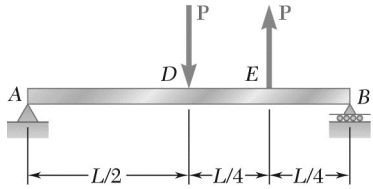
(c) Deflection at midpoint C.

$$y_A = y_C + \frac{L}{2}\theta_A + t_{C/A}$$

$$t_{C/A} = \left(\frac{L}{6}\right)A_1 + \left(\frac{L}{8}\right)A_2 = \frac{wL^4}{128EI}$$

$$y_C = 0 + \left(\frac{L}{2}\right)\left(-\frac{9wL^3}{256EI}\right) + \frac{wL^4}{128EI} = -\frac{5wL^4}{512EI}$$

$$y_C = \frac{5wL^4}{512EI} \downarrow \blacktriangleleft$$



PROBLEM 9.141

For the beam and loading of Prob. 9.125, determine the magnitude and location of the largest downward deflection.

SOLUTION

$$+\circlearrowleft \Sigma M_B = 0: -R_A L + \frac{PL}{2} - \frac{PL}{4} = 0 \quad R_A = \frac{1}{4}P \uparrow$$

$$+\circlearrowleft \Sigma M_A = 0: -\frac{PL}{2} + P\frac{3L}{4} + R_B L = 0 \quad R_B = \frac{1}{4}P \downarrow$$

Draw V (shear) diagram and M/EI diagram.

$$A_1 = \frac{1}{2} \left(\frac{1}{8} \frac{PL}{EI} \right) \left(\frac{L}{2} \right) = \frac{1}{32} \frac{PL^2}{EI}$$

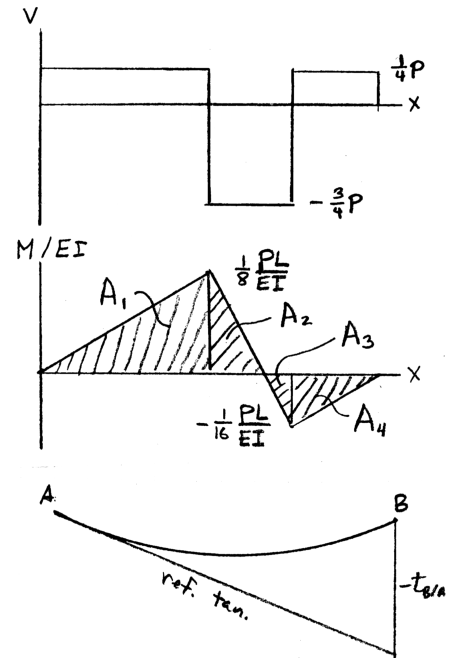
$$A_2 = \frac{1}{2} \left(\frac{1}{8} \frac{PL}{EI} \right) \left(\frac{L}{6} \right) = \frac{1}{96} \frac{PL^2}{EI}$$

$$A_3 = \frac{1}{2} \left(-\frac{1}{16} \frac{PL}{EI} \right) \left(\frac{L}{12} \right) = -\frac{1}{384} \frac{PL^2}{EI}$$

$$A_4 = \frac{1}{2} \left(-\frac{1}{16} \frac{PL}{EI} \right) \left(\frac{L}{4} \right) = -\frac{1}{128} \frac{PL^2}{EI}$$

Place reference tangent at A .

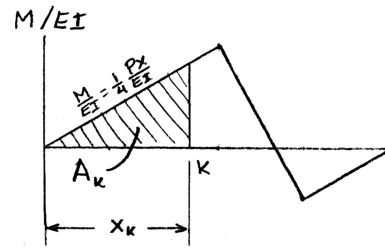
$$\begin{aligned} t_{B/A} &= \left(\frac{1}{32} \frac{PL^2}{EI} \right) \left(\frac{2L}{3} \right) + \left(\frac{1}{96} \frac{PL^2}{EI} \right) \left(\frac{L}{2} - \frac{1}{3} \frac{L}{6} \right) \\ &\quad + \left(-\frac{1}{384} \frac{PL^2}{EI} \right) \left(\frac{L}{4} + \frac{1}{3} \frac{L}{12} \right) + \left(-\frac{1}{128} \frac{PL^2}{EI} \right) \left(\frac{2}{3} \cdot \frac{L}{4} \right) \\ &= \left(\frac{1}{48} + \frac{1}{216} - \frac{5}{6912} - \frac{1}{768} \right) \frac{PL^3}{EI} = \frac{3}{128} \frac{PL^3}{EI} \\ \theta_A &= -\frac{t_{B/A}}{L} = -\frac{3}{128} \frac{PL^2}{EI} \end{aligned}$$



PROBLEM 9.141 (Continued)

Let point K be the location of $|y_m|$.

$$\begin{aligned}\theta_K &= \theta_A + \theta_{K/A} \\ &= -\frac{3}{128} \frac{PL^2}{EI} + A_K \\ &= -\frac{3}{128} \frac{PL^2}{EI} + \frac{1}{2} \left(\frac{1}{4} \frac{Px_K}{EI} \right) x_K \\ &= \frac{P}{EI} \left(-\frac{3}{128} L^2 + \frac{1}{8} x_K^2 \right) = 0\end{aligned}$$



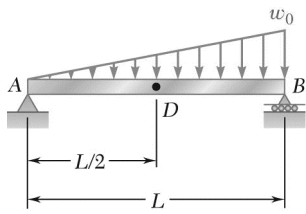
$$x_K = \sqrt{\frac{3}{16}} L = \frac{1}{4} \sqrt{3} L$$

$$x_K = 0.433 L \quad \blacktriangleleft$$

$$t_{K/A} = A_K \left(\frac{1}{3} x_K \right) = \frac{1}{2} \left(\frac{1}{4} \frac{Px_K^2}{EI} \right) \frac{x_K}{3} = \frac{1}{24} \frac{Px_K^3}{EI} = \frac{\sqrt{3}}{512} \frac{PL^3}{EI}$$

$$y_K = t_{K/A} - \frac{x_K}{L} t_{B/A} = \frac{\sqrt{3}}{512} \frac{PL^3}{EI} - \left(\frac{1}{4} \sqrt{3} \right) \frac{3}{128} \frac{PL^3}{EI} = -\frac{\sqrt{3}}{256} \frac{PL^3}{EI}$$

$$y_K = 0.00677 \frac{PL^3}{EI} \downarrow \quad \blacktriangleleft$$



PROBLEM 9.142

For the beam and loading of Prob. 9.127, determine the magnitude and location of the largest downward deflection.

SOLUTION

From Prob. 9.127:

$$\theta_A = -\frac{7w_0L^3}{360EI}$$

$$A_1 = \frac{1}{2} \left(\frac{w_0L}{6EI} x_m \right) (x_m) = \frac{w_0Lx_m^2}{12EI}$$

$$A_2 = \frac{1}{4} \left(-\frac{w_0x_m^3}{6EIL} \right) (x_m) = -\frac{w_0x_m^4}{24EIL}$$

Maximum deflection occurs at K, where $\theta_K = 0$.

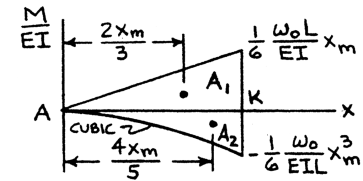
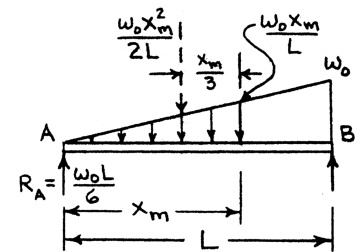
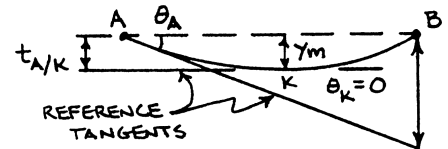
$$\theta_K = \theta_A + \theta_{K/A} = \theta_A + A_1 + A_2$$

$$0 = -\frac{7w_0L^3}{360EI} + \frac{w_0Lx_m^2}{12EI} - \frac{w_0x_m^4}{24EIL}$$

Rearranging:

$$0 = \frac{w_0L^2}{360EI} \left[-7 + 30 \left(\frac{x_m}{L} \right)^2 - 15 \left(\frac{x_m}{L} \right)^4 \right]$$

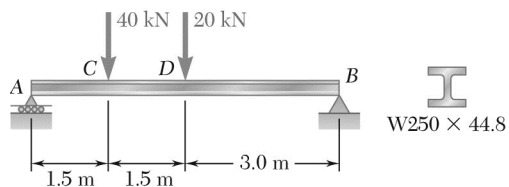
Solving biquadratic: $\left(\frac{x_m}{L} \right)^2 = 0.26970 \quad x_m = 0.51933L$



y_m is 0.519L from A. ◀

$$\begin{aligned} t_{A/K} &= A_1 \frac{2x_m}{3} + A_2 \frac{4x_m}{5} = \left(\frac{w_0Lx_m^2}{12EI} \right) \frac{2x_m}{3} + \left(-\frac{w_0x_m^4}{24EIL} \right) \frac{4x_m}{5} \\ &= \frac{w_0L^4}{90EI} \left[5 \left(\frac{x_m}{L} \right)^3 - 3 \left(\frac{x_m}{L} \right)^5 \right] = \frac{w_0L^4}{90EI} [5(0.51933)^3 - 3(0.51933)^5] \\ &= 0.0065222 \frac{w_0L^4}{EI} \quad y_m = |t_{A/K}| \end{aligned}$$

$$y_m = 6.52 \times 10^{-3} \frac{w_0L^4}{EI} \quad \blacktriangleleft$$



PROBLEM 9.143

For the beam and loading of Prob. 9.129, determine the magnitude and location of the largest downward deflection.

SOLUTION

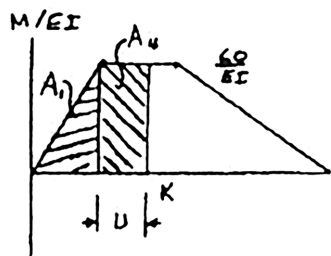
Referring to the solution to Prob. 9.129,

$$EI = 14,160 \text{ kN} \cdot \text{m}^2$$

$$R_A = 40 \text{ kN}, \quad A_1 = \frac{45}{EI}$$

$$t_{B/A} = \frac{742.5}{EI} \text{ m}$$

$$\theta_A = -\frac{123.75}{EI}$$



Let K be the location of the maximum deflection. Assume that K lies between C and D .



$$\begin{aligned} \theta_K &= \theta_A + \theta_{K/A} \\ &= -\frac{123.75}{EI} + A_1 + A_4 \\ &= -\frac{123.75}{EI} + \frac{45}{EI} + \frac{60u}{EI} = 0 \end{aligned}$$

$$u = \frac{123.75 - 45}{60} = 1.3125 \text{ m}$$

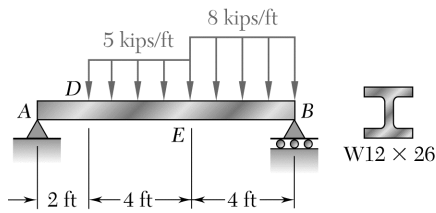
$$x_K = 1.5 + u = 2.8125 \text{ m}$$

$$\begin{aligned} t_{K/A} &= A_1(u + 0.5) + A_4\left(\frac{1}{2}u\right) \\ &= \frac{45}{EI}(1.8125) + \frac{(60)(1.3125)\left(\frac{1}{2}\right)(1.3125)}{EI} = \frac{133.242}{EI} \end{aligned}$$

$$\begin{aligned} y_K &= t_{K/A} - \frac{x_K}{L} t_{B/A} \\ &= \frac{133.242}{EI} - \frac{2.8125}{6} \left(\frac{742.5}{EI} \right) = -\frac{214.80}{EI} = -\frac{214.80}{14,160} \\ &= -15.17 \times 10^{-3} \text{ m} \end{aligned}$$

$$y_K = 15.17 \text{ mm} \downarrow \blacktriangleleft$$

$$x_K = 2.81 \text{ m} \blacktriangleleft$$



PROBLEM 9.144

For the beam and loading of Prob. 9.131, determine the magnitude and location of the largest downward deflection.

SOLUTION

From the solution to Prob. 9.130,

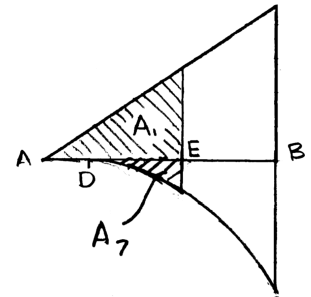
$$EI = 41083 \text{ kip} \cdot \text{ft}^2$$

$$R_A = 18.4 \text{ kips}$$

$$A_1 = 331.2 \text{ kip} \cdot \text{ft}^2$$

$$A_3 = -53.333 \text{ kip} \cdot \text{ft}^3$$

$$\theta_A = -5.3096 \times 10^{-3}$$



Slope at E.

$$\theta_E = \theta_A + \theta_{E/A}$$

$$\theta_{E/A} = \frac{1}{EI} \{A_1 + A_3\} = \frac{278.767}{41083} = 6.7855 \times 10^{-3}$$

$$\theta_E = 1.4759 \times 10^{-3}$$

Since $\theta_E > 0$, the point K of zero slope lies to the left of point E. Let x_K be the coordinate of point K.

$$A_6 = \frac{1}{2} R_A x_K^2 = 9.2 x_K^2$$

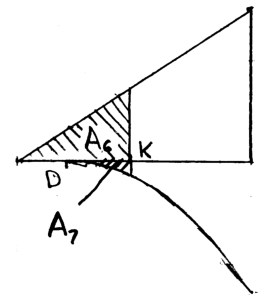
$$A_7 = -\frac{1}{6} (5)(x_K - 2)^3$$

$$\theta_K = \theta_A + \theta_{K/A} = \theta_A + \frac{1}{EI} \{A_6 + A_7\} = 0$$

$$A_6 + A_7 + EI\theta_A = 0$$

$$f(x_K) = 9.2x_K^2 - \frac{5}{6}(x_K - 2)^3 - 218.134 = 0$$

$$\frac{df}{dx_K} = 18.4x_K - 2.5(x_K - 2)^2$$



PROBLEM 9.144 (Continued)

Solve for x_K by iteration.

$$x_K = (x_K)_0 - \frac{f}{df/dx_K}$$

x_K	5	5.1473	5.1525	$x_K = 5.1525 \text{ ft}$
f	-10.634	-0.362	0.001	
df/dx_K	72.2	70.131		

$$A_6 = 244.244 \text{ kip} \cdot \text{ft}^2, \quad A_7 = -26.108 \text{ kip} \cdot \text{ft}^2$$

Maximum deflection.

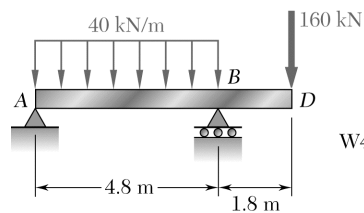
$$y_A = y_K + t_{A/K} = 0 \quad y_K = -t_{A/K}$$

$$\bar{x}_6 = \frac{2}{3}x_K \quad \bar{x}_7 = 2 + \frac{3}{4}(x_K - 2) = \frac{3x_K + 2}{4}$$

$$y_7 = -\frac{1}{EI} \{A_6 \bar{x}_6 + A_7 \bar{x}_7\} = -\frac{725.033}{41083} = -17.648 \times 10^{-3} \text{ ft}$$

$$y_K = 0.212 \text{ in.} \downarrow \blacktriangleleft$$

$$x_K = 5.15 \text{ ft} \blacktriangleleft$$



PROBLEM 9.145

For the beam and loading of Prob. 9.136, determine the largest upward deflection in span AB .

SOLUTION

Units: Forces in kN; lengths in meters.

$$I = 462 \times 10^6 \text{ mm}^4 = 462 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(462 \times 10^{-6})$$

$$= 92.4 \times 10^6 \text{ N} \cdot \text{m}^2 = 92400 \text{ kN} \cdot \text{m}$$

$$+\circlearrowleft M_B = 0: -4.8R_A + (40)(4.8)(2.4) - (160)(1.8) = 0$$

$$R_A = 36 \text{ kN}$$

$$A_1 = \frac{1}{2}x(36x) = 18x^2$$

$$A_2 = \frac{1}{3}x(-20x^2) = -\frac{20}{3}x^3$$

Place reference tangent at A .

$$y_B = y_A + L\theta_A + t_{B/A} = 0$$

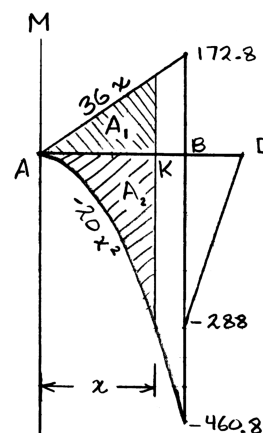
$$\theta_A = -\frac{t_{B/A}}{L}$$

$$(A_1)_B = (18)(4.8)^2 = 414.72 \text{ kN} \cdot \text{m}^2$$

$$(A_2)_B = \left(\frac{20}{3}\right)(4.8)^3 = -737.28 \text{ kN} \cdot \text{m}^2$$

$$\theta_A = -\frac{1}{EIL} \left\{ (A_1)_B \left(\frac{1}{3}\right)(4.8) + (A_2)_B \left(\frac{1}{4}\right)(4.8) \right\}$$

$$= -\frac{-221.184}{(92400)(4.8)} = 0.49870 \times 10^{-3}$$



PROBLEM 9.145 (Continued)

Locate Point K of maximum deflection.

$$\begin{aligned}\theta_K &= \theta_A + \theta_{K/A} = 0 \\ EI\theta_A + A_1 + A_2 &= 0 \\ f &= 46.08 + 18x_K^2 - \frac{20}{3}x_K^3 = 0 \quad \frac{df}{dx} = 36x_K - 20x_K^2\end{aligned}$$

Solve by iteration.

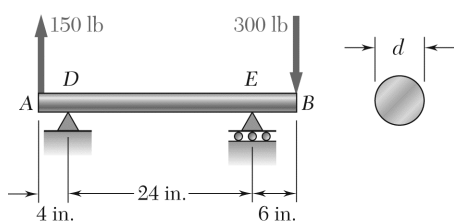
$$x_K = (x_K)_0 - \frac{f}{df/dx}$$

x_K	3	3.39	3.327	3.3251	3.32514 ←
df/dx	-72	-107.8	-101.6	-101.42	
f	28.08	-6.78	-0.188	0.005	

Place reference tangent at K .

$$\begin{aligned}y_A &= y_K + t_{A/K} \\ y_A - y_K &= -t_{A/K} \\ &= -\frac{1}{EI} \left\{ (A_1) \left(\frac{2}{3} x_K \right) + A_2 \left(\frac{3}{4} x_K \right) \right\} = -\frac{1}{EI} \{ 12x_K^3 + 5x_K^4 \} \\ &= -\frac{170.064}{92400} = -1.841 \times 10^{-3} \text{ m}\end{aligned}$$

$$y_K = 1.841 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 9.146

For the beam and loading of Prob. 9.137, determine the largest upward deflection in span DE .

SOLUTION

Units: Forces in lbs; lengths in inches.

From the solution to Prob. 9.137,

$$EI = 450.4 \times 10^3 \text{ lb} \cdot \text{in}$$

$$\frac{M_1}{EI} = 1.33215 \times 10^{-3} \text{ in}^{-1}$$

$$\frac{M_2}{EI} = -3.99645 \times 10^{-3} \text{ in}^{-1}$$

$$\theta_D = 5.3286 \times 10^{-3}$$

Location of maximum deflection.

$$\frac{M_3}{EI} = \frac{M_1}{EI} \left(1 - \frac{u}{24} \right)$$

$$\frac{M_4}{EI} = \frac{M_2}{EI} \frac{u}{24}$$

$$A_5 = \frac{1}{2} \frac{M_1}{EI} \cdot u = 0.666075 \times 10^{-3} u$$

$$A_6 = \frac{1}{2} \frac{M_3}{EI} \left(\frac{u}{24} \right) = 0.666075 \times 10^{-3} \left(1 - \frac{u}{24} \right) u$$

$$A_7 = \frac{1}{2} \frac{M_4}{EI} \frac{u}{24} = -1.998225 \times 10^{-3} \left(\frac{u}{24} \right) u$$

$$\theta_K = \theta_D + A_5 + A_6 + A_7 = 0$$

Multiply by 10^3 .

$$5.3286 + 0.666075u + 0.666075 \left(1 - \frac{u}{24} \right) u - (1.998225) \frac{u}{24} u = 0$$

$$5.3286 + 1.33215u - 0.1110125u^2 = 0$$

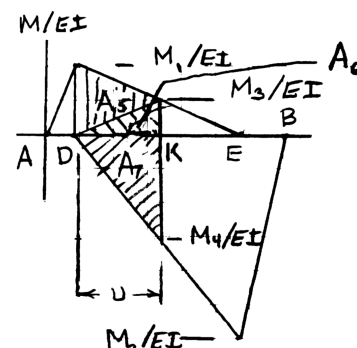
$$u = 15.16515 \text{ in.}$$

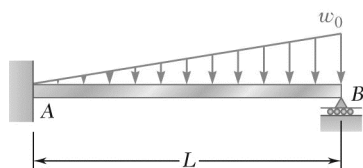
$$A_5 = 10.10113 \times 10^{-3}, \quad A_6 = 3.71842 \times 10^{-3}, \quad A_7 = -19.14814 \times 10^{-3}$$

Maximum deflection in portion DE . $y_D = y_K + t_{D/K} = 0$

$$\begin{aligned} y_K = -t_{D/K} &= - \left\{ A_5 \left(\frac{u}{3} \right) + A_6 \left(\frac{2u}{3} \right) + A_7 \left(\frac{2u}{3} \right) \right\} \\ &= - \{ -0.1049 \} \end{aligned}$$

$$y_K = 0.1049 \text{ in.} \quad \blacktriangleleft$$





PROBLEM 9.147

For the beam and loading shown, determine the reaction at the roller support.

SOLUTION

Remove support B and treat R_B as redundant.

Replace loading by equivalent shown at left.

Draw M/EI diagram for load w_0 and R_B .

Use parts as shown.

$$A_1 = \frac{1}{2} \left(\frac{R_B L}{EI} \right) (L) = \frac{1}{2} \frac{R_B L^2}{EI}$$

$$M_2 = -\frac{1}{2} w_0 L^2$$

$$A_2 = \frac{1}{3} \left(-\frac{1}{2} \frac{w_0 L^2}{EI} \right) L = -\frac{1}{6} \frac{w_0 L^3}{EI}$$

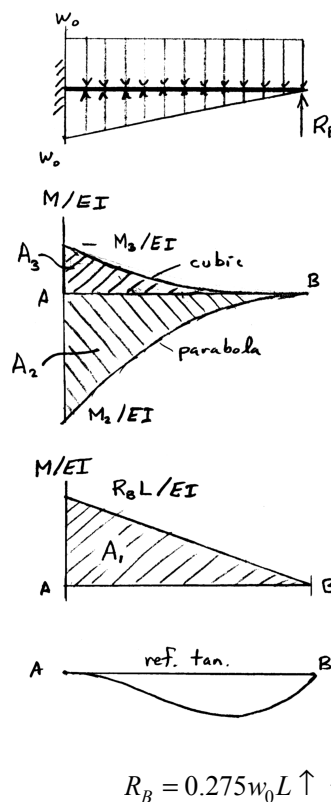
$$M_3 = \frac{1}{6} \frac{w_0}{L} L^3 = \frac{1}{6} w_0 L^2$$

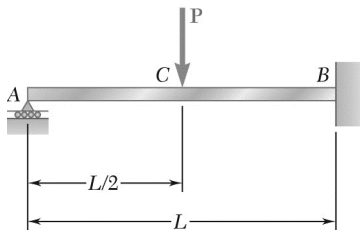
$$A_3 = \frac{1}{4} \left(\frac{1}{6} \frac{w_0 L^2}{EI} \right) L = \frac{1}{24} \frac{w_0 L^3}{EI}$$

Place reference tangent at A .

$$\begin{aligned} t_{B/A} &= A_1 \left(\frac{2}{3} L \right) + A_2 \left(\frac{3}{4} L \right) + A_3 \left(\frac{4}{5} L \right) \\ &= \frac{1}{3} \frac{R_B L^3}{EI} - \frac{1}{8} \frac{w_0 L^4}{EI} + \frac{1}{30} \frac{w_0 L^4}{EI} = 0 \end{aligned}$$

$$R_B = \frac{11}{40} w_0 L \uparrow$$





PROBLEM 9.148

For the beam and loading shown, determine the reaction at the roller support.

SOLUTION

Remove support A and treat R_A as redundant.

Draw the M/EI diagram by parts.

$$A_1 = \frac{1}{2} L \frac{R_A L}{EI} = \frac{R_A L^2}{2EI}$$

$$A_2 = -\frac{1}{2} \frac{L}{2} \frac{PL}{2} = -\frac{PL^2}{8EI}$$

Place reference tangent at B .

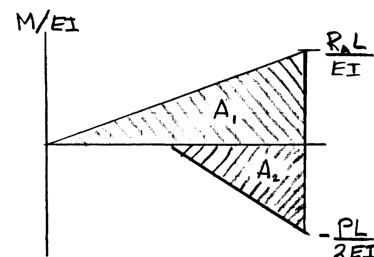
$$y_A = y_B - \theta_B L + t_{A/B} = 0$$

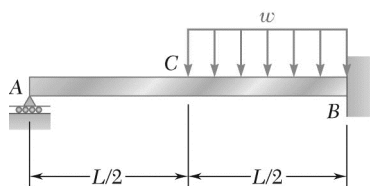
$$t_{A/B} = 0$$

$$A_1 \left(\frac{2L}{3} \right) + A_2 \left(\frac{L}{2} + \frac{L}{3} \right) = 0$$

$$\frac{R_A L^3}{3EI} - \frac{5PL^3}{48EI} = 0$$

$$R_A = \frac{5}{16} P \uparrow \blacktriangleleft$$





PROBLEM 9.149

For the beam and loading shown, determine the reaction at the roller support.

SOLUTION

Remove support A and treat R_A as redundant.

Draw M/EI diagram for loads R_A and w .

$$M_2 = -\frac{1}{2}w\left(\frac{L}{2}\right)^2 = -\frac{1}{8}wL^2$$

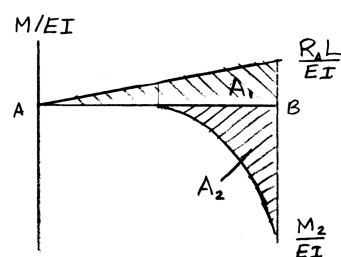
$$A_1 = \frac{1}{2}\left(\frac{R_AL}{EI}\right)L = \frac{1}{2}\frac{R_AL^2}{EI}$$

$$A_2 = \frac{1}{3}\left(-\frac{1}{8}\frac{wL^2}{EI}\right)\left(\frac{L}{2}\right) = -\frac{1}{48}\frac{wL^3}{EI}$$

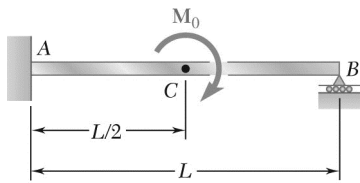
Place reference tangent at B .

$$\begin{aligned} t_{A/B} &= A_1\left(\frac{2}{3}L\right) + A_2\left(\frac{L}{2} + \frac{3}{4}\frac{L}{2}\right) \\ &= \frac{1}{3}\frac{R_AL^3}{EI} - \frac{7}{384}\frac{wL^4}{EI} = 0 \end{aligned}$$

$$R_A = \frac{7}{128}wL$$



$$R_A = \frac{7}{128}wL \uparrow \blacktriangleleft$$



PROBLEM 9.150

For the beam and loading shown, determine the reaction at the roller support.

SOLUTION

Remove support B and treat R_B as redundant.

Draw M/EI diagram.

$$A_1 = \frac{1}{2} L \frac{R_B L}{EI} = \frac{R_B L^2}{2EI}$$

$$A_2 = \frac{L}{2} \cdot \frac{M_0 L}{EI} = \frac{M_0 L^2}{2EI}$$

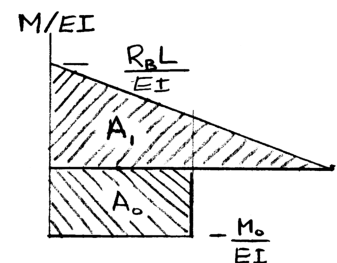
Place reference tangent at A .

$$y_B = y_A + L\theta_A + t_{B/A} = 0$$

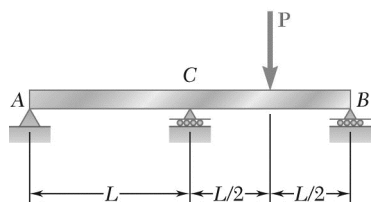
$$t_{B/A} = 0$$

$$A_1 \left(\frac{2L}{3} \right) + A_2 \left(\frac{L}{2} + \frac{L}{4} \right) = 0$$

$$\frac{R_B L^3}{3EI} - \frac{3M_0 L^2}{8EI} = 0$$



$$R_B = \frac{9}{8} \frac{M_0}{L} \uparrow \blacktriangleleft$$



PROBLEM 9.151

For the beam and loading shown, determine the reaction at each support.

SOLUTION

Remove support C and add reaction R_C .

Draw M/EI due to each of the loads P and R_C .

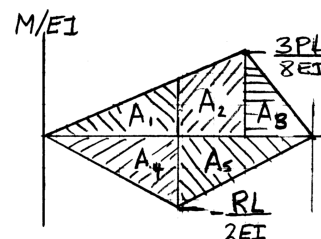
$$A_1 = \frac{1}{2} \cdot L \cdot \frac{2}{3} \cdot \frac{3PL}{8EI} = \frac{PL^2}{8EI}$$

$$A_1 + A_2 = \frac{1}{2} \cdot \frac{3L}{2} \cdot \frac{3PL}{8EI} = \frac{9PL^2}{32EI}$$

$$A_3 = \frac{1}{2} \cdot \frac{L}{2} \cdot \frac{3PL}{8EI} = \frac{3PL^2}{32EI}$$

$$A_4 = \frac{1}{2} \cdot L \left(-\frac{R_C L}{2EI} \right) = -\frac{R_C L^2}{4EI}$$

$$A_4 + A_5 = \frac{1}{2} (2L) \left(-\frac{R_C L}{2EI} \right) = -\frac{R_C L^2}{2EI}$$



Place reference tangent at A . $y_A = 0$

$$y_C = L\theta_A + t_{C/A} = 0 \quad \theta_A = -\frac{t_{C/A}}{L}$$

$$y_B = 2L\theta_A + t_{B/A} = 0 \quad -2t_{C/A} + t_{B/A} = 0$$

$$-2 \left[A_1 \frac{L}{3} + A_4 \cdot \frac{L}{3} \right] + \left[(A_1 + A_2) \left(\frac{L}{2} + \frac{2}{3} \cdot \frac{3L}{2} \right) + A_3 \cdot \frac{2}{3} \cdot \frac{L}{2} + (A_4 + A_5) \cdot L \right] = 0$$

$$-2 \left[\frac{PL^3}{24EI} - \frac{R_C L^3}{12EI} \right] + \left[\frac{9PL^3}{32EI} + \frac{PL^3}{32EI} - \frac{R_C L^3}{2EI} \right] = 0$$

$$-\frac{R_C L^3}{3EI} + \frac{11PL^3}{48EI} = 0$$

$$R_C = \frac{11}{16} P \uparrow \blacktriangleleft$$

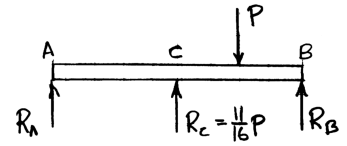
PROBLEM 9.151 (Continued)

$$+\curvearrowright \Sigma M_B = 0: -2LR_A - LR_C + \frac{L}{2}P = 0$$

$$R_A = \frac{P}{4} - \frac{1}{2}R_C = -\frac{3}{32}P$$

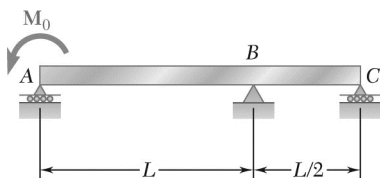
$$+\curvearrowright \Sigma M_A = 0: 2LR_B + LR_C - \frac{3L}{2}P = 0$$

$$R_B = \frac{3P}{4} - \frac{1}{2}R_C = \frac{13}{32}P$$



$$R_A = \frac{3}{32}P \downarrow \blacktriangleleft$$

$$R_B = \frac{13}{32}P \uparrow \blacktriangleleft$$



PROBLEM 9.152

For the beam and loading shown, determine the reaction at each support.

SOLUTION

Choose $R_B \downarrow$ as the redundant reaction.

Draw M/EI diagram for the loads R_B and M_0 .

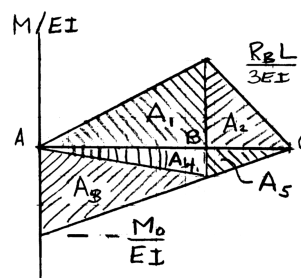
$$A_1 = \frac{1}{2}(L)\left(\frac{R_B L}{3EI}\right) = \frac{R_B L^2}{6EI}$$

$$A_2 = \frac{1}{2}\left(\frac{L}{2}\right)\left(\frac{R_B L}{3EI}\right) = \frac{R_B L^2}{12EI}$$

$$A_3 = \frac{1}{2}(L)\left(-\frac{M_0}{EI}\right) = -\frac{M_0 L}{2EI}$$

$$A_4 = \frac{1}{2}(L)\left(\frac{1}{3}\right)\left(-\frac{M_0}{EI}\right) = -\frac{M_0 L}{6EI}$$

$$A_3 + A_4 + A_5 = \frac{1}{2}\left(\frac{3L}{2}\right)\left(-\frac{M_0}{EI}\right) = -\frac{3M_0 L}{4EI}$$



$$y_B = y_A + L\theta_A + t_{B/A} \quad \theta_A = -t_{B/A}/L$$

$$y_C = y_A + \frac{3L}{2}\theta_A + t_{C/A} = 0 \quad -\frac{3}{2}t_{B/A} + t_{C/A} = 0$$

$$t_{B/A} = (A_1)\left(\frac{L}{3}\right) + A_3\left(\frac{2L}{3}\right) + A_4\left(\frac{L}{3}\right) = \frac{R_B L^3}{18EI} - \frac{7M_0 L^2}{18EI}$$

$$t_{C/A} = (A_1)\left(\frac{L}{2} + \frac{L}{3}\right) + A_2\left(\frac{L}{3}\right) + (A_3 + A_4 + A_5)(L) = \frac{R_B L^3}{6EI} - \frac{3M_0 L^2}{4EI}$$

$$-\frac{3}{2}t_{B/A} + t_{C/A} = \frac{R_B L^3}{12EI} - \frac{M_0 L^2}{6EI} = 0$$

$$R_B = \frac{2M_0}{L} \downarrow \blacktriangleleft$$

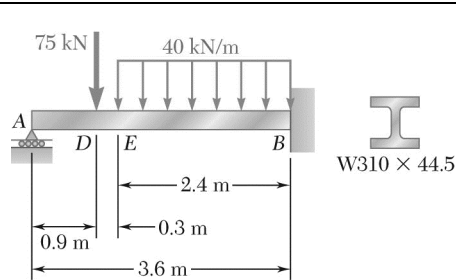
$$+\circlearrowleft \Sigma M_C = 0: \quad M_0 + \frac{L}{2}R_B - \frac{3L}{2}R_A = 0$$

$$R_A = \frac{2}{3L}[M_0 + M_0]$$

$$R_A = \frac{4M_0}{3L} \uparrow \blacktriangleleft$$

$$+\uparrow \Sigma F_y = 0: \quad R_A + R_B + R_C = 0 \quad \frac{4}{3}\frac{M_0}{L} - \frac{2M_0}{L} + R_C = 0$$

$$R_C = \frac{2M_0}{3L} \uparrow \blacktriangleleft$$



PROBLEM 9.153

Determine the reaction at the roller support and draw the bending moment diagram for the beam and loading shown.

SOLUTION

Units: Forces in kN; lengths in meters.

Let R_A be the redundant reaction.

Remove support at A and add reaction $R_A \uparrow$.

Draw bending moment diagram by parts.

$$M_1 = 3.6 R_A \text{ kN} \cdot \text{m}$$

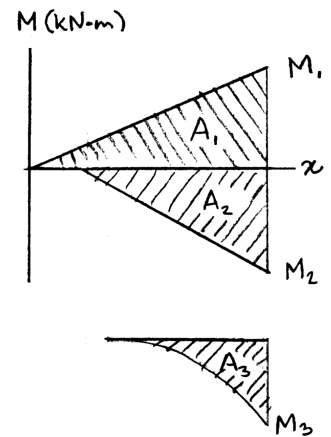
$$M_2 = -(75)(0.3 + 2.4) = -202.5 \text{ kN} \cdot \text{m}$$

$$M_3 = -\frac{1}{2}(40)(2.4)^2 = -115.2 \text{ kN} \cdot \text{m}$$

$$A_1 = \frac{1}{2}(3.6)(3.6 R_A) = 6.48 \text{ kN} \cdot \text{m}^2$$

$$A_2 = \frac{1}{2}(2.7)(-202.5) = -273.375 \text{ kN} \cdot \text{m}^2$$

$$A_3 = \frac{1}{3}(2.4)(-115.2) = -92.16 \text{ kN} \cdot \text{m}^2$$



Place reference tangent at B , where

$$\theta_B = 0 \quad \text{and} \quad y_B = 0.$$

Then

$$y_A = t_{A/B} = 0$$

$$\begin{aligned} t_{A/B} &= \frac{1}{EI} \left\{ \left(\frac{2}{3} \cdot 3.6 \right) A_1 + \left(0.9 + \frac{2}{3} \cdot 2.7 \right) A_2 + \left(0.9 + 0.3 + \frac{3}{4} \cdot 2.4 \right) A_3 \right\} \\ &= \frac{1}{EI} \{ 15.552 R_A - 1014.5925 \} = 0 \end{aligned}$$

$$R_A = 65.24 \text{ kN} \uparrow \quad \blacktriangleleft$$

PROBLEM 9.153 (Continued)

Draw shear diagram.

$$A \text{ to } D: V = R_A = 65.24 \text{ kN}$$

$$D \text{ to } E: V = 65.24 - 75 = -9.76 \text{ kN}$$

$$E \text{ to } B: V = -9.76 - 40(x - 1.2) \text{ kN}$$

$$\text{At } B, V_B = -105.76 \text{ kN}$$

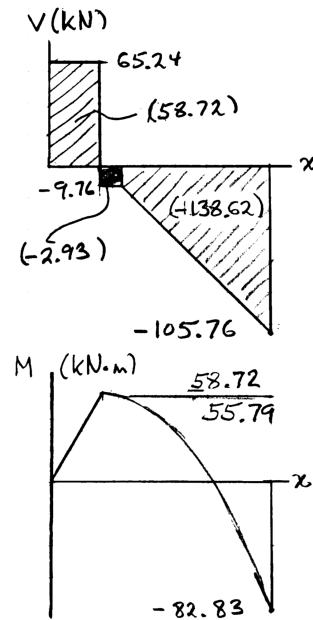
Bending moment diagram.

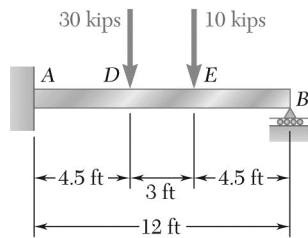
$$M_A = 0$$

$$M_D = M_A + 58.72 = 58.72 \text{ kN} \cdot \text{m}$$

$$M_E = 58.72 - 2.93 = 55.79 \text{ kN} \cdot \text{m}$$

$$M_B = 55.79 - 138.62 = -82.83 \text{ kN} \cdot \text{m}$$





PROBLEM 9.154

Determine the reaction at the roller support and draw the bending moment diagram for the beam and loading shown.

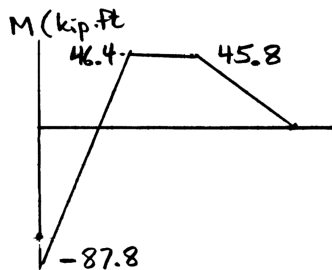
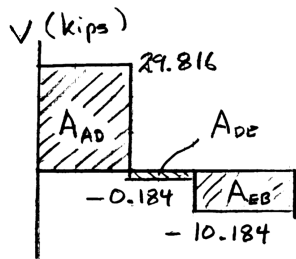
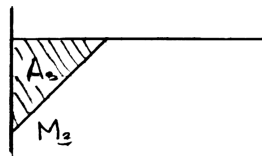
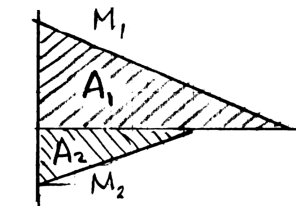
SOLUTION

Units: Forces in kips; lengths in feet.

Let R_B be the redundant reaction.

Remove support B and add load R_B .

Draw bending moment diagram by parts.



$$M_1 = 12R_B \text{ kip} \cdot \text{ft}$$

$$M_2 = -(4.5 + 3)(10) = -75 \text{ kip} \cdot \text{ft}$$

$$M_3 = -(4.5)(30) = -135 \text{ kip} \cdot \text{ft}$$

$$A_1 = \frac{1}{2}(12)(12R_B) = 72R_B \text{ kip} \cdot \text{ft}^2$$

$$A_2 = \frac{1}{2}(7.5)(-75) = -281.25 \text{ kip} \cdot \text{ft}^2$$

$$A_3 = \frac{1}{2}(4.5)(-135) = -303.75 \text{ kip} \cdot \text{ft}^2$$

$$y_B = y_A + 12\theta_A + t_{B/A} = 0$$

$$t_{B/A} = 0$$

$$t_{B/A} = \frac{1}{EI} \{ (72R_B)(8) + (-281.25)(4.5 + 5) + (-303.75)(7.5 + 3) \} = 0$$

$$576R_B - 5861.25 = 0$$

$$R_B = 10.18 \text{ kips} \uparrow \blacktriangleleft$$

Draw shear diagram working from right to left.

$$B \text{ to } E: V = -R_B = -10.176 \text{ kips}$$

$$E \text{ to } D: V = -10.184 + 10 = -0.176 \text{ kips}$$

$$D \text{ to } A: V = -0.176 + 30 = 29.824 \text{ kips}$$

PROBLEM 9.154 (Continued)

Areas of shear diagram.

$$A_{AD} = (4.5)(29.824) = 134.21 \text{ kip} \cdot \text{ft}$$

$$A_{DE} = (3)(-0.176) = -0.53 \text{ kip} \cdot \text{ft}$$

$$A_{EB} = (4.5)(-10.176) = 45.79 \text{ kip} \cdot \text{ft}$$

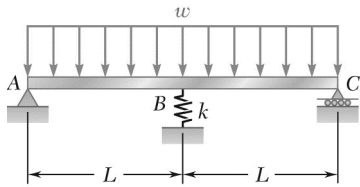
Bending moments.

$$M_A = M_1 + M_2 + M_3 = -87.89 \text{ kip} \cdot \text{ft} \quad \blacktriangleleft$$

$$M_D = M_A + A_{AD} = 46.32 \text{ kip} \cdot \text{ft} \quad \blacktriangleleft$$

$$M_E = M_D + A_{DE} = 45.79 \text{ kip} \cdot \text{ft} \quad \blacktriangleleft$$

$$M_B = M_E + A_{EB} = 0 \quad \blacktriangleleft$$



PROBLEM 9.155

For the beam and loading shown, determine the spring constant k for which the force in the spring is equal to one-third of the total load on the beam.

SOLUTION

Symmetric beam and loading:

$$R_C = R_A$$

Spring force:

$$F = \frac{1}{3}(2wL) = \frac{2}{3}wL$$

$$+\uparrow \Sigma F_y = 0: R_A + F - 2wL + R_C = 0$$

$$R_A = R_C = \frac{2}{3}wL$$

Draw M/EI diagram by parts.

$$A_1 = \frac{1}{2} \left(\frac{2}{3} \frac{wL^2}{EI} \right) L = \frac{1}{3} \frac{wL^3}{EI}$$

$$A_2 = -\frac{1}{3} \left(\frac{1}{2} \frac{wL^2}{EI} \right) L = -\frac{1}{6} \frac{wL^3}{EI}$$

Place reference tangent at B .

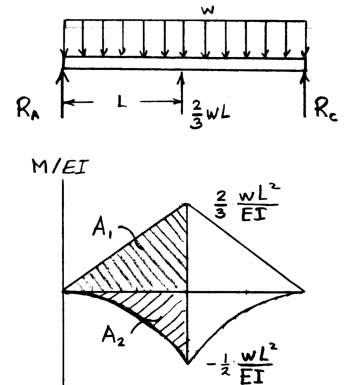
$$\theta_B = 0$$

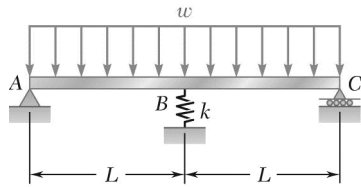
$$\begin{aligned} y_B &= -t_{A/B} \\ &= - \left(A_1 \cdot \frac{2}{3}L + A_2 \cdot \frac{3}{4}L \right) \\ &= -\frac{7}{72} \frac{wL^4}{EI} \end{aligned}$$

$$F = -ky_B$$

$$k = -\frac{F}{y_B} = \frac{\frac{2}{3}wL}{\frac{7}{72} \frac{wL^4}{EI}}$$

$$k = \frac{48}{7} \frac{EI}{L^3} \quad \blacktriangleleft$$





PROBLEM 9.156

For the beam and loading shown, determine the spring constant k for which the bending moment at B is $M_B = -wL^2/10$.

SOLUTION

Using free body AB ,

$$+\circlearrowleft M_B = 0: -R_A L + (wL)\left(\frac{L}{2}\right) - \frac{1}{10}wL^2 = 0$$

$$R_A = \frac{2}{5}wL \uparrow$$

Symmetric beam and loading:

$$R_C = R_A$$

Using free body ABC ,

$$+\uparrow \Sigma F_y = 0: \frac{2}{5}wL + F + \frac{2}{5}wL - 2wL = 0$$

$$F = \frac{6}{5}wL$$

Draw M/EI diagram by parts.

$$A_1 = \frac{1}{2} \left(\frac{2}{5} \frac{wL^2}{EI} \right) L = \frac{1}{5} \frac{wL^3}{EI}$$

$$A_2 = -\frac{1}{3} \left(\frac{1}{2} \frac{wL^2}{EI} \right) L = -\frac{1}{6} \frac{wL^3}{EI}$$

Place reference tangent at B .

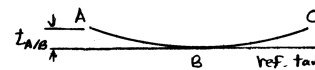
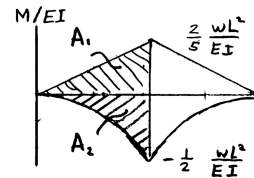
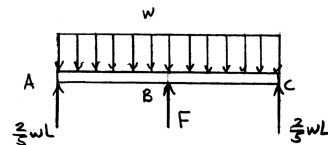
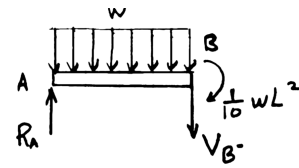
$$\theta_B = 0$$

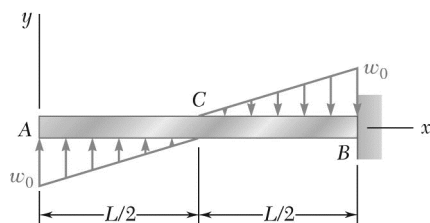
$$\begin{aligned} y_B &= -t_{A/B} \\ &= -\left(A_1 \cdot \frac{2}{3}L + A_2 \cdot \frac{3}{4}L \right) \\ &= -\frac{1}{120} \frac{wL^4}{EI} \end{aligned}$$

$$F = -ky_B$$

$$k = -\frac{F}{y_B} = \frac{\frac{6}{5}wL}{-\frac{1}{120} \frac{wL^4}{EI}}$$

$$k = 144 \frac{EI}{L^3} \blacktriangleleft$$





PROBLEM 9.157

For the loading shown, determine (a) the equation of the elastic curve for the cantilever beam AB , (b) the deflection at the free end, (c) the slope at the free end.

SOLUTION

$$w(x) = \frac{2w_0}{L}x - w_0$$

$$V(x) = -\int w(x)dx = -\int \left(\frac{2w_0}{L}x - w_0 \right) dx = -\frac{w_0}{L}x^2 + w_0x + C_1$$

$$[x = 0, V = 0] \quad 0 = 0 + 0 + C_1 \quad \therefore C_1 = 0$$

$$M(x) = \int V(x)dx = \int \left(-\frac{w_0}{L}x^2 + w_0x \right) dx = -\frac{w_0}{3L}x^3 + \frac{w_0}{2}x^2 + C_2$$

$$[x = 0, M = 0] \quad 0 = 0 + 0 + C_2 \quad \therefore C_2 = 0$$

$$EI \frac{d^2y}{dx^2} = M = -\frac{w_0}{3L}x^3 + \frac{w_0}{2}x^2$$

$$EI \frac{dy}{dx} = -\frac{w_0}{12L}x^4 + \frac{w_0}{6}x^3 + C_3$$

$$\left[x = L, \frac{dy}{dx} = 0 \right] \quad 0 = -\frac{w_0L^3}{12} + \frac{w_0L^3}{6} + C_3 \quad \therefore C_3 = -\frac{w_0L^3}{12}$$

$$EIy = -\frac{w_0}{60L}x^5 + \frac{w_0}{24}x^4 - \frac{w_0L^3}{12}x + C_4$$

$$[x = L, y = 0] \quad 0 = -\frac{w_0L^4}{60} + \frac{w_0L^4}{24} - \frac{w_0L^4}{12} + C_4 \quad \therefore C_4 = \frac{7w_0L^4}{120}$$

(a) Elastic curve.

$$y = -\frac{w_0}{120EI} (2x^5 - 5Lx^4 + 10L^4x - 7L^5) \quad \blacktriangleleft$$

(b) y at x = 0.

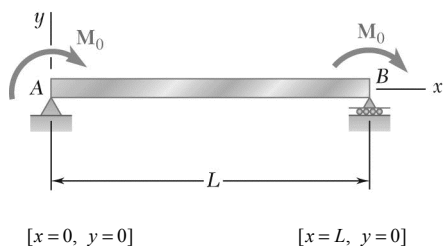
$$y_A = +\frac{7w_0L^4}{120EI}$$

$$y_A = \frac{7w_0L^4}{120EI} \uparrow \quad \blacktriangleleft$$

(c) $\frac{dy}{dx}$ at x = 0.

$$\left. \frac{dy}{dx} \right|_A = -\frac{w_0L^3}{12EI}$$

$$\theta_A = \frac{w_0L^3}{12EI} \searrow \quad \blacktriangleleft$$



PROBLEM 9.158

- (a) Determine the location and magnitude of the maximum absolute deflection in AB between A and the center of the beam.
 (b) Assuming that beam AB is a W18×76 rolled shape, $M_0 = 150 \text{ kip} \cdot \text{ft}$, and $E = 29 \times 10^6 \text{ psi}$, determine the maximum allowable length L so that the maximum deflection does not exceed 0.05 in.

SOLUTION

Using AB as a free body,

$$\Sigma M_B = 0: -2M_0 - R_A L = 0$$

$$R_A = -\frac{2M_0}{L}$$

Using portion AJ as a free body,

$$\Sigma M_J = 0: -M_0 + \frac{2M_0}{L}x + M = 0$$

$$M = \frac{M_0}{L}(L - 2x)$$

$$EI \frac{d^2 y}{dx^2} = \frac{M_0}{L}(L - 2x)$$

$$EI \frac{dy}{dx} = \frac{M_0}{L}(Lx - x^2) + C_1$$

$$EI y = \frac{M_0}{L} \left(\frac{1}{2} Lx^2 - \frac{1}{3} x^3 \right) + C_1 x + C_2$$

$$[x = 0, y = 0] \quad 0 = 0 - 0 + 0 + C_2$$

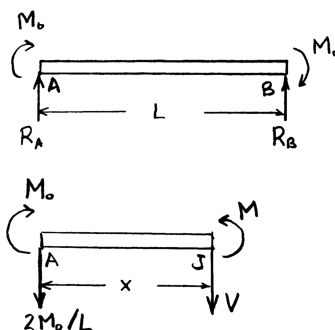
$$C_2 = 0$$

$$[x = L, y = 0] \quad 0 = \frac{M_0}{L} \left(\frac{1}{2} L^3 - \frac{1}{3} L^3 \right) + C_1 L + 0$$

$$C_1 = -\frac{1}{6} M_0 L^2$$

$$y = \frac{M_0}{EIL} \left(\frac{1}{2} Lx^2 - \frac{1}{3} x^3 - \frac{1}{6} L^2 x \right)$$

$$\frac{dy}{dx} = \frac{M_0}{EIL} \left(Lx - x^2 - \frac{1}{6} L^2 \right)$$



PROBLEM 9.158 (Continued)

To find location of maximum deflection, set $\frac{dy}{dx} = 0$.

$$x_m^2 - Lx_m - \frac{1}{6}L^2 = 0$$

$$x_m = \frac{L - \sqrt{L^2 - (4)\left(\frac{1}{6}L^2\right)}}{2}$$

$$= \frac{1}{2}\left(1 - \frac{\sqrt{3}}{3}\right)L = 0.21132L$$

$$x_m = 0.211L \quad \blacktriangleleft$$

$$y_m = \frac{M_0 L^2}{EI} \left\{ \left(\frac{1}{2}\right)(0.21132)^2 - \left(\frac{1}{3}\right)(0.21132)^3 - \left(\frac{1}{6}\right)(0.21132) \right\}$$

$$= -0.0160375 \frac{M_0 L^2}{EI}$$

$$|y_m| = 0.0160375 \frac{M_0 L^2}{EI}$$

$$|y_m| = 0.01604 \frac{M_0 L^2}{EI} \quad \blacktriangleleft$$

Solving for L ,

$$L = \left\{ \frac{EI|y_m|}{0.0160375 M_0} \right\}^{1/2}$$

Data: $E = 29 \times 10^6 \text{ psi}$ For W18×76 beam, $I = 1330 \text{ in}^4$

$$|y_m| = 0.05 \text{ in.}$$

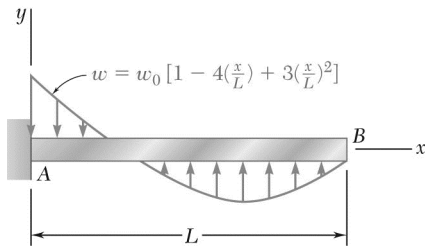
$$M_0 = 150 \text{ kip} \cdot \text{ft} = 150 \times 10^3 \text{ lb} \cdot \text{ft}$$

$$= 1.800 \times 10^6 \text{ lb} \cdot \text{in}$$

$$L = \left\{ \frac{(29 \times 10^6)(1330)(0.05)}{(0.0160375)(1.800 \times 10^6)} \right\}^{1/2} = 258.5 \text{ in.}$$

$$L = 21.5 \text{ ft} \quad \blacktriangleleft$$

PROBLEM 9.159



For the beam and loading shown, determine (a) the equation of the elastic curve, (b) the deflection at the free end.

SOLUTION

Boundary conditions are shown at right.

$$[x = 0, y = 0] \quad [x = L, V = 0]$$

$$[x = 0, \frac{dy}{dx} = 0] \quad [x = L, M = 0]$$

$$\frac{dV}{dx} = -w = -w_0 \left[1 - 4\left(\frac{x}{L}\right) + 3\left(\frac{x}{L}\right)^2 \right]$$

$$V = -w_0 \left[x - \frac{2x^2}{L} + \frac{x^3}{L^2} \right] + C_V$$

$$[x = L, V = 0]: 0 = -w_0 [L - 2L + L] + C_V = 0 \quad C_V = 0$$

$$\frac{dM}{dx} = V = -w_0 \left[x - \frac{2x^2}{L} + \frac{x^3}{L^2} \right]$$

$$M = -w_0 \left[\frac{x^2}{2} - \frac{2x^3}{3L} + \frac{x^4}{4L^2} \right] + C_M$$

$$[x = L, M = 0]: 0 = -w_0 \left[\frac{1}{2}L^2 - \frac{2}{3}L^2 + \frac{1}{4}L^2 \right] + C_M \quad C_M = \frac{1}{12}w_0L^2$$

$$EI \frac{d^2y}{dx^2} = M = -w_0 \left[\frac{1}{2}x^2 - \frac{2}{3}\frac{x^3}{L} + \frac{1}{4}\frac{x^4}{L^2} - \frac{1}{12}L^2 \right]$$

$$EI \frac{dy}{dx} = -w_0 \left[\frac{1}{6}x^3 - \frac{1}{6}\frac{x^4}{L} + \frac{1}{20}\frac{x^5}{L^2} - \frac{1}{12}L^2x \right] + C_1$$

$$[x = 0, \frac{dy}{dx} = 0] \quad C_1 = 0$$

$$EIy = -w_0 \left[\frac{1}{24}x^4 - \frac{1}{30}\frac{x^5}{L} + \frac{1}{120}\frac{x^6}{L^2} - \frac{1}{24}L^2x^2 \right] + C_2$$

$$[x = 0, y = 0] \quad C_2 = 0$$

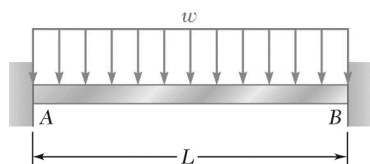
PROBLEM 9.159 (Continued)

(a) Elastic curve.

$$y = -\frac{w_0}{EIL^2} \left(\frac{1}{24} L^2 x^4 - \frac{1}{30} L x^5 + \frac{1}{120} x^6 - \frac{1}{24} L^4 x^2 \right) \blacktriangleleft$$

(b) Deflection at $x = L$.

$$y_B = -\frac{w_0}{EIL^2} \left(\frac{1}{24} L^6 - \frac{1}{30} L^6 + \frac{1}{120} L^6 - \frac{1}{24} L^6 \right) = -\frac{w_0 L^4}{40EI} \quad y_B = \frac{w_0 L^4}{40EI} \downarrow \blacktriangleleft$$



PROBLEM 9.160

Determine the reaction at A and draw the bending moment diagram for the beam and loading shown.

$$[x = 0, y = 0]$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right]$$

$$[x = L, y = 0]$$

$$\left[x = L, \frac{dy}{dx} = 0 \right]$$

SOLUTION

Reactions are statically indeterminate.

By symmetry, $R_B = R_A$; $M_B = M_A$

$$\frac{dy}{dx} = 0 \quad \text{at} \quad x = \frac{L}{2}$$

$$+\uparrow \Sigma F_y = 0: R_A + R_B - wL = 0$$

$$R_B = R_A = \frac{1}{2}wL \quad \uparrow \quad \blacktriangleleft$$

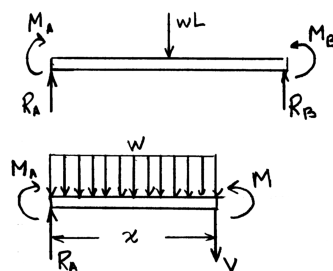
Over entire beam, $M = M_A + R_A x - \frac{1}{2}wx^2$

$$EI \frac{d^2 y}{dx^2} = M_A + \frac{1}{2}wLx - \frac{1}{2}wx^2$$

$$EI \frac{dy}{dx} = M_A x + \frac{1}{4}wLx^2 - \frac{1}{6}wx^3 + C_1$$

$$\left[x = 0, \frac{dy}{dx} = 0 \right] \quad 0 + 0 - 0 + C_1 = 0 \quad C_1 = 0$$

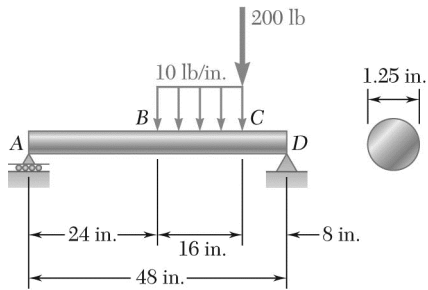
$$\left[x = \frac{L}{2}, \frac{dy}{dx} = 0 \right] \quad \frac{1}{2}M_A L + \frac{1}{16}wL^3 - \frac{1}{48}wL^3 + 0 = 0$$



$$M_A = -\frac{1}{12}wL^2 \quad \blacktriangleleft$$

$$M = -\frac{1}{12}wL^2 + \frac{1}{2}wLx - \frac{1}{2}wx^2$$

$$M = w[6x(L-x) - L^2]/12 \quad \blacktriangleleft$$



PROBLEM 9.161

For the beam and loading shown, determine (a) the slope at end A, (b) the deflection at point B. Use $E = 29 \times 10^6$ psi.

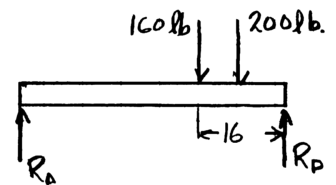
SOLUTION

Units: Forces in lbs; lengths in inches.

$$c = \frac{1}{2}d = \left(\frac{1}{2}\right)(1.25) \\ = 0.625 \text{ in.}$$

$$I = \frac{\pi}{4}c^4 = \frac{\pi}{4}(0.625)^4 \\ = 119.84 \times 10^{-3} \text{ in}^4$$

$$EI = (29 \times 10^6)(119.84 \times 10^{-3}) \\ = 3.4754 \times 10^6 \text{ lb} \cdot \text{in}^2$$



Use entire beam ABCD as free body.

$$+\circlearrowleft \Sigma M_D = 0: -48R_A + (16)(160) + (8)(200) = 0 \quad R_A = 86.667 \text{ lb}\uparrow$$

$$w(x) = 10\langle x - 24 \rangle^0 - 10\langle x - 40 \rangle^0 \quad \text{lb/in}$$

$$\frac{dV}{dx} = -w = -10\langle x - 24 \rangle^0 + 10\langle x - 40 \rangle^0 \quad \text{lb/in}$$

$$\frac{dM}{dx} = V = -10\langle x - 24 \rangle^1 + 10\langle x - 40 \rangle^1 + 86.667 - 200\langle x - 40 \rangle^0 \quad \text{lb} \cdot \text{in}$$

$$EI \frac{d^2y}{dx^2} = M = -5\langle x - 24 \rangle^2 + 5\langle x - 40 \rangle^2 + 86.667x - 200\langle x - 40 \rangle^1 \quad \text{lb} \cdot \text{in}$$

$$EI \frac{dy}{dx} = -\frac{5}{3}\langle x - 24 \rangle^3 + \frac{5}{3}\langle x - 40 \rangle^3 + 43.333x^2 - 100\langle x - 40 \rangle^2 + C_1 \quad \text{lb} \cdot \text{in}^2$$

$$EIy = -\frac{5}{12}\langle x - 24 \rangle^4 + \frac{5}{12}\langle x - 40 \rangle^4 + 14.4444x^3 - \frac{100}{3}\langle x - 40 \rangle^3 + C_1x + C_2 \quad \text{lb} \cdot \text{in}^3$$

PROBLEM 9.161 (Continued)

$$[x = 0, y = 0] \quad -0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = 48, y = 0] \quad -\left(\frac{5}{12}\right)(24)^4 + \left(\frac{5}{12}\right)(8)^4 + (14.4444)(48)^3 \\ -\left(\frac{100}{3}\right)(8)^3 + 48C_1 = 0$$

$$C_1 = -30.08 \times 10^3 \text{ lb} \cdot \text{in}^2$$

(a) Slope at end A. $\left(\frac{dy}{dx} \text{ at } x = 0\right)$

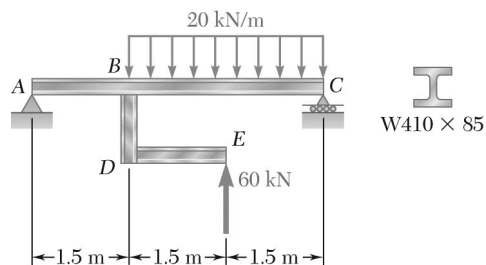
$$EI \left(\frac{dy}{dx}\right)_A = -0 + 0 + 0 + C_1$$

$$\left(\frac{dy}{dx}\right)_A = \frac{C_1}{EI} = \frac{-30.08 \times 10^3}{3.4754 \times 10^6} = -8.66 \times 10^{-3} \quad \theta_A = 8.66 \times 10^{-3} \text{ rad} \quad \nwarrow \blacktriangleleft$$

(b) Deflection at point B. $(y \text{ at } x = 24 \text{ in.})$

$$EI y_B = -0 + 0 + (14.4444)(24)^3 - 0 + (-30.08 \times 10^3)(24) \\ = -522.24 \times 10^3 \text{ lb} \cdot \text{in}^3$$

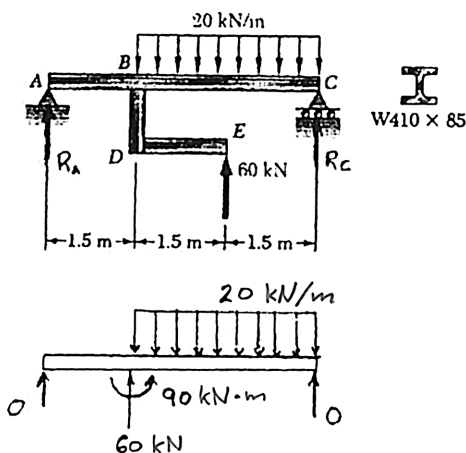
$$y_B = \frac{-522.24 \times 10^3}{3.4754 \times 10^6} = -0.1503 \text{ in.} \quad y_B = 0.1503 \text{ in.} \downarrow \blacktriangleleft$$



PROBLEM 9.162

The rigid bar BDE is welded at point B to the rolled-steel beam AC . For the loading shown, determine (a) the slope at point A , (b) the deflection at point B . Use $E = 200$ GPa.

SOLUTION



$$+\circlearrowleft M_C = 0:$$

$$-4.5R_A + (20)(3)(1.5) - (60)(1.5) = 0 \quad R_A = 0$$

Units: Forces in kN; lengths in m.

$$EI \frac{d^2 y}{dx^2} = M = 60\langle x - 1.5 \rangle^1 - 90\langle x - 1.5 \rangle^0 - \frac{1}{2}(20)\langle x - 1.5 \rangle^2$$

$$EI \frac{dy}{dx} = 30\langle x - 1.5 \rangle^2 - 90\langle x - 1.5 \rangle^1 - \left(\frac{1}{6}\right)(20)\langle x - 1.5 \rangle^3 + C_1$$

$$EI y = 10\langle x - 1.5 \rangle^3 - 45\langle x - 1.5 \rangle^2 - \frac{1}{24}(20)\langle x - 1.5 \rangle^4 + C_1 x + C_2$$

Boundary conditions:

$$[x = 0, y = 0] \quad 0 + 0 + 0 + 0 + C_2 = 0 \quad C_2 = 0$$

$$[x = 4.5, y = 0] \quad (10)(3)^3 - (45)(3)^2 - \frac{1}{24}(20)(3)^4 + 4.5C_1 + 0 = 0 \quad C_1 = 45 \text{ kN} \cdot \text{m}^2$$

Data: $E = 200 \times 10^9 \text{ Pa}$, $I = 316 \times 10^6 \text{ mm}^4 = 316 \times 10^{-6} \text{ m}^4$

$$EI = (200 \times 10^9)(316 \times 10^{-6}) = 63.2 \times 10^6 \text{ N} \cdot \text{m}^2 = 63,200 \text{ kN} \cdot \text{m}^2$$

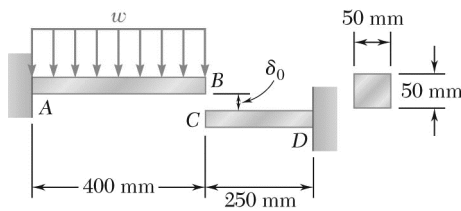
(a) Slope at A. $\left(\frac{dy}{dx} \text{ at } x = 0\right) \quad EI\theta_A = C_1 = 45 \text{ kN} \cdot \text{m}^2$

$$\theta_A = \frac{45}{63,200} = 0.712 \times 10^{-3} \text{ rad} \quad \theta_A = 0.712 \times 10^{-3} \text{ rad} \swarrow \blacktriangleleft$$

(b) Deflection at B. $(y \text{ at } x = 1.5)$

$$EI y_B = (C_1)(1.5) = (45)(1.5) = 67.5 \text{ kN} \cdot \text{m}^3$$

$$y_B = \frac{67.5}{63,200} = 1.068 \times 10^{-3} \text{ m} \quad y_B = 1.068 \text{ mm} \uparrow \blacktriangleleft$$



PROBLEM 9.163

Before the uniformly distributed load w is applied, a gap, $\delta_0 = 1.2$ mm, exists between the ends of the cantilever bars AB and CD . Knowing that $E = 105$ GPa and $w = 30$ kN/m, determine (a) the reaction at A , (b) the reaction at D .

SOLUTION

$$I = \frac{1}{12}(50)(50)^3 = 520.833 \times 10^3 \text{ mm}^3 = 520.833 \times 10^{-9} \text{ m}^4$$

$$EI = (105 \times 10^9)(520.833 \times 10^{-9}) = 54.6875 \times 10^3 \text{ N} \cdot \text{m}^2 \\ = 54.6875 \text{ kN} \cdot \text{m}^2$$

Units: Forces in kN; lengths in meters.

Compute deflection at B due to w . Case 8 of Appendix D.

$$(y_B)_1 = -\frac{wL^4}{8EI} = -\frac{(30)(0.400)^4}{(8)(54.6875)} \\ = -1.75543 \times 10^{-3} = -1.7553 \text{ mm}$$

The displacement is more than δ_0 , the gap closes.

Let P be the contact force between points B and C .

Compute deflection of B due to P . Use Case 1 of Appendix D.

$$(y_B)_2 = \frac{PL^3}{3EI} = \frac{P(0.4)^3}{(3)(54.6875)} \\ = 390.095 \times 10^{-6} P \quad \text{m}$$

Compute deflection of C due to P .

$$y_C = -\frac{PL^3}{3EI} = -\frac{P(0.25)^3}{(3)(54.6875)} = -95.238 \times 10^{-6} P \quad \text{m}$$

Displacement condition:

$$y_B + \delta_0 = y_C$$

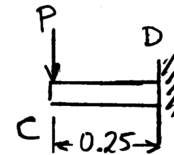
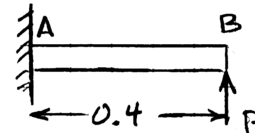
Using superposition,

$$(y_B)_1 + (y_B)_2 - \delta_0 = y_C$$

$$-1.75543 \times 10^{-3} + 390.095 \times 10^{-6} P + 1.2 \times 10^{-3} = -95.238 \times 10^{-6} P$$

$$485.333 \times 10^{-6} P = 0.55543 \times 10^{-3}$$

$$P = 1.14443 \text{ kN}$$



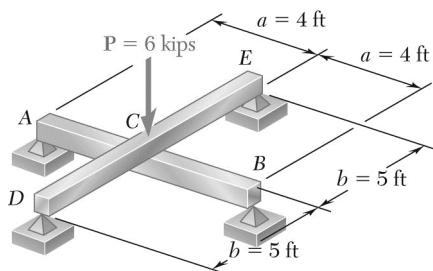
PROBLEM 9.163 (Continued)

(a) Reaction at A. $+ \uparrow \Sigma F_y = 0: R_A - 12 + 1.14443 = 0$
 $R_A = 10.86 \text{ kN } \uparrow \blacktriangleleft$

$+ \curvearrowright \Sigma M_A = 0: M_A - (0.2)(12) + (0.4)(1.14443) = 0$
 $M_A = 1.942 \text{ kN} \cdot \text{m } \curvearrowright \blacktriangleleft$

(b) Reaction at D. $+ \uparrow \Sigma F_y = 0: R_D - 1.14443 = 0$
 $R_D = 1.144 \text{ kN } \uparrow \blacktriangleleft$

$+ \curvearrowright \Sigma M_D = 0: -M_D + (0.25)(1.14443) = 0$
 $M_D = 0.286 \text{ kN} \cdot \text{m } \curvearrowright \blacktriangleleft$



PROBLEM 9.164

For the loading shown, and knowing that beams AB and DE have the same flexural rigidity, determine the reaction (a) at B , (b) at E .

SOLUTION

Units: Forces in kips; lengths in ft.

For beam ACB , using Case 4 of Appendix D:

$$(y_C)_1 = -\frac{R_C(2a)^3}{48EI}$$

For beam DCE , using Case 4 of Appendix D:

$$(y_C)_2 = \frac{(R_C - P)(2b)^3}{48EI}$$

Matching deflections at C ,

$$-\frac{R_C(2a)^3}{48EI} = \frac{(R_C - P)(2b)^3}{48EI}$$

$$R_C = \frac{Pb^3}{a^3 + b^3} = \frac{(6)(5)^3}{4^3 + 5^3} = 3.968 \text{ kips}$$

$$P - R_C = 6 - 3.968 = 2.032 \text{ kips}$$

Using free body ACB ,

$$+\circlearrowleft M_A = 0: 2aR_B - aR_C = 0$$

(a) Reaction at B . $R_B = \frac{1}{2} R_C$

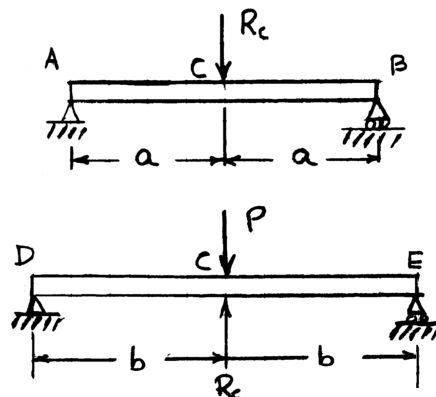
$$R_B = 1.984 \text{ kips } \uparrow \blacktriangleleft$$

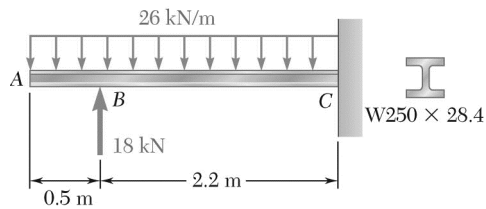
Using free body DCE ,

$$+\circlearrowleft M_D = 0: 2bR_E - b(P - R_C) = 0$$

(b) Reaction at E . $R_E = \frac{1}{2} (P - R_C)$

$$R_E = 1.016 \text{ kips } \uparrow \blacktriangleleft$$

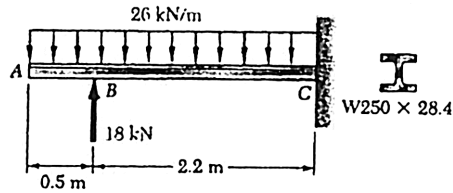




PROBLEM 9.165

For the cantilever beam and loading shown, determine (a) the slope at point A, (b) the deflection at point A. Use $E = 200 \text{ GPa}$.

SOLUTION



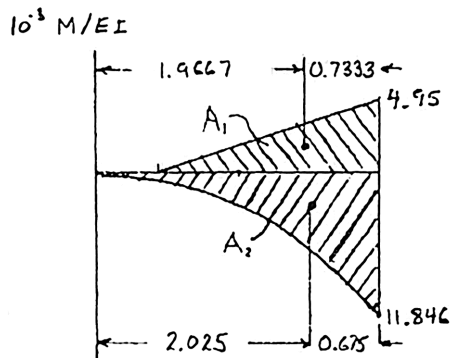
Units: Forces in kN; lengths in m.

$$E = 200 \times 10^9 \text{ Pa}$$

$$I = 40.1 \times 10^6 \text{ mm}^4 = 40.1 \times 10^{-6} \text{ m}^4$$

$$EI = (200 \times 10^9)(40.1 \times 10^{-6}) = 8.02 \times 10^6 \text{ N} \cdot \text{m}^2 \\ = 8020 \text{ kN} \cdot \text{m}^2$$

Draw M/EI diagram by parts.



$$\frac{M_1}{EI} = \frac{(18)(2.2)}{8020} = 4.9377 \times 10^{-3} \text{ m}^{-1}$$

$$A_1 = \frac{1}{2}(4.9377 \times 10^{-3})(2.2) = 5.4315 \times 10^{-3}$$

$$\bar{x}_1 = \frac{1}{3}(2.2) = 0.7333 \text{ m}$$

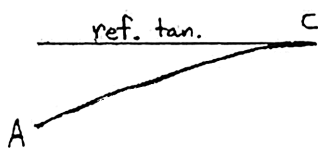
$$\frac{M_2}{EI} = -\frac{(26)(2.7)^2}{(2)(8020)} = -11.8167 \times 10^{-3} \text{ m}^{-1}$$

$$A_2 = \frac{1}{3}(-11.8167 \times 10^{-3})(2.7) = -10.6350 \times 10^{-3}$$

$$\bar{x}_2 = \frac{1}{4}(2.7) = 0.675 \text{ m}$$

Draw reference tangent at C.

$$\theta_C = \theta_A + \theta_{C/A} = \theta_A + A_1 + A_2 = 0$$



(a) Slope at A.

$$\theta_A = -A_1 - A_2 = -5.4315 \times 10^{-3} + 10.6350 \times 10^{-3} \\ = 5.20 \times 10^{-3} \text{ rad}$$

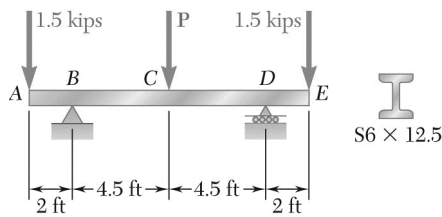
$$\theta_A = 5.20 \times 10^{-3} \text{ rad} \quad \swarrow \blacktriangleleft$$

PROBLEM 9.165 (Continued)

(b) Deflection at A.

$$\begin{aligned}y_A &= y_C - \theta_C L + t_{A/C} \\&= 0 - 0 + A_1 \bar{x}_1 + A_2 \bar{x}_2 \\&= 0 - 0 + (5.4315 \times 10^{-3})(1.9667) - (10.6350 \times 10^{-3})(2.025) \\&= -10.85 \times 10^{-3} \text{ m}\end{aligned}$$

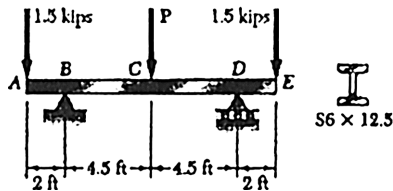
$$y_A = 10.85 \text{ mm} \downarrow \blacktriangleleft$$



PROBLEM 9.166

Knowing that the magnitude of the load **P** is 7 kips, determine (a) the slope at end *A*, (b) the deflection at end *A*, (c) the deflection at midpoint *C* of the beam. Use $E = 29 \times 10^6$ psi.

SOLUTION



Use units of kips and ft. $P = 7$ kips

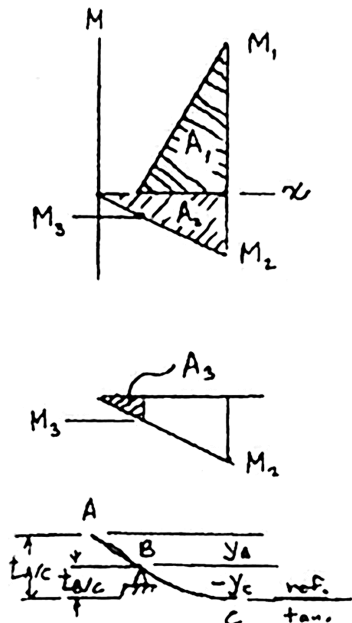
For $S6 \times 12.5$, $I = 22.0 \text{ in}^4$

$$EI = (29 \times 10^6)(22.0) = 638 \times 10^6 \text{ lb} \cdot \text{in}^2 \\ = 4430.6 \text{ kip} \cdot \text{ft}^2$$

Symmetric beam with symmetric loading. Place reference tangent at midpoint *C* where $\theta_C = 0$.

$$R_B = R_D = \frac{1}{2}(1.5 + 7 + 1.5) = 5 \text{ kips } \uparrow$$

Draw the bending moment diagram by parts for the left half of the beam.



$$M_1 = (4.5)(5) = 22.5 \text{ kip} \cdot \text{ft}$$

$$A_1 = \frac{1}{2}(4.5)(22.5) = 50.625 \text{ kip} \cdot \text{ft}^2$$

$$M_2 = -(2 + 4.5)(1.5) = -9.75 \text{ kip} \cdot \text{ft}$$

$$A_2 = \frac{1}{2}(6.5)(-9.75) = -31.6875 \text{ kip} \cdot \text{ft}^2$$

$$M_3 = -(2)(1.5) = -3 \text{ kip} \cdot \text{ft}$$

$$A_3 = \frac{1}{2}(2)(-3) = -3 \text{ kip} \cdot \text{ft}^2$$

Formulas: $\theta_A = -\theta_{C/A}$, $y_A - y_C = t_{A/C}$, $y_C = y_A - t_{A/C}$

$$y_B = y_A - 2\theta_A + t_{B/A} = 0, \quad y_A = -2\theta_A - t_{B/A}$$

$$\theta_{C/A} = \frac{1}{EI}(A_1 + A_2) = \frac{50.625 - 31.6875}{4430.6} = 4.27425 \times 10^{-3}$$

$$t_{A/C} = \frac{1}{EI} \left\{ (2 + 3)A_1 + \frac{2}{3}(6.5)A_2 \right\} = \frac{115.8125}{4430.6} = 26.1392 \times 10^{-3} \text{ ft}$$

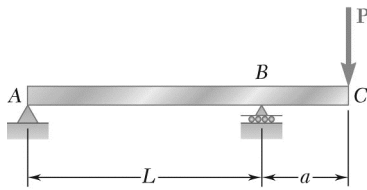
$$t_{B/A} = \frac{1}{EI} \left\{ \frac{1}{3}(2)A_3 \right\} = \frac{-2}{4430.6} = -0.45141 \times 10^{-3} \text{ ft}$$

PROBLEM 9.166 (Continued)

(a) Slope at end A. $\theta_A = -4.27 \times 10^{-3} \text{ rad}$ $\theta_A = 4.27 \times 10^{-3} \text{ rad}$ 

(b) Deflection at A. $y_A = -(2)(-4.27425 \times 10^{-3}) - (-0.45141 \times 10^{-3})$
 $= 8.9999 \times 10^{-3} \text{ ft}$ $y_A = 0.1080 \text{ in.}$ 

(c) Deflection at C. $y_C = 8.9999 \times 10^{-3} - 26.1392 \times 10^{-3} = -17.1393 \times 10^{-3} \text{ ft}$
 $y_C = -0.206 \text{ in.}$ $y_C = 0.206 \text{ in.}$ 



PROBLEM 9.167

For the beam and loading shown, determine (a) the slope at point C, (b) the deflection at point C.

SOLUTION

$$A_1 = -\frac{PaL}{2EI},$$

$$A_2 = -\frac{Pa^2}{2EI}$$

$$t_{A/B} = A_1 \left(\frac{2}{3}L \right) = -\frac{PaL^2}{3EI}$$

$$\theta_B = \frac{t_{A/B}}{L} = -\frac{PaL}{3EI}$$

(a) Slope at C.

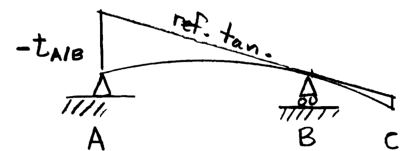
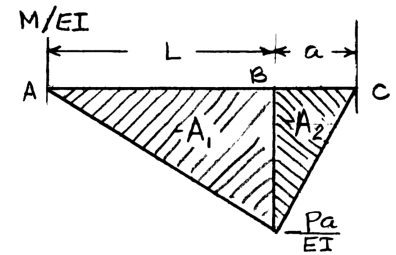
$$\theta_C = \theta_B + \theta_{C/B}$$

$$\begin{aligned} \theta_C &= -\frac{PaL}{3EI} - \frac{Pa^2}{2EI} \\ &= -\frac{Pa(2L+3a)}{6EI} \end{aligned}$$

(b) Deflection at point C.

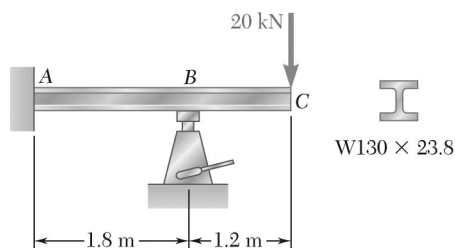
$$y_C = a\theta_B + t_{C/B}$$

$$\begin{aligned} y_C &= -\frac{Pa^2L}{3EI} + \left(-\frac{Pa^2}{2EI} \right) \left(\frac{2}{3}a \right) \\ &= -\frac{Pa^2(L+a)}{3EI} \end{aligned}$$



$$\theta_C = \frac{Pa(2L+3a)}{6EI} \quad \swarrow \blacktriangleleft$$

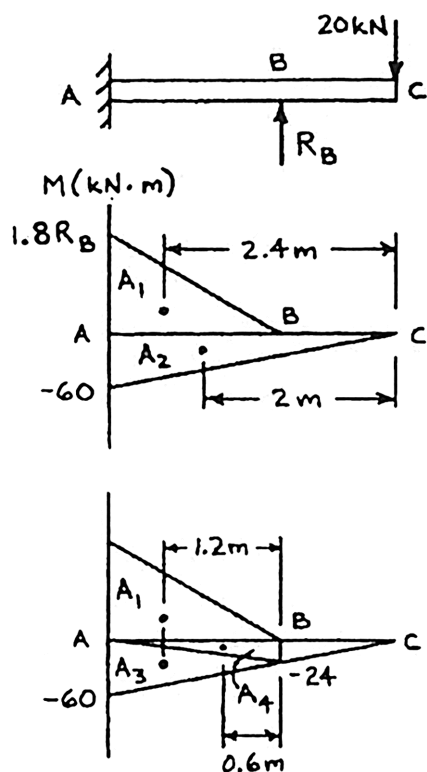
$$y_C = \frac{Pa^2(L+a)}{3EI} \quad \downarrow \blacktriangleleft$$



PROBLEM 9.168

A hydraulic jack can be used to raise point B of the cantilever beam ABC . The beam was originally straight, horizontal, and unloaded. A 20-kN load was then applied at point C , causing this point to move down. Determine (a) how much point B should be raised to return point C to its original position, (b) the final value of the reaction at B . Use $E = 200$ GPa.

SOLUTION



For $W130 \times 23.8$, $I_x = 8.91 \times 10^6 \text{ mm}^4$

$$EI = (200 \times 10^6 \text{ kPa})(8.91 \times 10^{-6} \text{ m}^4) = 1782 \text{ kN} \cdot \text{m}^2$$

Let R_B be the jack force in kN.

$$A_1 = \frac{1}{2}(1.8 R_B)(1.8) = 1.62 R_B$$

$$A_2 = \frac{1}{2}(-60)(3) = -90 \text{ kN} \cdot \text{m}^2$$

$$EI t_{C/A} = (2.4)A_1 + (2)A_2$$

$$0 = (2.4)(1.62 R_B) + (2)(-90)$$

$$R_B = 46.296 \text{ kN}$$

$$A_1 = 75 \text{ kN} \cdot \text{m}^2$$

$$A_3 = \frac{1}{2}(-60)(1.8) = -54 \text{ kN} \cdot \text{m}^2$$

$$A_4 = \frac{1}{2}(-24)(1.8) = -21.6 \text{ kN} \cdot \text{m}^2$$

$$EI t_{B/A} = (1.2)A_1 + (1.2)A_3 + (0.6)A_4$$

$$= (1.2)(75) + (1.2)(-54) + (0.6)(-21.6)$$

$$= 12.24 \text{ kN} \cdot \text{m}^2$$

(a) Deflection at B . $y_B = t_{B/A} = \frac{EI t_{B/A}}{EI} = \frac{12.24}{1782} = 6.8687 \times 10^{-3} \text{ m}$ $y_B = 6.87 \text{ mm} \uparrow \blacktriangleleft$

(b) Reaction at B . $R_B = 46.3 \text{ kN} \uparrow \blacktriangleleft$